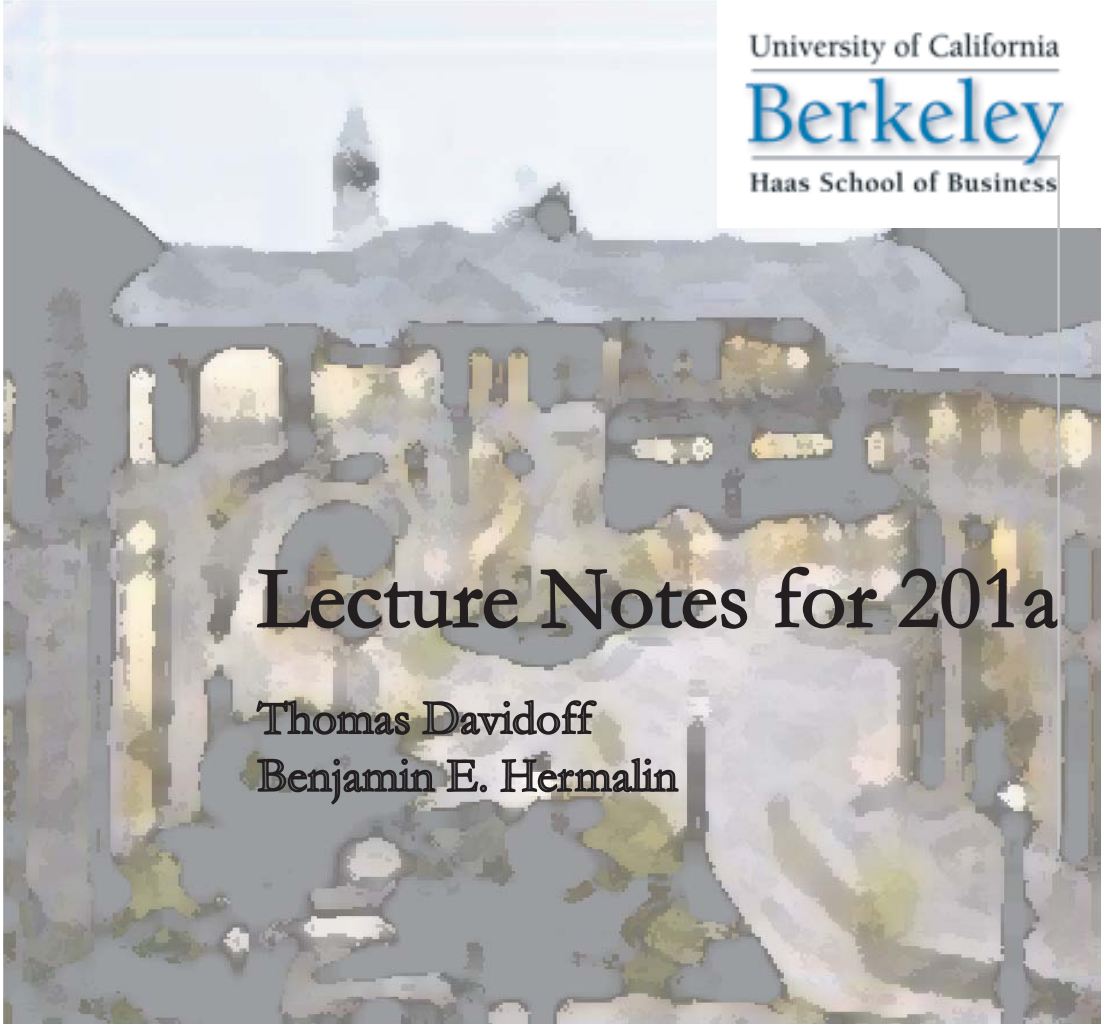




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# Lecture Notes for 201a

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# Preface

These lecture notes are intended to supplement the lectures and other materials in the Microeconomics for Business Decision Making course at the Haas School of Business.

## A Word on Notation

Various typographic conventions are used to help guide you through these notes.

Text that *looks like this* is an important definition. On the screen or printed using a color printer, such definitions should appear blue.

As one of the authors is an old Fortran programmer, variables  $i$  through  $n$  inclusive will generally stand for integer quantities, while the rest of the alphabet will be continuous variables (*i.e.*, real numbers). The variable  $t$  will typically be used for time, which can be either discrete or continuous.

The  $\textcircled{X}$  symbol in the margin denotes a paragraph that may be hard to follow and, thus, requires particularly close attention (not that you should read any of these notes without paying close attention).

The symbol  $\int dx$  in the margin denotes a section that uses calculus. Observe that such sections are also indented relative to the rest of the text. Some technical footnotes are also prefaced with  $\int dx$ . We recognize that not everyone is comfortable with calculus and for those who aren't we simply ask that you skim those sections and make note of the main conclusions. You may also wish to read through Appendix A3. There is no expectation that every student will fully grasp the details of the analysis in those sections that employ calculus and certainly we won't expect you to repeat the analysis on an exam.

When using calculus, we will use primes to indicate the derivatives of functions. Hence, for instance, the derivative of  $f(\cdot)$  would be represented at  $f'(\cdot)$ . Occasionally we will make use of the  $dy/dx$  style notation for derivatives.

**Notes in margins:**  
*These denote important "takeaways."*

## A Word on Currency

As a rule we will refer to monetary amounts as dollars. Unless we're citing actual data, there is nothing in these notes specific to dollars. The analysis is the same whether the monetary amounts are dollars, euros, pesos, pounds, yen, or what have you.

For those who feel we're being too American-centric in using dollars, we note that we don't specify which dollars. For all you know, we have Australian, Canadian, or Hong Kong dollars in mind.

## A Word on the Appendices

There are two sets of appendices. The "A" appendices review basic algebra, solving equations, and calculus. The "B" appendices consider basic probability. Reading them is optional, although we doubt anyone will be harmed by reading them. Even though reading them is optional, this does not mean that using the mathematics they contain is necessarily optional. Therefore, if you feel rusty with respect to your math or probability skills and knowledge, you would be well advised to read through them.

# Fundamentals of Decision Theory

# 1

A few years back, Greg Wolfson, one of our former students, and his wife were in the Caribbean as Hurricane Andrew approached. Understandably, they were nervous about the possibility of being stuck on the Turks and Caicos Islands during one of the worst storms of the century. Should they stay on the Islands or should they try to make it to Miami on route back home? If they stayed and the hurricane hit the Islands, then they faced having their vacation ruined or worse. If they left, then they gave up the rest of their vacation, incurred additional costs of getting last-minute plane tickets, and ran the risk of being caught in the hurricane while in Miami. Fortunately, Greg had studied *decision theory*. Decision theory helped Greg and his wife to think *systematically* through their decision problem—stay or flee—and reach their best decision. This turned out to be “stay,” and a good thing too: While Miami was being battered by Hurricane Andrew, Greg and his wife were on a Hobie Cat, sailing the lovely turquoise waters off the Turks and Caicos Islands—another MBA 201A success story!

Decision theory is a set of tools for deciding “which.” For example, Greg and his wife were deciding *which* would be the better course of action for them, stay or flee. These tools help by *formalizing* your decision making. They help you recognize the alternatives available to you; they help you see what additional information would be useful in reaching a decision; and they make you aware of the assumptions or conditions that are critical to the decision you make.

*A word of caution:* As powerful as these tools are, they are *not* a substitute for your own thinking. Rather, they are aids to your thinking. Put another way, they are not magic formulæ that can make your decisions for you (which is just as well, since otherwise someone would program a computer with them, which would likely do you out of a job).

## Some Basic Issues

## 1.1

When you make a decision, you are choosing among *alternatives*. Your objective is to choose the alternative that is *best*, where “best” depends on what your goals are. Indeed, the first rule of decision making is to know what your goals are. For example, if your decision problem is which movie to see at the multiplex, then “best” means “most entertaining” (assuming being entertained is your goal).

Although the first rule of decision making may strike you as obvious, you would be surprised how often people start making decisions without thinking through what their goals are. For instance, obeying the first rule can often be a

**The first rule of  
decision making:**  
*Know your goals  
(objectives).*

problem when a committee makes a decision, because committee members can have different goals. Sometimes the committee members recognize their differences in advance, but sometimes they are unspoken. Occasionally committee members believe they are in agreement with respect to their goals when, in fact, they are not (you've likely had conversations that began "it just didn't occur to me that you wanted ..."). Psychology also plays a role here. You may not, for example, want to admit to yourself what your true goals are—perhaps because they are socially unacceptable—so you convince yourself that your goals are something else. Unfortunately, it is beyond these notes to make sure that you obey the first rule. All they can do, as they have just done, is point out that obeying the first rule is not as easy as it may at first seem.

Having identified your goals, you next have to identify your alternatives. For some decision-making problems, your alternatives are obvious. For instance, if you are deciding which movie at the multiplex to see, then your alternatives are the movies playing plus, possibly, not seeing any movie at all. For other problems, however, identifying your alternatives is more difficult. For instance, if you are deciding which personal computer to buy, then it can be quite difficult to identify *all* your alternatives (*e.g.*, you may not know all the companies that make computers or all the optional configurations available). Fortunately, there are ways to overcome, at least partially, such difficulties, as we will see later. We will even study the *decision* of whether you should expend resources expanding your list of alternatives later in these notes.

Your choice of alternative will lead to some consequence. Depending on the decision-making problem you face, the consequence of choosing a given alternative will be either known or uncertain. If you are driving in your neighborhood, then you know where you will end up if you turn left at a given intersection. If you are investing in the stock market, then you are uncertain about what returns you will earn. Typically, we will suppose that even if you are uncertain about which particular consequence will occur, you know the set of possible consequences. For instance, although you don't know what your stock price will be a year from now, you do know that it will be some non-negative number. Moreover, you likely know something about which stock prices are more or less likely. For example, you may believe that it is more likely that your stock's price will change by 20% or less than it will change by 21% or more.

Sometimes, however, you may not know what all the possible consequences are. That is, some possible consequences could be *unforeseen*. To give an example, one of the authors once met a vineyard owner who was proud of his "green" farming techniques. Unlike many of his fellow vintners, he used pesticides that killed only the "bad" bugs, leaving the "good" bugs—those that ate the bad bugs—alive. A consequence of this, which was unforeseen by the vintner, was that if he successfully killed the bad bugs, then the good bugs would be left with nothing to eat and would starve.

By their very nature, unforeseen consequences are difficult to identify prior to making your decisions. And for the same reason, it is difficult to predict which consequences will be unforeseen by others. As a practical manner, one way to *help* identify unforeseen consequences in your own decision making is

to think about what your “un-goals” are; that is, the consequences you would like *not* to happen. For instance, an un-goal of the vintner was to kill the good bugs. Another way to identify unforeseen consequences is to reframe your way of thinking about your goals. For instance, instead of thinking about not killing the good bugs, think instead of helping the good bugs to survive. Reframed in this way, the adverse consequence of killing the good bugs’ food supply might be more apparent.

## Decision Making Under Certainty

# 1.2

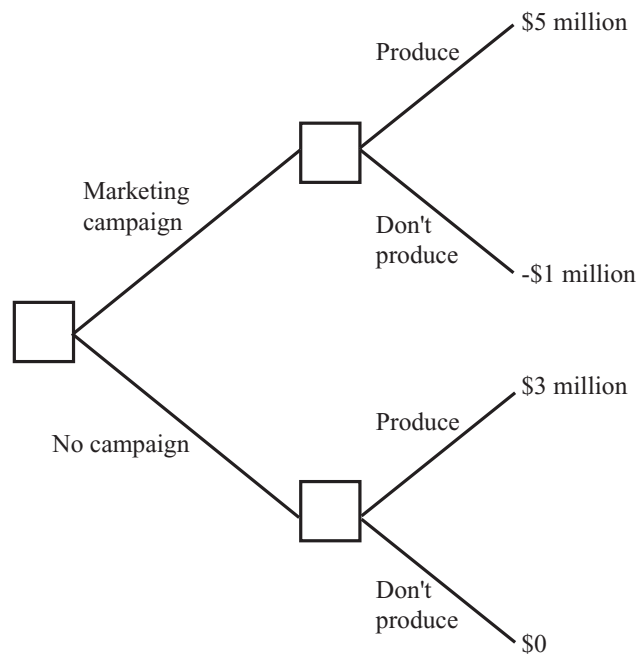
**Decision tree:** A graphical representation of a decision problem as a series of branching alternatives.

To represent, in a schematic way, a decision problem, we draw a *decision tree*. An example of such a tree is shown in Figure 1.1. It represents the following problem: A firm is considering producing a new product. Prior to producing, the firm can conduct a marketing campaign (*e.g.*, advertise heavily) or not. Suppose that a marketing campaign costs \$1 million. Suppose that if the product is marketed and produced, it will generate revenues of \$8 million while costing \$2 million to produce; so profit will be \$5 million ( $= \$ (8 - 2 - 1)$  million). If the product is marketed, but not produced, it will generate no revenue and no production cost; so profit will be  $-\$1$  million (*i.e.*, just the cost of the marketing campaign). If the product is not marketed, but produced, it will generate a revenue of \$4 million but cost \$1 million to produce; so profit will be \$3 million. Finally, if the product is neither marketed, nor produced, then profit will be \$0.

Each square in Figure 1.1 is a *decision node*. A decision node indicates that the decision maker (in this case the firm) has a decision to make. The alternatives available to him or her at that decision node are represented by the *branches* that stem from the right-side of the decision node. For instance, the alternatives available to the firm at the left-most—first—decision node are “marketing campaign” and “no campaign.” Note that the tree is read from left to right; moreover, going from left to right is meant to represent the sequence of decisions. For example, the firm must first decide whether to have a marketing campaign before it decides whether to produce the product. At the end of the tree are the consequences or *payoffs* from the sequence of decisions. For instance, if the firm chose “no campaign” and, then, “produce,” it would make a profit of \$3 million. Payoffs are expressed in terms relevant to the decision maker’s goals. Here, the goal is to make money, so they are represented in monetary terms.

Having represented a decision problem by a tree, the next step is to solve it. Solving a tree means determining which decisions will best accomplish the decision maker’s goals. If, as in Figure 1.1, the goal is to make money, then this means determining the decisions that will lead to the most money. Trees are solved by working *backwards*, a procedure known as *backwards induction*: Start at the *rightmost* decision nodes and select the branches that give the decision maker the largest payoffs. For the tree in Figure 1.1, this means choosing “produce” at the top rightmost decision node—since a \$5 million gain is bet-

**Backwards induction:** Trees are solved by working backwards



**Figure 1.1:** A decision tree for a firm that must first decide whether to have a marketing campaign for a new product or to have no campaign. The firm's subsequent decision is whether or not to produce the product. The squares are *decision nodes*. From them stem *branches*. At the end of the tree are the *payoffs*.

ter than a \$1 million loss—and choosing “produce” at the bottom rightmost decision node—since a \$3 million gain is better than \$0. Next move *left* to the preceding decision nodes. Again, select the branches that give the decision maker the largest payoffs *taking into account, if necessary, the future decisions that will be made*. In Figure 1.1, this means choosing “marketing campaign” at the first node because “market” ultimately leads to a payoff of \$5 million, while “no marketing campaign” ultimately leads to a payoff of \$3 million. Were there any decision nodes to the left of the “marketing campaign/no marketing campaign” node (*i.e.*, were there decisions to be taken prior to the decision of whether to market), then you would choose the alternatives at those nodes taking into account your decision to have a marketing campaign at the “marketing campaign/no marketing campaign” node.

The tree in Figure 1.1 is straightforward, so solving it is fairly straightforward as well. For more complicated trees, however, it is important to keep track of where you are as you work backwards. Two devices for keeping track are **arrowing** the correct decision and **valuing** the intermediate decision nodes (*i.e.*, the decision nodes other than the first). Arrowing means putting a little arrow (or other mark) on the correct decision. When you’re done arrowing, the arrows will “guide” you through the tree. For instance, in Figure 1.1, you would put an arrow on the “marketing campaign” branch, because that is the correct alternative to choose. Valuing a decision node means writing the payoff from making the correct decision at that decision node. Figure 1.2 shows Figure 1.1 with arrows and values.

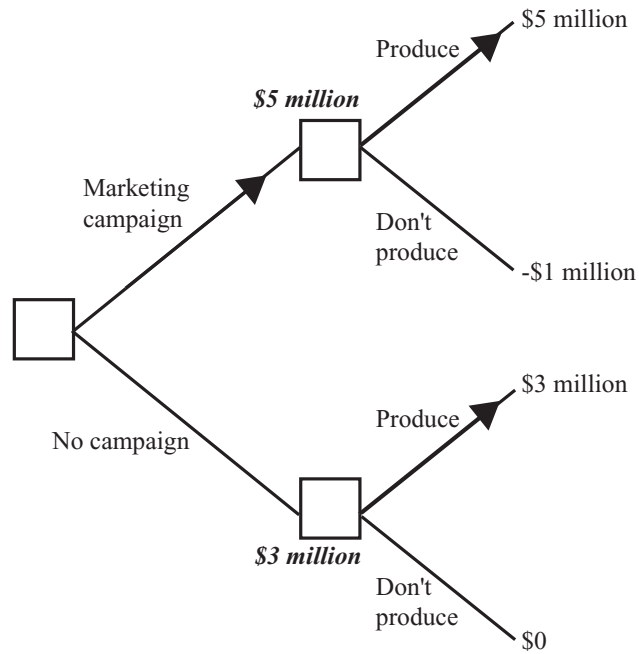
## Decision Making Under Uncertainty

# 1.3

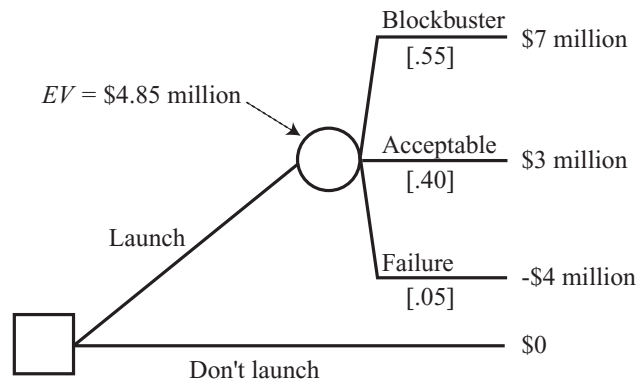
[Before reading this section you may wish to review the material in Appendices B1 and B3.]

Many decisions are made in situations of uncertainty. To represent such decision problems, we use a second kind of node: a **chance node**. A chance node, drawn as a circle, indicates that what follows is uncertain. Each branch stemming from the chance node shows a possible outcome of the random process that the chance node represents. For instance, Figure 1.3 represents the decision tree associated with launching a new product.

In Figure 1.3, the firm can either launch or not launch a new product. This is a decision of the firm, so represented by a decision node (square). If it doesn’t launch, then its payoff is \$0. If it does launch, its payoff is uncertain. Hence, the “launch” branch leads to a chance node (circle). There are three possible outcomes if the firm launches. The new product can be very successful—a *blockbuster*; or it do acceptably well; or it can fail. The profits associated with each outcome are indicated at the ends of the respective branches (*e.g.*, acceptable performance yields \$3 million in profit). The probabilities of each of the possible outcomes are shown in square brackets. Thus, for instance, the probability of failure is .05. Note the probabilities at any chance node must sum



**Figure 1.2:** The correct branches to follow are noted with arrows (▲). The values of the intermediate decision nodes are also noted.



**Figure 1.3:** A firm faces the decision of launching a new product or not launching it. There are three possible outcomes if it launches: blockbuster, acceptable, and failure. The payoff from each is shown at the end of the tree and the probability of each is indicated in brackets.

to 1.

Recall that the *expected value of a gamble* when there are  $N$  possible outcomes, denoted  $EV$ , is given by the formula

$$EV = p_1x_1 + \cdots + p_Nx_N,$$

where  $p_n$  is the probability of the  $n$ th outcome and  $x_n$  is the payoff should the  $n$ th outcome occur (see expression (B3.1) on page 211). For example, the expected value of the gamble shown in Figure 1.3 is, in millions,

$$\$4.85 = .55 \times \$7 + .4 \times \$3 + .05 \times (-\$4).$$

It might strike you as odd that \$4.85 million is called the “expected” value of the gamble in Figure 1.3: How could \$4.85 million be “expected” if the only possible values are \$7 million, \$3 million, and −\$4 million? There are two justifications for the term “expected.” First, suppose that we repeated the gamble many times, say 10,000 times. Your average winnings—that is, the total of your winnings for the 10,000 repetitions divided by 10,000—would very likely be very close to \$4.85 million.<sup>1</sup> In other words, we would *expect* your average winnings to be \$4.85 million.

As a second justification, suppose that you ran a life insurance company. Then, for each insured, you are essentially gambling on when he or she will die (*i.e.*,  $x$  would be age at death). The expected value would, thus, refer to the expected age of death of an insured. If you insured a large enough population, then the average age at death among your insureds would be close to the expected age of death of a single insured.

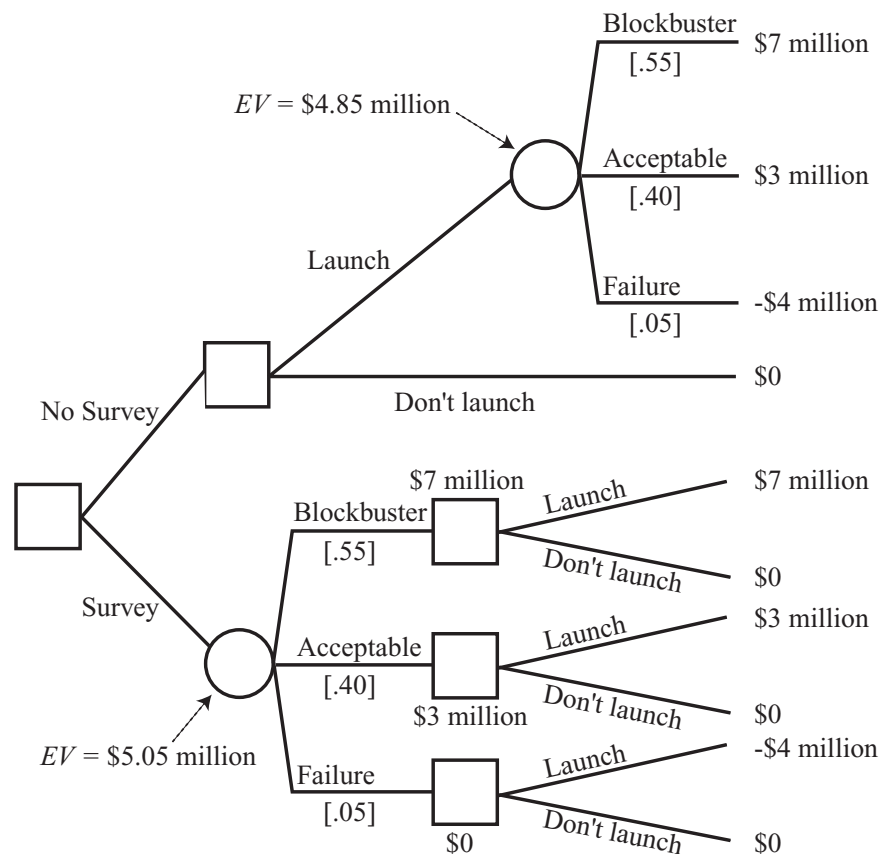
Expected value is useful for decision theory because many decision makers are *expected-value maximizers*:

**Definition 1** *A decision maker is an expected-value maximizer if he chooses among his alternatives the one that yields the greatest expected value.*

To better understand an expected-value maximizer’s behavior, consider the decision problem shown in Figure 1.3. As noted above, the expected value of launching is \$4.85 million. Because \$4.85 million is greater than \$0 (the firm’s payoff if it doesn’t launch), the firm would choose to launch if it is an expected-value maximizer.

Note that we solve the tree in Figure 1.3 in the same way we solved the others we encountered—in fact, the way we solve all trees—by working backwards: We started at the rightmost node, in this case a chance node, calculated the value of that node (*i.e.*, its expected value), and then moved left to the preceding node (*i.e.*, the launch/no launch decision node).

<sup>1</sup>For instance, there is approximately an 80% probability that your average winnings would fall between \$4.75 and \$4.95 million and a 95% probability that your average winnings would fall between \$4.7 and \$5 million. If you have had an advanced course in probability, you may recognize that this is nothing more than the law of large numbers.



**Figure 1.4:** Now, the firm can conduct a survey, if it wishes, prior to making its launch decision.

To extend this example, suppose that prior to making the launch/no launch decision, the firm could commission a survey that would tell it how successful the product will be. Of course, whether to conduct the survey is itself a decision, so it must be added to the tree. Figure 1.4 revises the tree in Figure 1.3 to reflect these changes.

As always, we solve the tree by working backwards. The top of tree is the same as Figure 1.3, so we know from our previous analysis that the firm would choose to launch. Its expected payoff is \$4.85 million. At the bottom of the tree, the rightmost nodes are all decision nodes. Note that they *follow* the chance node because, if the firm does a survey, it will know how successful its new product will be *at the time* it decides whether to launch. At the top two of these decision nodes, the firm would launch—positive amounts of money beat nothing. At bottommost node it would not launch—\$0 beats suffering a

loss. Note that the values of these three decision nodes have been appropriately labeled. When the firm decides to take a survey, it doesn't know what it will learn, so what it will learn is uncertain. This is reflected by the chance node that precedes the decision nodes. The expected value at that node is

$$\$5.05 \text{ million} = .55 \times \$7 \text{ million} + .4 \times \$3 \text{ million} + .05 \times \$0.$$

Comparing \$5.05 million to \$4.85 million, it follows that the firm would prefer to undertake the survey than make a decision without it.

This last example also illustrates how we can calculate the value of this survey: The value of the survey is the difference in the expected payoff with the survey, \$5.05 million, and the expected payoff without the survey, \$4.85 million; that is, the survey is worth \$200,000. The firm would pay up to \$200,000 to have this survey conducted.

To summarize:

#### Solving trees for expected-value maximizers

1. For each of the rightmost nodes proceed as follows:
  - (a) If the node is a decision node, determine the best alternative to take. The payoff from this alternative is the *value of this decision node*. Arrow the best alternative.
  - (b) If the node is a chance node, calculate the expected value. This expected value is the *value of this chance node*.
2. For the nodes one to the left proceed as follows:
  - (a) If the node is a decision node, determine the best alternative to take, using, as needed, the values of future nodes (nodes to the right) as payoffs. The payoff from this alternative is the value of this decision node. Arrow the best alternative.
  - (b) If the node is a chance node, calculate the expected value, using, as needed, the values of future nodes (nodes to the right) as payoffs. The expected value is the value of this chance node.
3. Repeat Step 2 as needed, until the leftmost node is reached. Following the arrows from *left to right* gives the sequence of appropriate decisions to take.

Often when we make a decision under uncertainty we would like more information about the uncertainty we face. You, for example, might read the prospectus for a security that you are considering purchasing, or you may ask a friend or

**Perfect information:**  
*Information that completely reveals what will happen; that is, information that once learned means there is no longer uncertainty about a given event.*

**Imperfect information:**  
*Information that improves a decision maker's ability to predict the outcome of a future event, but which does not completely reveal what will happen.*

a stock broker for advice about that security. We have, in fact, already seen a situation of information gathering. Recall Figure 1.4. In that tree, a firm was deciding whether to launch a new product or not. Prior to making this decision, the firm could, if it wished conduct a survey that would *perfectly* reveal how successful the product would be. Without conducting a survey, the firm would have to make its launch decision “in the dark.” This is an example of a firm deciding whether to acquire additional information before making a decision. Note that whether to acquire additional information is, itself, a decision; one that we will study in this section.

Information is divided into two classes: perfect and imperfect. *Perfect information*, like that in Figure 1.4, is information that completely reveals in advance the outcome of some future uncertain event. In Figure 1.4, for instance, the survey completely revealed how successful the launch would be.

In other circumstances, we might expect to receive *imperfect information*. Imperfect information helps us to have a better idea of what will happen in the future, but it does not completely reveal what will happen.

As an example, consider Figure 1.5, which illustrates the following situation: A firm uses sheet metal in making its product (*e.g.*, car parts). It is concerned with whether a recent shipment of sheet metal is up to its standards. It can use the sheet metal and do an entire production run, return the sheet metal to the supplier, or do a test run and then decide whether to do a production run or return the sheet metal. Unfortunately, a test run is not definitive as to whether the metal is up to the firm's standards, although it gives indications. Let those indications be summarized as “likely okay” and “likely not okay.” If the test run comes back “likely okay,” then the probability that the metal is high quality is  $p$  and, hence the probability that the metal is low quality is  $1 - p$ . If the test run comes back “likely not okay,” then the probability that the metal is high quality is  $1 - p$  and the probability that it is low quality is  $p$ . Absent a test run, the probability that the metal is high quality is  $1/2$  and, hence, the probability that it is low quality is  $1/2$ .

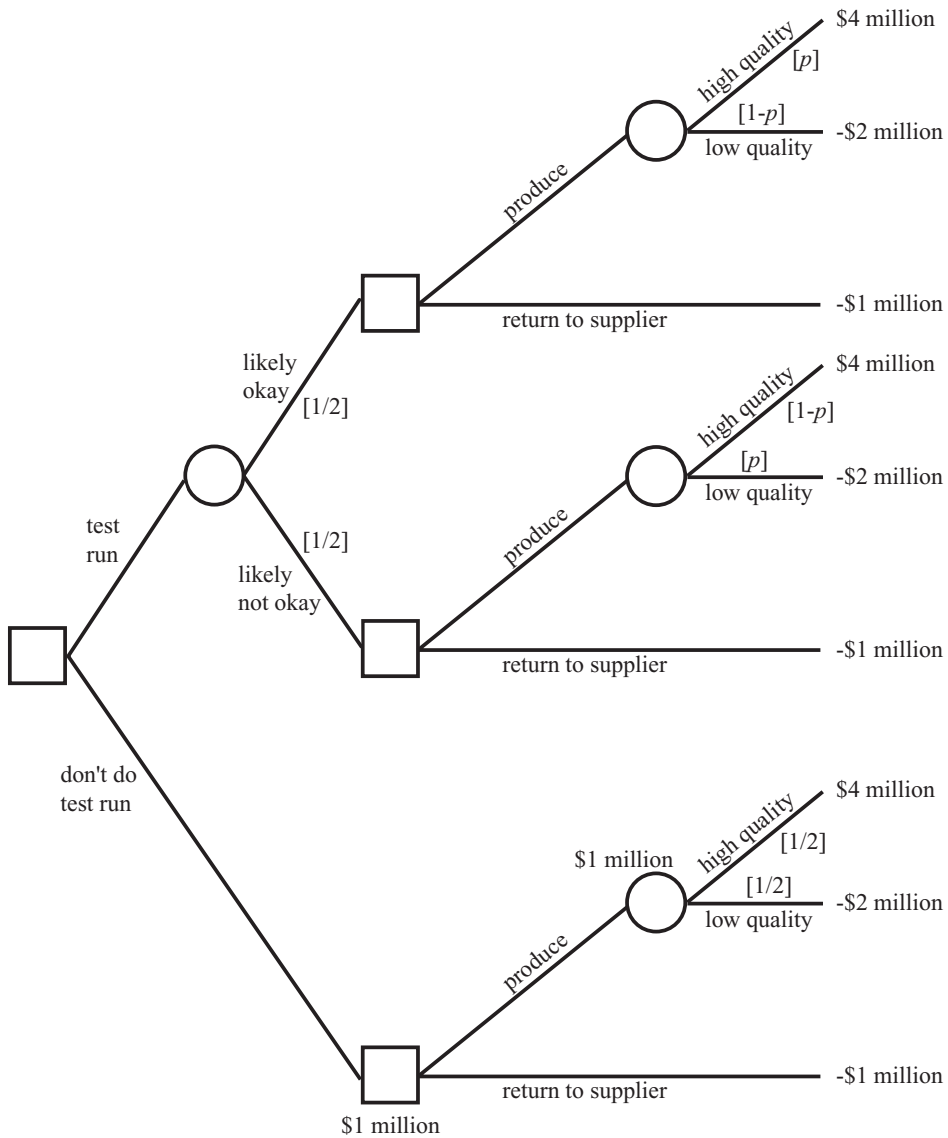
For our model of information to be consistent, it must be that that the expected probability of high quality *prior* to doing a test run is  $1/2$  (taking an action can't change the underlying probabilities). Given our assumption about what the indicators mean (*i.e.*, our assumptions on the  $ps$ ), this consistency can be achieved only if the probability of getting the “likely okay” indicator is  $1/2$  and, thus, the probability of the “likely not okay” indicator is  $1/2$  (remember the probabilities at any chance node must sum to one).<sup>2</sup>

<sup>2</sup>If you want to see this demonstrated formally, let  $\alpha$  be the probability of the “likely okay” indicator and, thus,  $1 - \alpha$  the probability of “likely not okay.” The ultimate probability that the metal is high quality *prior* to any testing is  $\alpha p + (1 - \alpha)(1 - p)$  (recall the rule of total probability, Proposition 12 on page 206). So

$$\alpha p + (1 - \alpha)(1 - p) = \frac{1}{2}$$

or, rewriting,

$$\frac{1}{2} = 2p\alpha - \alpha - p + 1 = 2\left(p - \frac{1}{2}\right)\alpha - p + 1.$$



**Figure 1.5:** An example of *imperfect* information. By doing a test run, the firm gets information about how likely it is that the sheet metal is high quality versus low quality.

If the test run is informative, which we assume it is, then  $p > 1/2$ ; that is, if the test run comes back “likely okay,” then the firm’s updated probability that the metal is high quality is greater than if it had received no information. Similarly, if the test run comes back “likely not okay,” then the firm’s updated probability that the metal is high quality is lower than if it had received no information.

The variable  $p$  is, therefore, a measure of how informative the test run is. The closer  $p$  is to  $1/2$ , the less informative the test run is. Indeed, were  $p = 1/2$ , then there would be no information in a test run, because the outcome of a test run would not change the probability that metal is high quality. Conversely, the farther  $p$  is from  $1/2$  (equivalently, the closer it is to 1), the more informative the test run is. Indeed, were  $p = 1$ , then we would have perfect information like we had in Figure 1.4.

We solve the tree in Figure 1.5 like any other tree—starting at the right and moving left. The top rightmost chance node yields an expected value of

$$EV_{\text{top}} = p \times \$4 \text{ million} + (1 - p) \times (-\$2) \text{ million}.$$

Because  $p > \frac{1}{2}$ ,  $EV_{\text{top}} > \$1$  million. Consequently, at the top right decision node the decision would be to produce, and the value of this decision node would be  $EV_{\text{top}}$ .

Going down to the bottom rightmost chance node, we see it has an expected value of

$$\frac{1}{2} \times \$4 \text{ million} + \frac{1}{2} \times (-\$2) \text{ million} = \$1 \text{ million}$$

(note this is indicated on the tree). Given that a gain of \$1 million beats a loss of \$1 million, the correct decision at the preceding decision node is to produce; hence, the value of this decision node is \$1 million.

Now consider the middle rightmost chance node. It has an expected value of

$$\begin{aligned} EV_{\text{middle}} &= (1 - p) \times \$4 \text{ million} + p \times (-\$2) \text{ million} \\ &= \$4 \text{ million} - p \times \$6 \text{ million}. \end{aligned}$$

The decision to take at the preceding decision node depends on the value of  $p$ : If  $EV_{\text{middle}} \geq -\$1$  million, then the firm should produce. If  $EV_{\text{middle}} < -\$1$  million, then the firm should not produce. When is  $EV_{\text{middle}} \geq -\$1$  million? Answer: when

$$\$4 \text{ million} - p \times \$6 \text{ million} \geq -\$1 \text{ million};$$

---

Adding  $p - 1$  to both sides yields

$$p - \frac{1}{2} = 2\alpha \left( p - \frac{1}{2} \right),$$

which entails that  $\alpha = 1/2$  as required.

or, dividing through by  $-\$1$  million (remember this reverses the inequality sign), when

$$6 \times p - 4 \leq 1;$$

or, adding 4 to both sides and, then, dividing both sides by 6, when

$$p \leq \frac{5}{6}.$$

So when  $p \leq 5/6$ —it is not super likely the metal is low quality even conditional on an indication of “likely not okay”—then the firm should produce even though the result of the test run is not promising. On the other hand, if  $p > 5/6$ —it is very likely that the metal is low quality conditional on the test run returning “likely not okay”—then the firm should not produce if the result of the test run is not promising.

Working back to the first chance node, we find that the expected value of that node is

$$\frac{1}{2} \times EV_{\text{top}} + \frac{1}{2} \times \begin{cases} EV_{\text{middle}} & \text{if } p \leq \frac{5}{6} \\ -\$1 \text{ million} & \text{if } p > \frac{5}{6} \end{cases}.$$

Finally, we can decide what to do at the first decision node; that is, we can determine whether it is worthwhile to do a test run or not. Suppose first, that a test run is not especially informative, which, here, means  $p \leq 5/6$ . The value of doing the test run is, then,

$$\begin{aligned} \frac{1}{2} \times EV_{\text{top}} + \frac{1}{2} \times EV_{\text{middle}} &= \frac{1}{2} \times (p \times \$4 \text{ million} + (1 - p) \times (-\$2) \text{ million}) \\ &\quad + \frac{1}{2} \times ((1 - p) \times \$4 \text{ million} + p \times (-\$2) \text{ million}) \\ &= \frac{1}{2} \times \$4 \text{ million} + \frac{1}{2} \times (-\$2) \text{ million} \\ &= \$1 \text{ million.} \end{aligned}$$

In which case, there is *no* additional value to doing a test run.

Suppose, in contrast, that a test run is very informative, which, here, means  $p > 5/6$ . The value of doing the test run is, then,

$$\begin{aligned} \frac{1}{2} \times EV_{\text{top}} + \frac{1}{2} \times (-\$1) \text{ million} &= \frac{1}{2} \times (p \times \$4 \text{ million} + (1 - p) \times (-\$2) \text{ million}) \\ &\quad + \frac{1}{2} \times (-\$1) \text{ million} \\ &= p \times \$3 \text{ million} - \$1.5 \text{ million} \\ &> \$1 \text{ million (because } p > \frac{5}{6} \text{).} \end{aligned}$$

In this case, there is value to doing a test run.

Why were the two cases so different? That is, why was there value to doing a test run when  $p > 5/6$  but not otherwise? The answer has to do with the difference in how the firm responds to “likely not okay” in the two cases. When

$p \leq 5/6$ , the firm produces even if the result from the test run is “likely not okay.” This is also the action it takes absent any information (*i.e.*, at the bottom decision node) and if the result is “likely okay.” That is, when  $p \leq 5/6$ , the information learned from the test run has *no* potential to affect the firm’s action—when  $p \leq 5/6$ , the firm always produces. In contrast, when  $p > 5/6$ , then the firm does not produce if the result from the test run is “likely not okay.” That is, when  $p > 5/6$ , the information learned from the test run *has* the potential to affect the firm’s action—the firm takes *different* actions depending on the result of the test run. This reflects a general result about the value of information:

**Result 1 (The fundamental rule of information)** *Information has value only if it has the potential to affect a decision maker’s choice of action.*

When  $p \leq 5/6$ , the information has *no* potential to affect the firm’s decision, but when  $p > 5/6$ , the information *has* the potential to affect the firm’s decision. This is why the information is valueless when  $p \leq 5/6$ , but valuable when  $p > 5/6$ .

How valuable is the information when it is valuable? To find out, we subtract the expected value of not doing the test run from the expected value of doing the test run:

$$\underbrace{p \times \$3 \text{ million} - \$1.5 \text{ million}}_{EV \text{ if do test run}} - \underbrace{\$1 \text{ million}}_{EV \text{ if don't do test run}} \\ = p \times \$3 \text{ million} - \$2.5 \text{ million}$$

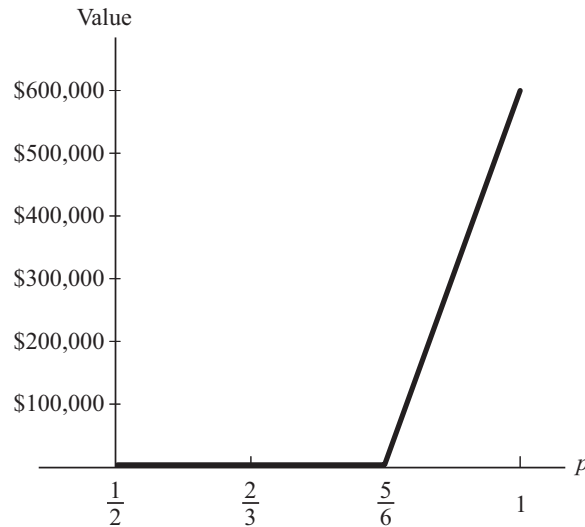
(for  $p > \frac{5}{6}$ ). Figure 1.6 plots the value of information for  $p$  between  $1/2$  and  $1$ . Note that the value is zero for  $p$  between  $1/2$  and  $5/6$  and is increasing (upward sloping) for  $p$  between  $5/6$  and  $1$ . The information is most valuable when  $p = 1$ , which makes sense: We would expect *perfect* information (which is what  $p = 1$  represents) to be more valuable than *imperfect* information. This, too, is a general result:

**Result 2** *Perfect information is always at least as valuable as imperfect information.*

Figure 1.7 repeats Figure 1.5, except now the payoff if the metal is returned to the supplier is left as a variable,  $z$ , and the value of  $p$  is fixed at  $5/6$ .

What we want to do now is comparative statics with respect to  $z$ . In particular, we want to see how the value of information changes as  $z$  changes. Note that the expected values have been written in for each of the rightmost chance nodes. From this information, we see that we want to divide our analysis into four regions:

1.  $z \leq -\$1 \text{ million}$ ;
2.  $-\$1 \text{ million} < z \leq \$1 \text{ million}$ ;



**Figure 1.6:** A plot of the value of information. Information is valueless for  $p \leq 5/6$ . It is valuable for  $p > 5/6$ . Perfect information (*i.e.*,  $p = 1$ ) maximizes the value of the information.

3. \$1 million  $< z \leq$  \$3 million; and

4. \$3 million  $< z$ .

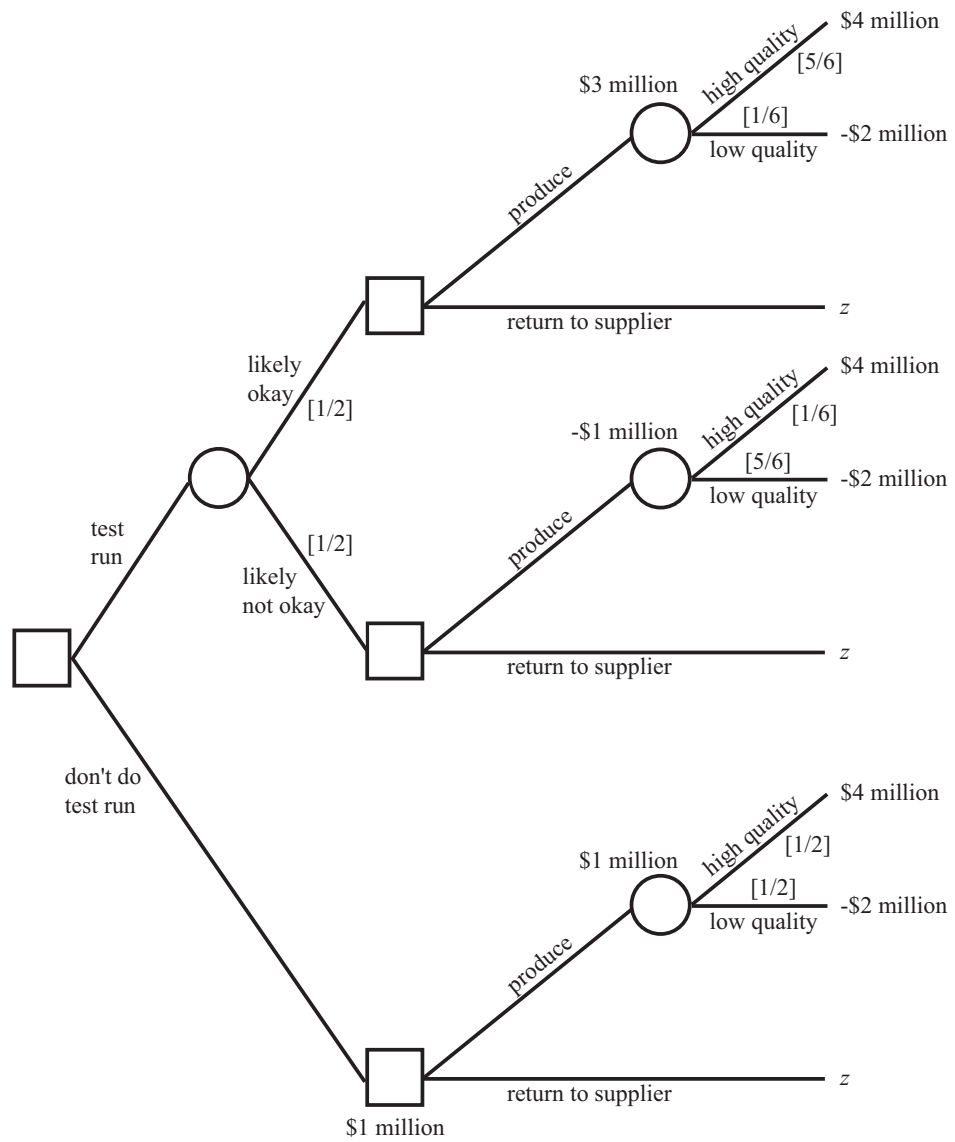
In region 1, the decision to make at each of the three rightmost decision nodes is “produce.” Moreover, this is the decision regardless of the information learned. Since the information, therefore, has no potential to change the firm’s action, the information must be worthless (this is just the Fundamental Rule of Information).

In region 2, the firm produces at the top and bottom decision node, but returns the metal to the supplier at the middle node. The information has, therefore, the potential to change the firm’s action, so we know it has value. Specifically,

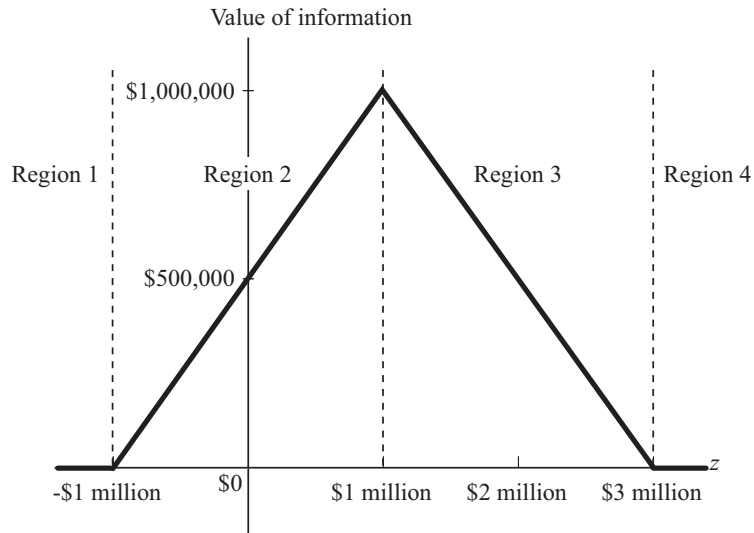
$$\begin{aligned} \text{Value of information in region 2} &= \frac{1}{2} \times \$3\text{mil.} + \frac{1}{2} \times z - \$1 \text{ mil.} \\ &= \frac{1}{2} \times z + \$500,000. \end{aligned}$$

Note that, in region 2, the value of information is increasing in  $z$ .

In region 3, the firm produces only at the top decision node, but returns the metal to the supplier at the bottom two nodes. Again, the information has the



**Figure 1.7:** More on the value of information.



**Figure 1.8:** A plot of the value of information against the value of returning the metal to the supplier,  $z$ . (Note horizontal and vertical scales are different.)

potential to change the firm's action, so we know it has value. Specifically,

$$\begin{aligned} \text{Value of information in region 3} &= \frac{1}{2} \times \$3\text{mil.} + \frac{1}{2} \times z - z \\ &= \$1.5 \text{ mil.} - \frac{1}{2} \times z. \end{aligned}$$

Note that, in region 3, the value of information is *decreasing* in  $z$ .

Finally, in region 4, the firm always does best to return the metal to the supplier. Moreover, this is the decision regardless of the information learned. Since the information has, therefore, no potential to change the firm's action, the information must be worthless.

Figure 1.8 plots the value of the information as a function of  $z$ . From the figure, it is clear that the information is most valuable when  $z = \$1$  million. What's significant about \$1 million? It's the value of  $z$  that makes the firm indifferent between producing and returning the metal when it has *no* information. This is not a fluke, but an illustration of a general result:

**Result 3** *The value of information is maximized when the decision maker would, absent the information, view alternative actions as equally attractive.*

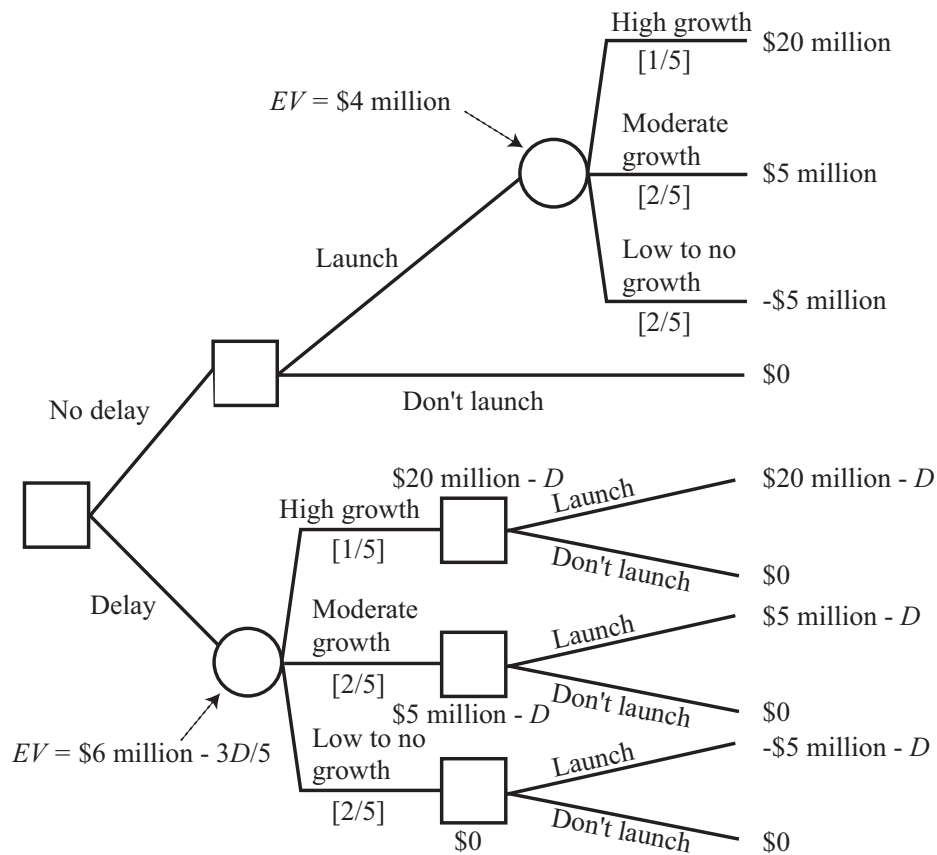


Figure 1.9: A firm has the option of delay.

## Real Options

# 1.5

In our discussion of information, we have so far treated information as something actively sought by the decision maker (*e.g.*, by doing a survey or a test run). Another way a decision maker can acquire information is through experience; that is, the passage of time resolves some of the uncertainty. The ability to use information learned over time can create certain options for the decision maker. To distinguish these options from financial options, they are often referred to as *real options*.

The main idea is that before you commit to an irreversible action, you have the option to commit or not to commit (just as with an in-the-money call option, where you have option to exercise or wait).

Figure 1.9 illustrates a real option (note its resemblance to Figure 1.4). A

firm can delay its launch of a new product or not delay. The success of the product depends on the growth rate of the economy. If the economy enjoys a high rate of growth, then the firm stands to earn \$20 million. If the rate of growth is moderate, then the firm makes \$5 million. Finally if there is low or no growth, then firm will *lose* \$5 million. Observe from the figure that delay allows the firm to learn the state of the economy before making its launch decision. Delay is not without cost, however. If it delays, then the payoffs should it launch are all reduced by  $D \geq 0$ .

From Figure 1.9, we see the firm should delay if

$$\$6 \text{ million} - \frac{3}{5}D \geq \$4 \text{ million};$$

or, solving for  $D$ , if

$$\$3\frac{1}{3} \text{ million} \geq D.$$

Observe there is a difference between information gained by waiting and information bought prior to a decision (*e.g.*, a survey). Suppose that rather than the delay/no delay decision, the initial decision was forecast/no forecast, where forecast is an econometric forecast of the economy's future state. Let  $F$  be the cost of the survey. Because  $F$  is spent regardless of whether the firm ultimately launches or not, the expected value of doing the survey is \$6 million  $- F$ . So it pays to do the forecast only if it costs no more than \$2 million. In other words, because the cost of delay is not always borne (*e.g.*, it is not borne if the firm decides not to launch), whereas the cost of *ex ante* information (*e.g.*, a forecast) is always borne, the cost of delay that a decision maker will accept is greater than the cost he or she will accept to obtain the information in advance (*e.g.*, through a survey or forecast).

## Summary

# 1.6

We began this lecture note by considering decision making under certainty. Such decision problems could be represented by *decision trees* consisting of decision nodes (squares), branches for the alternatives, and payoffs. These trees, *like all trees*, were solved by working from right to left (although the tree is read left to right).

Next we added the possibility of uncertainty. We indicated uncertainty in our trees by using chance nodes (circles). The branches stemming from a chance node are the possible outcomes of the random event that the chance node represents. We practiced solving trees with uncertainty under the assumption that the decision makers were *expected-value maximizers*.

We next analyzed the value of information in decision-making problems. Three points were made: (1) Whether or not to seek more information is, itself, a decision that needs to be considered as part of the overall decision making

process; (2) information is valuable only if it has the potential to change decisions; (3) information is maximally valuable when, absent the information, the decision maker is indifferent among his options.

Finally, we observed that delay is often a way to obtain information. In many ways it is similar to obtaining information *ex ante* (consider, *e.g.*, the similarities between Figures 1.4 and 1.9). However, because the cost of delay is not always borne, decision makers will generally be willing to pay a higher cost in terms of delay than they will to obtain the information *ex ante*.

## Risk Aversion

# 2

Consider the following situation. You own a “lottery” ticket with the following properties. Tomorrow, a fair coin will be flipped. If it lands heads up, you will receive \$1 million. If it lands tails up, you will receive nothing. The ticket is transferable; that is, you can give or sell it to another person, in which case this other person is entitled to the winnings from the lottery ticket (if any). Between today and tomorrow, people may approach you about buying your ticket. What is the *smallest* price that you would accept in exchange for your ticket?

A possible answer is \$500,000 because that is the expected value of this lottery. Many people, however, would be willing to accept less than \$500,000. You, for instance, might be willing to sell the ticket for as little as \$400,000, under the view that a certain \$400,000 was worth the same as a half chance at \$1 million. Moreover, even if you are unwilling to sell your ticket for \$400,000, many people in your situation would be.

Selling your ticket for less than \$500,000 is, however, *inconsistent* with being an expected-value maximizer, because you wouldn’t be choosing the alternative that yielded you the greatest expected value. So, if you would, in fact, accept \$400,000, then you are *not* an expected-value maximizer. Moreover, even if *you* are (*i.e.*, *you* wouldn’t sell for less than \$500,000), others definitely aren’t. As this discussion makes clear, we need some way to model decision makers who aren’t expected-value maximizers. That is the task of this lecture note.

## Attitudes Toward Risk

# 2.1

One reason that someone is not an expected-value maximizer is that he is concerned with the riskiness of the gambles he faces. For instance, a critical difference between an *expected* payoff of \$500,000 and a *certain* payoff of \$400,000 is that there is considerable risk with the former—you might win \$1 million, but you also might end up with nothing—but no risk with the latter. Most people don’t like risk, and they are willing, in fact, to pay to avoid it. Such people are called [risk averse](#). By accepting less than \$500,000 for the ticket, you are effectively paying to avoid risk; that is, you are behaving in a risk-averse fashion. When you buy insurance, thereby reducing or eliminating your risk of loss, you are behaving in a risk-averse fashion. When you put some of your wealth in low-risk assets, such as government-insured deposit accounts, rather than investing it all in high-risk stocks with greater expected returns, you are behaving in a risk-averse fashion.

**Risk aversion:** A distaste for risk.

To define risk aversion more formally, we begin with the concept of a *certainty-equivalent value*:

**Definition 2** *The certainty-equivalent value of a gamble is the minimum payment a decision maker would accept, if paid with certainty, rather than face a gamble. The certainty-equivalent value is often abbreviated CE.*

For example, if \$400,000 is the smallest amount that you would accept in exchange for the lottery ticket discussed previously, then your certainty-equivalent value for the gamble is \$400,000 (*i.e.*,  $CE = \$400,000$ ).

We can now define risk aversion formally:

**Definition 3** *A decision maker is risk averse if his or her certainty value for any given gamble is less than the expected value of that gamble. That is, we say an individual is risk averse if  $CE \leq EV$  for all gambles and  $CE < EV$  for at least some gambles.*

**Risk neutral:**  
*Unaffected by risk.*

In contrast, an expected-value maximizer is *risk neutral*—his or her decisions are unaffected by risk. Formally,

**Definition 4** *A decision maker is risk neutral if the certainty-equivalent value for any gamble is equal to the expected value of that gamble; that is, he or she is risk neutral if  $CE = EV$  for all gambles.*

In rare instances, a decision maker *likes* risk; that is, he or she would be willing to pay to take on more risk (or, equivalently, require compensation to part with risk). For instance, if someone's certainty-equivalent value for the aforementioned lottery ticket were \$600,000 (*i.e.*, she had to be compensated for giving up the risk represented by the ticket), then she would be called *risk loving*:

**Definition 5** *A decision maker is risk loving if the certainty-equivalent value for any gamble exceeds the expected value of that gamble; that is, he or she is risk loving if  $CE \geq EV$  for all gambles and  $CE > EV$  for at least some gambles.*

It is worth emphasizing that risk-loving behavior is fairly rare.<sup>1</sup>

At this point you might ask: When is it appropriate (*i.e.*, reasonably accurate) to assume a decision maker is risk neutral and when is it appropriate to assume he is risk averse? Some answers:

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<sup>1</sup>Although it is true that a number of people like to pay to gamble on occasion (*e.g.*, they visit casinos or play state lotteries), their more typical behavior can be described as risk neutral or risk averse (*e.g.*, even people who visit casinos typically purchase homeowner's insurance). Moreover, it is not clear that people gamble because they love the risk *per se*; they may simply like the excitement of the casino or like to dream about what they would do if they won the next Powerball drawing.

**SMALL STAKES VERSUS LARGE STAKES:** If the amounts of money involved in the gamble are small *relative* to the decision maker's wealth or income, then his behavior will tend to be approximately risk neutral. For example, for gambles involving sums less than \$10, most people's behavior is approximately risk neutral. On the other hand, if the amounts of money involved are large *relative* to the decision maker's wealth or income, then his behavior will tend to be risk averse. For example, for gambles involving sums of more than \$10,000, most people's behavior exhibits risk aversion.

**SMALL RISKS VERSUS LARGE RISKS:** If the possible payoffs (or at least the most likely to be realized payoffs) are close to the expected value, then the risk is small and the decision maker's behavior will be approximately risk neutral. For instance, if the gamble is heads you win \$500,001, but tails you win \$499,999, then your behavior will be close to risk neutral since both payoffs are close to the expected value (*i.e.*, \$500,000). On the other hand, if the possible payoffs are far from the expected value, then the risk is greater and the decision maker's behavior will tend to be risk averse. For instance, we saw that we should expect risk-averse behavior when the gamble was heads you win \$1 million, but tails you win \$0.

**DIVERSIFICATION:** So far the question of whether someone takes a gamble has been presented as an all-or-nothing proposition. In many instances, however, a person purchases a portion of a gamble. For example, investing in General Motors (or any other company) is a gamble, but you don't have to buy all of General Motors to participate in that gamble. Moreover, at the same time you buy stock in General Motors, you can purchase other securities, giving you a *portfolio* of investments. If you choose your portfolio wisely, you can *diversify* away much of the risk that is unique to a given company. That is, the risk that is unique to a given company in your portfolio no longer concerns you—you are *risk neutral* with respect to it. Consequently, you would like your firm to act as an expected-value maximizer. As we discuss in the next section, diversified decision makers are risk neutral (or approximately so), while undiversified decision makers are more likely to be risk averse.

## Diversification

# 2.2

To clarify the issue of diversification, consider the following example. There are two companies in which you can invest. One sells ice cream. The other sells umbrellas. Ice cream sales are greater on sunny days than on rainy days, while umbrella sales are greater on rainy days than on sunny days. Suppose that, on average, one out of four days is rainy; that is, the probability of rain is  $\frac{1}{4}$ . On

a rainy day, the umbrella company makes a profit of \$100 and the ice cream company makes a profit of \$0. On a sunny day, the umbrella company makes a profit of \$0 and the ice cream factory makes a profit of \$200. Suppose you invest in the umbrella company only; specifically, suppose you own all of it. Then you face a gamble: on rainy days you receive \$100 and on sunny days you receive nothing. Your expected value is

$$\$25 = \frac{1}{4} \times \$100 + \frac{3}{4} \times \$0.$$

Suppose, in contrast, that you sell three quarters of your holdings in the umbrella company and use some of the proceeds to buy one eighth of the ice cream factory. Now on rainy days you receive \$25 from the umbrella company (since you can claim one quarter of the \$100 profit), but nothing from the ice cream company (since there are no profits). On sunny days you receive \$25 from the ice cream company (since you can claim one eighth of the \$200 profit), but nothing from the umbrella company (since there are no profits). That is, rain or shine, you receive \$25—your risk has disappeared! Your expected value, however, has remained the same (*i.e.*, \$25). This is the magic of diversification.

Moreover, once you can diversify, you want your companies to make expected-value-maximizing decisions. Suppose, for instance, that the umbrella company could change its strategy so that it made a profit of \$150 on rainy days, but lost \$10 on sunny days. This would increase its daily *expected* profit by \$5—the new *EV* calculation is

$$\frac{1}{4} \times \$150 + \frac{3}{4} \times (-\$10) = \$30.$$

It would also, arguably, increase the riskiness of its profits by changing its strategy in this way. Suppose, for convenience, that 100% of a company trades on the stock exchange for 100 times its expected daily earnings.<sup>2</sup> The entire ice cream company would, then, be worth \$15,000 ( $= 100 \times (\frac{1}{4} \times \$0 + \frac{3}{4} \times \$200)$ ) and the entire umbrella company would, then, be worth \$3000. To return to your position of complete diversification and earning \$25 a day, you would have to reduce your position in the umbrella company to hold one sixth of the company and you would have to increase your holdings of the ice cream company to  $\frac{2}{15}$ th

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<sup>2</sup>The price-to-earnings ratio is 100 here, but the value of the price-to-earnings ratio does not matter for the conclusions reached here. If the ratio were  $r$ , then decreasing your holdings of the umbrella company to  $\frac{1}{6}$ th of the company and increasing your holdings of the ice cream company to  $\frac{2}{15}$ th would yield a trading profit of

$$\frac{30r}{12} - \frac{150r}{120} = \frac{5}{4}r > 0.$$

Now you might wonder whether it is appropriate to use the same price-to-earnings ratio for both firms. In this case it is, at least if you believe that the stock price is driven by fundamentals (that is, future profits).

of the company:

$$\text{Earnings on a rainy day : } \frac{1}{6} \times \$150 + \frac{2}{15} \times \$0 = \$25; \text{ and}$$

$$\text{Earnings on a sunny day : } \frac{1}{6} \times (-\$10) + \frac{2}{15} \times \$200 = \$25.$$

Going from holding one fourth of the umbrella company to owning one sixth of the umbrella company means selling  $\frac{1}{12}$ th of the umbrella company,<sup>3</sup> which would yield you \$250 ( $= \frac{1}{12} \times \$3000$ ). Going from holding one eighth of the ice cream company to owning  $\frac{2}{15}$ ths means buying an additional  $\frac{1}{120}$ th of the ice cream company,<sup>4</sup> which would cost you \$125 ( $= \frac{1}{120} \times \$15,000$ ). Your profit from these stock market trades would be \$125. Moreover, you would still receive a riskless \$25 per day. So because you can diversify, you benefit by having your umbrella company do something that increases its expected value, *even if it is riskier*.

## Modeling Risk-Averse Decision Makers

# 2.3

In this section, we will explore one way to model risk-averse decision makers. Specifically, we will assume a decision maker is motivated not by the money he may receive, but rather by the happiness that the money represents. In the parlance of economics, the decision maker is motivated by the *utility* the money represents and not the money itself. “Utility” is the word economists use when sensible people would use the word “happiness”;<sup>5</sup> that is, “utility” and “happiness” are synonyms.

**Utility:** a synonym for happiness or well-being.

Different situations, goods, and services, give us different amounts of happiness (or utility). You can, for instance, undoubtedly think of many situations in which you’ve used the phrase “I’m happier . . .,” thereby confirming that your happiness (utility) can vary. Imagine some fantastic machine—something that might appear in Dr. Frankenstein’s laboratory—with electrodes that can be attached to your scalp (or, if the thought of your being wired to this apparatus gives you the willies, some hapless victim’s scalp). Imagine that in the middle of this machine is a giant meter with a needle. Imagine further that this needle moves clockwise the happier you are (alternatively, the happier your victim is) and it moves counter-clockwise the less happy you are (alternatively, your victim is). Finally, imagine that there are no marks on the dial of this meter. By giving whomever is wired up to this machine things he likes or dislikes, we can see the *relative* happiness that these things provide him. Moreover, we can begin to graduate the dial. For instance, we can write “0” on the dial where the

<sup>3</sup>Because  $\frac{1}{4} - \frac{1}{6} = \frac{3}{12} - \frac{2}{12} = \frac{1}{12}$ .

<sup>4</sup>Because  $\frac{2}{15} - \frac{1}{8} = \frac{16}{120} - \frac{15}{120} = \frac{1}{120}$ .

<sup>5</sup>A lamentable characteristic pertaining to economists is a penchant for eschewing the quotidian term when an obfuscating term can be substituted.

needle points when our subject is “at rest” (*i.e.*, hasn’t been given anything) and we can write a “4” on the dial where the needle points after we give him a \$100. If the needle advances *twice* as far from the “rest” point when we give him \$300, then we can write a “8” on the dial where the needle now points. In this way we can arrive at a happiness or utility scale for *this* person.

Some points to note. First, this scale is specific to the person who we strapped into the machine. That is, if we strap someone else in and give her something that causes the needle to move to 4, we can’t say that this person is as happy as our first victim was when we gave him a \$100. Second, the first two numbers we write on to the dial are arbitrary. We could just as easily have written “-2.5” on the dial for the rest point and “ $\pi$ ” on the dial where the needle pointed after giving our subject a \$100. Thereafter, everything is determined by those first two markings. For instance, we would then have to write  $2 \cdot \pi + 2.5$  on the dial where the needle points after giving our subject \$300 since  $\pi$  is half way between -2.5 and  $2 \cdot \pi + 2.5$ . In this sense, our meter is very much like a thermometer: For instance, Gabriel D. Fahrenheit (1686–1736) essentially took a plain glass tube containing mercury and stuck it into water as it was just freezing and wrote “32” on the side of the tube and, similarly, stuck the same tube into water as it was just beginning to boil and wrote “212” on the side.<sup>6</sup> In contrast, Anders Celsius (1701–1744) took his mercury tube and stuck it into water as it was just freezing and wrote “0” on the side of the tube and, similarly, stuck the same tube into water as it was just beginning to boil and wrote “100” on the side. In other words, the numbers are arbitrary, only the *relative* relationship among the numbers has any meaning.

**Utility function:** A means of providing a score of the utility a person gets from what he or she receives.

Such a fantastic machine does not exist—too bad, it would certainly enliven lectures—but we can proceed as if such a machine existed. Specifically, we can imagine that each person is endowed with a *utility function*, which tells us the “reading” that we would have gotten from the machine (if it existed) for different amounts of money that we might give this person. As with our machine, the numbers refer only to the individual in question and the numbers themselves are arbitrary. An *example* of a utility function might be  $U(y) = \sqrt{y}$ ; that is,  $y$  dollars translates into a “reading” (or score) of the square-root of  $y$ . For instance,  $U(100) = 10$ .

Although philosophers may claim that money does not buy happiness, we, being somewhat more worldly, believe that there is a positive relationship between money and happiness. In particular, a reasonable assumption from our perspective is that the more money a person receives, the greater is his or her utility.<sup>7</sup> In other words, we assume

<sup>6</sup>To be precise, he stuck the tube into a *salt* water mixture formed from equal parts (by weight) of snow and salt and wrote “0” on the side of the tube (salt water, recall, freezes at a lower temperature than fresh water). He then wrote “30” on the tube for the freezing point of fresh water (this was subsequently revised to 32°).

<sup>7</sup>Note that this is *not* the same thing as saying that a person with more money is happier than another person with less money—recall we are interested in relative comparisons for the *same* person only.

**Assumption 1** *If  $y > y'$ , then a person's utility from  $y$  dollars is greater than his or her utility from  $y'$  dollars; that is,  $U(y) > U(y')$ .*

Because people do not eat, watch, wear, or play with money (at least not normally), money *per se* does not yield them utility. Rather they derive utility from the goods and services they purchase with this money. Therefore, we can suppose that decision makers are motivated by the utility a given amount of money yields them and not the money itself. Hence, a plausible assumption is that they seek to maximize their *expected utility* in situations of uncertainty:

**Definition 6** *A decision maker is an expected-utility maximizer if he or she chooses among his or her alternatives the one that yields the greatest expected utility.<sup>8</sup>*

**Definition 7** *The expected utility from a gamble is the expected value calculated over the utilities that each outcome yields. Specifically, if  $U(y)$  is the decision maker's utility function;  $y_1, \dots, y_N$  are the possible monetary prizes from the gamble; and  $p_1, \dots, p_N$  are the corresponding probabilities, then his or her expected utility (denoted  $EU$ ) is*

$$\text{Expected utility} = EU = p_1 \cdot U(y_1) + \dots + p_N \cdot U(y_N).$$

For example, suppose the gamble is win \$100 if a coin lands heads and win nothing if it lands tails. Suppose, too, that the decision maker's utility function is  $U(y) = \sqrt{y}$ . Then her expected utility would be

$$EU = \frac{1}{2} \times \sqrt{100} + \frac{1}{2} \times \sqrt{0} = \frac{1}{2} \times 10 + \frac{1}{2} \times 0 = 5.$$

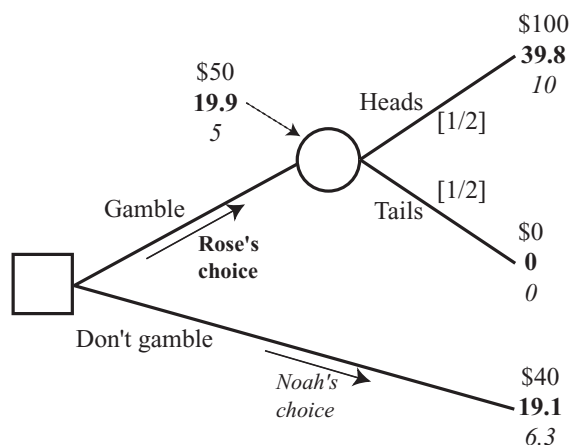
*A note of caution:* Expected utility is generally *not* the same as the utility from the expected monetary payoff. That is,  $EU \neq U(EV)$  generally. As an example, we have just seen that the expected utility from the coin-flip gamble is 5. The expected (monetary) value of this gamble is \$50. The utility from \$50,  $\sqrt{50}$ , is 7.07, which is not 5.

**Caution: Generally,**  
 $EU \neq U(EV)$ .

Figure 2.1 illustrates decision making by two expected-utility maximizers, Rose and Noah, each of whom is deciding whether to take \$40 with certainty or to play the coin-flip gamble just discussed. Suppose that Rose's utility function is  $U(y) = y^{\frac{4}{5}}$  and Noah's utility function is  $V(y) = \sqrt{y}$  ( $= y^{\frac{1}{2}}$ ). In Figure 2.1, monetary payoffs are indicated in plain text (like this), Rose's utility in bold (**like this**), and Noah's utility in italics (*like this*). Rose's expected utility from taking the gamble is

$$\frac{1}{2} \times 100^{\frac{4}{5}} + \frac{1}{2} \times 0^{\frac{4}{5}} = 19.9.$$

<sup>8</sup>Although expected-utility maximization is a reasonable assumption, it should be noted that it does rely on some important and, in some people's eyes, not necessarily innocuous assumptions about behavior.



**Figure 2.1:** Two expected-utility-maximizing decision makers, Rose and Noah, decide whether to get \$40 with certainty or to gamble. The monetary payoffs are shown in plain text, Rose's utility and expected utility are shown in **bold**, and Noah's utility and expected utility are shown in *italics*.

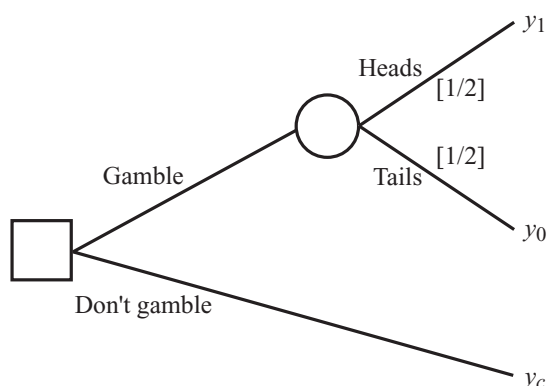
Rose's utility from \$40 is 19.1 ( $= 40^{\frac{4}{5}}$ ). Because  $19.9 > 19.1$ , Rose prefers the gamble to getting \$40 with certainty. In contrast, Noah's expected utility from taking the gamble is

$$\frac{1}{2} \times \sqrt{100} + \frac{1}{2} \times \sqrt{0} = 5.$$

His utility from \$40 is 6.3 ( $= \sqrt{40}$ ). Because  $5 < 6.3$ , Noah prefers getting \$40 with certainty to taking the gamble. Note that, for both Rose and Noah, their expected utility from the gamble is *not* equal to their utility from the expected monetary payoff from the gamble; that is,  $19.9 \neq 22.9 (= 50^{\frac{4}{5}})$  and  $5 \neq 7.07 (= \sqrt{50})$ .

You no doubt noticed that the reason why Rose and Noah make different decisions in Figure 2.1 is that they have different utility functions. This, in turn, raises some questions. First, Rose's utility function gives her higher utility scores than Noah's gives him. Does this mean Rose is happier than Noah? The answer is no. Utility cannot be used for comparisons across people. A second, more fundamental, question is how does one know what a decision maker's utility function is? Although, there are ways to derive utility functions, the truth is that generally we cannot know what another person's utility function is. Fortunately, there are, as we shall see, *general* properties of expected-utility maximization that make it a useful model for studying economic situations.

To summarize about expected-utility maximizers:



**Figure 2.2:** A decision maker has a choice between  $y_c$  with certainty and a gamble over  $y_0$  and  $y_1$ .

### Solving trees for expected-utility maximizers

1. For each of the rightmost nodes proceed as follows:
  - (a) If the node is a decision node, determine the best alternative to take. The utility from this alternative is the *value of this decision node*. Arrow the best alternative.
  - (b) If the node is a chance node, calculate the expected utility. This expected utility is the *value of this chance node*.
2. For the nodes one to the left proceed as follows:
  - (a) If the node is a decision node, determine the best alternative to take, using, as needed, the values of future nodes (nodes to the right) as the utilities. The utility or expected utility from this alternative is the value of this decision node. Arrow the best alternative.
  - (b) If the node is a chance node, calculate the expected utility, using, as needed, the values of future nodes (nodes to the right) as the utilities. The expected value is the value of this chance node.
3. Repeat Step 2 as needed, until the leftmost node is reached. Following the arrows from *left to right* gives the sequence of appropriate decisions to take.

## Risk Aversion, Expected Utility, and the Certainty- Equivalent Value

# 2.4

Consider Figure 2.2, which is similar to Figure 2.1: A decision maker can get a certain payoff,  $y_c$ , or face a gamble over two amounts  $y_0$  and  $y_1$ . Suppose that the decision maker likes these two alternatives equally well. This means that

$$\begin{aligned} U(y_c) &= EU \\ &= \frac{1}{2} \cdot U(y_1) + \frac{1}{2} \cdot U(y_0) . \end{aligned}$$

It follows that if we reduced  $y_c$  by a little bit (to, say,  $y_c - \varepsilon$ ), then the decision maker would strictly prefer the gamble (since  $U(y_c - \varepsilon) < U(y_c)$  by Assumption 1). Conversely, if we increased  $y_c$  by a little bit (to, say,  $y_c + \varepsilon$ ), then the decision maker would strictly prefer not to gamble. Putting all this math together, we can conclude that  $y_c$  is the smallest amount that the decision maker would be willing to accept instead of facing the gamble. But from Definition 2, on page 22, this just says that  $y_c$  is the certainty-equivalent value of the gamble. More generally, we have

**Result 4** *The utility from the certainty-equivalent value of a gamble,  $CE$ , equals the expected utility of the gamble; that is,  $EU = U(CE)$ .*

Recall that if an individual is risk averse, then his certainty-equivalent value for a gamble is less than the gamble's expected (monetary) value; that is,  $CE < EV$ . Given Assumption 1, this means that

$$U(CE) < U(EV)$$

for a *risk-averse* decision maker. Given Result 4, this yields

**Result 5** *If a decision maker is risk averse, then his expected utility from a gamble is less than his utility from the expected value of the gamble; that is,  $EU < U(EV)$ .*

Result 5 can also serve as an alternative, but equivalent, definition of risk aversion: A decision maker is risk averse if  $EU < U(EV)$ .

Recall that if an individual is risk neutral, then his certainty-equivalent value for a gamble equals the gamble's expected value; that is,  $CE = EV$ . It follows that

$$U(CE) = U(EV)$$

for a *risk-neutral* decision maker. It also follows that if a decision maker is risk neutral, then his expected utility from a gamble is equal to his utility from the expected value of the gamble; that is,  $EU = U(EV)$  for a *risk-neutral* decision maker.

Recall Noah of Figure 2.1 fame. His expected utility from taking the gamble was 5. What is his certainty-equivalent value for the gamble? We know

$$\begin{aligned} U(CE) &= EU = 5 \text{ and} \\ U(y) &= \sqrt{y}. \end{aligned}$$

It follows, then, that

$$\begin{aligned} \sqrt{CE} &= 5 \text{ or, squaring both sides,} \\ CE &= 5^2 = 25; \end{aligned}$$

that is, the minimum that Noah would be willing to accept instead of the gamble shown in Figure 2.1 is \$25. Note, as it had to be, this is consistent with the fact that Noah prefers \$40 with certainty to the gamble in Figure 2.1; since \$40 > \$25, Noah is getting *more* than the minimum amount he would accept rather than face the gamble. This method of solving for the certainty-equivalent value generalizes:

**Result 6** *The certainty-equivalent value of a gamble can be calculated by applying the inverse of the utility function to the expected utility; that is, if  $U^{-1}(\cdot)$  is the inverse of the utility function (e.g., if  $U(y) = \sqrt{y}$ , then  $U^{-1}(u) = u^2$ ), then  $CE = U^{-1}(EU)$ .*

## Risk Neutrality Again

# 2.5

Consider a decision tree and imagine that we change all the payoffs in the *same* way, where payoffs are in utility terms if the decision maker is an expected-utility maximizer or are in monetary terms if the decision maker is an expected-value maximizer. Would a change of this sort cause the decision maker to alter the decision he or she would have made before the change? That is the question we first address in this section.

To develop an answer, we begin with the following: A transformation of one function,  $f(\cdot)$ , to another,  $g(\cdot)$ , is a **positive affine transformation** if, for all  $x$ ,

$$g(x) = A \cdot f(x) + B,$$

where  $A$  and  $B$  are constants and  $A > 0$ .

An important property of affine transformations is the following.

**Result 7** *Let  $x$  be a random variable, then*

$$A\mathbb{E}\{x\} + B = \mathbb{E}\{Ax + B\}.$$

**Proof:** Recall that the expectation of a function,  $f(\cdot)$ , of a random variable,  $x$ , is

$$\mathbb{E}\{f(x)\} = p_1 f(x_1) + \cdots + p_N f(x_N)$$

(see expression (B3.3) in Appendix B3). Hence,

$$\begin{aligned}\mathbb{E}\{Ax + B\} &= p_1 \cdot \underbrace{(Ax_1 + B)}_{f(x_1)} + \cdots + p_N \cdot \underbrace{(Ax_N + B)}_{f(x_N)} \\ &= A \sum_{n=1}^N p_n x_n + B \sum_{n=1}^N p_n \quad (\text{using the distributive rule}) \\ &= A\mathbb{E}\{x\} + B,\end{aligned}$$

where the last line follows from the definition of expectation and because probabilities sum to one. ■

It is important to note that affine transformations are the only class of functions for which  $\mathbb{E}\{f(x)\} = f(\mathbb{E}\{x\})$ ; in general (*i.e.*, for other classes of functions) this relation does *not* hold.

Let  $\pi_1, \dots, \pi_N$  be the  $N$  possible payoffs in a decision tree. Note, if the decision maker is an expected-utility maximizer, then the  $\pi$ s are in terms of utility. If he or she is an expected-value maximizer, then the  $\pi$ s are in monetary terms. Assume that there is a single optimal course of action for this tree. That is, if  $\Pi^*$  is the ultimate (possibly expected) payoff from this optimal course of action, then

$$\Pi^* > \Pi, \quad (2.1)$$

where  $\Pi$  is the ultimate (possibly expected) payoff from any other course of action.



Consider a positive affine transformation of the payoffs. That is, let the  $n$ th payoff,  $\tilde{\pi}_n$ , now be  $A\pi_n + B$ . We claim that the optimal course of action to take is the same whether the payoffs are the  $\pi$ s or the  $\tilde{\pi}$ s. To establish this claim, suppose it were false. That is, when the payoffs are the  $\tilde{\pi}$ s suppose a different course of action is optimal to pursue than would be optimal were the payoffs the  $\pi$ s. Observe that each course of action defines a density over the payoffs.<sup>9</sup> This is true even if there is no uncertainty in the tree or the course of action doesn't encompass any chance nodes (the density in this case simply puts all weight on one payoff—assigns it probability 1—and no weight on any of the other payoffs—assigns them probabilities 0). Let  $\{p_n\}$  be the density induced by the uniquely optimal course of action when the payoffs are the  $\pi$ s and let  $\{\tilde{p}_n\}$  be the density induced by the optimal action when the payoffs are the  $\tilde{\pi}$ s. Observe, by the definition of  $\Pi^*$ ,

$$\Pi^* = p_1\pi_1 + \cdots + p_N\pi_N.$$

Let

$$\tilde{\Pi} = \tilde{p}_1\pi_1 + \cdots + \tilde{p}_N\pi_N;$$

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<sup>9</sup>See Appendix B3 to review the concept of a density.

that is,  $\tilde{\Pi}$  is the ultimate payoff (possibly expected) if the decision maker pursues the course of action associated with the density  $\{\tilde{p}_n\}$  when the payoffs are the  $\pi$ s. From expression (2.1), we know that

$$\Pi^* > \tilde{\Pi}. \quad (2.2)$$

From the definition of the  $\tilde{p}_n$ s (*i.e.*, the optimality of the corresponding course of action), we know that

$$\tilde{p}_1 \tilde{\pi}_1 + \cdots + \tilde{p}_N \tilde{\pi}_N \geq p_1 \pi_1 + \cdots + p_N \pi_N.$$

But  $\tilde{\pi}_n$  is the positive affine transformation  $A\pi + B$ . So, from Result 7, we can rewrite this last expression as

$$A \sum_{n=1}^N \tilde{p}_n \pi_n + B \geq A \sum_{n=1}^N p_n \pi_n + B,$$

which implies

$$\tilde{\Pi} \geq \Pi^*. \quad (2.3)$$

But expression (2.3) contradicts expression (2.2), hence our original supposition that different courses of actions would be chosen must have been wrong.<sup>10</sup> We have established:

**Result 8** *A positive affine transformation of the payoffs to a decision tree cannot change what the optimal course of action is.*

*Note* Result 8 is a statement about positive affine transformations only. It doesn't apply to arbitrary transformations. In fact we know that some transformations *can* change the optimal decision: Recall Noah from Figure 2.1, he doesn't choose the gamble. In contrast, an expected-value maximizer would choose the gamble. But observe Noah's payoffs (his utility) is the transformation  $\sqrt{y}$  of the payoffs faced by an expected-value maximizer. In other words, as Figure 2.1 illustrates, the transformation square root, for instance, *can* change what the optimal course of action is.

Earlier, we wrote that an expected-value maximizer is risk neutral. How did we know this? To answer, begin by assuming that the utility that an individual gets from a monetary payoff of  $y$ ,  $U(y)$ , is  $Ay + B$ ,  $A$  and  $B$  constants, where

<sup>10</sup>To better understand the logic of this argument, observe that if statement A implies statement B and we show statement B to be false, then statement A must also be false (if it were true, then, because it implies B is true, we would have a logical contradiction). In our argument, statement B is expression (2.3), which we know is false because expression (2.1) is true. Statement A is the claim that a positive affine transformation leads to different courses of action. We showed that A implies B. But B is false, so A must be false as well; that is, a positive affine transformation does *not* lead to different courses of action. This style of argument is sometimes called *proof by contradiction* or a *reductio ad absurdum* argument.

$A > 0$  (consistent with Assumption 1). Observe this utility function is a positive affine transformation of the monetary payoff.

An individual with such a utility function is risk neutral. How do we know? Recall from Definition 4 that an individual is risk neutral if  $CE = EV$ . Recall from Result 6 that  $CE = U^{-1}(EU)$ . For the individual in question,

$$U^{-1}(u) = \frac{u - B}{A}.$$

So we have,

$$\begin{aligned} CE = U^{-1}(EU) &= \frac{EU - B}{A} \\ &= \frac{\mathbb{E}\{Ay + B\} - B}{A} \\ &= \frac{A\mathbb{E}\{y\} + B - B}{A} \quad (\text{from Result 7}) \\ &= \mathbb{E}\{y\} = EV, \end{aligned}$$

where the last equality follows from the definition of  $EV$  (*i.e.*, it is the expected value of the monetary payoffs of the gamble). So  $CE = EV$ ; that is, as claimed, the individual in question is risk neutral.

**Result 9** *A person whose utility function is a positive affine transformation of monetary payoffs is risk neutral.*

A utility function that is a positive affine transformation of monetary payoffs is a positive affine transformation of payoffs. Hence, by Result 8, a decision maker whose utility function is a positive affine transformation of monetary payoffs—that is, a risk-neutral decision maker—must take the same actions as an expected-value maximizer.<sup>11</sup> In other words, being a risk-neutral decision maker is equivalent to being an expected-value maximizer.



### Risk Neutrality Implies $U(\bullet)$ is a Positive Affine Transformation

Above, we established that a person whose utility function was of the form  $Ay + B$ ,  $A$  and  $B$  constants and  $A > 0$ , was risk neutral. Here, we will establish that a risk-neutral individual must have a utility function of the form  $Ay + B$ .

Remember that, for a risk-neutral individual,  $EU = U(EV)$  for *all* gambles. Consider the gamble in which, with probability  $p$ , the individual receives  $y$ , and in which, with probability  $1 - p$ , he or she receives  $z$ . Hence,

$$pU(y) + (1 - p)U(z) = U(py + (1 - p)z)$$

<sup>11</sup>Technically, we need to establish that being risk neutral implies that your utility function is a positive affine transformation of the monetary payoffs; we take up that issue in the next subsection.

Because this holds for *all*  $y$ ,  $p$ , and  $z$ , we can differentiate both sides with respect to  $y$  to derive:<sup>12</sup>

$$pU'(y) = pU'(py + (1 - p)z) \quad (2.4)$$

Because this expression holds for all  $p$ ,  $y$ , and  $z$ , expression (2.4) tells us that  $U'(\cdot)$  must be a constant (*i.e.*, the same for all values). Let  $A$  be that constant value. By Assumption 1, we know  $A > 0$ . So we have,

$$U'(y) = A.$$

Integrating both sides, we get

$$U(y) = Ay + B,$$

where  $B$  is the constant of integration.

## Summary | 2.6

We began this lecture note by observing that most individuals dislike risk; that is, they are risk averse. We defined risk aversion in terms of the certainty equivalent value ( $CE$ ) of a gamble; that is, the minimum amount that, if paid with certainty, would induce the individual to part with the gamble. If a decision maker is risk averse, then his or her  $CE$  is less than the expected (monetary) value ( $EV$ ) of any gamble.

Those individuals for whom  $CE = EV$  are risk neutral. As we saw, a risk-neutral individual's utility from an amount of money,  $y$ , is  $Ay + B$ , where  $A$  and  $B$  are constants,  $A > 0$ . Because, as we saw, positive affine transformations of the payoffs of a decision tree leave the optimal course of action unaffected, risk-neutral decision makers are expected-value maximizers.

We also introduced the idea of expected-*utility* maximization.

We considered circumstances in which risk neutrality or risk aversion were the more sensible assumption about a decision maker's attitudes toward risk. One circumstance in which risk neutrality is appropriate is when the decision maker is well diversified. We considered diversification in some detail through an example.

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<sup>12</sup>You might wonder how we know  $U(\cdot)$  is differentiable. There is a theorem of real analysis that tells us that any monotonic function must be differentiable almost everywhere (*i.e.*, we can essentially ignore any points at which it is not differentiable).



## Costs

# 3

What is the cost of an activity (*e.g.*, providing a good or service)? The obvious answer might seem “the money spent on that activity.” The purpose of this lecture note is to convince you that, like so many “obvious” answers, this is not (necessarily) the correct answer.

## Opportunity Cost

# 3.1

The correct way to think about cost from the perspective of making good business decisions is to consider the cost of some activity (alternatively, good, decision, service) to be the value of the most highly valued forgone activity (*i.e.*, the value of the best alternative decision). Economists describe this way of viewing cost as considering the *opportunity cost* of an activity or decision. In other words, the cost of something is the value of the best opportunity given up.

**Opportunity cost:**  
*The value of the most highly valued forgone activity or use of a good.*

In many circumstances, the opportunity cost notion coincides with what might be termed the naïve view of cost, namely its your expenditure on the activity in question. If, for instance, you purchase three tons of grapes for your wine-making business, then the cost of the grapes is the amount you pay for those grapes.

This naïve view, focusing on expenditures, can, however, lead one to make bad decisions. For instance, suppose you purchased 1000 liters of a chemical that is used in your production process. Suppose you paid a price of \$10 per liter. Suppose, however, that after purchase, but before use, the price of this chemical jumps to \$11 on the open market. The cost if you use this chemical is not your expenditure, \$10,000, but rather \$11,000, the current value of the 1000 liters. Why? Because your next best use of the chemical is to sell it on the open market and the value of this next-best activity is \$11,000. You can see this by supposing that the products you would make from the 1000 liters would sell for a total of \$10,500. If you produce, you’ll have \$10,500 in the bank and you might, incorrectly, suppose yourself to have made a \$500 profit. In fact, you’ve suffered a \$500 *loss*; had you sold the chemical, you would have had \$11,000 in the bank and \$11,000 beats \$10,500. From this, we see both the danger of the naïve view and the benefit of adopting the opportunity-cost view.

This scenario also illustrates two concepts that derive from opportunity cost, *imputed cost* and *sunk expenditure*.

An imputed cost (*e.g.*, the \$11,000 in the above scenario) is a cost not

**Imputed cost:** *The imputed value of a forgone opportunity.*

**Sunk expenditure:** *An expenditure is sunk if it cannot be recovered or avoided over the relevant decision-making horizon. A sunk expenditure is not a cost.*

associated with an expenditure; that is, you don't have to pay anyone \$11,000 to use the chemical, but it's a cost of your using the chemical nonetheless.

A sunk expenditure is an expenditure that has been made—sunk—in the past or an expenditure that one will make regardless of which of the relevant alternatives is chosen (*e.g.*, because of a previously made commitment).<sup>1</sup>

In particular, what you paid for machinery, property, services, inputs, or any other item in the past is *not* a cost *today*—it is sunk. Moreover, a past expenditure is *irrelevant* to any decision you are making now. As the old saying goes, “there is no point crying over spilled milk.” Or, in terms of decision trees, one can view a past expenditure as a constant amount subtracted from *all* the payoffs. As we saw previously (Result 8, page 33), subtracting a constant amount from all the payoffs cannot affect the decision to be made.

Note, however, an expense doesn't have to have been made in the past to be sunk. Any expense, even one not yet paid, is sunk if it cannot be avoided given the relevant set of actions from which the decision maker must choose.

Some examples will further illustrate the benefits of adopting the opportunity-cost view.

**Example 1:** Your store is situated in rented space. Your lease expires in six months time and you cannot break it prior to then. Monthly rent is \$5000. Your other monthly expenditures (*e.g.*, inventory, sales help, etc.) are \$3000, for total monthly expenditures of \$8000. Your monthly revenue (sales) total \$6000. You are considering whether to shutdown effective immediately. This might seem an attractive course of action: After all it would seem you're “losing” \$2000 a month. This, however, is an incorrect assessment of the situation. What is the *opportunity cost* of staying open? Well, if you stay open, your expenditures will be \$8000 per month. If you shutdown, your expenditures will be \$5000 a month (remember you can't break your lease). Thus, the true (opportunity) cost of staying open is \$3000 ( $= \$8000 - \$5000$ ): Relative to your best alternative (shutting down), you are only forgoing \$3000 by staying open. Given that your *true* costs are half of your monthly revenues, you would choose to remain open over the next six months. To see it another way, note that your bank account will be \$30,000 poorer at the end of six months if you shutdown ( $6 \times -\$5000$ ); whereas, if you remain open, your bank account will be only \$12,000 poorer at the end of six months ( $6 \times [\$6000 - \$5000 - \$3000]$ ).

By employing the concept of opportunity cost you saved yourself \$18,000 in the preceding example. This is because you recognized that the rent over the next six months was *not* a cost, but rather a sunk expenditure.

**Example 2:** You run a theater. At the moment, the show that you're running sells on average 100 tickets per night (you are risk neutral). The ticket price is \$50 per ticket. The nightly expenses of the show are \$3000 and these are essentially independent of the number of people who attend.

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<sup>1</sup>An expenditure that is sunk is sometimes called a *sunk cost*. This, however, is unfortunate terminology, because if an expenditure is sunk then it is not a cost with regard to the decision at hand.

A local civic group approaches you about purchasing all the tickets in the house (300) for a given evening. Because they are a charitable group, they ask if you might be willing to sell them the tickets at cost. What price would you quote them if you are indeed willing to sell them at cost?

The answer is *not* \$3000. Why? Well whether or not you sell to this group, the show will run. The expenses of the show are, thus, sunk, and, thus, not a cost. The cost, remember, is the value of what you give up. What you give up are the 100 tickets sold at \$50 per ticket; that is, \$5000. So the correct price to quote the group is \$5000.

Observe that, in both examples, the answer is dependent on the decision problem you face. In Example 1, over the *relevant decision-making horizon*—the next six months—the monthly rent is sunk. Were you, however, deciding to renew the lease at the end of the six months, then the rent would be a cost because you could avoid it by shutting down (*i.e.*, not renewing the lease). Likewise, in Example 2, you are already committed to running the show. Hence, any expenses associated with running the show are sunk with respect to other decisions, such as whether to sell to the civic group or not. Had, however, the decision been a different one, specifically whether or not to close the show, then the nightly expenses of running the show would be a cost because they could then be avoided. Both of these examples illustrate one test for whether an expense is a cost: Is there a branch (outcome) of the relevant decision tree in which you avoid the expense? If the answer is *no*, then the expense cannot be a cost.

**Result 10** *If, within the set of relevant decisions, an expense is unavoidable, then it is not a cost.*

**Example 3:** Your company owns a warehouse. Currently, the warehouse is in such poor condition that it cannot be used. Your company would have to spend \$500,000 to make it usable. If made usable, the value to your company, if it uses it over its productive life, is \$300,000. You are also aware that another company, in a different industry than you, is looking to purchase a warehouse and would certainly purchase your warehouse *if it were usable* (note, then, this other company would *not* buy your warehouse “as is”). In thinking about what sales bid to make to this other company, is the \$500,000 a cost; that is, relevant to your decision? What would your answer be if the value to *your* company of a usable warehouse was \$600,000?

Under the original conditions (*i.e.*, the value to you is \$300,000), the \$500,000 is most certainly a cost. If you sell your warehouse you will have to spend \$500,000. If, instead, you pursue your next best alternative—which is do nothing with the warehouse—you don’t spend the \$500,000. On the other hand, if the value to your company was \$600,000, then the \$500,000 is not a cost; you will incur the expense if you sell the warehouse or if you pursue your next best alternative, which, now, is to use the warehouse yourself.

Example 3 suggests another test to see whether an expense is a cost, namely the question of *cost causation*:

**Result 11 (Cost causation)** *If a decision causes you to incur an expense that you wouldn't incur under the next best alternative, then that expense is a cost; it has been caused by your decision. Conversely, if an expense would have also been incurred under the next best alternative, then it is not a cost of your decision.*

As an additional example, consider the following.

**Example 4:** You may have noticed or read that when oil prices jump up (say because of Mideast unrest), prices at your local gas station quickly follow suit. Yet, as newspaper articles are quick to point out, the gasoline in your local gas station's tanks was produced from oil purchased at the old, lower, prices. Is your local gas station engaged in price gouging, as some allege, or is there a sensible opportunity-cost explanation?

It won't surprise you that the answer is the latter. One way to think about it is as follows. The wholesale price of new gasoline will be higher than the wholesale price previously paid.<sup>2</sup> "Old" and "new" gasoline are perfect substitutes, so your local gas station could, in theory, resell the gas in its tanks to other stations in the wholesale market. In other words, the situation is like the chemical example considered at the beginning of this section.

Even if reselling the gasoline in the wholesale market isn't feasible, the opportunity cost of selling the gasoline already in the tanks has increased. A rise in oil prices must, ultimately, lead to an increase in gasoline prices. Gasoline today is a substitute for gasoline tomorrow. So an alternative available to your local station is to choose not to sell gasoline now and to wait until the price goes up, at which time it can sell the gasoline it holds at the higher price.

Example 4 illustrates an example of decision makers' thinking about costs correctly. Yet, there are many cases in which business people fail to do so. One situation is when business people consider historic prices, such as those paid for inputs, as relevant for current decision making; that is, failing to recognize them as sunk. For instance, a well-known computer manufacturer used the price it had paid for the memory chips in its inventory when it priced its computers, ignoring the fact the price of these memory chips had fallen considerably from the time they had been purchased. By using historic price, rather than the chips' current value, the manufacturer overestimated its true costs. This, in turn, caused it to overprice its computers. As a consequence, it lost a considerable amount of market share and profit. In another example, a well-known car manufacturer, seeking a competitive advantage, hired people to forecast metal prices. The idea was that it would buy and stockpile metals whose prices were forecast to increase. Then, later, it would have "cost advantage" over its rivals when it used that metal in manufacturing. While speculating in commodity markets, such as metals, might be an okay investment strategy for the firm, it can't provide any cost advantage if the metal is ultimately used in manufacturing—the cost

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<sup>2</sup>The wholesale price is what the gas station pays its supplier for the gasoline it purchases to sell to consumers.

of the metal is its market value at the time it is used, not what was paid for it in the past.

As a final example, consider the following.

**Example 5:** A large conglomerate knows now that it will need to send 50 mid-level executives to a two-day convention in Miami in February.<sup>3</sup> That is high season and hotels are routinely filled to capacity. Any hotel suitable for these executives charges \$250 a night per guest. Seeking to avoid paying \$25,000, someone in the conglomerate notes that it happens to own a suitable hotel in Miami. Said person proposes that the 50 executives just stay at that hotel for free; further claiming that the cost will be only the cost of room maintenance, which is \$50 per night; that is, cost will be only \$5000.

Whatever the advantages of a conglomerate, this is not one. This someone has made a fundamental error. Recall that hotels sell out. Hence, what the company is forgoing by lodging its executives at one of its hotels is not \$50 per night (it would pay that whether an executive is lodged in a room or an outside guest is), but rather \$250, the forgone revenue from renting the room to an outside guest. That is, the cost per night is still \$250—no savings can be achieved by this proposal.

Example 5 illustrates a rather common mistake, assuming that because one owns an asset, its use is free. Typically, unless there is no alternative use of the asset, its use is not free. The cost of using it is the value of its next best use (*e.g.*, in Example 5 renting the rooms to outside guests).

A formula that can help relate expenditures to costs is the following:

$$\text{Cost} = \text{Expenditures} - \text{Sunk Expenditures} + \text{Imputed Costs}.$$

**Cost:** *Expenditures plus imputed costs less sunk expenditures.*

## Cost Concepts | 3.2

Consider a firm that makes a single product. Suppose that every morning, prior to the start of production, the machines used for production must be maintained. Suppose this costs \$100 per day. Suppose each unit of the product requires \$5 worth of raw materials, \$8 worth of labor, and will cost \$2 to ship. What is the cost of selling a single unit of the product? The answer depends on the decision-making horizon: If it is mid-day and the firm has been producing all morning, then the cost of a unit is \$15 ( $= \$5 + \$8 + \$2$ ); if, however, it is the *first* unit of the morning, then the cost is \$115.

To understand these answers, recognize that the cost incurred by maintaining the machines is different than the costs incurred by producing an additional unit. Maintenance costs are, here, costs that are incurred once and that do not depend on the number of units produced. That is, they are incurred when the firm decides to produce that day (go from the 0th unit to the 1st unit). A cost that is incurred when a firm decides to produce rather than shutdown over the

<sup>3</sup>This highly disguised example is based on an actual incident.

**Overhead cost:** A cost incurred by operating that does not vary with the number of units produced.

**Variable cost:** A cost that varies with the number of units produced.

**Average cost:** The total cost of production divided by the number of units produced.

relevant decision-making horizon (here, a day) and that does not vary with the total number of units produced is an *overhead cost*.<sup>4</sup>

In contrast, the expenditures on raw materials, labor, and shipping are, here, examples of *variable costs*; that is, costs that vary with each unit produced.

Now these distinctions might strike you as odd: Why not simply use the total amount spent on producing a day's output divided by a day's output as the cost per unit of output (this measure is called *average cost*)?<sup>5</sup> For example, if this firm produced 20 units today, why not simply treat the cost per unit as \$20 per unit ( $= (20 \times \$15 + \$100) / 20$ )? There are two reasons why *this is a bad idea*. First, it depends on the number actually produced. If the firm's output varied, would you really want to think that its unit costs are varying even though the technology and input prices (*e.g.*, the cost of raw materials) had remained unchanged? The second, and more important reason is that this approach violates the fundamental notion of cost: *The cost of an activity* (*e.g.*, producing the 20th unit) *is the value of the next best alternative* (*e.g.*, stopping at the 19th unit). Here, if you stopped at the 19th unit you would save just the additional \$15 that you would need to spend to produce a 20th unit. Over the relevant decision-making horizon—make or don't make a 20th unit *given* that 19 units have already been made—the daily maintenance cost is sunk and, thus, not a cost *over this decision-making horizon*. To make this more concrete, consider the following example.

**Example 6:** Suppose the firm under consideration gets an order to produce 20 units for \$21 per unit. The firm will accept this order because

$$\underbrace{20 \times \$21}_{\text{revenue}} - \underbrace{(20 \times \$15 + \$100)}_{\text{cost}} = \$20;$$

that is, the firm will make a positive profit. Suppose, *after the firm has started production*, someone else calls in an order for 10 units at \$18 per units. Should the firm accept this second order? Well if it *mistakenly* used average cost, \$20, as its cost per unit, it would reject the offer because it would “lose” \$2 per unit. If, however, it used the correct cost per unit, namely \$15, it would accept the offer because it would make a \$3 profit per unit. To verify that this is, indeed, the correct cost concept—*i.e.*, leads to the correct decision—observe that

$$\underbrace{(20 \times \$21 + 10 \times \$18)}_{\text{revenue}} - \underbrace{(30 \times \$15 + \$100)}_{\text{cost}} = \$50 > \$20.$$

**Marginal cost:** The marginal cost of the *n*th unit is the additional cost incurred by producing the *n*th unit.

This last example shows the importance of using *marginal cost*—the additional cost incurred by producing one more unit—in deciding *how much* to produce. In Example 6, the marginal cost of the 20th unit is \$15 because, as

<sup>4</sup>Overhead costs are sometimes referred to as *fixed costs*. The latter is, however, unfortunate terminology: If some expenditure is *truly* fixed (*i.e.*, immutable), then it cannot be a cost (it's sunk).

<sup>5</sup>Observe that average cost, abbreviated *AC*, is cost, *C*, divided by number of units produced, *x*. That is,  $AC = C/x$ .

previously argued, \$15 is the *additional* cost incurred by producing the 20th unit. Note that we can equivalently define marginal cost as

$$\begin{aligned} \text{Marginal cost of } n\text{th unit} &= \text{Cost of producing all } n \text{ units} \\ &\quad - \text{Cost of producing all } n - 1 \text{ units.} \end{aligned}$$

Rather than writing out “marginal cost of” and “cost of producing all,” we will use the notation  $MC(n)$  to denote the marginal cost of the  $n$ th unit and  $C(n)$  to denote the cost of producing all  $n$  units. We will also refer to  $C(n)$  as the *total cost* of producing  $n$  units.

Given the importance of the marginal cost concept, it is worth considering two more examples.

**Example 7:** Suppose the firm under consideration had to maintain the machines after every fifty units; that is, before producing the 1st, 51st, 101st, etc., units. Consequently, the marginal cost of the 1st unit is \$115, as is the marginal cost of the 51st unit, the 101st unit, etc. The marginal cost of all other units remains just \$15.

**Example 8:** Return to the assumption that the machinery only needs to be maintained once a day, first thing in the morning. Suppose, now, that if the firm produces 250 or more units, it needs to pay time-and-a-half overtime to its employees; that is, the cost of labor per unit rises to \$12 per unit. Then the marginal cost of the first unit is \$115. The marginal cost of the  $n$ th unit,  $1 < n < 250$ , is \$15. But now the marginal cost of the  $m$ th unit,  $m \geq 250$ , is \$19 (= \$5 in raw materials + \$2 in shipping + \$12 in labor).

## Relations Among Costs

# 3.3

In this section we explore the relations among total cost, average cost, and marginal cost from an algebraic perspective. Later, in Section 3.5, we consider a graphical interpretation.

First, observe that the total cost of producing nothing (*i.e.*,  $C(0)$ ) must be zero. Why? If you’re producing nothing, then you aren’t forgoing anything by producing, so the cost must be zero.

**Result 12**  $C(0) = 0$ .

Recall from the previous section that

$$MC(n) = C(n) - C(n - 1). \quad (3.1)$$

Using expression (3.1), observe that we can express total cost as follows.

$$\begin{aligned}
 C(n) &= C(n) - C(0) \text{ (recall } C(0) = 0) \\
 &= C(n) - \underbrace{C(n-1) + C(n-1) - \cdots - C(1) + C(1) - C(0)}_{+0} \\
 &= \underbrace{C(n) - C(n-1)}_{MC(n)} + \cdots + \underbrace{C(1) - C(0)}_{MC(1)} \text{ (associative rule)} \\
 &= MC(n) + \cdots + MC(1) = \sum_{j=1}^n MC(j).
 \end{aligned} \tag{3.2}$$

We can summarize this as

**Result 13** *The total cost of producing  $n$  units is the sum of the marginal costs of producing the first  $n$  units.*

Recall that average cost of producing  $n$  units is

$$AC(n) = \frac{C(n)}{n}; \tag{3.3}$$

that is, average cost is total cost divided by the number of units produced. Using Result 13, we can rewrite expression (3.3) as

$$AC(n) = \frac{\sum_{j=1}^n MC(j)}{n}. \tag{3.4}$$

A baseball analogy may help to explain expression (3.4). Think of  $MC(j)$  as being 1 if a batter gets a hit on his  $j$ th at bat and as being 0 if he does not (to simplify matters, assume no walks, no errors, no hit batters, etc.). A batter's average, recall, is just the number of hits he gets divided by his number of at bats. Given that  $MC(j) = 0$  if he doesn't get a hit, it follows that expression (3.4) is the formula for his batting average.

We can also use this baseball analogy to consider how  $AC$  changes with  $MC$ . If a batter gets a hit at his latest at bat, his average goes up (a hit is like batting 1.000, which is greater than his average to that point). If he doesn't get a hit, his average goes down (failing to hit is like batting .000, which is less than his average to that point). In other words, we have<sup>6</sup>

$$\begin{aligned}
 MC(n+1) &> AC(n), \text{ then } AC(n+1) > AC(n); \text{ and} \\
 MC(n+1) &< AC(n), \text{ then } AC(n+1) < AC(n).
 \end{aligned}$$

---

<sup>6</sup>Proof:

$$AC(n+1) = \frac{MC(n+1) + \sum_{j=1}^n MC(j)}{n+1} = \frac{MC(n+1)}{n+1} + \frac{n}{n+1}AC(n).$$

If  $MC(n+1) > AC(n)$ , then

$$AC(n+1) = \frac{MC(n+1)}{n+1} + \frac{n}{n+1}AC(n) > \frac{(1+n)AC(n)}{n+1} = AC(n).$$

The case in which  $MC(n+1) < AC(n)$  is established similarly.

Another way to state this conclusion is

**Result 14** *If average cost is declining, then marginal cost is less than average cost. If average cost is increasing, then marginal cost is greater than average cost.*

## Costs in a Continuous Context | 3.4

For some goods (*e.g.*, liquids) a unit is a fairly arbitrary notion. Not only could we have liters versus quarts, say, but we can also half liters, quarter liters, and so forth. Likewise we could have tons, half tons, quarter tons, and so on of coal or wheat. For such goods it is useful to treat costs as continuous functions of output. Even for goods for which a unit is clearly defined (*e.g.*, shirts), it can prove convenient to act as if a continuum of units can be produced.

In such contexts, we treat the cost function,  $C(x)$ , as being a continuous function of output,  $x$ .<sup>7,8</sup> That is, we can talk of the cost of producing one liter,  $C(1)$ , the cost of producing 2.5 liters,  $C(2.5)$ , and so forth.

We define average cost as before, namely total cost divided by units. That is,  $AC(x) = C(x)/x$ . The average cost of 2.5 liters,  $AC(2.5)$ , would, thus, be  $C(2.5)/2.5$ . Note that because  $C(\cdot)$  is a continuous function,  $AC(\cdot)$  is also a continuous function.

Marginal cost is a bit trickier. The incremental cost of producing  $h$  more units of a good is easy enough to calculate, it is  $C(x+h) - C(x)$ . Marginal cost, however, is a statement about the incremental cost *per unit*. That is, marginal cost can be seen, roughly, as the average of the incremental cost. That is,

$$MC(x) \approx \frac{C(x+h) - C(x)}{h}, \quad (3.5)$$

where the symbol  $\approx$  means approximately equal to.

An analogy may help. Suppose you have driven  $x$  hours. If you drive one hour *more*, then it is easy to calculate your speed:

$$\text{speed} = \frac{(D(x+1) - D(x)) \text{ km}}{1 \text{ hour}},$$

where  $D(\cdot)$  is your distance from your starting place as a function of time. Note the division by 1 hour, reflecting our desire to have speed measured as km per (whole) hour. Of course, we could also determine your speed after you've driven half an hour more:

$$\text{speed} = \frac{(D(x + \frac{1}{2}) - D(x)) \text{ km}}{\frac{1}{2} \text{ hour}}.$$

<sup>7</sup>The cost function is sometimes called the *cost schedule*. In general, the word “schedule” can be read as a synonym for “function.”

<sup>8</sup> $\int dx$  As a technical matter, we also take  $C(\cdot)$  to be differentiable except, possibly, at 0. That is,  $C'(x)$  is defined for all  $x > 0$ .

We divide by  $1/2$  hour because we want to express speed in terms of km per hour, not km per half hour. This explains why we divide by  $h$  in expression (3.5), we want to express the increment in cost per (whole) unit. Of course these speed calculations only tell us the *average* speed you travelled over the hour or half hour respectively. At some points, you might have gone faster and at some points you might have gone slower. For this reason, if we wanted to know your speed at a particular moment in time, we would ideally like to consider a very small time interval—indeed, the smaller the better. That is, if we want to know your speed at time  $x$ , the best measure would be

$$\text{speed} = \frac{(D(x+h) - D(x)) \text{ km}}{h \text{ hour}},$$

where  $h$  is some small fraction of an hour (say a second).



Likewise, we typically want to know not the average incremental cost over some interval, but the incremental cost (per whole unit) at a specific point. We can do this by calculating expression (3.5) with as small an  $h$  as possible. Indeed, we want to do it as  $h$  goes to zero. That is, for continuous cost functions we define  $MC$  by

$$MC(x) = \lim_{h \rightarrow 0} \frac{C(x+h) - C(x)}{h}, \quad (3.6)$$

where “lim” means the limit of that ratio as  $h$  goes toward zero.<sup>9</sup> For instance, suppose that  $C(x) = Ax$ , where  $A$  is a positive constant. Then

$$\begin{aligned} \frac{C(x+h) - C(x)}{h} &= \frac{A(x+h) - Ax}{h} \\ &= \frac{Ah}{h} \\ &= A. \end{aligned}$$

Clearly, as the expression doesn't depend on  $h$ , the limit as  $h$  goes to zero is  $A$ ; that is, if  $C(x) = Ax$ , then  $MC(x) = A$ .

As a second example, suppose  $C(x) = Ax^2 + Bx$ , where  $A$  and  $B$  are con-

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<sup>9</sup>For example, suppose  $C(x) = x^2$  and we wanted to know  $MC(1)$ . If  $h = 1$ , then our approximation is

$$\frac{2^2 - 1^2}{1} = 3.$$

If  $h = 1/2$ , then our approximation is

$$\frac{1.5^2 - 1^2}{.5} = 2.5.$$

If  $h = 1/4$ , the approximation is 2.25. If  $h = 1/8$ , then 2.125. If  $h = 1/1000$ , then 2.001. The limit is the number that this sequence approaches as  $h$  goes to zero, which looks to be 2. Indeed, as shown later (see Result 15), it is 2.

stants. Then

$$\begin{aligned}\frac{C(x+h) - C(x)}{h} &= \frac{A(x^2 + 2xh + h^2) + B(x+h) - Ax^2 - Bx}{h} \\ &= \frac{2Axh + Ah^2 + Bh}{h} \\ &= 2Ax + B + Ah.\end{aligned}$$

As  $h$  goes to zero, this last expression clearly approaches  $2Ax + B$ ; that is, if  $C(x) = Ax^2 + Bx$ , then we've shown that  $MC(x) = 2Ax + B$ .

Sometimes, there is an overhead cost associated with production. For instance, a chemical plant might have the following cost schedule:

$$C(x) = \begin{cases} 0, & \text{if } x = 0 \\ G(x) + F, & \text{if } x > 0 \end{cases};$$

that is, if the firm decides to produce at all, then it incurs an overhead cost of  $F$ ,  $F > 0$ , as well as a variable cost of  $G(x)$ . This cost function is discontinuous at  $x = 0$  and, thus, we cannot define  $MC(0)$ . Nevertheless, we can define  $MC(x)$  for  $x > 0$  using expression (3.6)—the  $F$  doesn't matter if  $x > 0$ :

$$MC(x) = \lim_{h \rightarrow 0} \frac{(G(x+h) + F) - (G(x) + F)}{h} = \lim_{h \rightarrow 0} \frac{G(x+h) - G(x)}{h}.$$

Observe, then, we can summarize our analysis of specific functional forms as follows:

**Result 15** *Let  $C(x) = Ax^2 + Bx + F$ , where  $A$ ,  $B$ , and  $F$  are non-negative constants (i.e., each is greater than or equal to zero). Then  $MC(x) = 2Ax + B$  for  $x > 0$ . If  $F = 0$ , then  $MC(0)$  is defined and is equal to  $B$ .*



### Marginal Cost as a Derivative

If you've had calculus, you no doubt recognize expression (3.6) as the definition of a derivative. That is, we have

$$MC(x) = \frac{d}{dx}C(x) = C'(x).$$

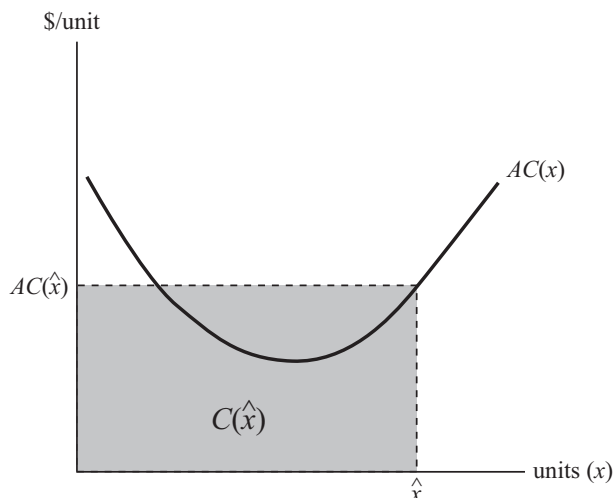
## A Graphical Analysis of Costs

# 3.5

We can also understand the relations among  $C(x)$ ,  $MC(x)$ , and  $AC(x)$  graphically.

For instance, because  $AC(x) = C(x)/x$ , it follows that

$$C(x) = x AC(x). \quad (3.7)$$



**Figure 3.1:** A graph of an average cost schedule. The total cost of  $\hat{x}$  units is the gray rectangle whose width is  $\hat{x}$  and whose height is  $AC(\hat{x})$ .

We can illustrate this graphically. See Figure 3.1. In this figure, we have plotted  $AC$  as a function of units,  $x$ . Because  $AC$  is in dollars per unit, the vertical axis in Figure 3.1 is in terms of \$/unit. Consider a specific number of units,  $\hat{x}$ . Notice that we can consider  $\hat{x}$  to be the width of a rectangle whose height is  $AC(\hat{x})$ . The area of this rectangle (shown in gray in Figure 3.1) is its width times height, or  $\hat{x} AC(\hat{x})$ , which, from expression (3.7), is  $C(\hat{x})$ . That is, the total cost of  $\hat{x}$  units can read off the graph of average cost as the area of the rectangle formed by  $\hat{x}$  and  $AC(\hat{x})$ .

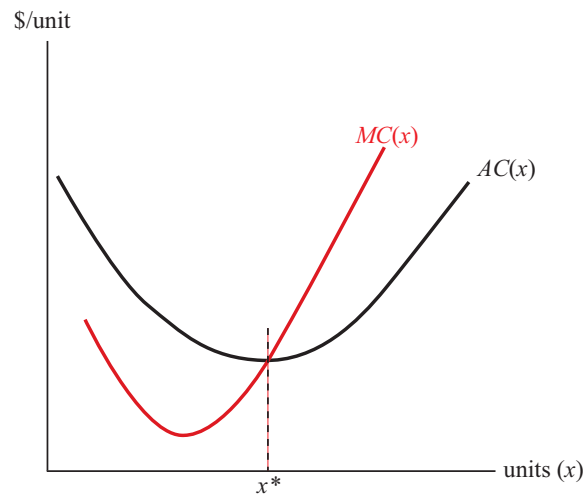
We can also relate  $MC$  and  $AC$ . Consider Figure 3.2. The average cost schedule is the same as in Figure 3.1. To this, we have added the marginal cost schedule shown in red. Consistent with Result 14, we see that  $MC(x) < AC(x)$  for  $x$  at which  $AC(\cdot)$  is decreasing (headed downward) and that  $MC(x) > AC(x)$  for  $x$  at which  $AC(\cdot)$  is increasing (headed upward). It follows, therefore, that at the point at which  $AC(x)$  is at a minimum,  $x^*$ ,  $MC$  and  $AC$  must coincide; that is,  $MC(x^*) = AC(x^*)$ .

**Result 16** *The marginal cost schedule intersects the average cost schedule at the minimum of average cost.*<sup>10</sup>

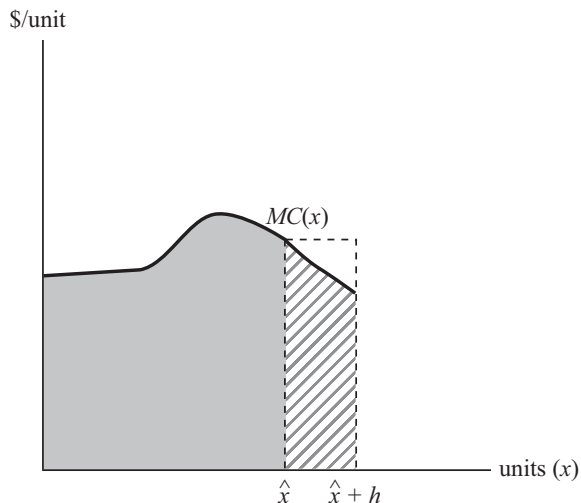
<sup>10</sup>  $\int dx$  Observe that, because  $AC(x) = C(x)/x$ , it follows that

$$AC'(x) = \frac{xC'(x) - C(x)}{x^2} = \frac{xMC(x) - C(x)}{x^2}.$$

When  $AC$  is at a minimum,  $AC'(x) = 0$ . From the last expression, this means the numerator



**Figure 3.2:** A graph of an average cost schedule (in black) and the corresponding marginal cost schedule (in red). Observe that AC is falling when it is above MC and is rising when it is below MC. At the minimum of AC, AC and MC are equal. .



**Figure 3.3:** A marginal cost schedule has been plotted. The area beneath the marginal cost schedule between two points corresponds to the difference in total costs between those two points.

Finally, we can relate  $MC$  and total cost graphically. Figure 3.3 shows a plot of the marginal cost schedule. Because  $MC$  is expressed in terms of dollars per unit, the vertical axis is labeled  $\$/unit$ . Consider the cross-hatched area beneath the marginal cost schedule between  $\hat{x}$  and  $\hat{x} + h$ . That area is approximately equal to the area of the dotted-line rectangle that surrounds it. The width of the rectangle is  $h$  and the height is  $MC(\hat{x})$ . So the area of that rectangle is  $h MC(\hat{x})$ . But from expression (3.5) on page 45, we know that

$$h MC(\hat{x}) \approx C(\hat{x} + h) - C(\hat{x}). \quad (3.8)$$

Moreover, this approximation (expression (3.8)) is more accurate the smaller is  $h$ .<sup>11</sup> But this implies that the area beneath the marginal cost schedule between  $x_1$  and  $x_2$  is  $C(x_2) - C(x_1)$ . Why? Because, if we define  $h = (x_2 - x_1)/N$  for a

of the last fraction equals zero:

$$xMC(x) - C(x) = 0.$$

Dividing both sides by  $x$  and rearranging we obtain

$$MC(x) = C(x)/x = AC(x).$$

Observe this argument entails that at *every* local maxima and minima of  $AC(\cdot)$ ,  $AC = MC$ .

<sup>11</sup>You may find it helpful to review the discussion in Appendix Sections A3.4 and A3.5 before reading the rest of this section.

very large  $N$  (so  $h$  is very small), we have

$$\begin{aligned}
 C(x_2) - C(x_1) &= C(x_2) + \underbrace{(C(x_1 + (N-1)h) - C(x_1 + (N-1)h))}_{+0} + \cdots \\
 &\quad \cdots + \underbrace{(C(x_1 + h) - C(x_1 + h))}_{+0} - C(x_1) \\
 &= \left( C(x_2) - (C(x_1 + (N-1)h)) \right) + \cdots + \left( C(x_1 + h) - C(x_1) \right) \quad (\text{rearranging terms}) \\
 &\approx h MC(x_1 + (N-1)h) + \cdots + h MC(x_1) \quad (\text{expression (3.8)}) \\
 &= \sum_{n=0}^{N-1} h MC(x_1 + nh),
 \end{aligned}$$

where the approximation gets better the larger is  $N$  (the smaller is  $h$ ). Because the last sum is the areas of a bunch of rectangles like the one shown in Figure 3.3, it approximates the area under the marginal cost curve between  $x_1$  and  $x_2$  and equals that area as we let the number of rectangles,  $N$ , get arbitrarily large (let  $h$  tend to zero). To conclude:

**Result 17** *The area beneath the marginal cost curve between two points,  $x_1$  and  $x_2$ ,  $0 < x_1 < x_2$ , equals  $C(x_2) - C(x_1)$ . If  $MC(0)$  is defined (i.e., there is no overhead cost), then we can let  $x_1 = 0$ ; hence, in this case,  $C(x)$  is the area beneath the marginal cost curve from 0 to  $x$ .<sup>12</sup>*

The last sentence of Result 17 is illustrated in Figure 3.3: As drawn,  $MC(0)$  is defined. Hence,  $C(\hat{x})$  is the solid gray area under the marginal cost schedule between 0 and  $\hat{x}$ .

It should be noted that Result 17 is a general result concerning the relation between marginal and total functions:

**Result 18** *The area beneath any marginal curve,  $MZ(\cdot)$ , between two points,  $x_1$  and  $x_2$ ,  $x_1 < x_2$ , equals  $Z(x_2) - Z(x_1)$ , where  $Z(\cdot)$  is the corresponding total schedule.<sup>13</sup>*

What if there is an overhead cost? Then  $MC(0)$  is not well defined, but we can consider the  $MC$  schedule as starting arbitrarily close to 0. The area to the left of our arbitrarily close starting point is negligible and can be ignored. Hence, the area under the marginal cost curve starting almost from zero to some point  $x$  must be the total *variable* cost of  $x$  units. Total cost is the sum of variable cost and overhead cost. Hence,  $C(x)$  is the area under the marginal cost schedule starting almost at 0 to  $x$  plus the overhead cost.

**Recall:** The sum of variable and overhead cost is total cost.

<sup>12</sup>Recall  $C(0) = 0$ .

<sup>13</sup> $\int dx$  This is just the fundamental theorem of calculus.

**Result 19** *If there is an overhead cost  $F > 0$ , then*

$$C(x) = (\text{Area under } MC \text{ from almost } 0 \text{ to } x) + F.$$

### $\int dx$ Total Cost as an Integral

Write

$$C(x) = \begin{cases} 0, & \text{if } x = 0 \\ G(x) + F, & \text{if } x > 0 \end{cases},$$

where  $F \geq 0$  is overhead cost and  $G(\cdot)$  is the *variable* cost function. Assume  $G(\cdot)$  is differentiable. By the definition of a variable cost, it must be that  $G(0) = 0$ .

Observe that for  $x > 0$ ,  $MC(x) = G'(x)$ . Because a single point cannot affect the value of an integral, there is no loss in our treating  $MC(0)$  as equaling  $G'(0)$ ; that is, it is immaterial how we define  $MC(0)$  with respect to integration, so we're free to define it as  $G'(0)$ .

By the fundamental theorem of calculus,

$$G(x_2) - G(x_1) = \int_{x_1}^{x_2} G'(x)dx = \int_{x_1}^{x_2} MC(x)dx. \quad (3.9)$$

Hence, the area under the marginal cost schedule between  $x_1$  and  $x_2$  is the difference in variable cost from producing  $x_2$  units as opposed to  $x_1$  units. For  $x_2 > x_1 > 0$ ,

$$C(x_2) - C(x_1) = (G(x_2) + F) - (G(x_1) + F) = G(x_2) - G(x_1).$$

Therefore, we have

$$C(x_2) - C(x_1) = \int_{x_1}^{x_2} MC(x)dx \quad (x_2 > x_1 > 0);$$

that is, Result 17.

If we let  $x_1 = 0$  in expression (3.9), we obtain

$$G(x_2) = \int_0^{x_2} MC(x)dx$$

(recall  $G(0) = 0$ ). Substituting into the definition of  $C(\cdot)$ , we arrive at

$$C(x_2) = G(x_2) + F = \int_0^{x_2} MC(x)dx + F.$$

## The Cost of Capital | 3.6

As we saw in Section 3.1, whether an expenditure is a cost or not (*i.e.*, sunk) can depend on the decision-making horizon. Recall, for instance, Example 1.

Once locked into a lease, rent payments are not a cost. But, at the time of lease renewal, they are a cost.

One *might* view capital expenditures (*e.g.*, machinery, vehicles, plant and facilities, etc.) similarly. That is, once purchased, the purchase price is a sunk expenditure. Prior to purchase, the purchase price is a cost. Yet, while it is definitely true that the price paid is sunk after purchase, this does not entail there is no cost to *using* the capital. It would be costless only if, as with the lease, there was nothing else that could be done. Typically, however, there is at least one thing else that can be done with capital assets: resale.

The ability to resell capital goods creates two sources of imputed cost. To identify them, consider a relevant decision-making horizon; namely, the time between points at which you can resell the capital good or asset. This might, for instance, be a day, a month, or longer depending on circumstances. Let  $V_0$  be the value—the resale price—for this asset at the beginning, time 0, of the time period and let  $V_1$  be its value—resale price—at the end of the period, time 1. Let  $r$  be the amount that the firm can earn per dollar during this time period on money optimally invested (this could just be the interest rate on funds banked). If the firm sold the asset at the beginning of the period, time 0, then, at time 1, it would have  $(1 + r)V_0$ . If it sells it at the end of the period, time 1, then it would have  $V_1$ . The difference,  $(1 + r)V_0 - V_1$ , is what is forgone by using the asset over the time interval; that is, it is the cost of using the asset.

Decompose this cost as follows:

$$\text{capital cost} = \underbrace{rV_0}_{\text{forgone return}} + \underbrace{(V_0 - V_1)}_{\text{depreciation}}. \quad (3.10)$$

**Depreciation** is the amount by which the resale value of an asset falls over time. Depreciation is driven by two factors. First, using an asset often means wear and tear, reducing the asset's productive life and, hence, its resale value. Second, changes that take place in technology over time can also lower resale value (consider, *e.g.*, that a computer with a 486-chip, even if never used, is worth far less today than when it was first sold).

Divide the cost of capital, expression (3.10), by  $V_0$ . This yields a capital-cost rate equal to  $r + \delta$ , where

$$\delta = \frac{V_0 - V_1}{V_0}$$

is the **rate of depreciation**. From our earlier discussion, observe that  $\delta$  is greater the greater is the rate of wear and tear and the greater is the rate of technology change.

**Example 9:** Suppose you are in the business of renting residential housing. In a stable housing market, the resale value of a house today is essentially equal to its resale value next month (at least assuming normal use). Ignoring tax considerations and the “joys of home ownership,” the value of a house to a rational consumer is equal to the discounted value of the rent payments she would have to make for equivalent housing; that

is,<sup>14</sup>

$$V = \sum_{t=1}^{\infty} \frac{\rho}{(1+r)^t} = \frac{\rho}{r},$$

where  $\rho$  is the rental price. If you and your tenants face the same rate of return (*i.e.*, interest rate), then, in a stable housing market, it will be the case that rental prices and house prices are related by the condition

$$\rho = rV.$$

In other words, rental prices should equal the cost of the capital that the house represents.

Let's continue this example by supposing that house prices appreciate suddenly. Then rents will go up, because  $V_1 > V_0$ , but those tenants *with* leases will find themselves paying below market rates. This explains why, in an appreciating housing market, average house prices are seen to be rising faster than average rents.

Further consider a housing market with a forecast appreciation rate of  $\alpha$ , driven, perhaps, by in migration due to a strong labor market. The rate  $\alpha$  is like a negative  $\delta$ , so the cost of renting a house over the relevant period is  $(r - \alpha)V$ . Consequently, you, as a landlord, make a profit for any rent  $\rho \geq (r - \alpha)V$ .

It is important to observe that depreciation (or appreciation) is determined by the resale value of the asset, which is a function of the actual wear and tear on the asset, as well as trends in the resale market (*e.g.*, technological changes). This should be contrasted with *accounting measures* of depreciation. Accounting measures employ formulæ that has little to no relation to actual changes in market value. For instance, the IRS states that cars and computers have a useful life of five years and allows them to be depreciated at the same rate. That is, the accounting procedure acts as if their market values fall at the same rate. Yet, experience tells us that the rates at which cars and computers actually lose market value are very different.

## Costs and Returns to Scale

# 3.7

A firm has *increasing returns to scale* if average cost,  $AC(x)$ , is decreasing in output,  $x$ . Decreasing average costs also tell us that total costs must increase less than proportionally with an increase in output; that is, if the firm enjoys increasing returns to scale, then

$$C(\beta x) < \beta C(x) \tag{3.11}$$

---

<sup>14</sup>We can treat the value of the house as a perpetuity because, even though any one owner won't live forever, when he or she sells it, he or she will capture its expected future value (including expected future resale value).

for all  $x > 0$  and all  $\beta > 1$ . To validate expression (3.11), divide both sides by  $\beta x$ :

$$\frac{C(\beta x)}{\beta x} < \frac{C(x)}{x} \quad \text{or, equivalently,} \\ AC(\beta x) < AC(x) \quad (\text{definition of } AC),$$

where the last line must be true because  $AC(\cdot)$  is decreasing if the firm has increasing returns to scale.

A firm has *decreasing returns to scale* if average cost is increasing in output. Similar to our conclusion about increasing returns to scale, decreasing returns to scale imply that total costs increase more than proportionally with an increase in output; that is, decreasing returns to scale imply

$$C(\beta x) > \beta C(x) \quad (3.12)$$

for all  $x > 0$  and all  $\beta > 1$ . To validate this last expression, again divide both sides by  $\beta x$ :

$$AC(\beta x) > AC(x),$$

which holds because  $AC(\cdot)$  is increasing if the firm has decreasing returns to scale.

A firm has *constant returns to scale* if average cost remains constant as output increases. This entails that total cost must rise proportionally with an increase in output; that is, constant returns to scale imply

$$C(\beta x) = \beta C(x) \quad (3.13)$$

for all  $x$  and all  $\beta$ . Expression (3.13) can be validated using the same steps as we employed to validate expressions (3.11) and (3.12).

What determines returns to scale? Returns to scale relate to the underlying production process and the markets for inputs. Some production processes require large overhead expenses (*e.g.*, building or maintaining a factory, engineering, corporate staff and other non-production-line workers, etc.) Because average cost is the sum of overhead and variable cost, average cost should decrease with output—the firm enjoy increasing returns to scale—if variable costs aren't rising too quickly with output. For instance, if  $C(x) = F + cx$  for  $x > 0$ , where  $F > 0$  is overhead and  $c > 0$  is a constant (note  $cx$  is the variable cost of  $x$  units), then

$$AC(x) = \frac{F}{x} + c,$$

which is clearly decreasing in  $x$ .

In other contexts, as a firm expands its output it finds it harder (*i.e.*, more expensive) to acquire the necessary inputs (*e.g.*, raw materials have to be shipped from farther away, new workers need to be trained, etc.). If the firm is large enough, its increased demand for some inputs can drive up the market price of those inputs (*e.g.*, General Motors could not double its production of cars

without driving up the price of steel). Expansion could also strain control mechanisms, leading to problems with quality (so number of saleable units per amount of inputs falls) or other inefficiencies. Running machines faster could increase their wear and tear (*i.e.*, rate of depreciation) and increase the amount of maintenance they require. In such cases, average cost would clearly be rising; that is, the firm suffers from decreasing returns to scale. For instance, if  $C(x) = x^2$ , then  $AC(x) = x$ , which is clearly increasing in output.

In many cases, a firm enjoys increasing returns to scale at low levels of output—dividing a large amount of overhead over more units dominates any other effect—but suffers decreasing returns to scale at high levels of output, perhaps for the reasons outlined in the previous paragraph. So initially, average cost is falling, then it is increasing. That is, it resembles the average cost curves shown in Figures 3.1 and 3.2. For obvious reasons, such average cost curves are called *U-shaped average cost curves*.



Finally, we observe that if a firm has constant returns to scale at all levels of output, then its total cost,  $C(x)$ , must equal  $cx$ , where  $c$ , a positive constant, is also marginal cost. To see this, observe that because  $AC(\cdot)$  is always the same, a constant, there must exist some number to which it is equal. Call that number  $c$ . So we have  $AC(x) = c$  for all  $x$ . Multiply both sides by  $x$ , so that  $x AC(x) = cx$ . Recall that  $C(x) = x AC(x)$ , so we see that  $C(x) = cx$  as claimed. Moreover, using Result 15 with  $A = F = 0$ , we have that  $MC(x) = c$ .

## Introduction to Cost Accounting

# 3.8

This section introduces and explains some cost-accounting terminology. As we saw with depreciation, accounting terminology overlaps considerably with economic terminology. The problem is that, although the words are the same, their meanings are not. What are deemed costs by accountants may not be costs as we've defined them or they may not be properly allocated for the purposes of making sound managerial decisions. In addition, some true costs can fail to be accounted for by the accounting system.

On the other hand, accounting information is valuable information—it is not to be disregarded. To exploit this information, however, one must learn to read the information correctly and not be misled by the terminology.

### The Principal Goal of Accounting

Although cost accounting serves many purposes, its principal purpose is to *account* for the consumption of resources by the firm. In essence, accounting seeks to determine where the money went. This is clearly a necessary activity in any firm.

### The Problem and an Example

A problem is that, in addition to *accounting* for expenditures, most accounting methods also *allocate* expenditures. This allocation can be at odds with what economics tells us.

To consider a simple scenario, suppose that a firm makes lefthanded scissors and righthanded scissors on the same machine. The labor required for each pair of scissors (unit) made is \$1. The raw inputs for each pair of scissors is \$1. In addition, each time the machine is switched from one type of scissors to the other, a set-up cost of \$200 is incurred. Suppose the machine is set-up once at the beginning of the day for righthanded scissors and, then, later in the day it is switched to lefthanded scissors. Finally, suppose 900 righthanded scissors and 100 lefthanded scissors are produced each day. Daily expenditure is, therefore, \$2400. A traditional accounting method for *allocating* these expenditures would call the \$2 per scissors incurred from labor and raw inputs a *direct* cost. Hence, righthanded scissors incur \$1800 per day in direct costs, while lefthanded scissors incur \$200 per day in direct costs. This traditional method would also treat the set-ups as *overhead* and allocate this overhead between the two types of scissors on the basis of units produced. Consequently, 90% of the \$400 in daily overhead would be allocated to righthanded scissors and 10% of the \$400 would be allocated to lefthanded scissors. The total “cost” of righthanded scissors would be \$2160 and the total “cost” of lefthanded scissors would be \$240 (note these sum to \$2400). The per-unit “cost” of righthanded scissors would, therefore, be \$2.40 and the per-unit “cost” of lefthanded scissors would also be \$2.40. This allocation of expenditures would make it seem that righthanded and lefthanded scissors were equally expensive to produce. This conclusion, however, is wrong!

To see why it’s wrong, suppose the firm in question made only righthanded scissors and was considering adding lefthanded scissors. How would expenditures change? Making only righthanded scissors, there are no set-ups (except on the very first day of production). So adding lefthanded scissors means incurring an additional \$400 per day in set-up costs plus \$200 in direct costs—a total of \$600. If the price at which scissors were sold was \$5 per pair, then it would not be profitable to add lefthanded scissors: total additional revenue = \$500 < \$600 = total additional cost. A conclusion that would have been missed if the traditional allocation had been used.<sup>15</sup>

What this example illustrates is the danger of assigning costs by any means other than cost causation.

**Moral:** Assign costs by cost causation.

### Accounting Terminology

*Basis of allocation* refers to how *shared overhead* (see below) is allocated among different products. In the example above, set-up, which was treated as shared

<sup>15</sup>To verify: Daily profits if the firm produces both types is \$2600 (= \$5000 – \$2400). Daily profit if it produces only righthanded scissors is \$2700 (= 4500 – \$1800).

overhead, was allocated on the basis of units produced. Other common bases of allocation are labor hours and machine hours. A synonym for basis is *overhead rate*.

**Direct costs** are expenditures that can be traced to a single product or activity. In the example above, the \$2 in labor and raw inputs are direct costs. See, also, *variable costs* below.

**Direct labor** is the compensation paid to employees whose time and effort can be traced to the product in question. The \$1 in labor of each unit in the example above would represent direct labor.

**Direct materials** are the raw inputs that can be traced directly to the product in question. The \$1 in raw inputs for each unit in the example above would represent direct materials.

**Fixed costs** are expenditures that do not vary with the number of units produced or that are not always increasing with the number of units produced.

**Manufacturing cost** is the sum of *direct labor*, *direct materials*, *manufacturing overhead*, and work beginning in process inventory minus work ending in process inventory.

**Manufacturing overhead** is all manufacturing expenditures except for *direct costs*. Included in manufacturing overhead are indirect materials, indirect labor, maintenance, utilities, rent, insurance, depreciation, and taxes. Manufacturing overhead is a *product cost* (see below). Manufacturing overhead is sometimes called *direct overhead* or *factory overhead*.

**Overhead** refers to expenditures that are not directly attributed to a given unit of output. For instance, in the example above, set-up costs are considered overhead because they are not directly attributed to a given unit of output.<sup>16</sup>

**Overhead pools** are different accounts into which *overhead* may be divided if the *basis of allocation* for these different accounts differ. For instance, if some overhead is to be allocated on the basis of labor hours and other overhead is to be allocated on the basis of machine hours, then the first set of overhead would be one pool and the second set of overhead would be another pool.

**Period costs** are those expenditures incurred during a given period of time (usually the period that an accounting statement covers). They are not allocated to the production process itself. If, in the above example, the firm in question spent \$100 per day marketing scissors, then this \$100 would be a period cost (assuming, unrealistically, that a daily accounting statement was produced for the entire firm). Period costs are necessarily *overhead*.

**Product costs** are those costs incurred during production that can be allocated to production itself. In the example above, all the expenditures would be treated

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<sup>16</sup>From an *economic* standpoint, however, we could attribute the set-up costs to a given unit, namely the first unit produced after the machine is switched. That is, for example, the *marginal cost* schedule for lefthanded scissors is  $MC(1) = \$202$  and  $MC(2) = \dots = MC(100) = \$2$ .

as product costs.

**Shared overhead** is overhead common to more than one product. In the example above, the salary of the factory manager would be shared overhead.

**Unit cost** is the accounting cost of a product divided by the number of units produced. In the example above, the unit cost of a pair of scissors was \$2.40. Note that the unit cost is *not* the same as *marginal cost*. When accounting cost equals economic cost, a synonym for unit cost is *average cost*.

**Variable costs** are expenditures that vary with each unit produced. See *direct costs*.

### An Additional Example: Red Pens and Blue Pens

**Note:** This subsection is taken from an old handout by Professor Hermalin, “The Parable of Red Pens and Blue Pens.”<sup>17</sup> It has been left in its original, parable, form.

Once upon a time a little firm made two products, red pens and blue pens. Each pen used 15 cents worth of labor and raw materials. Each pen was run through a machine, the daily cost of which was \$1000 regardless of how many pens were run through or their colors. The firm could sell the first 5000 red pens it made each day at 30 cents each; additional red pens, however, were sold at 20 cents per pen. The firm could sell all the blue pens it wanted at 25 cents per pen. The firm, however, could make no more than 8000 pens a day and it could not expand over its relevant decision-making horizon. The little firm chose to manufacture 5000 red pens and 3000 blue pens a day for a daily profit of

$$\$50 \left( = 5000 \times (.30 - .15) + 3000 \times (.25 - .15) - 1000 \right).$$

As this was greater than \$0, the little firm was happy to produce.

One day an evil accountant came along and said the little firm should adopt an accounting system that allocated shared overhead (e.g., the \$1000 for the aforementioned machine). Being naïve, the little firm went along. The accountant chose to allocate the shared overhead on the basis of output—thus, the red-pen line was billed \$625, which is five eighths of \$1000,<sup>18</sup> and the blue-pen line was billed the remaining \$375. The new accounting is shown in Table 3.1.

Upon examining the accounting data, the evil accountant snickered, “Aha! Your blue-pen line is unprofitable—you should shut it down.” Dutifully, the naïve little firm shut down its blue-pen line and switched over to producing nothing but red pens. Now the firm’s revenues were

$$\$2100 \left( = 5000 \times .3 + 3000 \times .2 \right).$$

<sup>17</sup>Copyright © 1996 Benjamin E. Hermalin. All rights reserved.

<sup>18</sup>Allocating on the basis of output means taking the output of the red-pen line, 5000 pens, and dividing it by total output, 8000 pens, to get the red-pen line’s share. Similarly, the blue-pen line’s share would be 3000/8000 or  $\frac{3}{8}$ .

|              | Pens | Revenue | Direct Cost | Shared Overhead | Total Expense | Profit |
|--------------|------|---------|-------------|-----------------|---------------|--------|
| Red Pens     | 5000 | \$1500  | \$750       | \$625           | \$1375        | \$125  |
| Blue Pens    | 3000 | \$750   | \$450       | \$375           | \$825         | -\$75  |
| <b>Total</b> | 8000 | \$2250  | \$1200      | \$1000          | \$2200        | \$50   |

**Table 3.1:** The Evil Accountant's New Accounting System

Because the blue-pen line was shut, the \$1000 cost of the machine was fully allocated to the red-pen line. The firm's new accounting is shown in Table 3.2.

|              | Pens | Revenue | Direct Cost | Shared Overhead | Total Expense | Profit |
|--------------|------|---------|-------------|-----------------|---------------|--------|
| Red Pens     | 8000 | \$2100  | \$1200      | \$1000          | \$2200        | -\$100 |
| Blue Pens    | 0    | \$0     | \$0         | \$0             | \$0           | \$0    |
| <b>Total</b> | 8000 | \$2100  | \$1200      | \$1000          | \$2200        | -\$100 |

**Table 3.2:** New Accounting After Blue-Pen Line Shut

Upon examining the accounting data, the evil accountant chortled, "Aha! Your entire company is unprofitable—you should shut down completely." Ductifully, the naïve little firm did, shutting its doors forever. So thanks to the evil accountant, the little firm went from making a tidy profit of \$50 a day to going out of business!

*Moral of the story:* Don't allocate shared overhead, it can only lead to dopey decisions.

In the lefthanded-righthanded-scissors example, the problem was that the accounting system missed that an overhead cost should properly be allocated fully to lefthanded scissors—this is a case where allocatable overhead is mistakenly treated as shared, with the mistake further compounded by allocating it on the basis of an *ad hoc* rule. In the red-pens-blue-pens example, the overhead cost shouldn't be allocated to either line—it is *true* shared overhead. In both examples, because the basis of allocation had nothing to do with cost causation, the accounting gave a misleading picture of what was going on.

**Shared overhead allocation:** Has nothing to do with good decision making.

## Summary | 3.9

The key takeaways of this lecture note are:

- To make the best business decisions, remember the cost of something is the value of the best forgone alternative. That is, use the notion of *opportunity cost*.
- $\text{Cost} = \text{Expenses} - \text{Sunk expenditures} + \text{Imputed costs}$ .

- 
- Sunk expenditures are *irrelevant* for decision making.
  - Understand marginal cost ( $MC$ ) and average cost ( $AC$ ), as well as their relations between each other and with total cost ( $C$ ).
  - Understand the cost of capital.
  - Returns to scale determine the shape of the average cost curve. Specifically, increasing returns to scale mean a decreasing average cost curve; constant returns to scale mean a flat average cost curve; and decreasing returns to scale mean an increasing average cost curve. Many firms have U-shaped average cost curves.
  - Cost causation is the right way to allocate costs. Allocation of overhead on any other basis will be misleading for decision making (recall lefthanded and righthanded scissors).
  - Shared overhead should not be allocated for the basis of decision making (recall red pens and blue pens).



# Introduction to Pricing

# 4

This lecture note describes how to figure out the optimal price and quantity if your firm is engaging in [simple pricing](#); that is, pricing in which the firm sets a given price per unit, which is paid by all customers, each of whom is free to buy as many units at that price as he or she desires.<sup>1</sup> After defining “optimal,” we discuss the concept of marginal revenue at length. While there is some involved analysis required, the topic justifies the pain. Optimal pricing and quantity setting in this context means making as much money as possible. The important takeaways from this note are

- Marginal revenue equals marginal cost at the optimal quantity produced (this equality may be approximate in the case of discrete goods).
- Marginal revenue comes from an underlying demand curve. You should know how to derive a marginal revenue curve from a demand curve assuming simple pricing.
- Demand curves themselves come from consumer preferences and from the prices of all goods. You should be able to evaluate the effects of changes in preferences and these other prices on your best choices.

## Simple Pricing Defined

## 4.1

This note discusses optimal pricing of a product by a firm that must charge the same price per unit to all of its consumers.

**Definition 8** *A firm engages in simple pricing for a particular product if that product is sold for the same price per unit no matter who the buyer is or how many units the buyer purchases.*

Observe that simple pricing is [nondiscriminatory pricing](#).

In subsequent notes, we will consider the possibility that a firm chooses to charge either different prices to different individuals or prices in such a way that a consumer’s expenditure is not a simple multiple of a per-unit price (*i.e.*, his or her expenditure is not proportional to quantity purchased). Simple pricing applies when the identity of the buyer cannot be observed or inferred at reasonable cost. It also applies when the seller cannot prevent [arbitrage](#) among buyers

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<sup>1</sup>This is sometimes called [linear pricing](#).

when buyers can purchase multiple units. Arbitrage means taking advantage of an ability to buy low and sell high. So, if one set of buyers faces a lower price than another set and the former can resell to the latter, then the consequent arbitrage opportunity will force a single price to hold. For example, if a chemical producer proposed to charge a different price for the same chemical depending on the buyer's industry, then those buyers in the industry facing a low price could resell to those buyers in the industry facing the high price and reap a profit. The producer would, effectively, be limited to a single price, the lowest one it sets.

## Profit Maximization

## 4.2

**Note:** The topics considered in Sections 4.2–4.4 are quite general and pertain not only to simple pricing, but other forms of pricing as well.

**Profit:** *Is revenue minus cost.*

A firm's *profit* is the revenue it takes in minus its cost. If we let  $R(x)$  be the revenue from selling  $x$  units, then its profit from selling  $x$  units,  $\pi(x)$ , is  $R(x) - C(x)$ , where  $C(x)$  is the total cost of  $x$  units.

If the firm sets a price of  $p$  per unit—engages in simple pricing—then  $R(x) = px$ .

In choosing the amount to produce and sell, the firm seeks to find the quantity,  $x$ , that maximizes profit,  $\pi(x)$ . Under the real-life conditions that govern revenue and cost, an  $x$  that maximizes  $\pi(x)$  must exist. Let's use an asterisk to denote that quantity (*e.g.*,  $x^*$ ,  $n^*$ , etc.).

Suppose, first, that we are considering a discrete good (*e.g.*, shirts) rather than a continuous good (*e.g.*, a liquid). Saying that  $n^*$  is the profit-maximizing amount is the same as saying that  $\pi(n^*) \geq \pi(n)$  for all other  $n$ . In particular, consider the quantities  $n^* - 1$  and  $n^* + 1$ . We know that

$$\pi(n^*) \geq \pi(n^* - 1) \text{ and} \quad (4.1)$$

$$\pi(n^* + 1) \leq \pi(n^*) . \quad (4.2)$$

Substituting  $R(n) - C(n)$  for  $\pi(n)$ , this yields

$$\begin{aligned} R(n^*) - C(n^*) &\geq R(n^* - 1) - C(n^* - 1) \text{ and} \\ R(n^* + 1) - C(n^* + 1) &\leq R(n^*) - C(n^*) , \end{aligned}$$

or, rearranging,

$$\begin{aligned} R(n^*) - R(n^* - 1) &\geq \underbrace{C(n^*) - C(n^* - 1)}_{MC(n^*)} \text{ and} \\ R(n^* + 1) - R(n^*) &\leq \underbrace{C(n^* + 1) - C(n^*)}_{MC(n^*+1)} . \end{aligned}$$

If we define  $MR(n) = R(n) - R(n - 1)$ , we can rewrite this as

$$MR(n^*) \geq MC(n^*) \text{ and} \quad (4.3)$$

$$MR(n^* + 1) \leq MC(n^* + 1). \quad (4.4)$$

What is  $MR(n)$ ? It is the change in revenue incurred by selling the  $n$ th unit rather than selling only  $n - 1$  units and is called **marginal revenue**.

Expression (4.3) tells us that for  $n^*$  to be the profit-maximizing quantity, then the marginal revenue from the  $n$ th unit needs to be at least as great as the marginal cost of the  $n$ th unit—which makes sense; if it *weren't* true (i.e.,  $MR(n^*) < MC(n^*)$ ), then, whatever the gain in revenue from the  $n^*$ th unit, it is outweighed by the additional cost of producing the  $n^*$ th unit. Hence, it wouldn't be sensible to produce  $n^*$  units ( $n^* - 1$  units would be better). Expression (4.4) tells us that for  $n^*$  to be the profit-maximizing quantity, then the marginal revenue from the  $n^* + 1$ st unit cannot exceed the additional cost incurred by producing the  $n^* + 1$ st unit; that is, that  $n^* + 1$  units cannot be better than  $n^*$  units.

**Marginal revenue:**  
The change in revenue from selling an additional unit.

**Result 20** A necessary condition for  $n^*$  to be the profit-maximizing output is that expressions (4.3) and (4.4) both hold true.

## The Continuous Case

# 4.3

As noted in Section 3.4, for some goods a unit is a fairly arbitrary notion and amounts of such goods can be treated as continuous. Moreover, even for discrete goods, it is simply easier sometimes to treat quantity as an approximately continuous variable.

As with cost, we need to define marginal revenue in the continuous context. To that end, by analogy to expressions (3.5) and (3.6) in Section 3.4, we define  $MR$  as

$$MR(x) = \lim_{h \rightarrow 0} \frac{R(x + h) - R(x)}{h}. \quad (4.5)$$

For example, if  $R(x) = Ax - Bx^2$ , where  $A$  and  $B$  are constants,  $A > 0$  and  $B \geq 0$ , then

$$\begin{aligned} MR(x) &= \lim_{h \rightarrow 0} \frac{(A(x + h) - B(x + h)^2) - (Ax - Bx^2)}{h} \\ &= \lim_{h \rightarrow 0} \frac{Ax + Ah - Bx^2 - 2Bxh - Bh^2 - Ax + Bx^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{Ah - 2Bxh - Bh^2}{h} \\ &= \lim_{h \rightarrow 0} A - 2Bx - Bh \\ &= A - 2Bx. \end{aligned}$$

It will turn out to be useful to memorialize this example:

**Result 21** If  $R(x) = Ax - Bx^2$ ,  $A > 0$  and  $B \geq 0$ , then  $MR(x) = A - 2Bx$ .

In the continuous case, we can rewrite the conditions for  $x^*$  to be optimal, expressions (4.1) and (4.2), as

$$\begin{aligned}\pi(x^*) &\geq \pi(x^* - h) \text{ and} \\ \pi(x^* + h) &\leq \pi(x^*).\end{aligned}$$

Substituting  $R(x) - C(x)$  for  $\pi(x)$  and rearranging yields

$$\begin{aligned}R(x^*) - R(x^* - h) &\geq C(x^*) - C(x^* - h) \text{ and} \\ R(x^* + h) - R(x^*) &\leq C(x^* + h) - C(x^*).\end{aligned}$$

Dividing both sides of these inequalities by  $h$  and taking limits as  $h$  goes to zero yields

$$\begin{aligned}MR(x^*) &\geq MC(x^*) \text{ and} \\ MR(x^*) &\leq MC(x^*).\end{aligned}$$

The only way both inequalities can be satisfied is if

$$MR(x^*) = MC(x^*). \quad (4.6)$$

We can conclude, therefore, as follows.

**Result 22 (MR = MC rule)** In the continuous case, a necessary condition for  $x^*$  to be the profit-maximizing output is that  $MR(x^*) = MC(x^*)$ .

### $\int dx$ Optimization

If you've had calculus, you no doubt recognize expression (4.5) as the definition of a derivative. That is, we have

$$MR(x) = \frac{d}{dx}R(x) = R'(x).$$

Consider maximizing profits. We know that a function (e.g.,  $\pi(\cdot)$ ) is increasing wherever its derivative is positive and decreasing wherever its derivative is negative. It follows, therefore, that at an optimum, the derivative must be zero (i.e., the top of a hill is flat). Hence, if  $x^*$  is the profit-maximizing quantity, then

$$\frac{d}{dx}\pi(x)|_{x=x^*} = 0;$$

that is,

$$\pi'(x^*) = 0,$$

or, substituting,

$$R'(x^*) - C'(x^*) = MR(x^*) - MC(x^*) = 0.$$

From which we again see that a necessary condition for  $x^*$  to be the profit-maximizing quantity is that  $MR(x^*) = MC(x^*)$ .

## Sufficiency and the Shutdown Rule

# 4.4

Results 20 and 22 are only necessary conditions; that is, they identify *possible* candidates for being the profit-maximizing quantity, but they do not guarantee that a given  $n$  or  $x$  that satisfies those conditions is the profit-maximizing quantity. This is similar to the fact that while being admitted to the Berkeley MBA program is necessary for being in the Blue cohort, knowing someone has been admitted to the Berkeley MBA program is not sufficient—does not guarantee—the he or she is in the Blue cohort. Fortunately, there is a condition that insures that, if the firm should be in business at all, the conditions stated in Results 20 and 22 are also sufficient (*i.e.*, identify the profit-maximizing quantity).

We will establish the sufficiency condition for the continuous case first. Let  $x^*$  be the candidate for the profit-maximizing quantity identified by expression (4.6). If  $MR(x) > MC(x)$  for all  $x < x^*$  and  $MR(x) < MC(x)$  for all  $x > x^*$ , then  $x^*$  must be the profit-maximizing quantity (assuming the firm should operate at all). To see why, observe that marginal profit,  $MR(x) - MC(x)$  is positive for all  $x < x^*$ ; that is, every additional unit in this region contributes positively to total profit. On the other hand, marginal profit is negative for all  $x > x^*$ ; that is, every additional unit in this region reduces total profit. Graphically, we're climbing up the total profit "hill" in the region  $x < x^*$  and we're descending the total profit "hill" in the region  $x > x^*$ . See Figure 4.1.<sup>2</sup>

We've established:

**Result 23** *If*

- (i)  $MR(x^*) = MC(x^*)$ ,
- (ii)  $MR(x) > MC(x)$ , all  $x < x^*$ , and
- (iii)  $MR(x) < MC(x)$ , all  $x > x^*$ ,

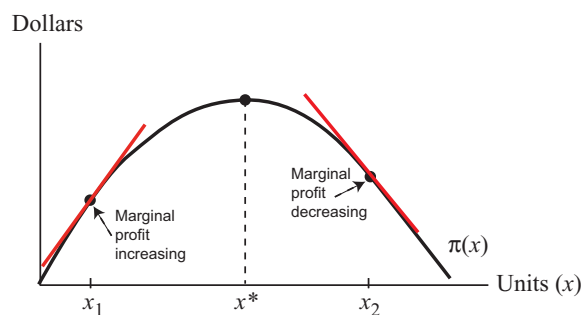
*then  $x^*$  is the profit-maximizing quantity for the firm to produce (if it should be in business at all).*

Another way to view Result 23 is that  $x^*$  is the profit-maximizing quantity if the marginal revenue schedule,  $MR(\cdot)$ , crosses the marginal cost schedule,  $MC(\cdot)$ , from above at  $x^*$ . See Figure 4.2 (note the two vertical scales—profits are in dollars, while marginal revenue and marginal cost are in dollars/unit). An equivalent way to state Result 23 is, thus,

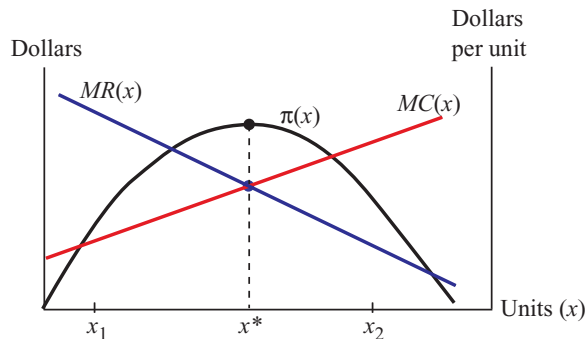
**Result 24** *If marginal revenue crosses marginal cost once at  $x^*$  and does so from above, then  $x^*$  is the profit-maximizing quantity (if the firm should be in business at all).*

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<sup>2</sup>  $\int dx$  See also Rule 36 in Appendix A3 and connected discussion.



**Figure 4.1:** Profits are increasing to the left of  $x^*$  and decreasing to the right of  $x^*$ . The slope of the leftmost red line (tangent) is equal to  $MR(x_1) - MC(x_1)$ . The slope of the rightmost red line (tangent) is  $MR(x_2) - MC(x_2)$ .



**Figure 4.2:** Marginal revenue (blue line) crosses marginal cost (red line) from above at the profit-maximizing quantity,  $x^*$ .

Now consider the discrete case. Observe that we could plot marginal revenue and marginal cost against output,  $n$ . If we connected the dots, then we would have, essentially, marginal revenue and marginal cost curves respectively. By Result 24, if the marginal revenue curve crosses marginal cost once from above at a given point, then that point approximates the profit-maximizing quantity. Why approximate? Well, remember, this is the discrete case and the curves might cross at a non-integer point. But, then, the profit-maximizing quantity would be the highest integer to the left of the cross; that is, an  $n^*$  such that  $MR(n^*) \geq MC(n^*)$  and  $MR(n^* + 1) < MC(n^* + 1)$ .

We can translate this analysis into a statement like Result 23. However, before we do so, we might question the likelihood that the marginal revenue schedule would cross the marginal cost schedule only once and from above in the discrete case. Remember, in the discrete case,  $MC(1)$  can be quite large because all the overhead costs are triggered by starting production (going from 0 to 1 unit). Hence,  $MC(1) \gg MR(1)$ . Fortunately, we can typically ignore that first unit for reasons that we will take up in a moment. We have, therefore,

**Result 25** *Consider the discrete case. If*

- (i)  $MR(n^*) \geq MC(n^*)$ ,
- (ii)  $MR(n) > MC(n)$ , all  $2 \leq n < n^*$ , and
- (iii)  $MR(n) < MC(n)$ , all  $n > n^*$ ,

*then  $n^*$  is the profit-maximizing quantity for the firm to produce (if it should be in business at all).*

The caveat “if it should be in business at all” has been used a number of times in this section. What do we mean by it? There is a final test that  $n^*$  (discrete case) or  $x^*$  (continuous case) must satisfy before we can conclude they’re the amounts the firm should produce. That test is *would the firm lose money at that level of output?* If the answer is no, then they’re the amounts that should be produced. If the answer is yes, then the firm should shutdown.

To go into detail, recall that  $C(0) = 0$ —if you’re not producing, then, by definition, you’re not forgoing the best alternative, so the cost must be zero. Not surprisingly, if you don’t produce, you don’t earn any revenue. That is,  $R(0) = 0$ . Hence, if the firm shuts down (produces nothing), then its profit is zero (*i.e.*,  $\pi(0) = R(0) - C(0) = 0$ ). We know that *if* the firm produces, the best it can do is produce  $n^*$  or  $x^*$ . So the alternatives are, in the discrete case, produce  $n^*$  or 0. The former is worse than the latter only if  $\pi(n^*) < \pi(0) = 0$ . Likewise, for the continuous case, the alternatives are produce  $x^*$  or 0. Again, the former is worse only if  $\pi(x^*) < \pi(0) = 0$ .

We can summarize this as

**Result 26 (Shutdown rule)** *If profit at the quantity determined by Results 23 or 25, as applicable, is non-negative, then the firm should produce that quantity. Otherwise, it should shutdown.*

Another way to describe this is in terms of average revenue and average costs. Let  $q^*$  be  $x^*$  or  $n^*$  as appropriate. Result 26 states that the firm should operate if and only if

$$R(q^*) - C(q^*) \geq 0.$$

Dividing through by  $q^*$  and rearranging, this yields

$$AR(q^*) \geq AC(q^*), \quad (4.7)$$

where  $AR(q) = R(q)/q$  is *average revenue*. In other words, the firm should operate if and only if  $AR(q^*) \geq AC(q^*)$ .

**Example 10:** A firm has a revenue,  $R(x)$ , given by

$$R(x) = 10x - \frac{1}{1000}x^2.$$

Its costs,  $C(x)$ , are given by

$$C(x) = \begin{cases} 0, & \text{if } x = 0 \\ 2x + F, & \text{if } x > 0 \end{cases},$$

where  $F$  is a non-negative constant. From Result 21,

$$MR(x) = 10 - 2\frac{1}{1000}x = 10 - \frac{1}{500}x.$$

From Result 15, we have that  $MC(x) = 2$  for  $x > 0$ . Setting  $MR(x) = MC(x)$  we have

$$10 - \frac{1}{500}x^* = 2,$$

which, solving for  $x^*$ , implies

$$x^* = \frac{10 - 2}{\frac{1}{500}} = 4000.$$

Observe that  $MR(x) > MC(x)$  for all  $x < x^*$  and  $MR(x) < MC(x)$  for all  $x > x^*$ . So, therefore, if the firm should produce, it should produce 4000 units to maximize profit. Should it produce? Observe

$$AR(x) = 10 - \frac{1}{1000}x \text{ and } AC(x) = 2 + \frac{F}{x}.$$

We have  $AR(x^*) \geq AC(x^*)$  if

$$10 - \frac{1}{1000}4000 \geq 2 + \frac{F}{4000};$$

that is, if

$$6 \geq 2 + \frac{F}{4000}, \text{ or}$$

$$16,000 \geq F.$$

So, provided  $F$  does not exceed 16,000, the firm should produce 4000 units. If  $F$  does exceed 16,000, then the firm should shutdown.

## Demand | 4.5

Where does the revenue function come from? The answer is it depends on how the firm prices and what its consumers' demands are for its product. In this section, we explore consumer demand and related issues.

### Individual demand

Consider an individual and a good (which may be a physical product or a service). If the consumer receives  $q$  units of this good, she enjoys some *benefit*,  $b(q)$ , from these  $q$  units. We can think of this benefit as being the monetary equivalent of the happiness she enjoys from the  $q$  units. That is, she would be indifferent between having the  $q$  units or having  $b(q)$  dollars in cash.

This benefit function,  $b(\cdot)$ , derives from the preferences, likes and dislikes, of the individual in question. It can also depend, as we will discuss later, on her income and the prices of other goods that she buys.

We can also define a *marginal benefit* schedule,  $mb(\cdot)$ , for this individual. In the discrete case, we have

$$mb(n) = b(n) - b(n - 1);$$

that is, as always with marginals, the marginal benefit is the increment in her total benefit from adding the  $n$ th unit.

In the continuous case, we have

$$mb(x) = \lim_{h \rightarrow 0} \frac{b(x + h) - b(x)}{h}.$$

Note the similarity of this definition to the definition of marginal cost and marginal revenue in the continuous case.

Regardless of the case, a property of marginal benefit schedules is the following.

**Observation 1** *Individuals' marginal benefits schedules are decreasing functions. That is, if  $q_1 > q_0$ , then  $mb(q_1) < mb(q_0)$ .*

Why is this? Well, it follows from two factors. First, the increase in most people's enjoyment or happiness from more of the same thing is diminishing in the amount they receive. For example, the additional enjoyment gained from a second candy bar is typically smaller than the enjoyment the first candy bar provided. Or, for those of you on the Atkins diet, the additional enjoyment gained from the second steak is typically smaller than the enjoyment the first steak provided. Economists refer to this property as *diminishing marginal utility*.

The second factor is simple "crowding out." If you eat one thing, you might not have room for something else. If you spend time in one activity, you might not have time for another activity. Hence, consumption of one thing often

**Slope of marginal benefit:** *Marginal benefit is a decreasing function.*

means forgoing consumption of something else. So, even if the marginal *gross* benefit of consumption isn't decreasing (*i.e.*, the effect on neural responses and neuro-transmitter production is constant), the overall marginal benefit would be decreasing because we are forgoing increasingly valued alternatives. That is, the *opportunity cost* is rising. For example, you might schedule your first hour of video game playing when there is nothing to watch on TV. If you play a second hour, however, it could mean forgoing shows you somewhat like. A third hour, could mean missing some of your favorite shows, etc.

Although we could do the analysis both for the discrete and the continuous case, it is far easier to do it for the continuous case. Hence, we will consider that case only. As, however, should become clear, marginal benefit is a lot like marginal revenue; so the analysis of marginal revenue and profit maximization in the discrete case carries over to marginal benefit and surplus maximization in the discrete case.

Goods are not typically given to us for free—usually, we have to pay for them. Because the money we spend on them could be spent on other things we like, we are always forgoing other things when we buy things. Opportunity cost! Hence, to determine how well off we are from buying  $q$  units of a good—how much we *profit*—we need to subtract the expenditure on the  $q$  units from our benefit,  $b(q)$ . Under simple pricing, the expenditure is  $pq$ , where  $p$  is the price per unit of the good. The individual's profit—called her *consumer surplus*—is

$$\begin{aligned} cs(x) &= b(x) - \text{expenditure on } x \\ &= b(x) - px \end{aligned} \quad (\text{under simple pricing}) \quad (4.8)$$

Using Result 15, the consumer's "marginal cost" is just  $p$ , the price. Note her marginal cost is a constant. If we assume  $mb(0) \geq p$ , then the  $mb(\cdot)$  schedule will cross the "marginal cost" schedule (*i.e.*, the horizontal line at height  $p$ ) once from above. We can then invoke Result 24 to conclude that the consumer maximizes her "profit"—that is, consumer surplus—by equating marginal benefit with price. In other words, she maximizes her consumer surplus by purchasing the amount  $x^*$  that solves

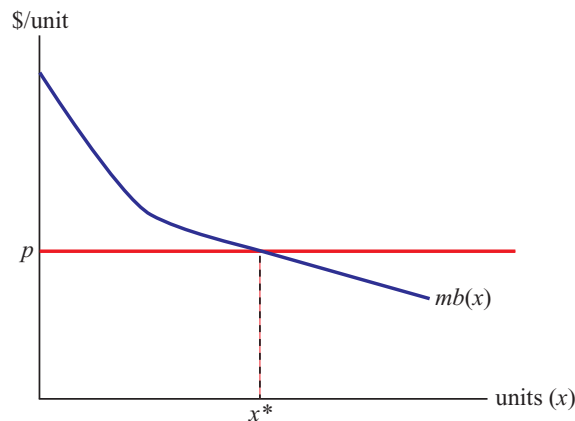
$$mb(x^*) = p. \quad (4.9)$$

Figure 4.3 illustrates.

If  $mb(0) < p$ , then the consumer does best not to purchase any units:

$$\begin{aligned} b(x) &= \text{Area under } mb(\cdot) \text{ curve from } 0 \text{ to } x \\ &\leq \text{Area under } mb(0) \text{ from } 0 \text{ to } x \\ &= mb(0)x \\ &\leq px, \end{aligned}$$

with the first inequality being strict (*i.e.*,  $<$ ) if  $x > 0$ . That is, if  $p > mb(0)$ , then we've shown  $b(x) < px$  for all  $x > 0$  and  $b(x) = px$  if  $x = 0$ . Hence, if  $p > mb(0)$ , the consumer maximizes her consumer surplus by purchasing zero units.



**Figure 4.3:** The marginal benefit schedule,  $mb(\cdot)$ , crosses the price line,  $p$ , once from above.

Because  $mb(\cdot)$  is decreasing, it is invertible (see Section A1.1). Hence, expression (4.9) can be read as defining  $x^*$  as a function of  $p$ . That is, if we vary  $p$ , we can see how the consumer's choice of optimal  $x$  varies using expression (4.9). Figure 4.4 illustrates. Note that we get a different  $x_t^*$  for each  $p_t \leq mb(0)$ . All  $p_t > mb(0)$  map to the same  $x^*$ , namely 0.

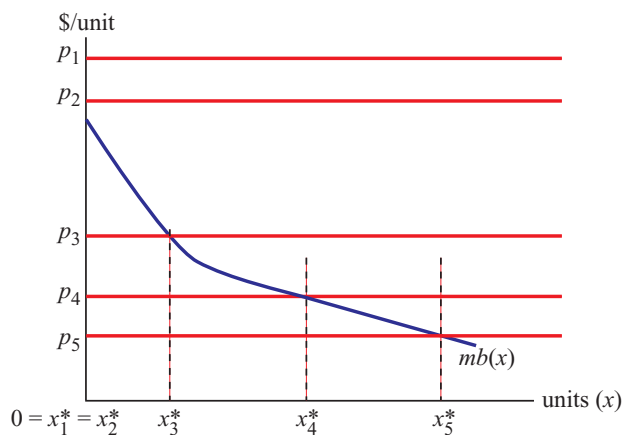
We can use the relation identified by Figure 4.4 to define a *demand curve* (alternatively, demand function or demand schedule) for the individual in question. Specifically, for  $p > mb(0)$ , we see that the quantity demanded (*i.e.*, that the consumer *wishes* to purchase) is 0. For  $p \leq mb(0)$ , we see that the quantity demanded (*i.e.*, that the consumer *wishes* to purchase) is the inverse of marginal benefit; that is, the amount demanded at  $p$  is  $mb^{-1}(p)$ . If we let  $d(\cdot)$  denote the demand function (*i.e.*, the amount the consumer in question wishes to purchase at a given price), then we have

$$d(p) = \begin{cases} 0, & \text{if } p > mb(0) \\ mb^{-1}(p), & \text{if } p \leq mb(0) \end{cases} \quad (4.10)$$

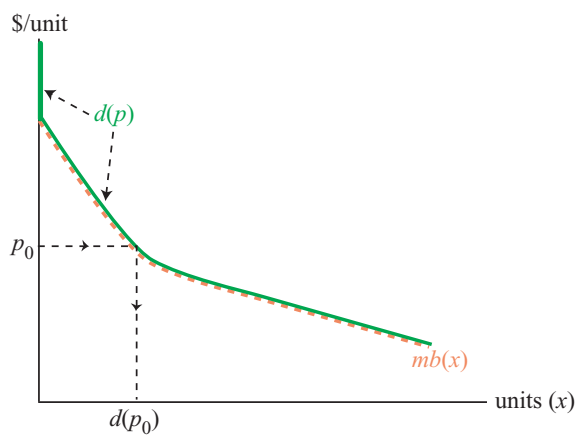
Figure 4.5 illustrates.

Observe, from Figure 4.5, that we determine the amount demanded at a price, say  $p_0$ , by reading across from that price until we hit the demand curve, then down from where we hit the demand curve. Hence, as illustrated,  $d(p_0)$ —a point on the horizontal axis—is the amount demanded at price  $p_0$ —a point on the vertical axis.

Observe, too, that because a demand curve is derived from the marginal



**Figure 4.4:** By varying the price, the marginal benefit curve can be used to determine the quantity the consumer wants (demands). Note that at high prices (e.g.,  $p_1$  and  $p_2$ ), demand is zero (i.e.,  $x_1^* = x_2^* = 0$ ).



**Figure 4.5:** The demand curve, shown as a solid green curve, corresponds to the marginal benefit curve (dashed curve in orange) for  $p \leq mb(0)$  and corresponds to the vertical axis for  $p > mb(0)$ .

benefit schedule, which is decreasing, the amount demanded is less the greater is the price. That is, if  $p$  and  $p'$  are two prices,  $p' > p$ , then

$$\begin{aligned} d(p') &< d(p) && \text{if } d(p) > 0; \text{ otherwise} \\ d(p') &= d(p) && \text{if } d(p) = 0. \end{aligned}$$

Consistent with Figure 4.5, we describe this by saying that demand curves slope down.<sup>3</sup>

**Note:** Demand curves slope down.

**Example 11:** Suppose that an individual's benefit schedule for a particular good is  $10x - x^2$ ; that is,

$$b(x) = 10x - x^2.$$

Using Result 21, we see that

$$mb(x) = 10 - 2x.$$

Observe  $mb(0) = 10$ . Let's derive this individual's demand curve. For  $p > 10$ , he purchases nothing; that is,  $d(p) = 0$  for all  $p > 10$ . For  $p \leq 10$ , we need to invert the marginal benefit schedule. Observe that if

$$p = mb(x) = 10 - 2x,$$

then

$$mb^{-1}(p) = \frac{10 - p}{2} = 5 - p/2 = x.$$

We can conclude that

$$d(p) = \begin{cases} 5 - p/2, & \text{if } p \leq 10 \\ 0, & \text{if } p > 10 \end{cases}.$$

### Properties of demand: Complements and substitutes

Our derivation of the individual's demand curve has held constant a number of factors. In particular, it has assumed that the prices of *other* goods has remained fixed. But what happens to demand for one good if the price of another good changes? The answer depends on whether the two goods are complements or substitutes.<sup>4</sup>

Two goods are *complements* if they are goods that tend to be consumed together. Examples of such pairs would be chips and salsa, PCs and operating systems, and whips and chains. Some pairs of goods are complements for some people (*e.g.*, orange juice and vodka) but not for other people (*e.g.*, vodka tonic drinkers).

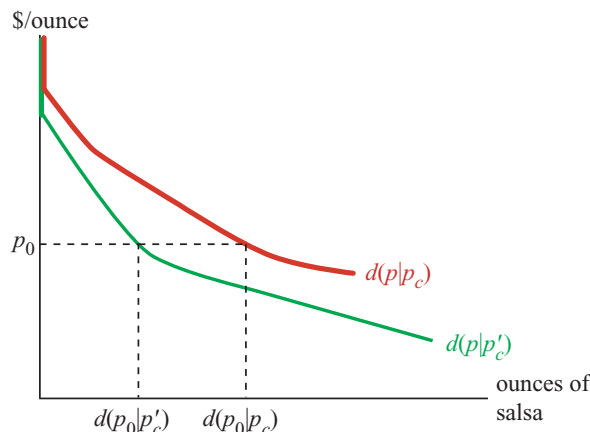
Two goods are *substitutes* if they are goods that tend to be seen as alterna-

**Complements:**  
*Goods consumed together.*

**Substitutes:**  
*Goods that are alternatives.*

<sup>3</sup>If you've taken economics before and heard of Giffen goods, forget about them—they are a theoretical possibility that have never been shown to exist in reality. If you don't know what a Giffen good is, consider yourself lucky.

<sup>4</sup>Point about spelling: Note that "complement" has no "i." Compliment with an "i" is a different word (as a verb it means to praise or give for free; as a noun it means a statement of praise).



**Figure 4.6:** As the price of chips, the complementary good, increases from  $p_c$  to  $p'_c$ , the demand curve for salsa shifts in (goes from being the red curve,  $p(\cdot|p_c)$ , to being the green curve,  $p(\cdot|p'_c)$ ).

tives. Examples would include DVDs and in-theater movies, beer and wine, and email and telephone calls.

If two goods, say chips and salsa, are complements and the price of one, say chips, goes up, then the demand for the other at a given price will tend to be less. That is, if  $p_c$  and  $p'_c$  are two prices for chips,  $p_c < p'_c$ , then

$$d(p_s|p'_c) \leq d(p_s|p_c),$$

where  $d(p_s|\hat{p}_c)$  means the amount of salsa demanded as a function of the price of salsa,  $p_s$ , given that the price of chips is  $\hat{p}_c$ . As illustrated in Figure 4.6, we can describe this as saying that an increase in the price of the complementary good causes the demand curve for the other good to shift in (equivalently, shift to the left). The reason for this is that the benefit of consuming salsa depends on the number of chips one consumes. If, because the price of chips has increased, one will consume fewer chips, then one has less demand for salsa. Another way to put this is that the marginal benefit of an ounce more salsa is less the fewer chips on which it can go.

If two goods, say beer and wine, are substitutes and the price of one, say beer, goes up, then the demand for the other at a given price will tend to be greater. That is, if  $p_b$  and  $p'_b$  are two prices for beer,  $p_b < p'_b$ , then

$$d(p_w|p'_b) \geq d(p_w|p_b),$$

where  $d(p_w|\hat{p}_b)$  means the amount of wine demanded as a function of the price of wine,  $p_w$ , given that the price of beer is  $\hat{p}_b$ . Thinking of Figure 4.6 “in reverse,” we can describe this as saying that an increase in the price of the substitute good causes the demand curve for the other good to shift out (equivalently, shift to the right). The reason for this is that the benefit of consuming wine depends on the amount of other alcohol one can consume. If one is buying less beer because the price of beer has gone up, then the demand for other alcohol (*e.g.*, wine) will go up.

### Properties of demand: other shifters

Changes in the prices of complements and substitutes are not the only factors that can shift demand. For example, when a hurricane is coming, the benefit of having bottled water, plywood, and canned foods increases. Because benefit is the area under the marginal benefit schedule (recall Result 18), a shift up in benefit must correspond to a shift up in marginal benefit. But, from Figure 4.6, it is clear that a shift up in marginal benefit is equivalent to a shift out in demand. That is why, for instance, we see more bottled water, plywood, and canned foods being demanded when a hurricane is coming.

Other changes, such as changes in technology, can also shift demand curves. The advent of the car, for instance, greatly lessened the benefit of buggy whips, which meant marginal benefit fell, which was equivalent to the demand for buggy whips shifting in. Likewise, the advent of USB-port jump (flash) drives has caused the demand for floppy diskettes to shift in.

Some technologically driven demand shifts, such as that caused by jump drives, can also be seen as examples of substitutes at work. If a product doesn't exist, that's equivalent to its price being infinity. The introduction of jump drives can be seen as a (dramatic) fall in the price of jump drives. Jump drives and floppies are substitutes. Hence, not surprisingly, the advent of jump drives caused demand for floppies to shift in.

### Properties of demand: income effects

Suppose your income doubled. One response might be that you eat more meals out than you did before. Of course, if you eat more meals out, then you are eating fewer meals at home, which means you're buying fewer items to cook at home (*e.g.*, frozen dinners). In terms of your demand curves, a doubling of your income causes your demand curve for restaurant meals to shift out and your demand curve for frozen dinners to shift in.

A good for which the demand curve shifts out as income rises is called a *normal good*. A good for which the demand curve shifts in as income rises is called an *inferior good*. Inferior goods are those goods that are seen as less desirable than certain substitutes for them (*e.g.*, a meal at Rivoli is better than a Swanson frozen dinner). The problem is that, at a given level of income, one can afford only so many restaurant dinners. At a higher level of income, one

can afford more, which crowds out the frozen dinners.<sup>5</sup>

This discussion of income effects highlights an omission in our discussion of demand to this point. What about the ability of an individual to afford things? That is, for example, if an individual's marginal benefit schedule intersects the price line at 100 units and a price of \$10, but she only has \$900 to spend, then it wouldn't make sense to say her demand is 100—she can't afford that many. The good news is that there is a way to accommodate the issue of affordability into the analysis. The bad news is that it is rather complicated (see the following subsection). Fortunately, for most goods the affordability issue is generally not that important. The number of candy bars, for instance, that you buy varies little or not at all with your income. Another way to say this is that, for most goods, income effects are negligible. In 201A, we will, therefore, ignore income effects.

### $\int dx$ Utility Maximization (*This section totally optional*)

We think of consumers as deriving utility from the consumption of different goods. A general way to write utility for a potential customer is to index all the different products, such as Cinnamon Apple Cheerios, sports cars, broccoli, Warriors tickets, etc., by  $i$  and write utility as

$$U(q_1, q_2, \dots, q_i, \dots, q_N),$$

where  $N$  is the total number of different goods. Consumers also have *budget constraints*. That is, a consumer's total expenditure cannot exceed his income,  $I$ . We write this as

$$p_1 q_1 + \dots + p_N q_N \leq I,$$

where  $p_i$  is the price of a unit of the  $i$ th good. Note the lefthand side could be written as  $\sum_{i=1}^N p_i q_i$ .

The consumer wishes to maximize his wellbeing—his utility—subject to his budget constraint. This can be done by Lagrange maximization: Define  $\lambda$  to be the Lagrange multiplier or shadow value of income. That is,  $\lambda$  is the value of the marginal dollar in terms of utility (*i.e.*, the increase in utility if income were increased by a dollar). Then this *constrained* maximization program can be written as

$$\max_{\{q_1, \dots, q_N\}} U(q_1, \dots, q_N) + \lambda \left( I - \sum_{i=1}^N p_i q_i \right).$$

Although we could solve this Lagrangean directly, it is easier to use some intuition. If someone is going to pay a dollar to consume

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<sup>5</sup>Alternatively, one might stay home, but have more steaks and fewer frozen dinners as income rises.

more of one good, it cannot be the case that spending that dollar on another good will yield more utility. The amount an additional dollar buys of good  $i$  is  $1/p_i$ . If  $MU_i$  is the marginal utility from the  $i$ th good (*i.e.*,  $MU_i = \partial U / \partial q_i$ ), then  $MU_i \times 1/p_i$  (marginal utility times quantity) is the amount of additional utility derived from a dollar spent on the  $i$ th good. If the consumer has maximized his utility, it cannot be that moving the marginal dollar from one good to another can increase his utility. That is, we have for any two goods  $i$  and  $j$ ,

$$MU_i \times 1/p_i \not> MU_j \times 1/p_j \text{ and} \\ MU_j \times 1/p_j \not> MU_i \times 1/p_i .$$

But this just says that for any pair of goods it must be that

$$\frac{MU_i}{p_i} = \frac{MU_j}{p_j} \quad (4.11)$$

if the individual is maximizing his utility subject to his budget constraint.<sup>6</sup> Equation (4.11) says that the benefit of consuming one good divided by the cost (sometimes called the “bang for the buck”) must be equal to the same ratio (bang for the buck) for any other good.

Your bang for the buck is the value, in terms of utility, of the marginal dollar of income. That is, it is the shadow value of income. We’ve thus derived:

$$\frac{MU_i}{p_i} = \lambda ,$$

or, rewriting,

$$MU_i = \lambda p_i . \quad (4.12)$$

Observe that we have a function of quantity only on the lefthand side and a function of price on the righthand side. If, as we assume, the consumer gets a diminishing marginal utility (or benefit) from each good, then  $MU_i$  is an invertible function of  $q_i$ . If we invert it, we get

$$q_i = MU_i^{-1}(\lambda p_i) . \quad (4.13)$$

This relates the amount the consumer wants to the price of the good. So, holding everything else constant, this is a demand curve.

To square this analysis with our earlier derivation of individual demand from marginal benefit schedules, we need to assume that the utility function has the form

$$U(q_1, \dots, q_N) = u(q_1, \dots, q_{N-1}) + q_N .$$

---

<sup>6</sup>To be precise, this holds only for goods for which the consumer actually purchases positive amounts.

There is no further loss of generality in normalizing the units of the  $N$ th good so that the price per unit is \$1; that is, so  $p_N = 1$ .<sup>7</sup> Clearly,  $MU_N = 1$ , hence, from expression (4.12) for  $i = N$ , we have

$$1 = \lambda \times 1;$$

that is, the shadow value of income,  $\lambda$ , is 1. Observe, therefore, that  $\lambda$  does not depend on  $I$ . Holding constant the amount of the goods other than  $i$  in expression (4.12), we have an expression that relates marginal *benefit* to price; that is, condition (4.9). Inverting—employing expression (4.13)—gives us

$$q_i = MU_i^{-1}(p_i) = d_i(p_i)$$

(recall  $\lambda = 1$ ). Observe this means that there are no income effects, therefore, in the demand for good  $i$ .

### From individual to aggregate demand

For simple pricing, what a firm cares about is not each individual's demand but *aggregate demand*, the amount that all individuals want at a given price. Aggregating all the individual demand curves gives the *aggregate demand curve*.

Aggregating demand is straightforward. If Rose and Noah are the only two consumers and Rose wants 3 units at a given price and Noah wants 5 units at a given price, then aggregate demand at that price is 8. Similarly, if Rose's demand curve is  $d_R(p)$  and Noah's is  $d_N(p)$ , then the aggregate demand curve,  $D(p)$ , is given by

$$D(p) = d_R(p) + d_N(p).$$

More generally, if we have  $N$  consumers, indexed by  $n$ , then the aggregate demand curve is given by

$$D(p) = d_1(p) + \cdots + d_N(p) = \sum_{n=1}^N d_n(p). \quad (4.14)$$

**Example 12:** Suppose that there are two types of consumers. There are 1000 people of type  $A$ , and these people have a demand for your product given by

$$d_A(p) = \begin{cases} 150 - 3p, & \text{if } p \leq 50 \\ 0, & \text{if } p > 50 \end{cases}.$$

There are an additional 2000 people of type  $B$ , with demand for your product given by

$$d_B(p) = \begin{cases} 1000 - 2p, & \text{if } p \leq 500 \\ 0, & \text{if } p > 500 \end{cases}.$$

---

<sup>7</sup>The  $N$ th good is the *numeraire* good.

The aggregate demand of type- $A$  consumers is

$$D_A(p) = \sum_{n=1}^{1000} d_A(p) = 1000d_A(p) = \begin{cases} 150,000 - 3000p, & \text{if } p \leq 50 \\ 0, & \text{if } p > 50 \end{cases}.$$

The aggregate demand of type- $B$  consumers is

$$D_B(p) = \sum_{n=1}^{2000} d_B(p) = 2000d_B(p) = \begin{cases} 2,000,000 - 4000p, & \text{if } p \leq 500 \\ 0, & \text{if } p > 500 \end{cases}.$$

So overall aggregate demand is

$$\begin{aligned} D(p) &= D_A(p) + D_B(p) \\ &= \begin{cases} 150,000 - 3000p + 2,000,000 - 4000p = 2,150,000 - 7000p, & \text{if } p \leq 50 \\ 0 + 2,000,000 - 4000p = 2,000,000 - 4000p, & \text{if } 50 < p \leq 500 \\ 0 + 0 = 0, & \text{if } 500 < p \end{cases}. \end{aligned}$$

## Demand Elasticity

# 4.6

For reasons that will become clear later, it is useful to define a quantity known as the *elasticity of demand*. Specifically, let  $D'(p)$  denote the slope of the demand curve at price  $p$ ,<sup>8</sup> then the elasticity of demand,  $\epsilon_D$ , is defined to be

$$\epsilon_D = D'(p) \frac{p}{D(p)}. \quad (4.15)$$

Because demand curves slope down,  $D'(p) < 0$ ; hence,  $\epsilon_D < 0$ .

What  $\epsilon_D$  is telling us is the percentage change in quantity demanded per percentage change in price. That is, it can be shown that,<sup>9</sup>

$$\epsilon_D = \left( \frac{dD(p)}{D(p)} \times 100\% \right) \div \left( \frac{dp}{p} \times 100\% \right), \quad (4.16)$$

where the  $d$  denotes an infinitesimally small change.

<sup>8</sup>  $\int dx$  That is,  $D'(p)$  is the derivative of the demand curve at  $p$ .

<sup>9</sup>  $\int dx$  Observe that equation (4.16) can be rewritten as  $dD(p)/dp \times p/D(p)$ . But  $dD(p)/dp = D'(p)$  by definition.

## Revenue and Marginal Revenue under Simple Pricing | 4.7

We can now go from demand to a revenue function for a firm that engages in simple pricing.

### Derivation of Revenue

Consider a firm that faces aggregate demand  $D(p)$ . What this says is that if it charges a price of  $p$ , then it will sell  $D(p)$  units. Its revenue is price times units sold, or  $pD(p)$ . Observe that if it were to raise its price, two things would happen. It would make more per unit (that's the good news). But it would sell fewer units—demand curves, recall, slope down (that's the bad news).

Our analysis of profit maximization was done in terms of quantity, whereas we have just derived an expression for revenue in terms of  $p$ . To utilize our earlier analysis, we need to convert  $pD(p)$  into an expression that is expressed in terms of quantity. Observe that  $D(p)$  is a quantity, call it  $q$ . Because demand curves slope down,  $D(\cdot)$  is invertible. We can, thus, invert the equation

$$q = D(p)$$

to get

$$D^{-1}(q) = p.$$

Rather than write  $D^{-1}(\cdot)$ , we use the notation  $P(\cdot)$  (*i.e.*,  $P(q) = D^{-1}(q)$ ). Because  $P(\cdot)$  is the inverse of the demand function, we call it the *inverse demand function* (alternatively, inverse demand curve or inverse demand schedule). So, substituting  $P(q)$  for  $p$  and  $q$  for  $D(p)$ , we can rewrite revenue in terms of quantity:

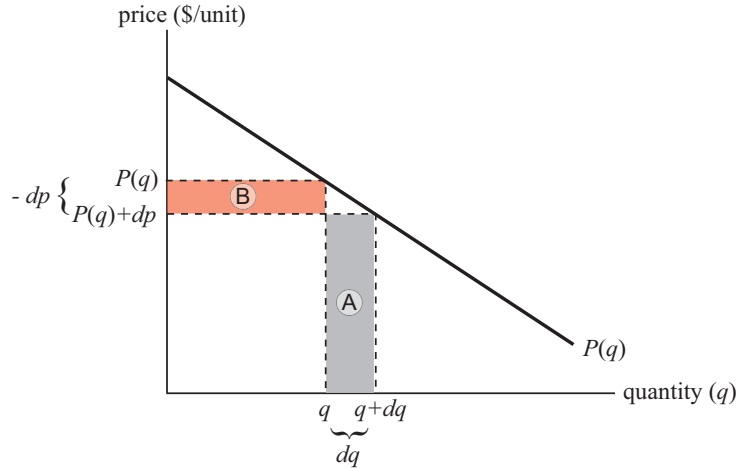
$$R(q) = P(q)q.$$

### Derivation of Marginal Revenue

To find the profit-maximizing quantity, we want to use the  $MR = MC$  rule. But this requires us to know  $MR$ .

If the firm wishes to increase  $q$ , it will have to lower price. Put another way, because demand curves slope down,  $P(\cdot)$  is a decreasing function—if  $q$  goes up, it goes down. We therefore face the same “good-news”-“bad-news” situation noted above. The good news from increasing  $q$  is that we sell more units. The bad news is that we can only do so at a lower price on *all* units sold. This second effect is sometimes referred to as the *driving-down-the-price* effect.

Consider a very small positive increment in the quantity to be sold,  $dq$ . As just discussed, this will result in a small change in price,  $dp$ . Because demand curves slope down,  $dp < 0$ . Figure 4.7 illustrates. The small change in revenue,  $dR$ , is captured by the two shaded areas. Because the firm is selling  $dq$  more units, it adds  $dq \times (P(q) + dp)$  to its revenue. This is the gray area labelled **A**. That's the good news. The bad news is the driving-down-the-price effect. Price



**Figure 4.7:** Increasing units sold by  $dq$  changes price by  $dp < 0$ . The firm gains, in revenue, the gray area, labelled A, but loses the red area, labelled B. The latter area represents the driving-down-the-price effect.

is changed by  $dp$ . So, on all the  $q$  units it would have sold had it not tried to increase sales, it gets  $-dp$  less (recall  $dp < 0$ ). So it loses revenue equal to the red area labelled B, which is  $-dp \times q$ . Hence, the change in revenue,  $dR$ , is given by

$$\begin{aligned} dR &= dq \times (P(q) + dp) - (-dp \times q) \\ &= dq(P(q) + dp) + dpq. \end{aligned}$$

If we divide both sides by  $dq$ , we get

$$\frac{dR}{dq} = P(q) + dp + \frac{dp}{dq}q.$$

Now a small change in revenue per unit change in quantity is just marginal revenue; that is,  $dR/dq = MR(q)$ . Similarly, a small change in the price per unit change in quantity is just the slope of the inverse demand curve,  $P'(q)$ . So we can rewrite this as

$$MR(q) = P(q) + dp + P'(q)q.$$

**MR under simple pricing:**  
 $P(q) + P'(q)q$ .

The quantity  $dp$  is very small, essentially zero, so it can be ignored. In fact, in the limit, as we consider an infinitesimally small change in quantity, it is zero. We've thus arrived at the formula for marginal revenue under simple pricing.<sup>10</sup>

**Result 27**

$$MR(q) = P(q) + P'(q)q. \quad (4.17)$$

Observe that the  $P'(q)q$  term in expression (4.17), which is negative because demand curves slope down, represents the driving-down-the-price effect.

**Example 13:** Suppose a firm faces demand given by

$$D(p) = 500,000 - 2500p.$$

To calculate marginal revenue, (i) we need to calculate *inverse* demand, then (ii) we need to employ expression (4.17).

Inverting demand, we have:

$$q = 500,000 - 2500P(q); \text{ hence,} \\ P(q) = 200 - \frac{q}{2500}.$$

Observe the second expression is a line in slope-intercept form (see Section A1.5) and the slope,  $P'(q)$ , is  $-1/2500$  (it is of course negative because demand curves slope down). Plugging into expression (4.17) yields:

$$\begin{aligned} MR(q) &= \underbrace{\left(200 - \frac{q}{2500}\right)}_{P(q)} + \underbrace{\left(-\frac{1}{2500}q\right)}_{P'(q)q} \\ &= 200 - \frac{2q}{2500} = 200 - \frac{q}{1250}. \end{aligned}$$

Observe the following, *all of which are general results*,

- If the inverse demand schedule is linear, then the marginal revenue schedule is also linear.
- The inverse demand schedule and the marginal revenue schedule share the same intercept (in this example, 200).
- If the inverse demand schedule is linear, then the slope of the marginal revenue schedule is twice the slope of the inverse demand schedule ( $-1/1250$  versus  $-1/2500$ , respectively, in this example).

<sup>10</sup>  $\int dx$  A quick derivation via calculus:  $R(q) = P(q)q$ . Differentiating, we have  $MR(q) = P(q) + P'(q)q$ , where the derivative of the righthand side follows by the product rule (see Rule 26).

### Marginal Revenue and Elasticity

The fact that marginal revenue under simple pricing has both a good-news term and a bad-news term raises the question of whether the bad-news term could dominate the good-news term. That is, could marginal revenue ever be negative? The answer is yes. Moreover, as we will show, the sign of marginal revenue is related to elasticity of demand,  $\epsilon_D$ .

Because price is positive,  $MR(q)/P(q)$  has the same sign as  $MR(q)$ . Observe, from expression (4.17), that

$$\frac{MR(q)}{P(q)} = 1 + P'(q) \frac{q}{P(q)}. \quad (4.18)$$

Recall that  $P'(q) = dp/dq$ ,  $q = D(p)$ , and  $P(q) = p$ . Hence, we can rewrite expression (4.18) as

$$\begin{aligned} \frac{MR(q)}{p} &= 1 + \frac{dp}{dq} \frac{D(p)}{p} \\ &= 1 + \frac{1}{\left[ \frac{dq}{dp} \frac{p}{D(p)} \right]}. \end{aligned}$$

$dq/dp$  is the change in quantity per unit change in price, that is the slope of the demand curve,  $D'(p)$ . So we have,

$$\begin{aligned} \frac{MR(q)}{p} &= 1 + \frac{1}{\left[ D'(p) \frac{p}{D(p)} \right]} \\ &= 1 + \frac{1}{\epsilon_D}, \end{aligned} \quad (4.19)$$

where the second equality follows from expression (4.15). Recalling that  $\epsilon_D < 0$  (demand curves slope down), expression (4.19) implies

**Result 28** *If  $\epsilon_D < -1$ , then marginal revenue is positive; if  $\epsilon_D = -1$ , then marginal revenue is zero; and if  $\epsilon_D > -1$ , then marginal revenue is negative.*

When  $\epsilon_D < -1$ , we say that we are on the *elastic* portion of demand. When  $\epsilon_D = -1$ , we say that we are on the *unitary elastic* portion (point) of demand. When  $\epsilon_D > -1$ , we say that we are on the *inelastic* portion of demand.

When  $\epsilon_D < -1$ , a one-percent change in price causes a more than one-percent change in quantity, which is why demand is called elastic in this case. Because revenue is price times quantity, this tells us that, on the elastic portion of demand, revenue goes up if we lower price (increase quantity); and revenue goes down if we raise price (decrease quantity). When  $\epsilon_D > -1$ , a one-percent change in price causes less than a one-percent change in quantity, which is why demand is called inelastic in this case. Moreover, this means that, on the inelastic portion of demand, revenue goes down if we lower price (increase quantity); and it goes up if we raise price (decrease quantity).

Observe that a firm engaged in simple pricing would never want to operate on the *inelastic* portion of demand. If it were on the inelastic portion, then it could raise price (equivalently, cutback output), which would raise revenue. Moreover, because it is producing less, it would also be reducing cost. Any move that both raises revenue and cuts cost is a winning move—so, at any point on the inelastic portion of demand, it would want to raise price, which means it would never operate on the inelastic portion.

**Result 29** *A firm engaged in simple pricing does not choose its price or output so as to be on the inelastic portion of its demand curve.*

## The Profit-Maximizing Price

# 4.8

We now have all the ingredients in place to determine the profit-maximizing price.

Under fairly general conditions, the marginal-revenue schedule under simple pricing is downward sloping. From Example 13, it will certainly be downward sloping if demand is linear.

Typically, we can expect the marginal-cost schedule to be either relatively flat or increasing. Either way, if the marginal-revenue schedule is decreasing, this means that, if the schedules cross at all, the marginal-revenue schedule crosses the marginal-cost schedule once and from above. Observe that, if  $P(0) > MC(0)$ , the curves will cross given their predicted slopes.<sup>11</sup>

Assuming, therefore, that the  $MR = MC$  rule is sufficient, as well as necessary, for determining the profit-maximizing quantity, we need to translate that into a price. This is straightforward. If  $q^*$  solves the expression  $MR(q^*) = MC(q^*)$ , then the profit-maximizing price is  $P(q^*)$ . Figure 4.8 illustrates.

The firm's profit is revenue minus cost. So the maximum profit is

$$\pi(q^*) = q^*P(q^*) - C(q^*).$$

We do need, of course, to check the shutdown rule; that is, make sure the firm should operate at all. Recall that, under simple pricing, average revenue is price. Hence, the condition for operating at all, expression (4.7), becomes

$$P(q^*) \geq AC(q^*).$$

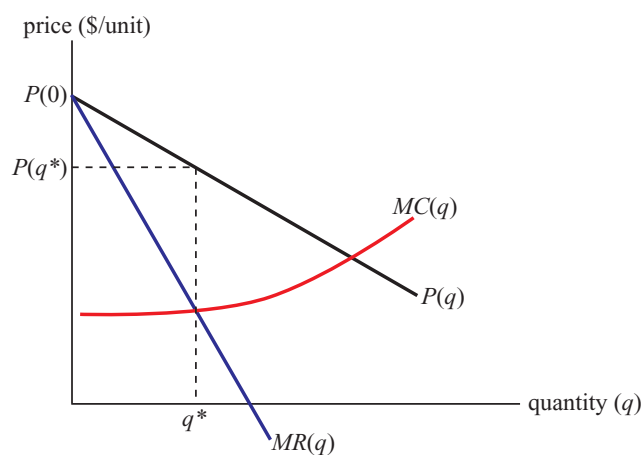
**Example 14 [From start to finish]:** Consider a firm that faces demand

$$D(p) = 1,000,000 - 50,000p.$$

Suppose its costs are given by

$$C(q) = \begin{cases} 0, & \text{if } q = 0 \\ 6q + 1,400,000, & \text{if } q > 0 \end{cases}.$$

<sup>11</sup>  $\int dx$  If, because of an overhead cost,  $MC(0)$  is not defined, then this condition can be restated as  $P(0) > \lim_{h \rightarrow 0} MC(h)$ .



**Figure 4.8:** The profit-maximizing quantity,  $q^*$ , is determined by the intersection of the marginal-revenue schedule (in blue) and the marginal-cost schedule (in red). Note that  $MR$  crosses  $MC$  once and from above. The profit-maximizing price,  $P(q^*)$ , is then read off the inverse demand curve (in black) at the profit-maximizing quantity,  $q^*$ .

For  $q > 0$ , it is readily seen that  $MC(q) = 6$  (recall Result 15). To derive  $MR(\cdot)$ , we need, first, to calculate inverse demand, and, then, marginal revenue using Result 27. Letting  $q = D(p)$  and  $p = P(q)$  we have:

$$\begin{aligned} D(p) &= 1,000,000 - 50,000p; \text{ hence,} \\ q &= 1,000,000 - 50,000P(q). \end{aligned}$$

So, solving for  $P(q)$ , we have

$$P(q) = 20 - \frac{q}{50,000}.$$

Plugging into Result 27, we have

$$\begin{aligned} MR(q) &= \left(20 - \frac{q}{50,000}\right) + \left(-\frac{1}{50,000}q\right) \\ &= 20 - \frac{q}{25,000}. \end{aligned}$$

Observe, as we knew had to be the case from Example 13, the  $MR$  schedule has the same intercept as the inverse demand curve and has twice its slope.

Because  $P(0) > 6$ ,  $MC(\cdot)$  is flat, and  $MR(\cdot)$  is downward sloping, we see that the  $MR$  schedule crosses the  $MC$  schedule once, from above. That is, the only candidate for the profit-maximizing quantity is the  $q$  that solves  $MR(q) = MC(q)$ . Let's find that  $q$ :

$$\begin{aligned} MR(q) &= MC(q) \\ 20 - \frac{q}{25,000} &= 6; \end{aligned}$$

hence, solving for  $q$ , we have

$$q = (20 - 6) \times 25,000 = 350,000.$$

That is,  $q^* = 350,000$ .

To get the profit-maximizing price, we stick that  $q^*$  into  $P(\cdot)$ :

$$P(q^*) = 20 - \frac{350,000}{50,000} = 13.$$

So the profit-maximizing price is \$13.

Should the firm operate? That is, is \$13 ( $= AR(q^*)$ ) greater than  $AC(q^*)$ . Observe

$$AC(q) = 6 + \frac{1,400,000}{q}.$$

Therefore,

$$AC(q^*) = 6 + \frac{1,400,000}{350,000} = 6 + 4 = 10;$$

so, yes,  $\$13 > AC(q^*)$ —the firm should produce.

Finally, its profit is revenue minus cost

$$\begin{aligned}\pi(350,000) &= \underbrace{13 \times 350,000}_{\text{revenue}} - \underbrace{(6 \times 350,000 + 1,400,000)}_{\text{cost}} \\ &= 1,050,000.\end{aligned}$$

That is, maximum profits are \$1,050,000.

## The Lerner Markup Rule

# 4.9

We can relate the price markup over marginal cost to the elasticity of demand. The *markup* is  $p^* - MC(q^*)$ , where we have used  $p^*$  to denote the profit-maximizing price (*i.e.*,  $p^* = P(q^*)$ ). What we want to do is determine what proportion of price is markup; that is, we want to calculate

$$\frac{p^* - MC(q^*)}{p^*}.$$

Recall, earlier, that we established that

$$MR(q) = p \times \left(1 + \frac{1}{\epsilon_D}\right) \quad (4.20)$$

(see expression (4.19) above). We also know that, at the profit-maximizing quantity,  $MR(q^*) = MC(q^*)$ . Hence, we have

$$\begin{aligned}\frac{p^* - MC(q^*)}{p^*} &= \frac{p^* - MR(q^*)}{p^*} && (MR = MC \text{ at profit-max'ing quantity}) \\ &= \frac{p^* - p^* \times \left(1 + \frac{1}{\epsilon_D}\right)}{p^*} && (\text{expression (4.20)}) \\ &= -\frac{1}{\epsilon_D} && (\text{simplifying})\end{aligned}$$

We have just established the Lerner markup rule:

**Result 30 (Lerner markup rule)** *Under simple pricing, at the profit-maximizing price and quantity, the proportion of the price that is markup over marginal cost is  $-1/\epsilon_D$ ; that is,*

$$\frac{p^* - MC(q^*)}{p^*} = -\frac{1}{\epsilon_D}.$$

The Lerner markup rule is useful for optimally adjusting price following small changes in cost.

**Example 15:** Due to a rise in the price of one of its raw materials, a company finds its marginal cost of production has increased from \$10 per unit to \$10.50 (a 5% increase). Data suggests that elasticity of demand at current price is -1.25. Assuming the company's old price was optimal, what should its new price be (approximately)?<sup>12</sup> Using the Lerner rule we have

$$\frac{p - 10.50}{p} = -\frac{1}{(-1.25)} = .8.$$

Solving for  $p$ , we have

$$p - 10.50 = .8p$$

or

$$p = 10.50/.2 = 52.50.$$

So the new price should be \$52.50. (Can you determine what the old price was?)

**Example 16:** Suppose that your firm's marginal cost increases by 2%. By what percentage (approximately) should you raise your price, assuming you were initially pricing optimally?

To determine this, let  $\delta$  denote the proportion by which you should increase your price; that is,  $\delta \times 100\%$  is the percentage by which you should increase your price. Let  $p$  and  $c$  denote your original price and marginal cost, respectively. The Lerner rule tells you that

$$\frac{p - c}{p} = -\frac{1}{\epsilon_D}.$$

Your new price, which is  $p + \delta p$ , must also satisfy this relationship when your marginal cost is  $c + .02c = 1.02c$ :

$$\frac{p + \delta p - 1.02c}{p + \delta p} = -\frac{1}{\epsilon_D}.$$

If two things equal the same third thing, then the two things equal each other:

$$\begin{aligned} \frac{p + \delta p - 1.02c}{p + \delta p} &= \frac{p - c}{p} \\ \frac{(1 + \delta)p - 1.02c}{(1 + \delta)p} &= \end{aligned}$$

Observe that if  $1 + \delta = 1.02$ , then we can cancel a 1.02 from numerator and denominator, which will make the lefthand side the same as the righthand side. Hence, the solution to this equation is  $1 + \delta = 1.02$  or  $\delta = .02$ . So you should raise your price by 2%. This is general: For *small* increases in marginal cost, the approximate best response is to increase price by the same percentage.

Finally, if elasticity were  $-1.5$ , then by increasing your price by 2%, you could expect an approximate reduction in units sold of 3%.

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<sup>12</sup>Why "approximately"? Because elasticity is typically not a constant. If we adjust the price, we're on a different part of the demand curve, so the elasticity will be different. For small movements in price, however, the change in elasticity is very small, so treating it as constant for small price changes yields a good approximation.

## Summary | 4.10

The focus of this section was on simple pricing; that is, charging the same price for all units and to all consumers. Before we could determine the profit-maximizing price, however, we needed to develop a suitable toolkit.

First, we had to work out how a firm maximizes profits. This involves a number of steps:

1. Determine the marginal-revenue and marginal-cost schedules.
2. Solve for the quantity that equates marginal revenue and marginal cost. This quantity is a *candidate* for being the profit-maximizing quantity. If the marginal-revenue schedule crosses the marginal-cost schedule once and from above, then this quantity is the profit-maximizing quantity if the firm should produce at all.
3. Check whether the firm should shutdown. That is, check that overhead costs are not so great as to make shutting down the preferable strategy.

Next, we investigated how individual preferences translated into demand. By maximizing their consumer surplus, individuals' determine how much they demand at any given price. Aggregating individual demand curves yields the firm's aggregate demand curve.

Knowing demand, we could then derive marginal revenue under simple pricing. Under simple pricing, marginal revenue has two components: a "good-news" component that points toward increased revenue if more units are sold; and a "bad-news" component—the driving-down-the-price effect—that arises because the price received on *all* units must be lowered to sell more units.

Once marginal revenue is determined, we can employ the steps outlined above to determine the profit-maximizing price.

Lastly, we considered the elasticity of demand and how it related to the profit-maximizing price via Lerner's markup rule.



## Advanced Pricing

# 5

In this lecture note, we consider other forms of pricing. Although simple pricing (sometimes called *linear pricing* or *uniform pricing*) is quite prevalent, it is neither the only nor the most desirable form of pricing. On your trips to the supermarket, you have no doubt noticed that the liter bottle of soda does not cost twice as much as the half-liter bottle of soda. In your purchases of airplane tickets, you have no doubt noticed that how much you pay is a function of what restrictions you are willing to accept (*e.g.*, staying over a Saturday night). Some goods and services you buy may have entry fees (*e.g.*, club membership) and per-use charges (*e.g.*, greens fees). All of these are examples of *price discrimination* or *nonlinear pricing*.

### What Simple Pricing Loses

## 5.1

To motivate other forms of pricing, it helps to know what simple pricing's deficiencies are. Basically, there are two. First, simple pricing leaves "money on the table," in a sense to be made formal below. Second, it leaves potential profit in the hands of consumers in the form of their consumer surplus.

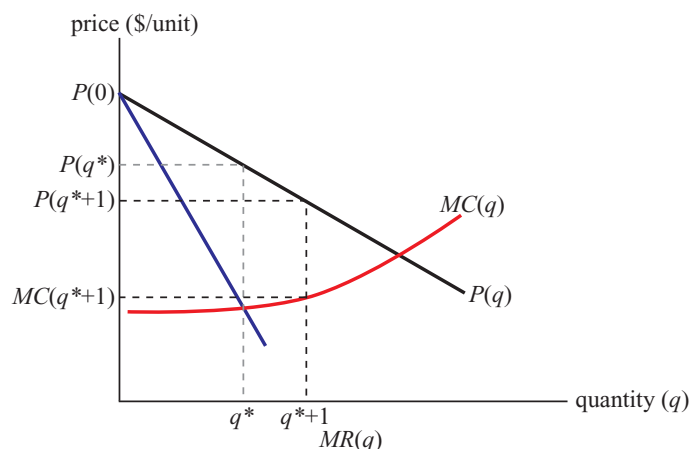
What do we mean that simple pricing "leaves money on the table"? Consider Figure 5.1. It reproduces Figure 4.8. Now suppose the firm in question wishes to sell one more unit. Observe that if it could sell that additional unit, the  $q^* + 1$ st unit, *without* having to lower the price on the other  $q^*$  units, then it would profit by doing so. Some consumer is willing to pay the firm  $P(q^* + 1)$  for the  $q^* + 1$ st unit and the incremental cost to the firm of producing that  $q^* + 1$ st unit is  $MC(q^* + 1)$ . As Figure 5.1 shows,

$$MC(q^* + 1) < P(q^* + 1).$$

In other words, if the firm could charge  $P(q^* + 1)$  for the  $q^* + 1$ st unit, but still charge  $P(q^*)$  for the other  $q^*$  units, it would gain

$$P(q^* + 1) - MC(q^* + 1)$$

in profit. Of course, under simple pricing, it *cannot* charge a price for the  $q^* + 1$ st unit that is different than the price it charges for the other  $q^*$  units. And the fact that it would, under simple pricing, have to lower the price on all  $q^*$  units to induce someone to buy the  $q^* + 1$ st unit, makes selling that  $q^* + 1$ st unit undesirable.

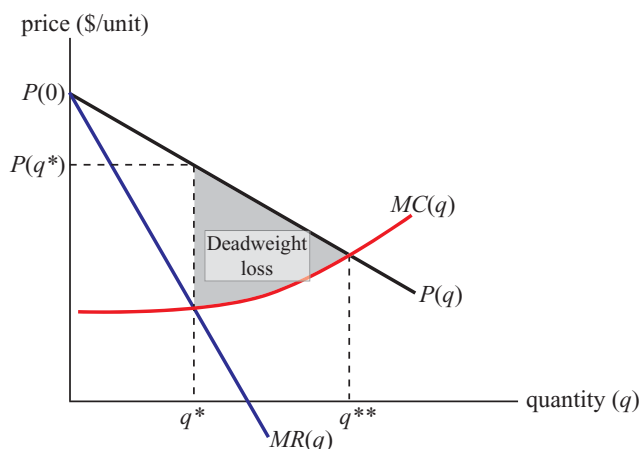


**Figure 5.1:** Were it *not* for the consequent driving-down-the-price effect, the firm would find it profitable to sell the  $q^* + 1$ st unit at price  $P(q^* + 1)$ .

In essence, then, the firm is leaving  $P(q^* + 1) - MC(q^* + 1)$  on the table; that is, it is a *potential* sale that *would* be profitable except for the driving-down-the-price effect of simple pricing. Observe, too, that we could repeat this analysis for the  $q^* + 2$ nd unit, the  $q^* + 3$ rd unit, and so forth until we reach the unit at which inverse demand falls below marginal cost (obviously, we would never want to sell the  $q$ th unit if  $P(q) < MC(q)$ ). Treating units as continuous, we have thus shown that the firm forgoes the area labelled deadweight loss in Figure 5.2. This region is called the *deadweight loss* because it corresponds to the potentially profitable sales that don't get made.

What these two figures show is that if a firm could devise a way to price that avoided the driving-down-the-price effect, then it could potentially increase its profits over those it can earn using simple pricing.

We also observed that simple pricing leaves profits, in the form of consumer surplus, in the hands of consumers. The firm would also like to capture some of this, as well. To understand this second “loss” from simple pricing, however, we will need digress and show how individual consumer surplus aggregates to overall consumer surplus.



**Figure 5.2:** The shaded triangular region is the *deadweight loss* induced by simple pricing.

## Aggregate Consumer Surplus

# 5.2

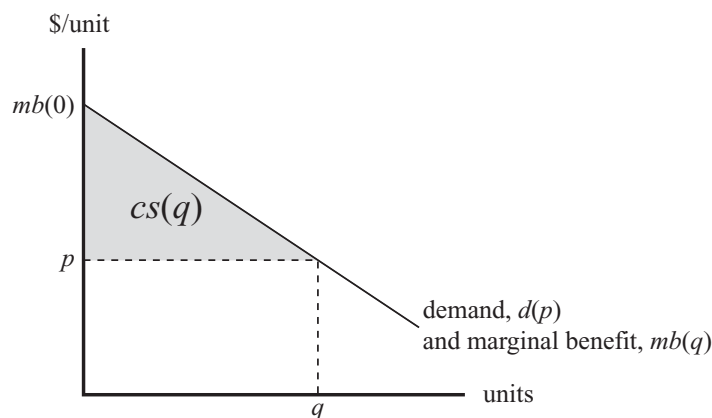
Our objectives in this section are (i) to reconsider individual consumer surplus and (ii) show how it aggregates into overall or total consumer surplus.

Recall, from our derivation of individual demand (see pages 71–75), that an individual's consumer surplus is

$$cs(q) = b(q) - pq$$

when she pays  $p$  per unit. The function  $b(\cdot)$ , recall, is her benefit function and we interpret  $b(q)$  to be her benefit, measured in dollars, of having  $q$  units of the good in question. Also remember that, for  $q > 0$ , her demand curve and her *marginal* benefit schedule are coincident (see Figure 4.5). For  $q = 0$ , her demand curve is coincident with the vertical (price) axis. Lastly, remember, Result 18: The area under a marginal curve is the total function. Observe, therefore, that a consumer's total benefit from  $q$  units,  $b(q)$ , is the area under her *marginal* benefit curve from 0 to  $q$  units.<sup>1</sup> Her marginal cost is just  $p$ , so her total cost of  $q$  units is the area under the price line (the horizontal line of height  $p$ ) from 0 to  $q$  units; that is,  $pq$ . Graphically, if we subtract the area under the price line

<sup>1</sup>In case you are wondering why we wrote  $b(q)$  and not  $b(q) - b(0)$ , the answer is that  $b(0) = 0$ .



**Figure 5.3:** If we subtract the area of the rectangle whose lower left corner is the origin and whose upper right corner is  $(q, p)$  from the area beneath the marginal benefit curve,  $mb(\cdot)$ , from 0 to  $q$  units, we're left with the shaded triangle, which is the individual's consumer surplus,  $cs(q)$ , from  $q$  units purchased at  $p$  per unit.

from the area under the marginal benefit curve, then we get the area under the marginal benefit curve but above the price line. In other words, the individual's consumer surplus is the area beneath her marginal benefit curve and above the price line from 0 to  $q$  units. Figure 5.3 illustrates.

**Consumer surplus:** *The consumer's value of the right to buy a given number of units at a given price per unit.*



One way to think about the individual's consumer surplus is the following. What is it worth to the consumer to be able to buy  $q$  units at a price of  $p$  per unit? Well her profit, her consumer surplus, is  $cs(q)$ . If she had to pay more than  $cs(q)$  for the ability to purchase  $q$  units at  $p$  per unit, she wouldn't do it—she would lose. If she had to pay less than  $cs(q)$ , she would do it because she'd still come out ahead. Therefore, we can consider  $cs(q)$  to represent the most that the consumer would be willing to pay for the right to buy  $q$  units at a price of  $p$ .

An area is an area. It doesn't matter, therefore, if we view the individual's consumer surplus as the area beneath the marginal benefit schedule, but above the price line, from 0 to  $q$  units or we view it as the area under her demand curve from  $p$  to  $mb(0)$ . Suppose, further, that we add to this area the area under her demand curve from  $p = mb(0)$  to infinity. This doesn't change the amount of the original area because there is no area under the demand curve from  $mb(0)$  to infinity (recall the demand line is coincident with the vertical axis in this interval). Hence, we don't change the area shown in Figure 5.3 if we add on the area under her demand curve from  $mb(0)$  to infinity (the reason for adding this zero will become clear later). We can, therefore, describe the individual's consumer surplus as follows.

**Result 31** *At price  $p$ , an individual's consumer surplus is the area under her demand curve from  $p$  to infinity.*

In light of this result, in addition to writing her consumer surplus as  $cs(q)$ , we will also write it as  $A_p^\infty(d)$ , where this notation means the area under demand from  $p$  to infinity.<sup>2</sup>

It makes sense to define aggregate consumer surplus as the sum of the individuals' consumer surpluses. Let  $CS$  denote aggregate consumer surplus. At a given price,  $p$ , we have, therefore, that

$$CS = A_p^\infty(d_1) + \cdots + A_p^\infty(d_N),$$

where  $d_i$  denotes the demand of the  $i$ th individual in a population of  $N$  consumers. The area operator,  $A_p^\infty(\cdot)$ , is linear, meaning that we can distribute it out of the summation. That is,

$$\begin{aligned} CS &= A_p^\infty(d_1) + \cdots + A_p^\infty(d_N) \\ &= A_p^\infty(d_1 + \cdots + d_N) \quad (\text{distributing out}) \\ &= A_p^\infty(D) \quad (\text{sum of individual demand is aggregate demand}). \end{aligned}$$

In other words, we've just established the following.<sup>3</sup>

**Result 32** *Aggregate consumer surplus,  $CS$ , at price  $p$  is the area under the aggregate demand curve,  $D(\cdot)$ , between  $p$  and infinity.*

We also know that

$$\begin{aligned} CS &= cs_1(q_1) + \cdots + cs_N(q_N) \\ &= (b_1(q_1) - pq_1) + \cdots + (b_N(q_N) - pq_N) \\ &= \sum_{i=1}^N b_i(q_i) - \sum_{i=1}^N pq_i \\ &= \sum_{i=1}^N b_i(q_i) - pQ, \end{aligned}$$

where  $Q$  is the total amount demanded at  $p$ ; that is,  $Q = D(p)$ . Hence,

$$A_p^\infty(D) = CS = \sum_{i=1}^N b_i(q_i) - pQ. \quad (5.1)$$

<sup>2</sup>  $\int dx$  Observe  $A_p^\infty(d) = \int_p^\infty d(z)dz$ , where  $z$  is a dummy of integration.

<sup>3</sup>  $\int dx$  Observe,

$$CS = \sum_{i=1}^N \int_p^\infty d_i(z)dz = \int_p^\infty \left( \sum_{i=1}^N d_i(z) \right) dz = \int_p^\infty D(z)dz,$$

where the first equals follows because the summation operator can be passed through the integral (*i.e.*, integration is a linear operation).

Observe that we can also view  $A_p^\infty(D) + pQ$  as the area under the aggregate *inverse* demand curve from 0 to  $Q$  units. From the last equation,

$$\text{Area under aggregate inverse demand} = \sum_{i=1}^N b_i(q_i) \equiv B(Q),$$

where we define  $B(Q)$ , aggregate benefit, to be the sum of the individual benefits. We have, thus, derived:

**Result 33** *The area under the inverse demand curve,  $P(\cdot)$ , from 0 to  $Q$  units is the total or aggregate benefit,  $B(Q)$ , enjoyed by consumers from the  $Q$  units.*

If the area under the aggregate inverse demand curve is aggregate benefit, then it must be that the curve itself is marginal aggregate benefit (recall Rule 18).<sup>4</sup> In other words, if  $MB(\cdot)$  is the marginal aggregate benefit schedule, then we've shown that

**Result 34**  $P(q) = MB(q)$  for all  $q > 0$ .

## The Holy Grail of Pricing | 5.3

Observe that the benefit generated by  $q$  units sold is  $B(q)$ . The cost to the firm of producing  $q$  units is  $C(q)$ . Hence, the total value created is  $B(q) - C(q)$ . This difference is called *total welfare* and denoted by  $W(q)$ ; that is,

$$W(q) = B(q) - C(q).$$

If a firm sold  $q$  units, it could never hope to gain more profits than  $W(q)$ . Why? Because  $W(q)$  is the total value created—the whole “pie” as it were—so if the firm were getting more than the whole pie, then consumers must be receiving less than zero. But consumers will choose not to buy rather than get less than zero. Therefore,  $W(q)$  represents the most that the firm could ever hope to capture.

Suppose the firm could capture all of  $W(q)$ . How much would it choose to produce? Looking at  $B(\cdot)$  as being like  $R(\cdot)$ , our earlier analysis of simple pricing tells us that the candidate for the welfare-maximizing amount,  $q^{**}$ , is the one that solves

$$MB(q) = MC(q).$$

From Result 34, this can be rewritten as

$$P(q^{**}) = MC(q^{**}).$$

We can summarize this as

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<sup>4</sup>  $\int dx$  This is also the fundamental theorem of calculus again (see Theorem 1).

**Result 35** *If the firm can capture all the welfare generated from selling  $q$  units (i.e.,  $W(q)$ ), then the firm will want to produce to the point at which inverse demand intersects marginal cost.*

Figure 5.2 (page 95) illustrates.

From Figure 5.2, we see that part of the gain the firm would enjoy, were it to capture maximized welfare, is the deadweight loss. That is, were it able to capture all the welfare, it would be able to pick up the money left on the table by simple pricing. Furthermore, because the firm is capturing all the gains from trade (i.e., all the value created), there can't be any left for the customers. That is the firm is also capturing all of the consumers' surplus as well.

As noted, producing  $q^{**}$  units and capturing all of welfare is the very best the firm could ever do. It is, therefore, the “Holy Grail” of pricing; it is what the firm would ideally like to do.

Because this outcome is so good, any form of pricing that achieves this Holy Grail is known as *perfect price discrimination*. For historic reasons, perfect price discrimination is also known as *first-degree price discrimination*.

## Two-part Tariffs

# 5.4

Can the firm ever obtain the Holy Grail? Generally, the answer is no. However, it can certainly get closer than it does by using simple pricing. In fact, if all consumers are the same in terms of their individual demand for the good, then the firm could actually obtain the Grail.

In this section, we explore a type of pricing called a *two-part tariff*. A two-part tariff can help get a firm closer to the Grail than can simple pricing. A two-part tariff is a pricing scheme (tariff) with, as the name indicates, two parts. One part is what's called the *entry fee*. It is the amount that the consumer must pay *before* she can buy any units at all. In this sense, it is like a consumer overhead charge. As a real-life example, consider it to be like the fee one pays to enter an amusement park.<sup>5</sup> The second part of a two-part tariff is the *per-unit charge*. It is the amount that the consumer must pay for each unit she chooses to purchase. At an amusement park, it would correspond to the price of each ride ticket. In some instances, as with some—but not all—amusement parks, the per-unit charge might even be set to zero.

Formally, consider a two-part tariff consisting of an entry fee,  $F$ , and a per-unit charge,  $p$ . A consumer's expenditure,  $T(q)$ , if she buys  $q$  units is

$$T(q) = \begin{cases} 0, & \text{if } q = 0 \\ pq + F, & \text{if } q > 0 \end{cases}.$$

The function  $T(\cdot)$  is the tariff.

How much will a consumer buy under this tariff? If she pays the entry fee, then it is sunk; and her decision on how many to buy is the same as under

**Two-part tariff:**  
*Pricing with an entry  
fee and per-unit  
charge.*

<sup>5</sup>In fact, for this reason, two-part tariffs are sometimes called “Disneyland” pricing.

simple pricing. That is, she chooses the  $q$  that equates her marginal benefit to  $p$  (i.e., the  $q$  such that  $mb(q) = p$ ). This is equivalent to saying that she buys  $d(p)$  units, where  $d(\cdot)$  is her individual demand schedule. We noted earlier that the value to a consumer of being *able* to buy  $q$  units at a price of  $p$  per unit was worth  $cs(q) = cs(d(p))$ . Hence, when she decides whether or not to pay the entry fee, she asks herself whether  $F$  exceeds  $cs(d(p))$ . If it does, then it isn't worth it to her to pay the fee; she's better off not buying at all. If it doesn't, then she should pay the fee. We can summarize the consumer's decision as

$$\text{buy} \begin{cases} 0 \text{ units if } F > cs(d(p)) \\ d(p) \text{ units if } F \leq cs(d(p)) \end{cases} . \quad (5.2)$$

The question of whether  $F$  is less than or greater than  $cs(d(p))$  is known as a *participation constraint*. It determines whether or not the consumer is willing to participate (i.e., buy) under the pricing scheme.

### All Consumers are Identical

In this section, we consider the case in which there are  $N$  consumers, each of whom has the same demand for the firm's product or service. Using expression (5.2), we see that each consumer buys  $d(p)$  units if  $F \leq cs(d(p))$  and 0 units otherwise. This means the firm's profits are

$$\pi = \begin{cases} 0, & \text{if } F > cs(d(p)) \\ N(F + pd(p)) - C(Nd(p)), & \text{if } F \leq cs(d(p)) \end{cases} , \quad (5.3)$$

where the  $Nd(p)$  term arises because if it sells  $d(p)$  units to each customer, it sells  $Nd(p)$  units in total.

We know the firm wants to operate, so it will set

$$F \leq cs(d(p)) .$$

From expression (5.3), the firm's profit is increasing in  $F$ , so it wants to make  $F$  as large as possible. But as just seen, the largest possible  $F$  is  $cs(d(p))$ . Therefore, the profit-maximizing entry fee is  $F = cs(d(p))$ . Using this, we can substitute out  $F$  in expression (5.3) and write

$$\pi = N \left( cs(d(p)) + pd(p) \right) - C(Nd(p)) .$$

Observe that  $Nd(p)$  is aggregate demand,  $D(p)$ . Let  $Q = D(p)$  and observe that  $p = P(Q)$ , where  $P(\cdot)$  is aggregate inverse demand. We can, therefore, rewrite the firm's profit as

$$\pi = N cs \left( \frac{Q}{N} \right) + P(Q)Q - C(Q) .$$

Recall from our discussion in Section 5.2 that aggregate consumer surplus,  $CS$ , is the sum of the individual consumer surpluses. Hence,

$$\pi = CS + P(Q)Q - C(Q) .$$

Observe, in this last expression, that profit is the sum of the consumers' surplus *and* the profits that the firm would have made under simple pricing. We can see already, therefore, that this two-part tariff yields greater profits than simple pricing would.

Finally, recall from expression (5.1) that  $CS$  is total benefit minus  $pQ$ . We have, therefore, that

$$CS = B(Q) - P(Q)Q;$$

hence,

$$\begin{aligned}\pi &= B(Q) - P(Q)Q + P(Q)Q - C(Q) \\ &= B(Q) - C(Q) = W(Q).\end{aligned}\tag{5.4}$$

We've thus shown that, if all the consumers are identical (homogeneous), profits under a two-part tariff achieve the Holy Grail! That is, a two-part tariff allows the firm to capture all the gains to trade. In other words, a two-part tariff with identical consumers achieves perfect or first-degree price discrimination.

From Result 35, we know that a firm that captures all the gains to trade maximizes profit by producing the amount that equates inverse demand and marginal cost. We can, therefore, conclude as follows.

**Result 36** *Suppose consumers have identical demands (are homogeneous). Under the profit-maximizing two-part tariff, the firm*

- (i) *produces  $q^{**}$  units, where  $P(q^{**}) = MC(q^{**})$ ;*
- (ii) *sets the per-unit charge,  $p$ , to equal  $P(q^{**})$ ; and*
- (iii) *sets the entry fee,  $F$ , to equal  $cs(d(p))$  (or, equivalently, sets it to equal  $CS/N$ ).*

**Example 17 [An amusement park]:** Consider an amusement park. Ignoring possible congestion costs, it seems reasonable to approximate the park's marginal cost of a seat on a ride as being 0. Suppose that all 50,000 patrons who come into the park have approximately the same demand for rides. Suppose that market research has revealed that demand to be

$$d(p) = 50 - 12.5p$$

if  $p \leq 4$  and  $d(p) = 0$  otherwise. What is the optimal two-part tariff for this amusement park to use.

From Result 36, the per-unit charge—that is, the price per ride—should be set equal to marginal cost, which in this case is zero. So  $p = 0$ . What about the entry fee? It should be set to  $cs(d(0))$ . We can calculate this quantity in two ways. First, we can calculate it as  $A_0^\infty(d)$ ; that is, as the area under individual demand from 0 to infinity. Alternatively, we can determine the marginal-benefit schedule and, then, calculate consumer surplus as the area under the marginal-benefit schedule and above the price line (here, the horizontal axis) from 0 to  $d(0)$  units. For purposes of illustration, we will do both.

At  $p = 4$ ,  $d(p) = 0$ . Hence, the area under the demand curve from 0 to infinity is equal to the area under the demand curve from 0 to 4. If you plot the demand curve, you will see that it is a right triangle with height 4 and, because  $d(0) = 50$ , width 50. Its area is, thus, 100.<sup>6</sup> So the entry fee is \$100.

Alternatively, we know

$$q = 50 - 12.5mb(q);$$

hence,

$$mb(q) = 4 - .08q.$$

Plotting this, we see it is a triangle with height 4 and width  $d(0) = 50$ . So its area is also 100 (of course, it had to be the same—these are equivalent procedures). So the entry fee is \$100.

**Example 18:** A firm's cost is  $C(q) = q^2/2000$ . It faces 1000 identical customers, each of whom has demand

$$d(p) = \begin{cases} 10 - p, & \text{if } p \leq 10 \\ 0, & \text{if } p > 10 \end{cases}.$$

What is the optimal two-part tariff for it to employ?

Unlike the amusement park example, here marginal cost is not a constant. Using Result 15, marginal cost is  $MC(q) = q/1000$ . We, therefore, need to calculate  $P(\cdot)$ . To do so, first we need to determine  $D(\cdot)$ , then invert:

$$D(p) = 1000d(p) = \begin{cases} 10,000 - 1000p, & \text{if } p \leq 10 \\ 0, & \text{if } p > 10 \end{cases}.$$

Where demand is positive, we have

$$q = 10,000 - 1000P(q);$$

hence,

$$P(q) = 10 - \frac{q}{1000}.$$

Equating  $P(q)$  and  $MC(q)$ , we have:

$$10 - \frac{q}{1000} = \frac{q}{1000}.$$

So  $q^{**} = 5000$ .  $P(q^{**}) = P(5000) = 5$ . So the per-unit charge is \$5.

What about the entry fee? The area under  $d(\cdot)$  from 5 to infinity is, because  $d(10) = 0$ , the same as the area under  $d(\cdot)$  from 5 to 10. Plotting, we see that this area corresponds to the area of the right triangle with vertices (0,5), (5,0), and (0,10). Its height is 5 and its width is 5, so its area is 12.5. Therefore, the entry fee is \$12.50.

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<sup>6</sup>The area of a triangle is  $bh/2$ , where  $b$  is the width of the base and  $h$  is the height of the triangle.

### Consumers are Heterogeneous

What if consumers don't all have the same demand curves? The firm can still profit from using a two-part tariff, but designing the optimal two-part tariff becomes much more complicated. Moreover, the elements of its design depend critically on a number of properties of the demand curves and how they vary across individuals (for example, the analysis is sensitive to whether individual demand curves cross each other or not). Because it is so involved, an investigation of two-part tariff design with heterogeneous customers is beyond the scope of an introductory course like 201A.<sup>7</sup>

One strategy, which we will explore in greater depth when we deal with third-degree price discrimination, would be to divide customers into homogeneous groups, then employ the optimal two-part tariff on each group. For example, an amusement park might conclude that seniors have very different demands for rides than others, so it has a different admission price for them.

### Real-life Use of Two-part Tariffs

Amusement parks have already been offered as a real-life example of firms that use two-part tariffs. Cover charges at bars, night clubs, and similar venues, are examples of entry fees, with the price per drink being the per-unit charge. Club stores (*e.g.*, Costco) that have membership fees are yet another example of two-part tariffs.

Because the per-unit charge could be zero, many two-part tariffs are hard to recognize initially. For example, service plans or tech-support plans are “disguised” two-part tariffs: The entry fee is the price of the plan and the per-unit charge is often zero. Of course, some plans like these have both a positive entry fee (plan price) and a positive per-unit charge (service-call charge).

The pricing of telephony is another example. For land-line phones, a common plan is one in which you pay an access fee per month plus some amount per call. Many calling plans are more complicated than this; they, for instance, charge a monthly fee, provide some minutes worth of calls at a very low fee (often zero), and then charge a different price per call for minutes beyond the provided minutes. These plans are examples of [multi-part tariffs](#). Some of our analysis of second-degree price discrimination touches on multi-part tariffs, but a general analysis of multi-part tariffs is beyond the scope of this course.

### What Limits Two-part Tariffs?

If two (or multi-) part tariffs are so great, why don't we see more of them? That is, why are amusement park rides and telephone minutes sold using such tariffs, while things such as cereal and soda are not (or at least appear not to be)?

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<sup>7</sup>For the motivated, a good book on nonlinear pricing is Robert B. Wilson's *Nonlinear Pricing*, Oxford: Oxford University Press, 1993. This book will only be accessible if you're very comfortable with calculus.

The answer has, in part, to do with arbitrage.<sup>8</sup> Suppose a grocery store attempted to sell soda using a two-part tariff. It would pay someone to break the scheme by paying the entry fee, then buying more soda than he wants in order to resell it to others. As long as his resale price exceeds his average cost,  $p + F/q$ , where  $q$  is the amount he is reselling, he is making a profit. Because they are avoiding the entry fee, his customers can find it cheaper to buy soda from him than from the grocery store.

As an example, recall Example 18. Suppose you ask a friend to buy 10 units instead of 5. Your friend pays a total of

$$F + 10p = 12.50 + 10 \times 5 = 62.50.$$

Suppose you buy 5 of his units for \$30. Your net surplus is<sup>9</sup>

$$b(5) - 30 = 37.50 - 30 = 7.50.$$

Your friend's net surplus is

$$b(5) + 30 - 62.50 = 5.$$

That is, you and your friend are up \$12.50—the amount of the saved entry fee.

So we see that a grocery store attempting to sell soda using a two-part tariff would find its scheme collapsing due to arbitrage. In the limit, it would sell lots of soda to just one enterprising customer.

Unlike soda, which is easily resold, items like amusement park rides, telephone calls from your house or cell phone, service calls to your place of business, etc., are difficult, if not impossible, to resell. There is little danger of arbitrage. Hence, we expect to see two-part tariffs with items that are difficult to resell and not to see such tariffs with items that are readily resold.

However, firms can attempt to defeat arbitrage. For instance, the grocery store could use a two-part tariff for some goods in the following way. Suppose that rather than being sold in packages, sugar were kept in a large bin and customers could take as much or as little as they like (*e.g.*, like fruit at most groceries or as is actually done at stores like Berkeley Natural Grocery). Suppose that, under the optimal two-part tariff, each consumer would buy two pounds of sugar at a price  $p$  per pound, which would yield him or her surplus of  $cs$ , which would be captured by the store using an entry fee of  $F = cs$ . Of course, arbitrage makes this infeasible. So suppose, in response, that rather than let consumers buy as much or as little sugar as they want, the store packaged sugar in two-pound bags, for which it charged  $2p + cs$  per bag. Each customer would face the decision of whether to have 0 pounds or 2 pounds. Each customer's

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<sup>8</sup>Some other reasons for not using two-part tariffs are the costs of monitoring entry and heterogeneity among customers.

<sup>9</sup>It is readily seen that your  $mb(\cdot)$  schedule is  $10 - q$ , so  $b(5)$  is the area under  $10 - q$  from 0 to 5. This can be shown to equal 37.5 (the area is the area of a trapezoid with height 5 and bases 10 and 5, so its area is  $5 \times (5 + 10)/2 = 37.5$ ; alternatively, it is  $\int_0^5 (10 - q)dq$ ).

total benefit from two pounds is  $2p + cs$ , so each would just be willing to pay  $2p + cs$  for the package of sugar. Because the entry fee is paid on every two-pound bag, the store has devised an arbitrage-proof (albeit disguised) two-part tariff. In other words, *packaging*—taking away customers ability to buy as much or as little as they wish—can represent an arbitrage-proof way of employing a two-part tariff.

### Metering

In addition to packaging and setting the per-unit price to zero, a third way in which two-part tariffs are disguised is via *metering*. Metering—also known as *tying*—has to do with situations in which the entry fee is the purchase price for a durable good (*e.g.*, an instant camera, a razor handle, a punch-card sorting machine, etc.) and the per-unit charge is the purchase price for a complementary good (*e.g.*, instant film, cartridge razor blades, punch cards, etc.). For example, one can view Gillette as using a two-part tariff to sell Mach 3 razor cartridges. The price of the handle (or a package with handle and some cartridges) represents the entry fee. The price of cartridge replacement packs is the per-unit charge.

Metering only works if the firm can keep others from making the complementary good. In the cases of Gillette or Polaroid, intellectual property rights prevent others from making cartridges that fit Gillette handles or instant film that works in Polaroid cameras. For commodity products, like punch cards and paper, firms used to attempt to prevent others from providing the complementary good through contracts. That is, IBM would, through various contract terms, require or strongly encourage users of its punch-card sorting machines to buy IBM punch cards. Xerox would likewise require or strongly encourage users of its photocopiers to use Xerox paper (as well as Xerox toner, etc.). In the US, such requirements or pressure via contract have been deemed to violate the antitrust laws (these are forms of illegal tying).

Although we have introduced metering as a way to engage in a two-part tariff, metering can also be useful for other forms of price discrimination (*e.g.*, second-degree price discrimination through quantity discounts). Metering can also be useful as a way of providing customer credit—the camera price is a down payment, the customer is “loaned” the rest of the camera’s full price, and the film price contains a repayment (principal and interest) portion on the original loan taken out by buying the camera. For instance, if the auto companies could compel you to buy gasoline from them only, then they could provide car financing through a low car price and a high gasoline price.

## Third-degree Price Discrimination

# 5.5

**Third-degree price discrimination:**  
Charging different prices on the basis of observed group membership.

In this section, we consider third-degree price discrimination.<sup>10</sup> *Third-degree price discrimination* means charging different tariffs to different consumers on the basis of identifiable characteristics. An example of third-degree price discrimination is a movie theater that charges a different price for children, a different price for seniors, and a different price for other adults.<sup>11</sup> The characteristics on which the theater distinguishes among patrons are readily identifiable, either by cashier observation or presentation of ID.

Senior-citizen discounts, child discounts, student discounts, and family discounts are all familiar forms of third-degree price discrimination. Ladies' nights at bars and similar venues are another form of third-degree price discrimination (albeit one also motivated by network externalities).<sup>12</sup> Sometimes third-degree price discrimination is on the basis of membership in certain organizations (*e.g.*, AAA discounts or discounts to KQED members).

Geography-based third-degree price discrimination is also quite prevalent. For example, a firm may quote different prices to buyers in one area than it does to buyers in another area. This happens, for instance, in air travel, where a round-trip ticket going city A to city B back to city A has a different price than one going B to A back to B (this is especially true if A and B are in different countries). A third-degree pricing example that has been much in the news of late is the different prices pharmaceutical companies charge for their drugs in different countries.

### The Simple Case

The most basic situation of third-degree price discrimination is when a firm engages in simple pricing to two distinct groups when the firm faces no capacity constraints.

Let  $P_i(\cdot)$  be the inverse aggregate demand of population  $i$ ,  $i = 1$  or  $2$ . For instance,  $P_1(\cdot)$  could be the inverse aggregate demand of students for a concert and  $P_2(\cdot)$  could be the inverse aggregate demand of non-students. Let  $C(\cdot)$  be the firm's cost function. Let  $q_i$  be the amount sold to population  $i$ . The firm's profit is its revenue from each population less its costs:

$$\pi(q_1, q_2) = q_1 P_1(q_1) + q_2 P_2(q_2) - C(q_1 + q_2).$$

Using expression (4.17), we can write

$$MR_i(q_i) = P_i(q_i) + P'_i(q_i)q_i.$$

<sup>10</sup>Yes, we can count. Regardless of the order implied by their names, it makes more sense pedagogically to consider third-degree price discrimination before second-degree price discrimination.

<sup>11</sup>If the over-60 set are senior citizens, does that make the rest of us "junior citizens"?

<sup>12</sup>Network externalities refer to situations in which one person's demand (*e.g.*, a guy looking to meet women) is a function of the demand of others (*e.g.*, how many women are at the bar).

We denote marginal cost in the usual way,  $MC(q_1 + q_2)$ .

How much should the firm sell to each population? From our earlier analysis, a good guess would be the amounts that equate the marginal revenues to marginal cost. This guess is, in fact, correct. Let's see why. Clearly, it cannot be optimal to produce so that

$$MR_i(q_i) < MC(q_1 + q_2)$$

because the firm could increase profits by

$$MC(q_1 + q_2) - MR_i(q_i)$$

if it sold one less unit to population  $i$ . What if

$$MR_i(q_i) > MC(q_1 + q_2)? \quad (5.5)$$

Then the firm could increase profits by

$$MR_i(q_i) - MC(q_1 + q_2)$$

if it sold one more unit to population  $i$ . So it cannot be optimal to produce so that expression (5.5) holds. By process of elimination, this leaves

$$MR_1(q_1^*) = MR_2(q_2^*) = MC(q_1^* + q_2^*) \quad (5.6)$$

at the profit-maximizing quantities,  $q_1^*$  and  $q_2^*$ . This analysis generalizes to  $N$  populations:

**Result 37** *A firm unconstrained in the amount it can produce and that can engage in simple pricing to  $N$  distinct populations maximizes its profit by selling  $q_i^*$  to population  $i$ , where*

$$MR_1(q_1^*) = \cdots = MR_N(q_N^*) = MC\left(\sum_{i=1}^N q_i^*\right).$$

Observe this expression can also be written as

$$\begin{aligned} MR_1(q_1^*) &= MC\left(\sum_{i=1}^N q_i^*\right) \\ &\vdots \\ MR_N(q_N^*) &= MC\left(\sum_{i=1}^N q_i^*\right). \end{aligned}$$

As with simple pricing, the price charged population  $i$  is  $P_i(q_i^*)$ .

Because a possible result with this type of third-degree price discrimination is that  $P_1(q_1^*) = \cdots = P_N(q_N^*)$ , optimal nondiscriminatory simple pricing (*i.e.*,

treating all populations the same) is a possible outcome of third-degree price discrimination. From this, we see that third-degree price discrimination can never do worse than nondiscriminatory simple pricing. Moreover, if we find  $P_i(q_i^*) \neq P_i(q_j^*)$  for two populations  $i$  and  $j$ , then it must be that third-degree price discrimination is generating strictly greater profits.

**Example 19:** Consider a firm (*e.g.*, a pharmaceutical company) that has the cost function  $C(Q) = Q^2/1000$ . Observe, using Result 15, that its marginal cost is  $Q/500$ . Suppose it faces two populations (*e.g.*, Americans and Canadians),  $i = 1, 2$ . Let their aggregate demands be

$$D_1(p) = \begin{cases} 100,000 - 1000p, & \text{if } p \leq 100 \\ 0, & \text{if } p > 100 \end{cases} \quad \text{and} \\ D_2(p) = \begin{cases} 150,000 - 2000p, & \text{if } p \leq 75 \\ 0, & \text{if } p > 75 \end{cases}.$$

Inverting these over the range of positive demand:

$$q_1 = 100,000 - 1000P_1(q_1) \quad \text{and} \\ q_2 = 150,000 - 2000P_2(q_2).$$

Therefore,

$$P_1(q_1) = 100 - \frac{q_1}{1000} \quad \text{and} \\ P_2(q_2) = 75 - \frac{q_2}{2000}.$$

Using expression (4.17), we have

$$MR_1(q_1) = 100 - \frac{q_1}{1000} + \left(-\frac{1}{1000}q_1\right) = 100 - \frac{q_1}{500} \quad \text{and} \\ MR_2(q_2) = 75 - \frac{q_2}{2000} + \left(-\frac{1}{2000}q_2\right) = 75 - \frac{q_2}{1000}.$$

We can now employ Result 37 to solve for  $q_1^*$  and  $q_2^*$ .

$$100 - \frac{q_1}{500} = \frac{q_1 + q_2}{500} \quad \text{and} \\ 75 - \frac{q_2}{1000} = \frac{q_1 + q_2}{500}.$$

Multiply the top equation through by 500 and the bottom equation through by 1000:

$$50,000 - q_1 = q_1 + q_2 \quad \text{and} \\ 75,000 - q_2 = 2q_1 + 2q_2.$$

The solution is  $q_1 = 18,750$  and  $q_2 = 12,500$ .<sup>13</sup> This yields prices of

$$P_1(18,750) = 81.25 \quad \text{and} \\ P_2(12,500) = 68.75.$$

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<sup>13</sup>Section A2.2 reviews how to solve two equations in two unknowns.

Observe the firm's profit is

$$18,750 \times 81.25 + 12,500 \times 68.75 - \frac{31,250^2}{1000} = 1,406,250.$$

Compare this profit to the profit the firm would make if restricted to nondiscriminatory simple pricing. Aggregate demand over both populations is

$$D(p) = \begin{cases} 250,000 - 3000p, & \text{if } p \leq 75 \\ 100,000 - 1000p, & \text{if } 75 < p \leq 100 \\ 0, & \text{if } p > 100 \end{cases}.$$

Considering only positive quantities, inverse demand,  $P(q)$ , solves

$$\begin{aligned} q &= 250,000 - 3000P(q) \text{ for } q \geq 25,000 \text{ and} \\ q &= 100,000 - 1000P(q) \text{ for } 0 \leq q < 25,000. \end{aligned}$$

We thus have

$$\begin{aligned} P(q) &= \frac{250}{3} - \frac{q}{3000} \text{ for } q \geq 25,000 \text{ and} \\ P(q) &= 100 - \frac{q}{1000} \text{ for } 0 \leq q < 25,000. \end{aligned}$$

We need to calculate marginal revenue over both segments of inverse demand. You should be able to verify that they are

$$\begin{aligned} MR(q) &= \frac{250}{3} - \frac{q}{1500} \text{ for } q \geq 25,000 \text{ and} \\ MR(q) &= 100 - \frac{q}{500} \text{ for } 0 \leq q < 25,000. \end{aligned}$$

If we equate the first expression to marginal cost,  $q/500$ , we get  $q = 31,250$ . If we equate the second expression to marginal cost, we get  $q = 25,000$ , which, because it is outside of the relevant range, is not a valid solution. This leaves us with  $q^* = 31,250$  and profits that are

$$31,250P(31,250) - \frac{31,250^2}{1000} = 31,250 \times 72.92 - \frac{31,250^2}{1000} = 1,302,080.$$

So the firm earns less relative to third-degree price discrimination if restricted to nondiscriminatory simple pricing. Of course, we knew the firm had to do better using third-degree price discrimination because the optimal scheme involved different prices for the two populations.

### The Capacity Constrained Firm

Consider a train company that is considering different prices to students and non-students for a very long trip (*e.g.*, a cross-country trip). Assume the company's cost function per train trip is

$$C(q) = \begin{cases} 0, & \text{if } q = 0 \\ q + 20,000, & \text{if } q > 0 \end{cases},$$

where  $q$  is the number of seats sold. Observe that marginal cost per seat is small, just \$1, while the overhead cost per trip is relatively high.

Unlike the situation in Example 19, the train company is constrained: It cannot sell more tickets than it has seats. This will pose a problem if the optimal third-degree pricing scheme indicates  $q_S^*$  tickets should be sold to students and  $q_N^*$  tickets should be sold to non-students, but

$$q_S^* + q_N^* > K, \quad (5.7)$$

where  $K$  is the number of seats on the train (the capacity).

Let's suppose that expression (5.7) is true. Then the firm's problem in designing the optimal third-degree price discrimination scheme is to choose the  $q_S$  and  $q_N$  that maximize profits,

$$q_S P_S(q_S) + q_N P_N(q_N) - \underbrace{(q_S + q_N + 20,000)}_{C(q_S + q_N)}$$

subject to the constraint

$$q_S + q_N \leq K.$$

By assumption, this constraint is binding; that is, we know  $q_S + q_N = K$  (if it weren't binding, then we could design the scheme ignoring it, but then we would get a solution,  $q_S^*$  and  $q_N^*$ , that violated the constraint). Substituting into our expression for profits, we find

$$q_S P_S(q_S) + q_N P_N(q_N) - \underbrace{(K + 20,000)}_{C(q_S + q_N)}.$$

Note that expenses,  $K + 20,000$ , are a constant; that is, our problem is independent of them. This makes sense from an opportunity-cost point of view: We've already decided we're selling all  $K$  seats—that decision is sunk—what we're deciding now is how to allocate those seats.

Thinking further along opportunity cost lines, we can solve for the optimal constrained allocation. What is the opportunity cost of selling the marginal seat to a student? It is the forgone value of selling that seat to a non-student. What's that value? It's the marginal revenue that would be realized by selling that seat to a non-student. In other words, we've just seen that the marginal cost of selling a seat to a student is the marginal revenue that seat would yield if sold to a non-student. Formally, we have

$$MC_S(q_S) = MR_N(q_N) = P_N(q_N) + P'_N(q_N)q_N,$$

where the second equality follows from the usual formula for marginal revenue, expression (4.17), and  $MC_S(\cdot)$  is the marginal cost schedule for seats sold to students.

Profit-maximization requires equating marginal revenue to marginal cost. So we know that

$$MR_S(q_S) = MC_S(q_S)$$

at the optimal number of seats to sell to students. Substituting for  $MC_S(q_S)$ , we have

$$MR_S(q_S) = MR_N(q_N) = P_N(q_N) + P'_N(q_N)q_N. \quad (5.8)$$

That is, given constrained capacity, the optimal quantities for the two populations will equate their marginal revenues. Expression (5.8) is one equation in two unknowns,  $q_S$  and  $q_N$ . Fortunately, we have a second equation, namely the constraint:

$$q_S + q_N = K. \quad (5.9)$$

We can conclude:

**Result 38** *If a firm faces a binding constraint of  $K$  units and it faces two populations,  $N$  and  $S$ , then the optimal quantities,  $\hat{q}_N$  and  $\hat{q}_S$ , to be sold to the two populations equate the two populations' marginal revenues and they sum to  $K$ ; that is, the quantities solve expressions (5.8) and (5.9).*

The prices to be charged to the two populations are  $P_S(\hat{q}_S)$  and  $P_N(\hat{q}_N)$  for students and non-students, respectively.

**Example 20:** Let's put some more numbers to our analysis of this train company's pricing problem. Specifically, suppose

$$P_S(q_S) = 401 - q_S \text{ and}$$

$$P_N(q_N) = 1001 - q_N.$$

Further suppose that  $K = 500$ .

If the number of seats on the train weren't constrained, the firm would determine its optimal allocation from  $MR_S(q_S) = MC(q_S + q_N)$  and  $MR_N(q_N) = MC(q_S + q_N)$  (see Example 19). Here, those expressions are

$$401 - 2q_S = 1 \text{ and}$$

$$1001 - 2q_N = 1,$$

respectively. The solutions are

$$q_S^* = 200 \text{ and } q_N^* = 500.$$

Because  $200 + 500 > 500$ , this solution doesn't work given the train's seating capacity.

Now employ Result 38: Equating the marginal revenues, we have

$$401 - 2q_S = 1001 - 2q_N.$$

We also have the constraint:

$$q_S + q_N = 500.$$

Solving these two equations yields<sup>14</sup>

$$\hat{q}_S = 100 \text{ and } \hat{q}_N = 400.$$

The ticket prices are

$$P_S(100) = 301 \text{ and } P_N(400) = 601$$

for students and non-students, respectively.

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<sup>14</sup>Section A2.2 reviews how to solve two equations in two unknowns.

### Third-degree Price Discrimination and Two-part Tariffs

If a firm can use two-part tariffs, then it may wish to offer different two-part tariffs to different populations. If marginal cost varies with output (*i.e.*, is not a constant), then the analysis is somewhat complex. For the sake of brevity, we will, therefore, skip a general analysis and just consider an example with constant marginal cost.

A pharmaceutical company wishes to sell a medicine to two different populations (*e.g.*, Americans and Canadians; or seniors and non-seniors). Let the marginal cost of a pill be \$5. Let the demand of *each* person in population #1 be  $d_1(p)$ . Let the demand of *each* person in population #2 be  $d_2(p)$ . We know from our study of optimal two-part tariffs that, with homogenous populations, the per-unit charge (*e.g.*, per-pill charge) equals marginal cost, \$5. So each member of population 1 buys  $d_1(5)$  pills and each member of population 2 buys  $d_2(5)$  pills. Assume, for prices that yield positive demand, we have

$$\begin{aligned}d_1(p) &= 25 - p \text{ and} \\d_2(p) &= 35 - p.\end{aligned}$$

The consumer surplus of a population-1 individual is the area under her demand curve from 5 to 25 (the latter because  $d(25) = 0$ ). This is a right triangle with base of  $d(5) = 20$  and height  $25 - 5 = 20$ , so  $cs_1 = 200$ .

The consumer surplus of a population-2 individual is the area under his demand curve from 5 to 35. This is a right triangle with base and height both equal to 30, so  $cs_2 = 450$ .

So the entry fee for population-1 individuals is \$200 and the entry fee for population-2 individuals is \$450.

Although the pharmaceutical company could, in theory at least, have its customers pay entry fees, then let them buy as many pills as they would like at \$5/pill, a more realistic strategy would be for it to utilize *packaging*. So it would sell bottles of 20 pills to population 1 consumers for  $20p + F_1 = 300$  dollars and it would sell bottles of 30 pills to population 2 consumers for  $30p + F_2 = 600$  dollars.

### Arbitrage

Because, under third-degree price discrimination, different populations face different tariffs, there is the opportunity for arbitrage. A student, for instance, could resell his ticket to a non-student. A population-1 patient could resell her bottle of pills to a population-2 patient. Hence, the ability to employ third-degree price discrimination effectively is dependent on an ability to prevent arbitrage (this is true, actually, of all discriminatory pricing).

In many instances, the ability to distinguish members of the different populations forecloses arbitrage. A non-senior with a senior ticket can be denied admission, for example. In other situations, especially when geography is used to identify different groups, it is much harder (*e.g.*, witness Americans going to

Canada to buy prescription medications). In some cases, transportation costs deter geographic arbitrage. This is why, for instance, a firm can charge very different prices in West Virginia and California, but needs to charge fairly similar prices in Alameda and Marin counties.

## Second-degree Price Discrimination | 5.6

As we've seen, third-degree price discrimination can yield greater profits than simple pricing. But it requires being able to identify the members of different populations (*e.g.*, students from non-students). What if you can't readily identify who's in what group? For instance, we know that business travellers are willing to pay more for plane tickets than vacationers. But mere inspection won't tell you which would-be flier is a business woman and which is a woman going on vacation.<sup>15</sup> Fortunately, in some circumstances, it is possible to *induce* customers to reveal the group to which they belong. Price discrimination via induced revelation of preferences is known as *second-degree price discrimination*.

**Second-degree price discrimination:**  
*Price discrimination via revealed preference.*

A well-known solution to the problem of not being able to distinguish business from leisure travellers by inspection is for the airlines to offer different kinds of tickets. For instance, because business travellers don't wish to stay over the weekend or often can't book much in advance, the airlines charge more for round-trip tickets that don't involve a Saturday-night stayover or that are purchased within a few days of the flight (*i.e.*, in the latter situation, there is a discount for advance purchase). Observe an airline still can't observe which type of traveler is which, but by offering different kinds of service it hopes to *induce revelation* of which type is which.

Restricted tickets are one example of second-degree price discrimination, specifically of second-degree price discrimination via *quality distortions*. Because a restricted ticket is less useful than an unrestricted ticket, a restricted ticket can be viewed as being lower quality than an unrestricted ticket. Other examples include:

- Different classes of service (*e.g.*, first and second-class carriages on trains). The classic example here is the French railroads in the 19th century, which removed the roofs from second-class carriages to create third-class carriages.
- Hobbling a product. This is popular in high-tech, where, for instance, Intel produced two versions of a chip by "brain-damaging" the state-of-the-art chip. Another example is software, where "regular" and "pro" versions (or "home" and "office" versions) of the same product are often sold.

<sup>15</sup>Although the two might dress differently, if airlines started charging different prices on the basis of what their passengers wore, then their passengers would all show up wearing whatever clothes get them the cheapest fare.

- Restrictions. Saturday-night stayovers and advance-ticketing requirements are a classic example. Another example is limited versus full memberships at health clubs.

The other common form of second-degree price discrimination is via *quantity discounts*. This is why, for instance, the liter bottle of soda is typically *less* than twice as expensive as the half-liter bottle. Quantity discounts can often be operationalized through multi-part tariffs, so many multi-part tariffs are examples of price discrimination via quantity discounts (*e.g.*, choices in calling plans between say a low monthly fee, few “free” minutes, and a high per-minute charge thereafter versus a high monthly fee, more “free” minutes, and a lower per-minute charge thereafter).

### Quality Distortions

Consider an airline facing  $N_b$  business travellers and  $N_\tau$  tourists on a given round-trip route. Suppose, for convenience, that the airline’s cost of flying are essentially all overhead costs (*e.g.*, the fuel, aircraft depreciation, wages of a *fixed*-size crew, etc.) and that the marginal costs per flier are effectively 0. Let there be two possible kinds of round-trip tickets,<sup>16</sup> restricted (*e.g.*, requiring a Saturday-night stayover) and unrestricted (*e.g.*, no stayover requirements); let superscripts  $r$  and  $u$  refer to these two kinds of tickets, respectively. Let  $\kappa$  denote an arbitrary kind of ticket (*i.e.*,  $\kappa = r$  or  $\kappa = u$ ).

A type  $\theta$ -flier ( $\theta = b$  or  $\tau$ ) has a valuation (benefit or utility) of  $v_\theta^\kappa$  for a  $\kappa$  ticket. Assume, consistent with experience, that

- $v_\theta^u \geq v_\theta^r$  for both  $\theta$  and *strictly* greater for business travellers. That is, fliers prefer unrestricted tickets all else equal and business travellers strictly prefer them.
- $v_b^\kappa > v_\tau^\kappa$  for both  $\kappa$ . That is, business travellers value travel more.

Clearly, if the airline offered only one kind of ticket, it would offer *unrestricted* tickets given that they cost no more to provide and they command greater prices. There are two pricing possibilities with only one kind of ticket (*i.e.*, with simple pricing): (i) sell to both types, which means  $p = v_\tau^u$ ; or (ii) sell to business travellers only, which means  $p = v_b^u$ . Option (i) is as good or better than option (ii) if and only if

$$N_b v_b^u \leq (N_b + N_\tau) v_\tau^u \text{ or, equivalently, } N_b \leq \frac{v_\tau^u}{v_b^u - v_\tau^u} N_\tau. \quad (5.10)$$

Alternatively, the airline could offer both kinds of tickets (price discriminate via quality distortion). In this case, the restricted ticket will be intended for one type of flier and the unrestricted ticket will be intended for the other. At some level, which kind of ticket goes to which type of flier needs to be determined,

<sup>16</sup> Assume all travel is round-trip.

but experience tells us, in this context, that we should target the unrestricted ticket to the business traveller and the restricted ticket to the tourist. Let  $p^\kappa$  be the price of a  $\kappa$  ticket.

The airline seeks to maximize the following in its choice of  $p^u$  and  $p^r$  (recall marginal cost is zero):

$$N_b p^u + N_\tau p^r. \quad (5.11)$$

It, of course, faces constraints. First, fliers need actually to purchase tickets. This means that buying the kind of ticket intended for you can't leave you with negative consumer surplus, because, otherwise, you would do better not to buy. Hence, we have *participation constraints*:

$$v_b^u - p^u \geq 0 \quad (5.12)$$

$$v_\tau^r - p^r \geq 0; \quad (5.13)$$

that is, the business traveller must do better buying than not buying (expression (5.12) must hold) and the tourist must do better buying than not buying (expression (5.13) must hold).

But we also have an additional set of constraints because the different types of flier need to be *induced* to reveal their types by purchasing the kind of ticket intended for them. Such *revelation* or *incentive compatibility* constraints are what distinguish second-degree price discrimination from other forms of price discrimination. The revelation constraints say that a flier type must do better (gain more consumer surplus) purchasing the ticket intended for her rather than purchasing the ticket intended for the other type:

$$v_b^u - p^u \geq v_b^r - p^r \quad (5.14)$$

$$v_\tau^r - p^r \geq v_\tau^u - p^u. \quad (5.15)$$

The first expression says that a business traveller must gain greater surplus from an unrestricted ticket than a restricted ticket. The second expression says that a tourist must gain greater surplus from a restricted ticket than an unrestricted ticket.

To summarize, the airline's problem is to maximize (5.11) subject to the four constraints, (5.12)–(5.15). This is a *linear* programming problem. To solve it, begin by noting that no solution would exist if

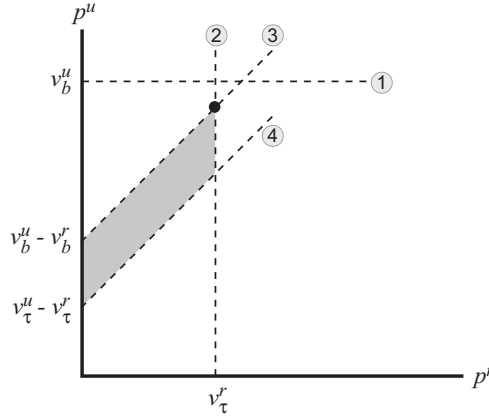
$$v_\tau^u - v_\tau^r > v_b^u - v_b^r$$

because, then, no price pair could satisfy (5.14) and (5.15). Hence, let's assume:

$$v_\tau^u - v_\tau^r \leq v_b^u - v_b^r. \quad (5.16)$$

This assumption (known as a *single-crossing condition*) tells us that the marginal gain in utility enjoyed by switching from a restricted to unrestricted ticket is less for a tourist than for a business traveller.

We can visualize the constraints in  $(p^r, p^u)$  space (*i.e.*, the space with  $p^r$  on the horizontal axis and  $p^u$  on the vertical axis). See Figure 5.4. There



**Figure 5.4:** The set of prices satisfying the four constraints, (5.12)–(5.15), is the gray region. Line ① corresponds to constraint (5.12). Line ② corresponds to constraint (5.13). Line ③ corresponds to constraint (5.14). Line ④ corresponds to constraint (5.15).

are two constraints on the maximum  $p^u$ : It must lie below the horizontal line  $p^u = v_b^u$  (line ① in Figure 5.4) and it must lie below the upward-sloping line  $p^u = p^r + v_b^u - v_b^r \equiv \ell(p^r)$  (line ③). The former condition is just (5.12) and the latter is (5.14). From (5.13), we know we must restrict attention to  $p^r \leq v_\tau^r$  (line ②). We also need to satisfy the tourist's incentive-compatibility constraint (5.15); that is, prices must lie above line ④. Hence the set of feasible prices is the gray region in Figure 5.4

How do we know line ① lies strictly above the gray region? We know this from the following. The tourist's participation constraint, (5.13), tell us  $p^r \leq v_\tau^r$ . Hence,

$$\underbrace{p^u \leq \overbrace{\ell(p^r)}^{\text{implied by constraint (5.13)}}}_{\text{constraint (5.14)}} \leq \ell(v_\tau^r) = v_b^u - (v_b^r - v_\tau^r) < v_b^u,$$

where the last inequality follows because  $v_b^u > v_\tau^u$ . From this, we see that the business traveller's participation constraint, (5.12), is slack (*i.e.*, is a strict inequality).

All else equal, the airline's profits are greatest the greater is  $p^u$  and the greater is  $p^r$ . Hence, its profits are maximized, subject to the constraints, at the highest point in the gray region. Observe this is where lines ② and ③ cross. Notice, therefore, that we can conclude that the tourist's participation constraint, (5.13), and the business traveller's incentive compatibility constraint are both binding (*i.e.*, are both equalities).

To summarize, we've shown, therefore, that at the profit-maximizing prices, the incentive compatibility constraint (5.14) for the business traveller is binding,

but her participation constraint (5.12) is not. We've also shown that the incentive compatibility constraint (5.15) for the tourist is slack, but his participation constraint (5.13) is binding.

If you think about it intuitively, it is the business traveller who wishes to keep her type from the airline. Her knowledge, that she is a business traveller, is valuable information because she's the one who must be induced to reveal her information; that is, she's the one who would have incentive to pretend to be the low-willingness-to-pay other type. Hence, it is not surprising that her revelation constraint is binding. Along the same lines, the tourist has no incentive to keep his type from the airline—he would prefer the airline know he has a low willingness to pay. Hence, his revelation constraint is not binding, only his participation constraint is. These are general insights:

**Result 39** *Under an optimal second-degree price discrimination scheme, the type who wishes to conceal information (has valuable information) has a binding revelation constraint and the type who has no need to conceal his information has just a binding participation constraint.*

As summarized above, we have just two binding constraints and two unknown parameters,  $p^r$  and  $p^u$ . We can, thus, solve the maximization problem by solving the two binding constraints (*i.e.*, determining the coordinates at which lines ② and ③ cross). This yields  $p_*^r = v_\tau^r$  and  $p_*^u = \ell(p_*^r) = v_b^u - (v_b^r - v_\tau^r)$ . Note that the tourist gets no surplus, but the business traveller enjoys  $v_b^r - v_\tau^r > 0$  of surplus. This is a general result: The type with the valuable information enjoys some return from having it. This is known as his or her *information rent*. A type whose information lacks any value fails, not surprisingly, to capture any return from it. Again this is general.

**Result 40** *Under optimal second-degree price discrimination, the type who wishes to conceal information (e.g., the business traveller) earns an information rent (i.e., retains some of his or her consumer surplus). The type who has no need to conceal his information (e.g., the tourist) earns no rent (i.e., retains no consumer surplus).*

Under price discrimination, the airline's profit is

$$N_b \underbrace{(v_b^u - (v_b^r - v_\tau^r))}_{p_*^u} + N_\tau \underbrace{v_\tau^r}_{p_*^r}. \quad (5.17)$$

It is clear that (5.17) is dominated by simple pricing if either  $N_b$  or  $N_\tau$  gets sufficiently small relative to the other. But provided that's not the case, then (5.17)—that is, second-degree price discrimination—yields greater profit. For instance, if  $v_b^u = 500$ ,  $v_b^r = 200$ ,  $v_\tau^u = 200$ , and  $v_\tau^r = 100$ , then  $p_*^r = 100$  and  $p_*^u = 400$ . If  $N_b = 60$  and  $N_\tau = 70$ , then the two possible nondiscriminatory prices,  $v_b^u$  and  $v_\tau^u$ , yield profits of \$30,000 and \$26,000, respectively; but price discrimination yields a profit of \$31,000.

**Example 21:** Consider a producer of computer software. The marginal cost of any copy of the software is negligible and can be considered zero. The company is considering selling an office version and a home version of its software. In the home version, the firm will remove  $h \times 100\%$  of the advanced features found in the office version. Market research indicates that office buyers have a value of \$500 for the full-featured office version and a value of  $500 - 100h$  dollars for the home version with  $h \times 100\%$  of the features removed. Home buyers have a value of \$350 for the full-featured office version and a value of  $350 - 25h$  dollars for the home version. Assume there are 100,000 office buyers and  $N_H$  home buyers. How many features should this company remove and what prices should it charge for the two versions?

As should be clear, the office buyers are the “high” types; that is, the types who get the high-quality version and earn an information rent. The home buyers are the low types—they get the low-quality version and earn no information rent.

The company’s profit (gross of overhead) is

$$100,000p_O + N_H p_H.$$

where  $p_O$  is the price of the office version and  $p_H$  is the price of the home version. The home buyers need to participate. Hence,

$$350 - 25h - p_H \geq 0.$$

The office buyers need to be induced to reveal their type. Hence,

$$\underbrace{500 - p_O}_{\text{cs if buy office ver.}} \geq \underbrace{500 - 100h - p_H}_{\text{cs if buy home ver.}}.$$

As derived earlier, both constraints are binding. Hence,

$$p_H = 350 - 25h$$

and

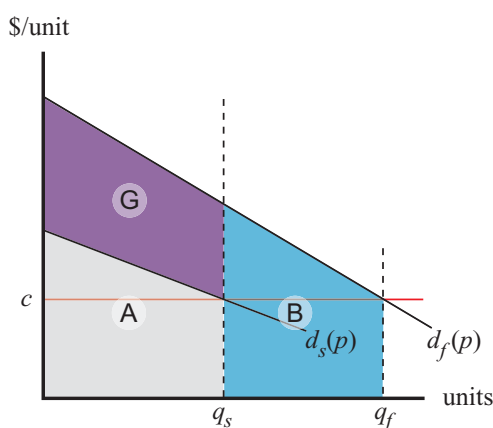
$$p_O = p_H + 100h = 350 + 75h.$$

Profits are

$$100,000(350 + 75h) + N_H(350 - 25h).$$

From this last expression, it is clear that, if  $N_H < 300,000$ , the firm does best to set  $h = 1$ ; that is, remove all the advanced features from the home version. If  $N_H > 300,000$ , then the firm does best to set  $h = 0$ ; that is, not offer two versions of the software. Its profit in this latter case is at least \$140,000,000.

We also need to check whether the firm wants to sell just the office version at a price of \$500 and forgo the home market. This means comparing \$50,000,000 (office only at \$500) to \$42,500,000 plus  $\$325N_H$  (office version at \$425 and home version at \$325). Provided the home market is bigger than 23,076 buyers (but fewer than 300,000 buyers), the firm does best by offering two versions.



**Figure 5.5:** The individual demands of the two types of consumers (family and single),  $d_f(\cdot)$  and  $d_s(\cdot)$ , respectively, are shown. Under the ideal *third-degree* price discrimination scheme, a single would buy a package with  $q_s$  units and pay an amount equal to area A (gray area). A family would buy a package with  $q_f$  units and pay an amount equal to the sum of all the shaded areas (A, B, and G).

### Quantity Discounts

The fact that different size containers of the same good often cost different amounts on a per-unit basis is well known. Typically, the larger the package, the less it costs per-unit; that is, for example, the liter bottle of Pepsi typically costs less than twice the price of the half-liter bottle of Pepsi. In this section we consider why such quantity discounts can be an effective form of second-degree price discrimination.

A fully general treatment can get quite complicated, so here we will restrict attention to a situation in which the firm has constant marginal cost,  $c$ .

Consider a product. Suppose the population of potential buyers is divided into families (indexed by  $f$ ) and single people (indexed by  $s$ ). Let  $d_f(\cdot)$  denote the demand of an *individual* family and let  $d_s(\cdot)$  denote the demand of an *individual* single. Figure 5.5 shows the two demands. Note that, at any price, a family's demand exceeds a single's demand.

The ideal would be if the firm could engage in *third-degree* price discrimination by offering two different two-part tariffs to the two populations. That is, if the firm could freely identify singles from families, it would sell to each member of each group the quantity that equated that member's relevant inverse demand to cost (*i.e.*,  $q_s$  or  $q_f$  in Figure 5.5 for a single or a family, respectively). It could make the per-unit charge  $c$  and the entry fee the respective consumer surpluses. Equivalently—and more practically—the firm could use packaging. The package for singles would have  $q_s$  units and sell for a single's total benefit,

$b(q_s)$ . This is the area labeled **A** in Figure 5.5. Similarly, the family package would have  $q_f$  units and sell for a family's total benefit of  $b(q_f)$ . This is the sum of the three labeled areas in Figure 5.5.

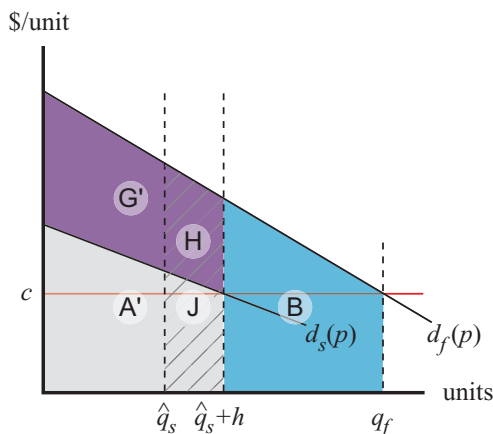
The ideal is not, however, achievable. The firm *cannot* freely distinguish singles from families. It must *induce* revelation; that is, it must devise a *second-degree* scheme. Observe that the third-degree scheme won't work as a second-degree scheme. Although a single would still purchase a package of  $q_s$  units at  $b(q_s)$ , a family would not purchase a package of  $q_f$  units at  $b(q_f)$ . Why? Well, were the family to purchase the latter package it would, by design, earn no consumer surplus. Suppose, instead, it purchased the package intended for singles. Its total benefit from doing so is the sum of areas **A** and **G** in Figure 5.5. It pays  $b(q_s)$ , which is just area **A**, so it would enjoy a surplus equal to area **G**. In other words, the family would deviate from the intended package, with  $q_f$  units, which yields it no surplus, to the unintended package, with  $q_s$  units, which yields it a positive surplus equal to area **G**.

Observe that the firm could induce revelation—that is, get the family to buy the intended package—if it cut the price of the  $q_f$ -unit package. Specifically, if it reduced the price to the sum of areas **A** and **B**, then a family would enjoy a surplus equal to area **G** whether it purchased the  $q_s$ -unit package (at price = area **A**) or it purchased the intended  $q_f$ -unit package (at price = area **A** + area **B**). Area **G** is a family's information rent.

Although that scheme induces revelation, it is not necessarily the *profit-maximizing* scheme. To see why, consider Figure 5.6. Suppose that the firm reduced the size of the package intended for singles. Specifically, suppose it reduced it to  $\hat{q}_s$  units, where  $\hat{q}_s = q_s - h$ . Given that it has shrunk the package, it would need to reduce the price it charges for it. The benefit that a single would derive from  $\hat{q}_s$  units is the area beneath its inverse demand curve between 0 and  $\hat{q}_s$  units; that is, the area labeled **A'**. Note that the firm is forgoing revenues equal to area **J** by doing this. But the surplus that a family could get by purchasing a  $\hat{q}_s$ -unit package is also smaller; it is now the area labeled **G'**. This means that the firm could raise the price of the  $q_f$ -unit package by the area labeled **H**. Regardless of which package it purchases, a family can only keep surplus equal to area **G'**. In other words, by reducing the quantity sold to the “low type” (a single), the firm reduces the information rent captured by the “high type” (a family).

Is it worthwhile for the firm to trade area **J** for area **H**? Observe that the *profit* represented by area **J** is rather modest: While selling the additional  $h$  units to a single adds area **J** in revenue it also adds  $ch$  in cost. As drawn, the profit from the additional  $h$  units is the small triangle at the top of area **J**. In contrast, area **H** represents pure profit—regardless of how many it intends to sell to singles, the firm is selling  $q_f$  units to each family (*i.e.*,  $cq_f$  is a sunk expenditure with respect to how many units to sell each single). So, as drawn, this looks like a very worthwhile trade for the firm to make.

One caveat, however: The figure only compares a single family against a single single. What if there were lots of singles relative to families? Observe



**Figure 5.6:** By reducing the quantity in the package intended for singles, the firm loses revenue equal to area J, but gains revenue equal to area H.

that the total net loss of reducing the package intended for singles by  $h$  is

$$(\text{area J} - ch) \times N_s,$$

where  $N_s$  is the number of singles in the population. The gain from reducing the package is

$$\text{area H} \times N_f,$$

where  $N_f$  is the number of families. If  $N_s$  is much larger than  $N_f$ , then this reduction in package size is not worthwhile. On the other hand if the two populations are roughly equal in size or  $N_f$  is larger, then reducing the package for singles by more than  $h$  could be optimal.

How do we determine the amount by which to reduce the package intended for singles (*i.e.*, the smaller package)? That is, how do we figure out what  $h$  should be? As usual, the answer is that we fall back on our  $MR = MC$  rule. Consider a small expansion of the smaller package from  $\hat{q}_s$ . Because we are using an implicit two-part tariff (packaging) on the singles, the change in revenue—that is, marginal revenue—is the change in a single's benefit (*i.e.*,  $mb(\hat{q}_s)$ ) times the number of singles. That is,

$$MR(\hat{q}_s) = N_s mb(\hat{q}_s).$$

Recall that the marginal benefit schedule is inverse demand. So if we let  $\rho_s(\cdot)$  denote the inverse *individual* demand of a single (*i.e.*,  $\rho_s(\cdot) = d_s^{-1}(\cdot)$ ), then we can write

$$MR(\hat{q}_s) = N_s \rho(\hat{q}_s). \quad (5.18)$$

What about  $MC$ ? Well, if we increase the amount in the smaller package we incur costs from two sources. First, each additional unit raises production costs

by  $c$ . Second, we increase each family's information rent (*i.e.*, area **H** shrinks). Observe that area **H** is the area between the two demand curves (thus, between the two *inverse* demand curves) between  $\hat{q}_s$  and  $\hat{q}_s + h$ . Using Result 18, this means that the marginal reduction in area **H** is

$$\rho_f(\hat{q}_s) - \rho_s(\hat{q}_s),$$

where  $\rho_f(\cdot)$  is the inverse demand of an individual family. Adding them together, and scaling by the appropriate population sizes, we have

$$MC(\hat{q}_s) = N_s c + N_f (\rho_f(\hat{q}_s) - \rho_s(\hat{q}_s)). \quad (5.19)$$

Some observations:

1. Observe that if we evaluate expressions (5.18) and (5.19) at the  $q_s$  shown in Figure 5.5, we have

$$\begin{aligned} MR(q_s) &= N_s \rho(q_s) \text{ and} \\ MC(q_s) &= N_s c + N_f (\rho_f(q_s) - \rho_s(q_s)). \end{aligned}$$

Subtract the second equation from the first:

$$\begin{aligned} MR(q_s) - MC(q_s) &= N_s (\rho(q_s) - c) - N_f (\rho_f(q_s) - \rho_s(q_s)) \\ &= -N_f (\rho_f(q_s) - \rho_s(q_s)) \\ &< 0, \end{aligned}$$

where the second equality follows because, as seen in Figure 5.5,  $\rho_s(q_s) = c$  (*i.e.*,  $q_s$  is the quantity that equates inverse demand and marginal cost). Hence, provided  $N_f > 0$ , we see that the profit-maximizing second-degree pricing scheme sells the low type (*e.g.*, singles) less than the welfare-maximizing quantity (*i.e.*, there is a deadweight loss of area **J** –  $ch$ ). This is a general result with respect to the low type known as *distortion at the bottom*.

2. How do we know we want the family package to have  $q_f$  units? Well, clearly we wouldn't want it to have more—the marginal benefit we could capture would be less than our marginal cost. If we reduced the package size, we would be creating deadweight loss. Furthermore, because we don't have to worry about singles' buying packages intended for families (that incentive compatibility constraint is slack) we can't gain by creating such a deadweight loss (unlike with the smaller package, where the deadweight loss is offset by the reduction in the information rent enjoyed by families). We can summarize this as there being no distortion at the top.
3. Do we know that the profit-maximizing  $\hat{q}_s$  is positive? That is, do we know that a solution to  $MR = MC$  exists in this situation? The answer is no. It is possible, especially if there are a lot of families relative to

singles, that it might be profit-maximizing to set  $\hat{q}_s = 0$ ; that is, sell only one package, the  $q_f$ -unit package, which only families buy. This will be the case if  $MR(0) \leq MC(0)$ . In other words, if

$$N_s(\rho(0) - c) - N_f(\rho_f(0) - \rho_s(0)) \leq 0 \quad (5.20)$$

4. On the other hand, it will often be the case that the profit-maximizing  $\hat{q}_s$  is positive, in which case it will be determined by equating expressions (5.18) and (5.19).

**Example 22:** Consider a cell-phone-service provider. It faces two types of customers, those who seldom have someone to talk to (indexed by  $s$ ) and those who frequently have someone to talk to (indexed by  $f$ ). Within each population, customers are homogeneous. The marginal cost of providing connection to a cell phone is 5 cents a minute (for convenience, all currency units are cents). Market research shows that a member of the  $s$ -population has demand:

$$d_s(p) = \begin{cases} 450 - 10p, & \text{if } p \leq 45 \\ 0, & \text{if } p > 45 \end{cases}.$$

Market research shows that a member of the  $f$ -population has demand:

$$d_f(p) = \begin{cases} 650 - 10p, & \text{if } p \leq 65 \\ 0, & \text{if } p > 65 \end{cases}.$$

There are 1,000,000  $f$ -type consumers. There are  $N_s$   $s$ -type consumers.

What is the profit-maximizing second-degree pricing scheme to use? How many minutes are in each package? What are the prices?

It is clear that the  $f$  types are the high types (like families in our previous analysis). There is no distortion at the top, so we know we sell an  $f$  type the number of minutes that equates demand and marginal cost; that is,

$$q_f^* = d_f(c) = 600.$$

We need to find  $q_s^*$ . To do this, we need to employ expressions (5.18) and (5.19). They, in turn, require us to know  $\rho_s(\cdot)$  and  $\rho_f(\cdot)$ . Considering the regions of positive demand, we have:

$$\begin{aligned} q_s &= 450 - 10\rho_s(q_s) \text{ and} \\ q_f &= 650 - 10\rho_f(q_f); \end{aligned}$$

hence,

$$\begin{aligned} \rho_s(q_s) &= 45 - \frac{q_s}{10} \text{ and} \\ \rho_f(q_f) &= 65 - \frac{q_f}{10}. \end{aligned}$$

Using expression (5.18), marginal revenue from  $q_s$  is, therefore,

$$MR(q_s) = N_s \rho_s(q_s) = N_s \times \left(45 - \frac{q_s}{10}\right).$$

Marginal cost of  $q_s$  (including forgone surplus extraction from the  $f$  type) is

$$\begin{aligned} MC(q_s) &= N_s c + N_f (\rho_f(q_s) - \rho_s(q_s)) \\ &= 5N_s + 1,000,000 \left( 65 - \frac{q_f}{10} - 45 + \frac{q_f}{10} \right) \\ &= 5N_s + 20,000,000. \end{aligned}$$

Do we want to shut out the  $s$ -types altogether? Employing expression (5.20), the answer is yes if

$$40N_s - 20,000,000 < 0;$$

that is, if

$$N_s < 500,000.$$

So, if  $N_s < 500,000$ , then  $q_s^* = 0$  and the price for 600 minutes (*i.e.*,  $q_f^*$ ) is  $b_f(600)$ , which is

$$\begin{aligned} b_f(600) &= \text{Area under } \rho_f(\cdot) \text{ from 0 to 600} \\ &= \text{Area of trapezoid with height 600 and bases 5 \& 65} \\ &= 600 \times \frac{5 + 65}{2} \\ &= 21,000 \end{aligned}$$

cents or \$210.<sup>17</sup>

Suppose that  $N_s \geq 500,000$ . Then, equating  $MR$  and  $MC$ , we have

$$N_s \times \left( 45 - \frac{q_s}{10} \right) = 5N_s + 20,000,000;$$

hence,

$$q_s^* = 400 - \frac{200,000,000}{N_s}.$$

The low type retains no surplus, so the price for  $q_s^*$  minutes is  $b_s(q_s^*)$ , which equals the area under  $\rho_s(\cdot)$  from 0 to  $q_s^*$ . This can be shown (see derivation of  $b_f(600)$  above) to be

$$\begin{aligned} b_s(q_s^*) &= q_s^* \times \frac{5 + 45}{2} \\ &= 25q_s^* \\ &= 25 \left( 400 - \frac{200,000,000}{N_s} \right). \end{aligned}$$

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<sup>17</sup>  $\boxed{\int dx}$  An alternative, but equivalent, derivation:

$$b_f(600) = \int_0^{600} \rho_f(z) dz = \int_0^{600} \left( 65 - \frac{z}{10} \right) dz = 65z - \frac{z^2}{20} \Big|_0^{600} = 21,000$$

cents or \$210.

The price charged the  $f$  types for their 600 minutes is  $b_f(600)$  less their information rent, which is the equivalent of area  $\mathbf{G}'$  in Figure 5.6. Because, in this example, the two demands are parallel (unlike what is shown in the figure), we can use the formula for the area of a parallelogram.<sup>18</sup> Hence, area  $\mathbf{G}'$  is  $20q_s^*$ , so the price charged for 600 minutes is  $21,000 - 20q_s^*$ .

To conclude: If  $N_s < 500,000$ , then the firm sells only a package with 600 minutes for \$210. In this case, only  $f$  types buy. If  $N_s \geq 500,000$ , then the firm sells a package with 600 minutes, purchased by the  $f$  types, for  $210 - .2q_s^*$  dollars; and it also sells a package with  $q_s^*$  minutes for a price of

$$\frac{1}{4} \left( 400 - \frac{200,000,000}{N_s} \right) = 100 - \frac{50,000,000}{N_s}$$

dollars. For example, if  $N_s = 5,000,000$ , then the two plans are (i) 600 minutes for \$138; and (ii) 360 minutes for \$90.

## Bundling | 5.7

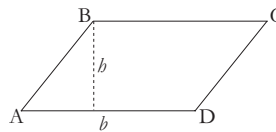
Often we see goods sold in packages. For instance, a CD often contains many different songs. A restaurant may offer a *prix fixe* menu that combines an appetizer, main course, and dessert. Theater companies, symphonies, and operas may sell season tickets for a variety of different shows. Such packages are called *bundles* and the practice of selling such packages is called *bundling*.

In some instances, the goods are available only in the bundle (*e.g.*, it may be impossible to buy songs individually). Sometimes the goods are also available individually (*e.g.*, the restaurant permits you to order *à la carte*). The former case is called *pure bundling*, the latter case is called *mixed bundling*.

Why bundle? One answer is it can be a useful competitive strategy; for instance, it is claimed that the advent of Microsoft Office, which bundled a wordprocessor, spreadsheet program, database program, presentation program, etc., helped Microsoft “kill off” strong competitor products that weren’t bundled (*e.g.*, WordPerfect, Lotus 123, Harvard Graphics, etc.).

Another answer, and one relevant to this lecture note, is that it can help price discriminate. To see this, suppose a Shakespeare company will produce two plays, a comedy and a tragedy, during a season. Type-C consumers tend to prefer comedies and, thus, value the comedy at \$40 and the tragedy at \$30. Hence, a type-C consumer will pay \$70 for a season ticket (*i.e.*, access to both shows). Type-D consumers tend to prefer dramas and, thus, value the comedy at \$25 and the tragedy at \$45. Hence, a type-D consumer will pay \$70 for a season ticket. Assume no capacity constraint (*i.e.*, the shows don’t sell out) and

<sup>18</sup> The area of a parallelogram (see figure at right) is its base,  $b$ , the distance from A to D, times its height,  $h$ . Here the height is  $q_s^*$  and the base is 20 ( $65 - 45$ ).



a constant marginal cost, which we can take to be negligible; that is,  $MC = 0$ . Let  $N_\theta$  denote the number of type- $\theta$  theater goers.

Suppose the company sold the shows separately. For the comedy, it can price at \$25 and attract all theater-goers; or it can price at \$40 and attract only the type-C theater-goers. Likewise, for the drama, it can price at \$30 and sell to all; or it can price at \$45 and sell to type-D theater-goers only. We can conclude, therefore, that if the company sold the shows separately, then its profit is

$$\underbrace{\max\{25(N_C + N_D), 40N_C\}}_{\text{profit from comedy}} + \underbrace{\max\{30(N_C + N_D), 45N_D\}}_{\text{profit from tragedy}},$$

where  $\max\{x, y\}$  means the larger of  $x$  and  $y$ . But observe that the profit shown in the last expression must be less than  $\$70(N_C + N_D)$ , which is what the theater company would earn if it sold season tickets only. This is an example in which pure bundling does better than selling the goods separately.

For an example where mixed bundling is optimal, change the assumptions so that type-D consumers are now only willing to pay \$20 for the comedy. If  $N_C > 4N_D$  (i.e., type-C consumers are more than 80% of the market), then the profit-maximizing solution is to sell season tickets for \$70, but now make the tragedy available separately for \$45.

Observe how the negative correlation between preferences for comedy versus tragedy helps the theater company price discriminate. Effectively, this negative correlation can be exploited by the company to induce the two types to *reveal* who they are for the purpose of price discrimination. It follows that bundling is related to the forms of second-degree price discrimination considered earlier.

## Summary | 5.8

In this lecture note we explored price discrimination. Ideally, a firm would like to capture all the gains to trade (welfare), which means leaving no profitable trades unmade (no deadweight loss) and capturing all of the consumers' surplus. This ideal, which we called the "Holy Grail" of pricing, is known as perfect or first-degree price discrimination.

If all consumers are identical, then perfect price discrimination can be achieved by a two-part tariff. One part, the per-unit charge, is set so as to equate inverse demand and marginal cost. The second part, the entry fee, is set equal to the consumer surplus that each consumer would realize were he or she able to buy as many units as he or she desired at that per-unit charge.

When consumers are not identical, then it is typically not possible to achieve perfect discrimination.

We saw that two-part tariffs are often disguised. Sometimes they are disguised because the per-unit charge is zero. Sometimes they are disguised through packaging; consumers have a choice between buying nothing or a package of fixed size, the price of which is set equal to a consumer's total benefit from the num-

ber of units in the package. Finally, metering is a way to execute a two-part tariff (although there are also other reasons to use metering).

When consumers are heterogeneous it is sometimes possible to divide them into different populations that are more homogeneous. If the firm can freely identify the population to which a consumer belongs (his or her “type”), then the firm can engage in third-degree price discrimination; that is, offer different tariffs to different populations.

In other circumstances, the firm cannot freely identify consumers’ types. It can, however, induce them to reveal their types. This form of price discrimination is known as second-degree price discrimination. Two prevalent forms of second-degree price discrimination are using quality distortions and quantity discounts. In any second-degree price discrimination scheme, the consumer type who values the good more (the “high type”) is the one who must be induced to reveal his or her type. Because this knowledge is valuable, the high-type consumer retains some of its value in the form of an information rent. On the other hand, there is no distortion in what is sold the high type (*e.g.*, she gets the unrestricted ticket or the welfare-maximizing quantity). The other type (the “low type”) doesn’t possess valuable information, so earns no information rent. Moreover, to reduce the rent earned by the high type, what is sold to the low type is distorted; either he gets a good of reduced quality or he gets less than the welfare-maximizing quantity (so-called distortion at the bottom). In some cases, it pays to shut out the low type altogether.

A final method of price discrimination explored was bundling. Bundling allows the firm to take advantage of the correlations that exist between consumers’ preferences for different products.

All discriminatory pricing is, in theory, vulnerable to arbitrage—the advantaged reselling to the disadvantaged. We considered when arbitrage might prevent the use of price discrimination and how firms might, in turn, seek to deter or reduce arbitrage.



## Transfer Pricing and Vertical Relations

# 6

Price serves as a powerful incentive. For instance, the higher the price of a good, the greater the cost that decision makers are willing to incur to supply the good. In particular, price serves as a means of inducing trade; those whose benefits exceed the price wish to buy and those whose costs are less than the price wish to sell. If set properly (*i.e.*, so that  $p = MB(x^*) = MC(x^*)$ ), a uniform price will lead to the welfare-maximizing level of trade (as seen in Section 5.3).

Because of the incentive properties of price, pricing *within* an organization—*transfer pricing*—is an important part of achieving internal efficiency.

Like almost anything powerful, pricing can also be misused. Surplus is transferred via the price (*e.g.*, an increase in a uniform price increases the amount of consumer surplus captured by the seller). Consequently, the private incentives of one party to capture the surplus of the other can interfere with the goal of using price to establish efficient trade.

In this lecture note we explore how pricing should be done internally and what can go wrong. This will also motivate a short analysis of the holdup problem.

### The Ideal

## 6.1

Do the following. Pick up a dollar bill with your left hand. Hold it a moment. Now pass it to your right hand. Are you any wealthier?

The answer, of course, is no. Both hands belong to you, so your wealth doesn't depend on which hand holds the bill. Now, instead of two hands, suppose we're talking about two divisions of the same firm. The same logic holds—money transferred between divisions does not increase the firm's income or wealth. Critically, though, *how* such transfers are arranged can have an effect on the firm's income.

To begin to understand this point, we start with the ideal way to set up transfer prices. Consider a firm with two divisions,  $u$  and  $d$ . Division  $u$  is the *upstream division*, which means the product it produces is an input used by division  $d$ , the *downstream division*. The terms “upstream” and “downstream” reflect a metaphor of the production process as a river; raw materials enter upstream and flow downstream through various stages of production and distribution until finally flowing to the consumers. Division  $u$ 's output, which is division  $d$ 's input, is called an *intermediate good*. Division  $d$ 's output, what is sold to customers, is called a *final good*.

To make our analysis more transparent, imagine that each unit of the *final* product produced by division  $d$  requires precisely one unit of division  $u$ 's output. To keep matters straightforward assume that  $d$ 's marginal cost is

$$MC_d = c + p_u,$$

where  $p_u$  is the price  $d$  pays  $u$  per unit used of  $u$ 's product and  $c$  is the cost per unit of output of the other inputs used. Let  $R_d(x)$  be the revenue earned by division  $d$  if it sells  $x$  units. Observe that division  $d$ 's profit is, thus,

$$\pi_d(x) = R_d(x) - (c + p_u)x.$$

For the timing being, assume that division  $u$  sells to division  $d$  only. Hence, its revenue from  $x$  units is  $p_u x$ . Let its cost be given by  $C_u(x)$ . Division  $u$ 's profit is, therefore,

$$\pi_u(x) = p_u x - C_u(x).$$

The firm's profit,  $\Pi(x)$ , is the sum of the two divisions' profits:

$$\begin{aligned} \Pi(x) &= \pi_d(x) + \pi_u(x) \\ &= R_d(x) - (c + p_u)x + p_u x - C_u(x) \end{aligned} \tag{6.1}$$

$$= R_d(x) - cx - C_u(x). \tag{6.2}$$

Observe that there is no  $p_u$  in this last expression. The value of  $p_u$  doesn't directly affect profits because, as seen in expression (6.1), the transfer of  $p_u x$  dollars between divisions is the equivalent of moving a dollar bill from your left to right hand.

The firm seeks to maximize (6.2). Under our usual assumptions, the firm would wish division  $d$  to sell (and, thus, division  $u$  to produce)  $x^*$  units, where

$$MR_d(x^*) = \underbrace{c + MC_u(x^*)}_{\text{overall } MC}. \tag{6.3}$$

Assume that the firm prefers to produce  $x^*$  than to shutdown.

If the top management of the firm could order the  $u$  division to produce the  $x^*$  of the intermediate good and the  $d$  division to sell  $x^*$  units of the final good, where  $x^*$  solved (6.3), then this is clearly what top management would do.

The problem is that top management typically doesn't possess this level of control over the divisions. For instance, top management might not know information known by division managers (*e.g.*, what  $R_d(\cdot)$  is). Alternatively, divisional costs will be greater than assumed here unless divisional managers are given incentives to maximize profits. Whatever the reason, suppose that the firm's top management can only intervene by setting the transfer price,  $p_u$ .

To begin, suppose that the upstream division must ship as many units as the downstream division demands at a price of  $p_u$  per unit, where  $p_u$  has been fixed by top management. How many units will the downstream division demand? Presuming the downstream division seeks to maximize its profit, the downstream division will select the  $\hat{x}$  that maximizes

$$R_d(x) - (c + p_u)x.$$

Equating divisional marginal revenue to divisional marginal cost, we find

$$MR_d(\hat{x}) = c + p_u. \quad (6.4)$$

Top management's goal is to set  $p_u$  so that the  $\hat{x}$  that solves expression (6.4) is  $x^*$ , where  $x^*$  is defined by expression (6.3). Observe that if  $p_u = MC_u(x^*)$ , then  $\hat{x} = x^*$ ; that is, we can transform expression (6.4) into expression (6.3) by requiring that  $p_u = MC_u(x^*)$ . We can restate this as follows.

**Result 41** *Efficiency (i.e., profit maximization) requires the upstream division transfer the input at its marginal cost.*

Observe that Result 41 is really just a restatement of the Holy Grail of pricing, Result 35. In this case, the firm captures all the welfare because it is both the seller (division  $u$ ) and the buyer (division  $d$ ). It follows, then, from Result 35, that it maximizes welfare (i.e., profit) by pricing so that the transfer price is marginal cost.

A possible objection to Result 41 is if top management doesn't know  $x^*$ —recall, if it did, it could just order that amount to be transferred—then how does it know  $MC_u(x^*)$ ? This is a good question, and one which we will consider further in Section 9.1. Notice, however, that we have a ready answer if  $MC_u(\cdot)$  is constant (i.e.,  $MC_u(x) = \gamma$  for all  $x$ ). When upstream marginal cost is constant (or approximately so), then top management can simply set the transfer price equal to this constant marginal cost (i.e.,  $p_u = \gamma$ ).

Another possible objection to Result 41 is that it is based on a number of simplifying assumptions (e.g., one unit of  $u$ 's output is needed as an input in producing  $d$ 's output). Fortunately, this objection is easily met. Again let  $x$  continue to denote the units that  $u$  transfers to  $d$ . Now, however, suppose that  $d$  produces  $Q_d(x, z_1, \dots, z_N)$  units, where  $z_n$  is the amount of input  $n$  (note there are a total of  $N + 1$  inputs, the input from division  $u$  and  $N$  others). Let  $p_n$  be the price of input  $n$ . The firm's total profit is

$$R_d(Q_d(x, z_1, \dots, z_N)) - p_1 z_1 - \dots - p_N z_N - C_u(x). \quad (6.5)$$

Observe there is, again, no  $p_u$  in this expression because it doesn't matter whether the dollar bill is in the left hand or the right. Suppose that  $z_1, \dots, z_N$  and  $x$  are chosen to maximize profits; that is, to maximize expression (6.5). Let an asterisk (i.e.,  $*$ ) denote the maximizing value. Define

$$\mathcal{R}(x) = R_d(Q_d(x, z_1^*, \dots, z_N^*)).$$

It follows that maximizing (6.5) with respect to  $x$  when  $z_n = z_n^*$  for all  $n$  is equivalent to maximizing

$$\mathcal{R}(x) - C_u(x). \quad (6.6)$$

Observe, with respect to the decision of how much  $x$  to transfer, the expenses on  $z_1^*, \dots, z_N^*$  are sunk, so don't enter into expression (6.6). The reason for choosing the notation  $\mathcal{R}(\cdot)$  should be apparent—it is the firm's revenue if  $x$

units are produced by the upstream division. The firm's cost of producing  $x$  units is  $C_u(x)$ . In other words, expression (6.6) is essentially our familiar expression of profit as revenue minus cost. Maximizing this means equating marginal "revenue" to marginal cost:

$$MR(x^*) = MC_u(x^*).$$

As before, we can induce the downstream division to purchase  $x^*$  units from the upstream division if we set  $p_u = MC_u(x^*)$ . That is, we see that Result 41 remains valid even in a fully generally setting.

### $\int dx$ An Alternative Route to Result 41

The profit-maximizing values of  $z_1, \dots, z_N$  and  $x$  can be found by differentiating expression (6.5) with respect to each of those arguments and setting the resulting derivatives equal to zero:

$$MR_d(Q_d(x^*, z_1^*, \dots, z_N^*)) \frac{\partial Q_d(x^*, z_1^*, \dots, z_N^*)}{\partial z_1} - p_1 = 0, \quad (6.7)$$

$\vdots$

$$MR_d(Q_d(x^*, z_1^*, \dots, z_N^*)) \frac{\partial Q_d(x^*, z_1^*, \dots, z_N^*)}{\partial z_N} - p_N = 0, \text{ and}$$

$$MR_d(Q_d(x^*, z_1^*, \dots, z_N^*)) \frac{\partial Q_d(x^*, z_1^*, \dots, z_N^*)}{\partial x} - MC_u(x) = 0. \quad (6.8)$$

The terms  $\partial Q_d / \partial z_n$  and  $\partial Q_d / \partial x$  are the *marginal products* of the inputs  $z_n$  and  $x$ , respectively. That is, they represent the increase in output (*i.e.*,  $Q_d$ ) per unit increase in the respective input (*i.e.*, input  $n$  or the input from the upstream division).  $MR_d(Q_d)$  is just marginal revenue. The product of  $MR_d(Q_d)$  and  $\partial Q_d / \partial z_n$  or  $\partial Q_d / \partial x$  is the *marginal revenue product* of that input; that is, the rate at which revenue increases per unit increase in the input.

Expression (6.7) tells us that the optimal amount of input 1 to employ equates the marginal revenue product of input 1 to its marginal cost—known as *marginal factor cost*—which here is just the price we pay per unit of input 1,  $p_1$ .<sup>1</sup> This makes perfect sense: If marginal revenue product were greater than marginal factor cost, then employing more of the input would increase revenues more than cost, which would increase profits. Conversely, if marginal revenue product were less than marginal factor cost, then employing *less* of the input would yield greater cost savings than we would lose in revenue. Hence, we can only be at an optimum where marginal revenue product equals marginal factor cost. Finally observe, from expression

<sup>1</sup>A synonym for input is *factor*.

(6.8) and by analogy to expression (6.7), that if the transfer price were  $p_u = MC_u(x^*)$ , then the downstream division would choose to purchase the profit-maximizing amount of the input,  $x^*$ . That is, we see that Result 41 holds.

## Vertical Relations When Upstream and Downstream Are Separate Firms

# 6.2

Often upstream and downstream are separate firms (or treated as such by central headquarters). In this section, we see what goes wrong with pricing relative to the ideal enunciated in the previous section, namely that the intermediate good be transferred at marginal cost. To keep the analysis straightforward, we will make three simplifying assumptions, none of which is critical to the conclusion:

- The downstream firm the upstream firm's only customer.<sup>2</sup>
- Each unit of the final good uses one (and only one) unit of the intermediate good (*i.e.*, the analysis is similar to the analysis that *preceded* Result 41.
- The intermediate good represents the only input (and, thus, the only source of costs) for the downstream firm.

Consider what happens if the upstream firm engages in simple pricing. Let  $p_u$  be the price it charges for the intermediate good. Because the intermediate good is the downstream firm's only input and each unit of the intermediate good is transformed into one unit of the final good, it follows that  $p_u$  is also the downstream firm's marginal cost. Let  $MR_d(\cdot)$  be the downstream firm's marginal revenue schedule. Like any profit-maximizing firm, the downstream firm chooses the amount to produce,  $x$ , such that it equates marginal revenue and marginal cost:

$$MR_d(x) = p_u. \quad (6.9)$$

Recall that every unit of the downstream firm's output corresponds to one unit of the intermediate good. Hence, expression (6.9) relates the price charged by the upstream firm to the number of units,  $x$ , it sells. In other words, expression (6.9) is *the upstream firm's inverse demand curve*.

To determine the upstream firm's pricing, we need to calculate its marginal revenue schedule from its inverse demand curve (*i.e.*, expression (6.9)). Recalling expression (4.17), marginal revenue is price less the driving-down-the-price effect. Here, the upstream firm's price, as a function of quantity, is  $MR_d(x)$ . The driving-down-the-price effect is, thus, the slope of  $MR_d(x)$  (denoted  $MR'_d(x)$ ) times the number of units,  $x$ . We have, therefore:

$$MR_u(x) = MR_d(x) + MR'_d(x)x, \quad (6.10)$$

---

<sup>2</sup>Normally, this would provide the downstream firm tremendous market power against the upstream firm. We will ignore the issue of downstream market power for the moment, however.

where  $MR_u(\cdot)$  is the upstream firm's marginal-revenue schedule. This method of deriving the upstream firm's marginal-revenue schedule is known as *double marginalization*. Next, the upstream firm equates its marginal revenue,  $MR_u(x)$ , to its marginal cost,  $MC_u(x)$ , to determine the profit-maximizing quantity to produce. Call that quantity  $\tilde{x}$ . Lastly, the upstream firm determines its price by plugging  $\tilde{x}$  back into its inverse demand function—that is, the *downstream* firm's marginal-revenue schedule—to obtain its profit-maximizing price:  $\tilde{p}_u = MR_d(\tilde{x})$ .

**Example 23:** Consider a downstream firm with demand  $100,000 - 2500p$ . Suppose the upstream firm's cost is  $x^2/5000$ . From this, we seek to determine (1) the price of the intermediate good,  $p_u$ , charged by the upstream firm; (2) the price, assuming simple pricing, charged by the downstream firm for the final good; and (3) compare the firms' profits as separate firms with the profit that a combined firm, which transferred at marginal cost, would earn.

Clearly, we need to know the upstream firm's marginal cost. Employing Result 15, we have

$$MC_u(x) = \frac{x}{2500}.$$

We need to derive the upstream firm's marginal-revenue schedule, which, from expression (6.10), requires us to determine the downstream firm's marginal-revenue schedule first. Observe we have

$$100,000 - 2500P_d(x) = x;$$

hence,

$$P_d(x) = 40 - \frac{x}{2500}.$$

Employing expression (4.17), we have

$$MR_d(x) = \underbrace{40 - \frac{x}{2500}}_{P_d(x)} + \underbrace{\left(\frac{-1}{2500}\right)x}_{P'_d(x)x} = 40 - \frac{x}{1250}.$$

To employ expression (6.10) we need to determine the slope of  $MR_d(x)$ . This is  $-1/1250$ . Using expression (6.10), we have

$$MR_u(x) = 40 - \frac{x}{1250} + \left(\frac{-1}{1250}\right)x = 40 - \frac{x}{625}.$$

The next step is to equate upstream marginal revenue with upstream marginal cost:

$$40 - \frac{x}{625} = \frac{x}{2500}$$

or, multiplying both sides by 2500,

$$100,000 - 4x = x.$$

Hence, the profit-maximizing number of units to produce,  $\tilde{x}$ , is 20,000. Hence, profit-maximizing price is

$$\tilde{p}_u = MR_d(\tilde{x}) = 40 - \frac{20,000}{1250} = 24$$

dollars per unit.

What about the price of the final good? That is, the price charged by the downstream firm? We know it produces 20,000 units (recall each unit of intermediate good produces one unit of final good). Plugging 20,000 into *its* inverse demand yields:

$$\tilde{p}_d = P_d(20,000) = 40 - \frac{20,000}{2500} = 32$$

dollars per unit.

The revenue of the downstream firm is

$$20,000 \times \$32$$

or \$640,000. Its cost is

$$20,000 \times \$24$$

or \$480,000. Its profit is, therefore, \$160,000.

The revenue of the upstream firm is the cost of the downstream firm, or \$480,000. Its cost is

$$\frac{20,000^2}{5000} = 80,000$$

dollars. The profits of the upstream firm are, thus, \$400,000. Observe the total of the two firms' profits when they are separate (*i.e.*, when there is double marginalization) is \$560,000.

Now suppose the two firms were merged. If the upstream division transfers at marginal cost, we have the following condition for profit maximization:

$$MR_d(x) = MC_u(x) \text{ or } 40 - \frac{x}{1250} = \frac{x}{2500} ;$$

multiplying both sides by 2500, this yields

$$100,000 - 2x = x .$$

Hence,  $x^* = 33,333$ . This yields a price for the final good of  $P_d(33,333)$  or \$26.66. Revenues are, thus, \$888,657.80. Cost is

$$\frac{33,333^2}{5000} = 222,217.80.$$

dollars. Profit is, therefore, \$666,440. By merging, the firms increase their aggregate profit by 19%.

Observe, from the example, that double marginalization yields less total profit than transferring at marginal cost. This is only to be expected: The upstream firm *cares* whether the dollar is in its hand or in the other firm's hand. Consequently, it prices so as to try to capture as much money as possible from the downstream firm. But, as we saw previously, there is no benefit to the firms in aggregate from moving money from one to the other. The consequence of the upstream firm pricing so as to maximize its profit alone is that the downstream

**Double marginalization:**

*Yields less total profit than if the intermediate good is transferred at marginal cost.*

firm is given the wrong signal with respect to how much of the intermediate good to buy. It buys too little of the intermediate good from the perspective of maximizing *joint* profits (*i.e.*, the profit of a merged firm).

The fact that a merged firm can earn more profit than two separate firms is one reason why firms wish to vertically integrate. *One note of caution:* Avoiding double marginalization only applies when *both* upstream and downstream face downward-sloping demand curves—that is, when both have some degree of market power. As we will see in Section 7.5, if either firm is in a perfectly competitive market, then avoiding double marginalization is no longer a motive for a vertical merger.

### An Upstream Division that Sells Both Internally and Externally

## 6.3

Consider an upstream division that sells (transfers) an intermediate good both externally—that is, to an *independent* downstream firm—and internally—that is, to another division of the same company. For example, a domestic pharmaceutical company could produce and market a drug in its home country but license another firm to market it abroad. Domestically, the production division sells to its internal marketing division, but sells to an external marketing firm overseas.

To avoid having to analyze the added complexities that arise when the internal downstream division competes with the external downstream firm, suppose—as in the pharmaceutical example—that the internal downstream division and the external downstream firm do not compete directly. For convenience, assume that each unit of the downstream firms' output requires one unit of the upstream division's output (alternatively, that the units of the upstream division's output have been normalized to match the downstream firms' units).

Define  $G_i(\cdot)$  to be the *gross-profit schedule* of the internal downstream division. That is,  $G_i(x)$  is the profit earned by the internal division from selling  $x$  units of the final good *before* subtracting the cost of the  $x$  units of the intermediate good. Likewise, define  $G_e(\cdot)$  to be the gross-profit schedule of the external downstream firm. Let  $MG_i(\cdot)$  and  $MG_e(\cdot)$  be the marginal gross-profit schedules of the internal division and the external firm, respectively.

We can think of the gross-profit schedules and the marginal gross-profit schedules as playing the same role as revenue and marginal-revenue schedules, respectively. After all, revenue is just profit before subtracting cost. Consequently, from our analysis of double marginalization, we know that the external inverse demand for the intermediate good is given by the external firm's marginal gross-profit schedule.

We can write the integrated firm's profit as

$$\underbrace{\overbrace{MG_e(x_e) \times x_e}^{\text{revenue from external firm}} + T(x_i) - C_u(x_e + x_i)}_{\text{upstream division's profit}} + \underbrace{G_i(x_i) - T(x_i)}_{\text{downstream division's profit}},$$

where

- $x_i$  is the number of units of the intermediate good sold internally.
- $x_e$  is the number of units of the intermediate good sold externally.
- $T(\cdot)$  is the tariff schedule that determines the amount the internal downstream division transfers to the upstream division.
- $C_u(\cdot)$  is the upstream division's cost schedule.

Note that the  $T(x_i)$  terms cancel—it doesn't matter whether the dollar is in the left hand or the right hand. Maximizing profits means equating the marginal revenue from selling externally to marginal cost and also equating marginal gross-profit from the internal division's sales to marginal cost:

$$MG_e(x_e) + MG'_e(x_e)x_e = MC_u(x_e + x_i) \text{ and} \quad (6.11)$$

$$MG_i(x_i) = MC_u(x_e + x_i). \quad (6.12)$$

Solving these two equations for  $x_e^*$  and  $x_i^*$  tells us how much the upstream division should produce for sale externally and for sale internally. Observe, from the second equation, that to provide the internal downstream division the proper incentives to purchase  $x_i^*$  units, the transfer price needs, again, to be set equal to marginal cost,  $MC_u(x_e^* + x_i^*)$ . The price externally is  $MG_e(x_e^*)$ .

**Example 24:** Consider a firm,  $A$ , with upstream and downstream divisions. The upstream division also sells externally to firm  $E$ . Let the cost of the upstream division be

$$C_u(x) = \frac{x^2}{5000}.$$

The internal downstream division's demand is

$$D_i(p_i) = 100,000 - 2500p_i.$$

Firm  $B$ 's demand is

$$D_e(p_e) = 200,000 - 5000p_e.$$

Each unit of final-good output requires one unit of the intermediate good (*i.e.*, what the upstream division produces). Assume the marginal cost of *other* inputs is \$5 for the downstream division and \$3 for firm  $E$ .

We need to determine the downstream division's gross profit function and firm  $B$ 's gross profit function. Consider the downstream division first. Its inverse demand is

$$P_i(x_i) = 40 - \frac{x_i}{2500}.$$

So its revenue less cost of *other* inputs—that is, its gross profit—is

$$G_i(x_i) = P_i(x_i)x_i - 5x_i = 35x_i - \frac{x_i^2}{2500}.$$

For firm  $E$ , we have

$$P_e(x_e) = 40 - \frac{x_e}{5000}$$

and, thus,

$$G_e(x_e) = 37x_e - \frac{x_e^2}{5000}.$$

Using Result 21, we have

$$MG_i(x_i) = 35 - \frac{x_i}{1250} \text{ and}$$

$$MG_e(x_e) = 37 - \frac{x_e}{2500}.$$

Observe the slope of  $MG_e(x_e)$  is  $-1/2500$ .

Employing Result 15, we have

$$MC_u(x_i + x_e) = \frac{x_i + x_e}{2500}.$$

Expressions (6.11) and (6.12) are, thus,

$$37 - \frac{x_e}{2500} + \left( \frac{-1}{2500} \right) x_e = \frac{x_i + x_e}{2500} \text{ and}$$

$$35 - \frac{x_i}{1250} = \frac{x_i + x_e}{2500}.$$

Multiplying both sides of both equations by 2500 yields

$$92,500 - 2x_e = x_i + x_e \text{ and}$$

$$87,500 - 2x_i = x_i + x_e.$$

So

$$x_i = 92,500 - 3x_e. \quad (6.13)$$

Substituting into the second equation, yields

$$87,500 = 3(92,500 - 3x_e) + x_e; \text{ hence,}$$

$$190,000 = 8x_e.$$

Solving, we have  $x_e^* = 23,750$ . Plugging back into (6.13), we find  $x_i^* = 21,250$ . The price at which the upstream division sells to firm  $E$  is

$$p_e = MG_e(x_e^*) = 27.50$$

dollars. Internal transfer is at marginal cost:

$$p_u = MC_u(x_e^* + x_i^*) = 18.00$$

dollars.

# Holdup | 6.4

**Holdup:** *The inability to capture enough of the future gains from trade to cover investments today.*

We have so far considered only pricing issues as a motive for merging upstream and downstream. In this section, we consider another motive: holdup. [Holdup](#) is what happens when one stream will be unable to capture enough of the surplus from trade in future negotiations to cover its investments today.

The topic of holdup and how to respond to it is large and we can only hope to touch on the most critical elements of here. To illustrate some of these elements, consider the following situation:

1. Initially, an upstream firm must make an investment of  $I$  dollars if it is going to trade at all. Assume, for convenience, that this investment is entirely sunk once made.<sup>3</sup>
2. Later, the upstream firm enters into negotiations with an independent downstream firm on supplying the latter with some number of units,  $x$ , of an intermediate good. Note, at this point, the investment in the previous stage is now *sunk* and, thus, irrelevant to decision making at this and subsequent stages. Let the downstream firm derive benefit  $B(x)$  from its acquisition of  $x$  units of the intermediate good. Let it cost the upstream firm  $C(x)$  to provide  $x$  units of the intermediate good. Because  $I$  is sunk at this stage, it does *not* enter into  $C(x)$ . Let  $x^*$  maximize  $B(x) - C(x)$ ; that is,  $x^*$  maximizes total surplus (the value created). Let  $B^* = B(x^*)$  and  $C^* = C(x^*)$ .
3. Although an analysis of bargaining theory is beyond the scope of these lecture notes, it should strike you as reasonable to assume (i) that, in these negotiations, the two firms would agree to fix  $x = x^*$ , so as to maximize the total surplus; and (ii) that they would divide this surplus between themselves somehow. Without going into an analysis of bargaining, we might reasonably suppose that the outcome of the bargaining is that they agree to split the surplus evenly. This can be thought of as their agreeing to a transfer  $T^*$  for the  $x^*$  units such that

$$T^* - C^* = \frac{B^* - C^*}{2} \text{ (i.e., upstream gets 1/2 the surplus) and}$$

$$B^* - T^* = \frac{B^* - C^*}{2} \text{ (i.e., downstream gets 1/2 the surplus).}$$

Note that this implies

$$T^* = \frac{B^* + C^*}{2}.$$

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<sup>3</sup>This is not critical. We could allow some of it to be salvageable, in which case the imputed salvage value times the cost of capital is captured in the cost,  $C$ .

4. Let  $r$  be the interest rate, so that net present profit today if the upstream firm invests  $I$  today is

$$\pi_u = \frac{1}{1+r} \frac{B^* - C^*}{2} - I.$$

Observe that there is a problem if  $\pi_u < 0$ ; the upstream firm won't invest because it fails to capture back enough of its investment in later bargaining. This is holdup. And the anticipation of holdup causes the upstream firm not to invest, so that, therefore, both upstream and downstream end up earning zero profit.

By merging, the two firms can correct the problem if

$$\frac{1}{1+r}(B^* - C^*) - I > 0. \quad (6.14)$$

If the firms are merged, then the future profit of the merged firm is  $B^* - C^*$ . From expression (6.14), it is a positive net-present-value investment to invest  $I$  today for  $B^* - C^*$  next period. Hence, by merging, the firms go from making no profit to the positive profit (in present-value terms) given in (6.14).

**Example 25:** Suppose that the upstream firm needs to invest \$5 million in order to be able to produce. The cost of the intermediate good is, then, \$5 per unit. The benefit of the intermediate good to the downstream division is  $B(x) = 45x - x^2/20,000$ . For purposes of illustration, take  $r = 0$ .

Equating marginal benefit to marginal cost to determine the value (surplus)-maximizing  $x$ , yields

$$45 - \frac{x}{10,000} = 5; \text{ so, thus,} \\ x^* = 400,000.$$

We have, therefore,

$$\begin{aligned} B^* &= 10,000,000 \\ C^* &= 2,000,000 \\ T^* &= 6,000,000 \end{aligned}$$

If the firms aren't merged, then upstream wouldn't invest:

$$\underbrace{6,000,000 - 2,000,000}_{T^* - C^*} - \underbrace{5,000,000}_I = -1,000,000$$

dollars. But if the firms merged, then the total surplus gained,  $B^* - C^*$ , exceeds the cost of investment (*i.e.*,  $\$8,000,000 > \$5,000,000$ ). So by merging, the firms go from a total profit of \$0 to a total profit of \$3,000,000.

Of course, at some level, holdup is the same problem as double marginalization. Both arise because the *way* in which transfers between the streams are carried out cause an inefficient distortion in the amount that is transferred. With double marginalization, too little is transferred. Similarly, with holdup, too little is transferred (indeed, as illustrated, perhaps nothing at all is transferred).<sup>4</sup>

## Summary | 6.5

This lecture note has touched on some of the important issues in transfer pricing and vertical relations. The principal message is that *how* transfers are made between upstream and downstream can affect the efficiency of those transfers. This insight translates into the message that, within the same firm, the transfer price should equal marginal cost.

One of the problems that can arise when the divisions are separate firms (or treated as such by headquarters) is *double marginalization*. Each stream marks up over marginal cost causing inefficiently little to be transferred relative to what should be transferred if the objective is to maximize the sum of the streams' profits. Merging the divisions into a single firm is a way to avoid double marginalization.

A related problem is *holdup*. Subsequent negotiations will cause the upstream firm to earn a return that is too small to cover its initial investment. Again, merger can be a way to eliminate this problem.

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<sup>4</sup>In more elaborate holdup models, one can have positive, but less than optimal, amounts transferred. Students wishing to learn more about holdup problems should consult Aaron S. Edlin and Benjamin E. Hermalin: "Contract Renegotiation and Options in Agency," *Journal of Law, Economics, & Organization*, Vol. 16 (2000), pp. 395–423. Students who don't feel comfortable with calculus might find the details of this article hard to follow. A more accessible, but somewhat dated and idiosyncratic, analysis can be found in Chapters 3–5 of Oliver E. Williamson: *The Economic Institutions of Capitalism*, New York: The Free Press, 1985.



## Perfect Competition

# 7

In our study of pricing so far, we have assumed that the firm faces a downward-sloping demand curve. Because demand is downward sloping, the firm has the ability to affect the price through its actions. For instance, when engaged in simple pricing, a firm that seeks to increase its sales must lower its price (the driving-down-the-price effect). Firms that can affect price are known as *price makers*.

Some firms, however, do not have the ability to affect price. They take the prevailing market price as *the* price, believing that

- they will lose all or almost all sales if they increase their price; and
- they can sell as many units as they wish at the prevailing market price; that is, they are not subject to the driving-down-the-price effect.

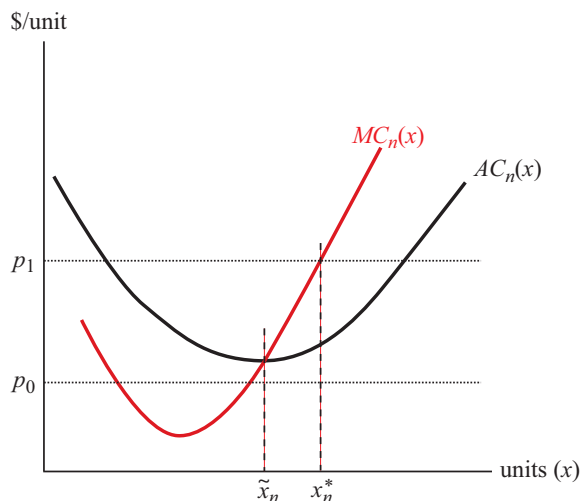
Given the first belief, such a firm has no incentive to raise its price above the prevailing price. Given the second belief, such a firm has no incentive to set its price below the prevailing price. Firms that don't believe they can affect price are known as *price takers*.

An example of a price taker (or *price-taking* firm) is the producer of a commodity product. For example, a given farmer cannot sell her produce (or much of it) at more than the market price. Similarly, her sales, even if she has a large farm, have no discernable effect on the market price for her produce. Another instance of a price taker would be a firm that sells to very price-sensitive consumers and that operates at such a small scale relative to the rest of the market that any increase in its output would have negligible effect on price. Finally, the sellers in many financial markets and exchanges are price takers—buyers of a shares in General Motors, say, only want to purchase it at as low a price as possible and don't care from whom they buy it. From the perspective of the market, those individuals selling General Motors stock are selling a commodity.

In this lecture note, we study industries in which the firms are price takers. Such industries are considered *perfectly competitive*. Among the questions we seek to answer are (i) how does the price get set in such industries? (ii) what establishes the number of firms in such an industry? and (iii) under what conditions can a firm earn a profit in such an industry? We will also derive something known as a *supply curve*, a relation between the prevailing price and the amount that an individual firm or firms in aggregate wish to sell.

**Price taker:** *A firm that believes it has no ability to affect price.*

**Perfect competition:** *When all firms in an industry are price takers.*



**Figure 7.1:** A price-taking firm shuts down if the prevailing price is  $p_0$ . It produces  $x_n^*$  units if the prevailing price is  $p_1$ . Note the firm never produces at less than minimum efficient scale (MES), which is  $\tilde{x}_n$ .

## The Price-Taking Firm in Isolation | 7.1

Consider a single price-taking firm. For reasons that will become clearer later, a price-taking firm must have a U-shaped average cost curve. If, instead, the firm enjoyed increasing returns to scale at all levels of output, then the firm would be a *natural monopoly*—it alone would dominate the industry. A dominant firm is a price maker, not a price taker. If, instead, the firm enjoyed decreasing returns to scale at all levels of output, then the firm could not operate profitably at a positive level of output in a perfectly competitive industry.

Let  $C_n(x)$  be this firm's cost function. Define  $\tilde{x}_n$  as the level of output that minimizes the firm's average cost function; that is,

$$AC_n(\tilde{x}_n) < AC_n(x) \text{ for all } x \neq \tilde{x}_n.$$

We know that marginal cost intersects average cost at the minimum of average cost (Result 16); hence,  $MC_n(\tilde{x}_n) = AC_n(\tilde{x}_n)$ . Because  $AC(\cdot)$  is U-shaped,  $AC(\cdot)$  is a decreasing function for  $x < \tilde{x}_n$ , which means that  $MC_n(x) < AC_n(x)$  for  $x < \tilde{x}_n$  (Result 14). The fact that  $AC(\cdot)$  is U-shaped also implies that  $AC(\cdot)$  is an increasing function for  $x > \tilde{x}_n$ . Therefore  $MC_n(x) > AC_n(x)$  for  $x > \tilde{x}_n$  (Result 14 again). The quantity  $\tilde{x}_n$  is known as the firm's *minimum efficient scale* or MES. Figure 7.1 illustrates.

The firm's revenue is  $px$ , where  $p$  is the prevailing market price. As a price taker, the firm's output decisions do not affect  $p$  (*i.e.*, there is no driving-down-the-price effect). Consequently, the firm's *marginal* revenue is just  $p$ , the prevailing price. Note, too, that the firm's *average* revenue is also  $p$ .

If the firm produces at all, it will produce at the point where marginal revenue—here, the prevailing price—crosses marginal cost from above. For example, in Figure 7.1, if the prevailing price is  $p_1$ , then marginal revenue (*i.e.*, the horizontal dotted line at height  $p_1$ ) crosses marginal cost from above at  $x_n^*$  units. The profit-maximizing quantity—assuming the firm produces at all—is  $x_n^*$  units.

Whether or not the firm should produce at all is determined by the Shutdown Rule (Result 26). Recall that the Shutdown Rule is equivalent to testing whether average revenue exceeds average cost at the proposed level of production. That is, repeating expression (4.7), if

$$AR_n(x^*) \geq AC_n(x^*).$$

With a price-taking firm (indeed, any firm engaged in linear pricing), average revenue is just price. Hence, the Shutdown Rule for a price-taking firm is produce if price exceeds average cost. For example, in Figure 7.1, at price  $p_1$ , the candidate profit-maximizing level of production,  $x_n^*$ , is such that

$$p_1 > AC_n(x_n^*).$$

In other words,  $x_n^*$  passes the Shutdown Rule—if the price is  $p_1$ , then the firm should produce  $x_n^*$  rather than shutdown.

Now suppose the prevailing price in Figure 7.1 is  $p_0$ . Consider any level of output,  $x$ . Observe

$$p_0 < AC_n(\tilde{x}_n) \leq AC_n(x).$$

This means that for *any* level of output, average cost would be *greater* than average revenue (*i.e.*,  $p_0$ ). Hence, for any level of output, the firm would always do better to shutdown than produce that level. But as this is true of *all* levels of output, the firm does better to shutdown than produce.

**Result 42** *A price-taking firm should shutdown if the prevailing price is less than its minimum average cost.*

Figure 7.1 also shows that a price-taking firm should never operate below minimum efficient scale. To see why, we employ a *reductio ad absurdum* argument: Suppose that the firm could profitably operate below MES. Consequently, there is some  $\hat{x}_n < \tilde{x}_n$  at which (i)  $p = MC_n(\hat{x}_n)$ ; and (ii)  $p \geq AC_n(\hat{x}_n)$ . In other words, for it to be profitable for the firm to operate at  $\hat{x}_n$ , then  $\hat{x}_n$  must satisfy both the  $MR = MC$  Rule and the Shutdown Rule, respectively. However, for  $x < \tilde{x}_n$ ,  $MC_n(x) < AC_n(x)$  (see Figure 7.1). Hence, conditions (i) and (ii) combined with this last fact imply

$$p = MC_n(\hat{x}_n) < AC_n(\hat{x}_n) \leq p.$$

No number can be strictly less than itself, so we have a contradiction. Therefore our original supposition, that the firm could profitably operate at below MES, must be false. We can, thus, conclude:

**Result 43** *A price-taking firm never operates below minimum efficient scale (i.e., it cannot be profit-maximizing to produce less than MES).*

Observe that we have concluded that

- If the prevailing price is less than the minimum of average cost, the firm produces nothing (shuts down);
- If the prevailing price is at or above the minimum of average cost, the firm produces the amount that equates price and marginal cost (e.g., produces  $x_n^*$  when price is  $p_1$  in Figure 7.1); and
- If the firm produces at all, then it will only produce at or above minimum efficient scale.

Using notation, we can summarize these points as

$$\text{Amount optimally produced} = \begin{cases} 0, & \text{if } p < AC_n(\tilde{x}_n) \\ x, & \text{where } p = MC_n(x) \text{ for } p \geq AC_n(\tilde{x}_n) \end{cases} \quad (7.1)$$

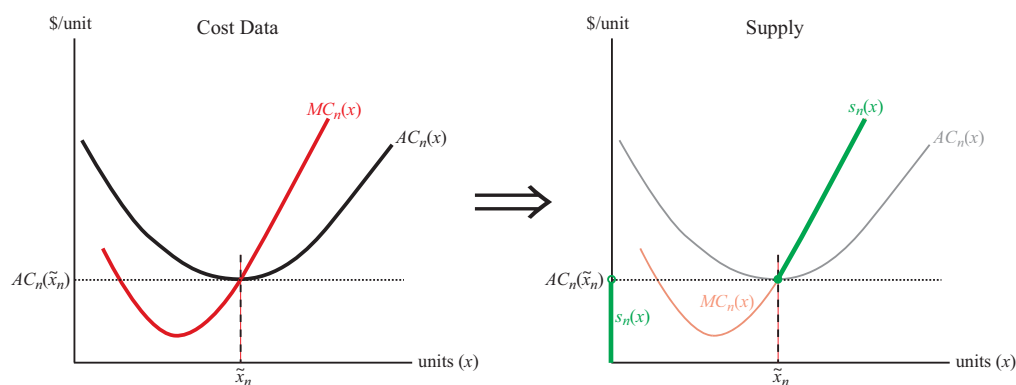
Observe that expression (7.1) is a relationship between the prevailing price,  $p$ , and the amount that the price-taking firm wishes to produce or *supply*. This relationship can be more succinctly captured by letting  $s_n(p)$  denote the amount that this firm optimally wishes to supply if the prevailing price is  $p$ . That is,

$$s_n(p) = \begin{cases} 0, & \text{if } p < AC_n(\tilde{x}_n) \text{ (i.e., the minimum of } AC) \\ MC_n^{-1}(p), & \text{if } p \geq AC_n(\tilde{x}_n) \end{cases}, \quad (7.2)$$

where  $MC_n^{-1}(\cdot)$  is the inverse of the marginal cost schedule (i.e.,  $MC_n^{-1}(p)$  gives the quantity,  $x$ , such that  $MC_n(x) = p$ ).<sup>1</sup> The function  $s_n(\cdot)$  is known as the firm's *supply curve* or schedule. As Figure 7.2 illustrates, the firm's supply curve corresponds to the vertical (price) axis for price less than the minimum of average cost and corresponds to the firm's marginal-cost schedule for price at or above the minimum of average cost. Observe that the firm's supply curve is increasing—slopes up—for prices at which the firm produces at all.

**Supply curves:**  
Are upward sloping.

<sup>1</sup>As a technical aside, observe that we require that  $MC_n(\cdot)$  be an invertible function over the range  $\{MC_n(x) | x \geq \tilde{x}_n\}$ . Observe this will be true if  $MC_n(\cdot)$  is everywhere increasing for  $x > \tilde{x}_n$  (e.g., as in Figure 7.1). If  $MC_n(\cdot)$  is not invertible, then it is still possible to derive the firm's supply curve or schedule; the analysis, however, is too specialized to warrant our attention here.



**Figure 7.2:** A price-taking firm's supply curve, shown in the right panel, is derived from the firm's costs as shown. The supply curve, in green, corresponds to the vertical access for  $p < AC_n(\tilde{x}_n)$  (the minimum of average cost) and corresponds to the firm's marginal cost schedule for  $p \geq AC_n(\tilde{x}_n)$ .

## Aggregate Supply and Short-Run Equilibrium | 7.2

Where does the prevailing price come from if the firm doesn't set it? To answer this question, we need to derive the *market equilibrium*. A system (*e.g.*, a market) is in equilibrium when there are no forces acting on the system to change its state. In the context of a market equilibrium, this means that the price is at a level such that there are no forces working to change the price. As we will see, the prevailing price is the price in equilibrium.

**Short run:** *Number of firms fixed.*

**Long run:** *Entry or exit is possible.*

In considering equilibria, we need to distinguish between *short-run* and *long-run* equilibrium. In a *short-run equilibrium*, the number of firms is fixed. That is, the short run is defined as a period of time too short for firms to either exit or enter an industry (*e.g.*, escape sunk commitments, such as leases, or build new facilities). A *long-run equilibrium* is one in which no firms wish to enter or exit the industry. That is, the long run is a period of time long enough for firms to exit or enter an industry. In this section, we will focus on short-run equilibria.

A perfectly competitive industry is one with many firms, which, in the short run, is fixed at some number,  $N$ . To determine the short-run equilibrium price, we need to derive the short-run aggregate supply curve. An aggregate supply curve tells us how many units, in total, the firms are willing to sell at each price. For instance, if Rose Inc. and Noahco are willing to sell 1000 and 1500 units, respectively, at a price of \$10 per unit, then aggregate supply (taking  $N = 2$ ) at a price of \$10 per unit is 2500. More generally, we have

$$\begin{aligned} S_{SR}(p) &= s_1(p) + \cdots + s_N(p) \\ &= \sum_{n=1}^N s_n(p), \end{aligned}$$

where  $S_{SR}(\cdot)$  is the aggregate short-run supply curve and  $s_n(\cdot)$  is the supply curve of the  $n$ th firm. Because the individual supply curves are upward sloping, so too is  $S_{SR}(\cdot)$ .

**Example 26:** There are two firms,<sup>2</sup> indexed 1 and 2. Our cost data on the two firms are

$$\begin{aligned} C_1(x) &= 1600 + \frac{x^2}{1600} \\ C_2(x) &= 5625 + \frac{x^2}{2500} \end{aligned}$$

---

<sup>2</sup>An industry with two firms only would be unlikely to be perfectly competitive. We fix on two simply for purposes of illustration.

(in both cases for  $x > 0$ ). Hence,

$$\begin{aligned} AC_1(x) &= \frac{1600}{x} + \frac{x}{1600} \\ MC_1(x) &= \frac{x}{800} \\ AC_2(x) &= \frac{5625}{x} + \frac{x}{2500} \\ MC_2(x) &= \frac{x}{1250}. \end{aligned}$$

We need, first, to determine the MES for both firms by equating average cost to marginal cost and solving for the  $\tilde{x}$  that equates them.

$$\begin{aligned} \frac{1600}{x} + \frac{x}{1600} &= \frac{x}{800} \Rightarrow \tilde{x}_1 = 1600 \\ \frac{5625}{x} + \frac{x}{2500} &= \frac{x}{1250} \Rightarrow \tilde{x}_2 = 3750. \end{aligned}$$

Observe

$$\begin{aligned} AC_1(1600) &= 2 \\ AC_2(3750) &= 3. \end{aligned}$$

Recall that a firm's supply curve is found by inverting its marginal cost schedule for price above the minimum of its average cost.

$$\begin{aligned} p &= \frac{s_1(p)}{800} \Rightarrow s_1(p) = 800p \quad (p \geq 2) \\ p &= \frac{s_2(p)}{1250} \Rightarrow s_2(p) = 1250p \quad (p \geq 3). \end{aligned}$$

Note  $s_1(p) = 0$  for  $p < 2$  and  $s_2(p) = 0$  for  $p < 3$ . Aggregating, we have

$$S_{SR}(p) = \begin{cases} 0, & \text{if } p < 2 \\ 800p, & \text{if } 2 \leq p < 3 \\ 2050p, & \text{if } 3 \leq p \end{cases}.$$

There is some aggregate demand for the product, given by  $D(p)$ . The short-run equilibrium occurs at the price,  $p$ , that equates aggregate demand with aggregate short-run supply. Why is this? Well at the price that equates  $D(p)$  and  $S_{SR}(p)$ , call it  $p^e$ , every unit that a seller wishes to sell has a buyer. Likewise for every unit that a buyer seeks, he or she can find a seller. Both sides—buyers and sellers—of the market have their desires satisfied at  $p^e$ . Because everyone is satisfied, there are no forces working to move the price away from  $p^e$  in the short run. The price  $p^e$ , thus, establishes an equilibrium.

Moreover,  $p^e$  is the only equilibrium price. To see why, suppose  $p < p^e$ . Then

$$D(p) > D(p^e) = S(p^e) > S_{SR}(p),$$

where the first inequality follows because demand curves slope down and the second inequality follows because supply curves slope up. But this expression

means that some buyers are unable to satisfy their demand at  $p$ . Unsatisfied buyers will be tempted to bid up the price; that is, there is a force working to change the price—this is, therefore, not an equilibrium. Similarly, suppose  $p > p^e$ . Then

$$D(p) < D(p^e) = S(p^e) < S_{SR}(p),$$

where the first inequality follows because demand curves slope down and the second inequality follows because supply curves slope up. But this expression means that some sellers are unable to find buyers at  $p$ . They will cut price in the hopes of inducing greater sales.<sup>3</sup> Again, there is a force working to change price; this is not an equilibrium either.

We can conclude:

**Result 44** *The unique equilibrium price in the short run is the price that equates aggregate demand and aggregate short-run supply (i.e.,  $p^e$  such that  $D(p^e) = S_{SR}(p^e)$ ).*

What is a firm's profit in equilibrium? Well, it produces  $s_n(p^e)$  units. Recalling that average revenue times units is revenue and average cost times units is cost, its profit is

$$\underbrace{p^e}_{AR} \times \underbrace{s_n(p^e)}_{\text{units}} - \underbrace{AC_n(s_n(p^e))}_{AC} \times s_n(p^e) = (p^e - AC_n(s_n(p^e))) s_n(p^e).$$

**Example 27:** Consider an industry that has, in the short run, 100 identical firms. Each firm's cost function is

$$C_n(x) = \begin{cases} 0, & \text{if } x = 0 \\ 40,000 + x^2/10,000, & \text{if } x > 0 \end{cases}.$$

Let  $D(p) = 15,000,000 - 1,000,000p$ .

We need, first, to derive the individual firms' supply curves. To this end, observe

$$AC_n(x) = \frac{40,000}{x} + \frac{x}{10,000}$$

$$MC_n(x) = \frac{x}{5000}.$$

Equating  $AC$  and  $MC$  to find MES, we have

$$\frac{40,000}{x} + \frac{x}{10,000} = \frac{x}{5000} \Rightarrow \tilde{x}_n = 20,000.$$

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<sup>3</sup>The idea that our price-taking sellers would cut price might seem at odds with the presumption that they are price takers. This apparent inconsistency can be resolved by noting that a price taker *believes* that it can sell as many units as it wishes at the prevailing price. If the seller finds that's not true (i.e., because  $p > p^e$ ), then it will have to revise its belief about what price it can charge. In other words, if you believe you can sell your entire stock at \$10 per item and you do, then you have no reason to question your belief. But if you have unsold inventory, then you would question your belief and, presumably, respond accordingly.

Minimum of average cost is, therefore, \$4/unit. Individual supply solves

$$p = \frac{s_n(p)}{5000} \text{ for } p \geq 4.$$

Hence,

$$s_n(p) = \begin{cases} 0, & \text{if } p < 4 \\ 5000p, & \text{if } p \geq 4 \end{cases}.$$

Aggregating, we have

$$S_{SR}(p) = \begin{cases} 0, & \text{if } p < 4 \\ 100 \times 5000p = 500,000p, & \text{if } p \geq 4 \end{cases}.$$

Equating demand with supply:

$$15,000,000 - 1,000,000p = 500,000p \Rightarrow p^e = 10.$$

At the equilibrium price, each firm's profit is

$$\underbrace{(10 - AC_n(5000 \times 10))}_{p^e - AC} \times \underbrace{(5000 \times 10)}_{\text{units}} = \$210,000.$$

## Entry, Exit, and Long-run Equilibrium | 7.3

In the long run, firms can enter and exit a perfectly competitive industry. Firms obviously will exit if they would lose money by staying in. Conversely, if there are profits to be made, firms will enter.

This last observation sometimes troubles students. Shouldn't firms enter only if the profits they will make are large enough? The answer is no: Firms should enter if they will make any profit. Why? Well, recall that we are measuring cost in terms of opportunity cost. That means that cost captures the value of the next best alternative (*i.e.*, not being in the industry). Therefore, if revenues from entry exceed cost, what that means is that the firm does better being in the industry than being out of it.

So, as long as positive profits can be made, firms will enter. If firms will suffer losses, firms will exit. This tells us that, if the industry is in *long-run equilibrium*, then no firm not in the industry would find it profitable to enter and no firm in the industry would find it better to exit.

What does this observation tell us? Well, suppose that the production technology is common across the industry and that all firms have access to the same set of inputs at the same prices, so that all firms have the same cost function. Because the firms are all identical, their profits in equilibrium must also be the same (recall Example 27). Now consider a long-run equilibrium. Because some firms are in, their profits cannot be negative. But some firms are out, they couldn't make a profit by entering. The only way this can be true with identical firms is that all firms that are in are earning zero profit and no additional firm could enter and make a profit.

In other words, if  $N^e$  firms are in the industry in equilibrium, then average revenue (*i.e.*,  $p^e$ ) must equal average cost. The latter is  $AC_n(D(p^e)/N^e)$  because a total of  $D(p^e)$  units are produced and traded in equilibrium and each firm gets one  $N^e$ th of the business. Now no additional firm can enter and make a profit. What does this tell us? Well, suppose a firm entered at minimum efficient scale (*i.e.*, entered and produced  $\tilde{x}_n$  units). It would make a profit if  $p^e > AC_n(\tilde{x}_n)$  (*i.e.*, if price exceeded the minimum of average cost). Because it can't make a profit by entering in equilibrium, it must be that  $p^e \leq AC_n(\tilde{x}_n)$  in equilibrium. But it can't be that  $p^e < AC_n(\tilde{x}_n)$  because, then, all firms should shutdown (exit). The only possibility remaining is that  $p^e = AC_n(\tilde{x}_n)$  in equilibrium. We can conclude:

**Flat long-run supply:** With identical firms, long-run supply is flat, with price equal to minimum of AC.

**Result 45** *If all firms are identical, then the long-run equilibrium price must equal the minimum of any given firm's average cost (*i.e.*,  $p^e = AC_n(\tilde{x}_n)$ ).*

Observe that Result 45 implies that, with identical firms, the long-run aggregate supply curve is a flat line at price equal to minimum of average cost.

Because each firm faces a price equal to  $AC_n(\tilde{x}_n)$  and  $AC_n(\tilde{x}_n) = MC_n(\tilde{x}_n)$ , it follows that each firm in the market produces  $\tilde{x}_n$  units.

**Example 28:** Consider the same industry studied in Example 27. Recall that each firm's minimum of average cost occurs at 20,000 units, at which level average cost is \$4/unit. Hence, in the long-run, the equilibrium price in this industry is \$4/unit. This yields a demand of

$$15,000,000 - 1,000,000 \times 4 = 11,000,000$$

units. Because each firm produces 20,000 units, it follows that the equilibrium number of firms,  $N^e$ , is 550.<sup>4</sup>

Observe the steps used in solving for *long-run* equilibrium with identical firms:

1. Identify the MES and the minimum of average cost.
2. The latter is the equilibrium price. Plug it into demand to determine the total amount traded in equilibrium.
3. To determine the equilibrium number of firms, divide the total amount traded by the MES.

What if all firms aren't identical? To determine long-run equilibrium in this case, imagine putting the firms in order by the minimum of their average costs. The first firm has the lowest minimum of average cost, the second firm

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<sup>4</sup>The astute reader might wonder, what would happen if the MES didn't divide evenly into the demand at a price equal to the minimum of average cost? There are a number of technical ways of dealing with this, none of which are worth pursuing here. The real answer is that the perfect competition model with absolutely identical firms is only an approximation to reality. Yet, while only an approximation, it does offer important insights and, thus, is useful.

the next lowest, and so forth. Assign each firm an index, so that firm  $i$  has a lower minimum of average cost than firm  $i + 1$ . As a convenient notation, let  $\widehat{AC}_n$  denote firm  $n$ 's minimum average cost. At a given price  $p$ , one has, in the long run, those firms for which  $\widehat{AC}_n \leq p$  in the industry and those for which  $\widehat{AC}_n > p$  out of the industry. Those in the industry supply  $s_n(p)$ . Hence,

$$S_{LR}(p) = \begin{cases} 0, & \text{if } p < \widehat{AC}_1 \\ s_1(p), & \text{if } \widehat{AC}_1 \leq p < \widehat{AC}_2 \\ \vdots & \\ \sum_{n=1}^N s_n(p), & \text{if } \widehat{AC}_N \leq p < \widehat{AC}_{N+1} \\ \vdots & \end{cases},$$

where  $S_{LR}(\cdot)$  denotes the long-run aggregate supply curve. Although  $S_{LR}(\cdot)$  and  $S_{SR}(\cdot)$  are both the sum of individual supply curves, there is an important difference. The former is the sum of the supply curves of the most efficient firms (*i.e.*, those with the lowest average costs), while the latter is the sum of the supply curves of those firms that happen to be in the industry at a moment in time.

Similar to short-run equilibrium, long-run equilibrium occurs at the price that equates demand and long-run aggregate supply; that is, if  $p^e$  is the long-run equilibrium price, then  $D(p^e) = S_{LR}(p^e)$ .

Observe that, in a long-run equilibrium with heterogeneous firms, the more efficient firms earn positive profits. Why? Because we know that, for the more efficient firms,  $\widehat{AC} < p^e$ . Hence, we know

$$0 < (p^e - \widehat{AC})\tilde{x} \leq \pi^*,$$

where  $\pi^*$  is the firm's profit at its profit-maximizing level of output given price  $p^e$ . We can conclude, therefore, that the way to make money in a perfectly competitive industry is to improve the efficiency of your operations relative to your competitors.

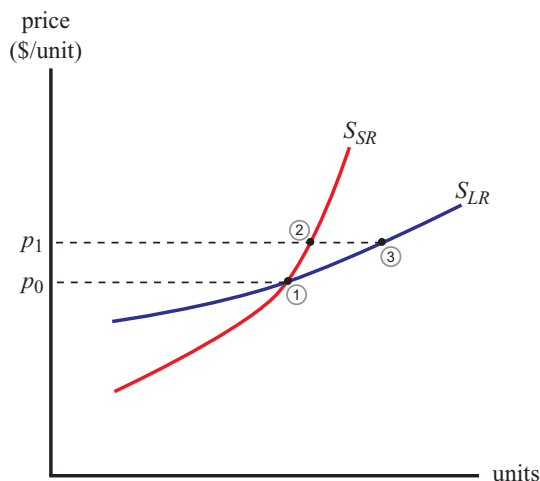
**Profit under perfect competition:** *To be profitable in a perfectly competitive industry, you must have a cost advantage.*

## Dynamics | 7.4

Using the analysis above, we can talk about what happens in the short and long run following an exogenous shock to the market.<sup>5</sup>

Observe, first, that the long-run supply curve is always flatter than any short-run supply curve. This is obviously the case when all firms are identical—a short-run supply curve is upward sloping, while the long-run supply is a flat line. It is also true even when the firms are heterogeneous. As price rises along short-run supply, it begins to rise above the minimum of average cost for some firms who were out, but who now enter. Their supply is added to the supply of

<sup>5</sup> *Exogenous* means external to the system.



**Figure 7.3:** At price  $p_0$  the long-run supply curve is flatter than the short-run supply curve through  $p_0$ . Were price to rise to  $p_1$ , supply would, first, increase from firms already in the market (*i.e.*, from ① to ②). But this higher price would induce entry, which would add to the amount supplied (*i.e.*, move the amount from ② to ③).

the firms initially in the industry, but this means that more in total is supplied. Figure 7.3 illustrates.

#### Demand shocks:

*Have a bigger effect in the short run than the long run.*

A consequence of this is that price reactions to demand shocks will be more severe in the short run than the long run. An example illustrates this point.

**Example 29 [Organic Produce]:** One of the authors' wives likes organic produce. She was anxious that the increased popularity of organic produce would, however, increase the price of organic produce. While her fear is well-founded in the short run, it is likely misplaced in the long run. Entry into the production of organic produce is relatively easy and there is no reason to believe that a new organic farm would be stuck with more expensive inputs than existing farms or would have access to less fertile land. Hence, one might reasonably view all organic farms as identical. Moreover, one would, then, conclude that the long-run supply curve of organic produce is essentially flat. Figure 7.4 illustrates: Demand shifts out, from demand curve  $D_0$  to curve  $D_1$ , due to the increased popularity of organic produce. This initially causes the price to rise from  $p_0$  to  $p_1$ . That is, the short-run equilibrium shifts from ① to ②. But this higher price attracts entry, pushing price back down to  $p_0$  in long-run equilibrium (*i.e.*, to ③).

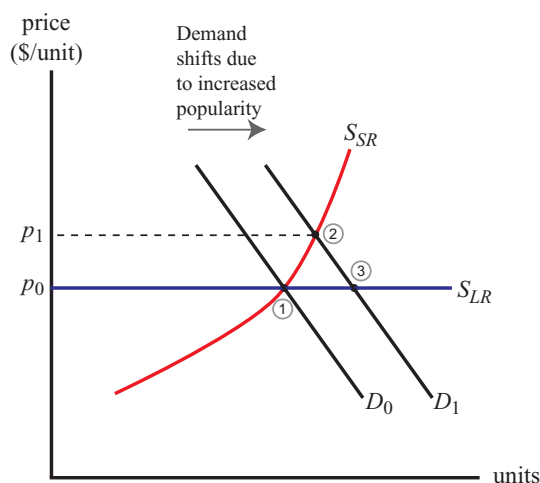


Figure 7.4: Illustration for Example 29.

## Vertical Integration with a Competitive Stream

## 7.5

We now employ our analysis of perfect competition to the question of vertical integration. Our principal finding will be that double marginalization no longer provides an incentive to merge the streams when one of the streams is in a perfectly competitive market.

### The Upstream Industry is Perfectly Competitive

Consider the situation in which the upstream industry is perfectly competitive. Because, as we've seen, perfect competition leads all firms to produce to the point that price equals their marginal costs, it follows that any such firm is selling the intermediate good in question at marginal cost. Double marginalization, recall, arose because the upstream firm was setting its price above marginal cost. In other words, there is no double marginalization—if the upstream is competitive, the downstream is purchasing the intermediate good at marginal cost.

### The Downstream Industry is Perfectly Competitive

Suppose, now, the upstream firm has market power. In fact, suppose it is a monopoly; that is, the only seller of the intermediate good. Suppose it sells to a perfectly competitive downstream market. For convenience, suppose that each unit of the final good requires exactly one unit of the intermediate good. To keep the analysis straightforward, suppose the downstream firms are identical.

Let  $moc_d(x) + p_u$  be the marginal cost of producing  $x$  units of the final good, where  $p_u$  is the price of the intermediate good in question and  $moc_d(\cdot)$  is the marginal cost arising from the *other* inputs. If  $C_d(\cdot)$  is the cost function of a downstream firm, then define

$$aoc_d(x) = \frac{C_d(x) - p_u x}{x};$$

that is,  $aoc_d(\cdot)$  is the average cost of the inputs (including possible overhead) other than the intermediate good in question. It follows that overall average cost is

$$AC_d(x) = aoc_d(x) + p_u.$$

Let  $\tilde{x}$  minimize  $aoc_d(x)$ . Then  $\tilde{x}$  also minimizes  $AC_d(x)$ ; that is,  $\tilde{x}$  is MES.

We know that, in long-run equilibrium, the equilibrium price of the final good will equal the minimum of average cost; that is,

$$p_d^e = AC_d(\tilde{x}) = aoc_d(\tilde{x}) + p_u.$$

Each firm will produce  $\tilde{x}$  units.

Let  $D_d(\cdot)$  be downstream demand. Because each unit of the final good uses one unit of the intermediate good, it follows that

$$D_d(aoc_d(\tilde{x}) + p_u)$$

is upstream demand (*i.e.*, number of units of the intermediate good demanded as a function of the intermediate good's price,  $p_u$ ). For example, if the downstream demand schedule is

$$D_d(p_d) = A - Bp_d,$$

then the upstream firm's demand schedule is

$$D_u(p_u) = D_d(aoc_d(\tilde{x}) + p_u) = (A - B aoc_d(\tilde{x})) - Bp_u. \quad (7.3)$$

To determine the upstream firm's profit-maximizing price, one proceeds as usual: Invert  $D_u(p_u)$ . Calculate marginal revenue from the inverse demand curve. Set marginal revenue equal to marginal cost. Solve for the profit-maximizing quantity. Then plug that back into inverse demand to determine the profit-maximizing price. For example, if  $D_u(\cdot)$  is given by expression (7.3), then inverse demand is

$$P_u(X) = \left( \frac{A - B aoc_d(\tilde{x})}{B} \right) - \frac{X}{B},$$

where  $X$  is the total amount of the intermediate good sold to downstream firms. Marginal revenue would be

$$MR_u(X) = \left( \frac{A - B aoc_d(\tilde{x})}{B} \right) - \frac{2X}{B}.$$

Equating this to marginal cost would, then, yield the profit-maximizing quantity.

**Example 30:** Downstream demand is

$$D_d(p_d) = 100,000 - 200p_d.$$

The upstream firm's marginal cost is \$3/unit.

Suppose that each downstream firm has the following cost:

$$C_d(x) = 50 + \frac{x^2}{200} + p_u x.$$

Observe,

$$aoc_d(x) = \frac{50}{x} + \frac{x}{200} \text{ and } moc_d(x) = \frac{x}{100}.$$

Equating the two to find the minimum of  $aoc_d(\cdot)$ , we have

$$\frac{50}{x} + \frac{x}{200} = \frac{x}{100} \Rightarrow \tilde{x} = 100.$$

Observe that  $aoc_d(\tilde{x}) = 1$  dollar/unit.

Upstream demand is

$$\begin{aligned} D_u(p_u) &= D_d(aoc_d(\tilde{x}) + p_u) \\ &= 100,000 - 200 \times (1 + p_u) \\ &= 99,800 - 200p_u. \end{aligned}$$

Inverting,

$$P_u(X) = 499 - \frac{X}{200}.$$

Hence,

$$MR_u(X) = 499 - \frac{X}{100}.$$

Equating with upstream marginal cost, \$3, we obtain  $X^* = 49,600$ . Plugging back into inverse demand,  $P_u(\cdot)$ , yields a price of \$251 per unit for the intermediate good. The price of the final good is  $AC_d(\tilde{x}) = 1 + 251 = 252$  dollars per unit. Observe that the upstream firm's profit (gross of any overhead) is

$$251 \times 49,600 - 3 \times 49,600 = 12,300,800$$

dollars.

Suppose that the upstream firm bought all 496 downstream firms.<sup>6</sup> From our earlier analysis (Result 41), we know that the upstream firm should transfer at marginal cost, here \$3. Given that firms have decreasing economies of scale for  $x > 100$  and the market is large, we know that the total amount produced,  $X$ , will be divided among the 496 downstream divisions evenly. So the cost of producing the final good for the integrated firm is

$$C_d(X) = 496 * \left( 50 + \frac{\left(\frac{X}{496}\right)^2}{200} \right) + 3X = 24,800 + \frac{X^2}{99,200} + 3X.$$

---

<sup>6</sup>We know there are 496, because each firm sells  $\tilde{x} = 100$  units and a total of 49,600 units are sold.

Marginal cost is, thus,

$$MC_d(X) = \frac{X}{49,600} + 3.$$

Inverse *downstream* demand is

$$P_d(X) = 500 - \frac{X}{200}.$$

Marginal revenue is

$$MR_d(X) = 500 - \frac{X}{100}.$$

Equating, we find  $X^* = 49,600$ —the same answer as when the firms were integrated. Profits of the integrated firm (gross of upstream overhead) are

$$\underbrace{3 \times 49,600 - 3 \times 49,600}_{\text{upstream profit}} + \underbrace{252 \times 49,600 - \left(24,800 + \frac{49,600^2}{99,200} + 3 \times 49,600\right)}_{\text{downstream profit}} = 12,300,800$$

dollars. Observe this is exactly what we got when the industry was unintegrated. In other words, there is no gain to the upstream firm integrating into a perfectly competitive downstream industry.

**Integrating into a competitive downstream:**  
*Cannot increase the profits of the integrated firm.*

The conclusion of this last example, that there is no point to vertically integrating into a perfectly competitive downstream industry is fully general. Intuitively, because an unintegrated downstream makes no profit in long-run equilibrium, there is no surplus to be extracted from the downstream firms. If the downstream firms capture no surplus, then the upstream firm must be capturing all the profit there is. There is nothing further to be gained by integrating into the downstream.

## Summary | 7.6

The main take-aways of this lecture note are:

- Perfectly competitive firms price at marginal cost.
- With identical firms, the long-run aggregate supply curve is a flat line at a height equal to the minimum of any given firm's average cost.
- With identical firms, no firm in a perfectly competitive market earns a profit in long-run equilibrium.
- The only way to make a profit in a perfectly competitive market is to have a cost advantage relative to other producers.
- The long-run aggregate supply curve is less steep than any short-run aggregate supply curve; consequently the effect of a demand shock on price is less severe in the long run than the short run.

- There is no gain to integrating up into a perfectly competitive upstream industry.
- There is no gain to integrating down into a perfectly competitive downstream industry.



## **Networks and Standards**

# **8**



## **Agency and Incentives**

**9**

### **Incentives for Efficient Divisional Transfer**

**9.1**



# **Mathematics Appendices**



## Algebra Review

# A1

This appendix reviews those aspects of algebra useful for successful completion of 201A.

## Functions

# A1.1

We begin with a quick review of functions.

**Definition 9** A function takes each element of one set of numbers and maps it to a unique number in a second set of numbers.

For example, the function  $f(x) = 5x + 3$  takes real numbers and maps them into real numbers. For instance,  $f(1) = 8$ ; that is, the function maps 1 into 8. Likewise  $f(3) = 18$ . Note that for any  $x$ ,  $f(x)$  is unique (*e.g.*, there is only one value for  $f(4)$ , namely 23).

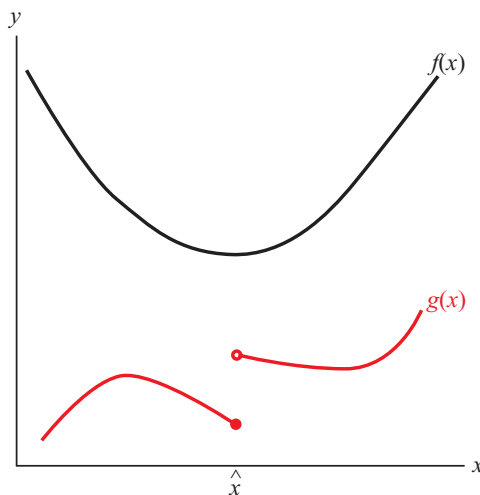
The set of numbers *taken* by the function—that is, the set of numbers for which the function is defined—is known as the *domain* of the function. The set of numbers to which this set is mapped is known as the *range* of the function. For example, because square root is defined for non-negative numbers only, the function  $g(x) = \sqrt{x}$  has a domain equal to  $\{x | 0 \leq x < \infty\}$  (*i.e.*, the set of numbers between 0 and infinity). Because, as discussed below,  $\sqrt{x} \geq 0$  for all  $x$  in the domain of  $\sqrt{\cdot}$ , we see that its range is also the set of non-negative numbers. As another example, the function  $h(x) = |x|$  has a domain equal to all real numbers and range consisting of all non-negative numbers.<sup>1</sup>

As a convention, we write  $f(\cdot)$  to denote the function and we write  $f(x)$  to denote a specific mapped value.

Although  $f(x)$  is a unique value, there is no guarantee that there is a unique  $x$  that solves the equation  $f(x) = y$  because more than one element of the domain can map to a single element of the range. For instance,  $|5| = |-5| = 5$ . So, if we define  $h(x) = |x|$ , then there is no single  $x$  that solves the equation  $h(x) = 5$ . In some cases, however, there is always a unique  $x$  that solves the equation  $f(x) = y$ . For instance, if  $f(x) = 5x + 3$ , then there is only one  $x$  for each  $y$ , namely  $x = (y - 3)/5$ . When there is always a unique  $x$  that solves  $f(x) = y$  for all  $y$ , we say that  $f(\cdot)$  is *invertible*. The standard notation for the inverse mapping is  $f^{-1}(\cdot)$ . So, for example, if  $f(x) = 5x + 3$ , then  $f^{-1}(y) = (y - 3)/5$ . Observe that the following rule holds for functions and their inverses:

---

<sup>1</sup>Recall that  $|x|$  denotes the absolute value of  $x$ . That is,  $|x| = x$  if  $x \geq 0$  and  $|x| = -x$  if  $x < 0$  (*e.g.*,  $|5| = 5$  and  $|-4| = 4$ ).



**Figure A1.1:** The function  $f(\cdot)$  is continuous. The function  $g(\cdot)$  is *not* continuous—it has a jump (point of discontinuity) at  $\hat{x}$ .

**Rule 1** For an invertible function  $f(\cdot)$ ,  $f(f^{-1}(y)) = y$  and  $f^{-1}(f(x)) = x$ .

A function,  $f(\cdot)$ , is *increasing* if, for all  $x_0$  and  $x_1$  in its domain,  $x_1 > x_0$  implies  $f(x_1) > f(x_0)$ . A function is *decreasing* if, for all  $x_0$  and  $x_1$  in its domain,  $x_1 > x_0$  implies  $f(x_1) < f(x_0)$ . Following the graph of a function from left to right, it is increasing if its graph goes up. It is decreasing if its graph goes down. The term *monotonic* will be used to refer to a function that either is an increasing function or is a decreasing function. Observe that a monotonic function must be invertible.

**Rule 2** *Monotonic functions are invertible.*

A function is *continuous* if its graph can be drawn without picking up your pen or pencil. See Figure A1.1. Technically, for any  $\hat{x}$  in the domain of the function  $f(\cdot)$  if we fix a  $\varepsilon > 0$  (but as small as we like), then there exists a  $\delta > 0$  such that

$$|f(x) - f(\hat{x})| < \varepsilon$$

if

$$|x - \hat{x}| < \delta.$$

Basically, this says that the graph can't jump at  $\hat{x}$ .

# Exponents | A1.2

Let  $n$  be a whole number (*i.e.*,  $n = 0$  or  $1$  or  $2 \dots$ ).<sup>2</sup> Let  $x$  be a real number. The expression  $x^n$  is shorthand for multiplying  $x$   $n$ -times by itself; that is,

$$x^n = \underbrace{x \times \cdots \times x}_{n \text{ times}}.$$

Note the convention that  $x^0 = 1$  for all  $x$ . The  $n$  term in  $x^n$  is the *exponent*. Sometimes  $x^n$  is described in words as “taking  $x$  to the  $n$ th power.”

## Addition and multiplication of exponents

One rule of exponents is

### Rule 3

$$x^n \times x^m = x^{n+m}.$$

For example,  $2^2 \times 2^3 = 2^{2+3} = 2^5$ . (As is readily verified:  $2^2 = 4$ ,  $2^3 = 8$ ,  $4 \times 8 = 32$ , and  $2^5 = 32$ .) Rule 3 follows because

$$\begin{aligned} x^n \times x^m &= \underbrace{x \times \cdots \times x}_{n \text{ times}} \times \underbrace{x \times \cdots \times x}_{m \text{ times}} \\ &= \underbrace{x \times \cdots \times x}_{(n+m) \text{ times}} \\ &= x^{n+m}. \end{aligned}$$

A second rule of exponents is

### Rule 4

$$(x^n)^m = x^{n \times m} = x^{nm}.$$

Observe the convention that the multiplication of two variables (*e.g.*,  $n$  and  $m$ ) can be expressed as  $nm$ . As an example,  $(2^2)^3 = 2^6$ . (As is readily seen:  $2^2 = 4$ ,

---

<sup>2</sup>Actually, one does *not* need to limit  $n$  to the whole numbers; it is certainly valid to let  $n$  be any real number. That is, for instance,  $x^{1/2}$  is a valid expression (it is the same as  $\sqrt{x}$ ). Even  $x^\pi$  is a valid expression. When, however,  $n$  is not a whole number, the interpretation of  $x^n$  is somewhat more involved. It is worth noting, however, that all the rules set forth in this section apply even if  $n$  is not a whole number.

$4^3 = 64$ , and  $2^6 = 64$ .) Rule 4 follows because

$$\begin{aligned}
 (x^n)^m &= \underbrace{x^n \times \cdots \times x^n}_{m \text{ times}} \\
 &= \underbrace{\underbrace{x \times \cdots \times x}_{n \text{ times}} \times \cdots \times \underbrace{x \times \cdots \times x}_{n \text{ times}}}_{m \text{ times}} \\
 &= \underbrace{x \times \cdots \times x}_{(n \times m) \text{ times}} \\
 &= x^{nm}.
 \end{aligned}$$

The rules for adding and multiplying exponents obey the usual rules of arithmetic; hence, for instance:

**Rule 5**  $x^{n+m} = x^n x^m = x^m x^n = x^{m+n}$ .

**Rule 6**  $x^{nm} = (x^n)^m = (x^m)^n = x^{mn}$ .

**Rule 7**  $x^{p(n+m)} = (x^{n+m})^p = (x^n x^m)^p = x^{np} x^{mp} = x^{np+mp}$ .

### More rules for exponents

**Rule 8**  $(xy)^n = x^n y^n$ .

**Rule 9**  $(x^n y^m)^p = x^{np} y^{mp}$ .

**Rule 10 (Rule of the Common Exponent I)**  $z^p x^n y^n = z^p (xy)^n$ .

**Rule 11 (Rule of the Common Exponent II)**  $z^{p+n} x^n = z^p (zx)^n$ .

**Rule 12 (Rule of the Common Exponent III)**  $z^{pn} x^n = (z^p x)^n$ .

### Some exponent no-noes

The following are common *mistakes* to be avoided:

**No no 1**  $(x + y)^n \neq x^n + y^n$ ; that is, exponentiation is not distributive over addition. For example,  $(2 + 3)^2 = 5^2 = 25$ , whereas  $2^2 + 3^2 = 4 + 9 = 13$ .

**No no 2**  $x^{pn} y^n \neq x^p (xy)^n$ . For example,  $2^6 \times 3^2 = 2^{3 \times 2} 3^2 = 64 \times 9 = 576$ , whereas  $2^3 (2 \times 3)^2 = 8 \times 36 = 288$ . (Also recall Rule 12.)

### Negative exponents

The notation  $x^{-n}$  means  $(1/x)^n$  (obviously  $x \neq 0$ ). Observe that, because

$$x^n x^{-n} = x^n \left(\frac{1}{x}\right)^n = \left(x \times \frac{1}{x}\right)^n = 1^n = 1 (= x^0)$$

(where the third equality follows from Rule 8), we have that  $x^{-n}$  is the multiplicative inverse<sup>3</sup> of  $x^n$ . It is also true that

$$x^n \times \frac{1}{x^n} = 1.$$

Because multiplicative inverses are unique, this means

**Rule 13**  $(1/x)^n = 1/x^n$ .

All of Rules 3–12 are valid with negative exponents (provided  $x$ ,  $y$ , or  $z$  are not zero).

It is readily seen that

**Rule 14**  $(x/y)^n = (y/x)^{-n}$ .

Note that

$$(x+y)^{-n} = \frac{1}{(x+y)^n}$$

not

$$\frac{1}{x^n} + \frac{1}{y^n};$$

that is,

**No no 3** *Observe*

$$(x+y)^{-n} \neq \frac{1}{x^n} + \frac{1}{y^n}.$$

For example,  $(2+3)^{-2} = 5^{-2} = 1/25$ , whereas  $1/2^2 + 1/3^2 = 1/4 + 1/9 = 13/36$ .

## Square Roots | A1.3

The positive **square root** of a number  $x \geq 0$  is denoted  $\sqrt{x}$ ; that is,  $\sqrt{x}$  is the positive number such that  $\sqrt{x} \times \sqrt{x} = x$ . Observe the convention that  $\sqrt{x} \geq 0$  (and equal to zero only if  $x = 0$ ). As a convention from now on, assume that, if we write  $\sqrt{x}$ , we are limiting attention to  $x \geq 0$ .

Because the product of two negative numbers is positive (e.g.,  $-2 \times (-2) = 4$ ), each number  $x \geq 0$  also has a *negative* square root. It is denoted  $-\sqrt{x}$ . Hence, as an example, the positive square root of 3 is denoted  $\sqrt{3}$  and the negative square root of 3 is denoted  $-\sqrt{3}$ .

<sup>3</sup>The **multiplicative inverse** of a number  $x$  is the number  $y$  such that  $xy = 1$ . For instance,  $1/2$  is the multiplicative inverse of 2. Note if  $y$  is the multiplicative inverse of  $x$ , then  $x$  is the multiplicative inverse of  $y$ .

### Rules for square roots

By definition,

**Rule 15**  $(\sqrt{x})^2 = x$  and  $\sqrt{x^2} = |x|$ . That is, squaring and taking square roots are inverse operations.

Also by definition, the square root of  $xy$  is  $\sqrt{xy}$ . But observe

$$(\sqrt{x}\sqrt{y})^2 = (\sqrt{x}\sqrt{y})(\sqrt{x}\sqrt{y}) = \underbrace{\sqrt{x}\sqrt{x}}_{=x} \underbrace{\sqrt{y}\sqrt{y}}_{=y} = xy,$$

where the second equality follows because multiplication is associative (*i.e.*, we are free to rearrange the order of the terms). But this says, then, that  $\sqrt{x}\sqrt{y}$  is also the square root of  $xy$ ; from which we can conclude:

**Rule 16**  $\sqrt{xy} = \sqrt{x}\sqrt{y}$ .

Rule 16 is useful for simplifying expressions. For instance,  $\sqrt{8} = \sqrt{4 \times 2} = \sqrt{4}\sqrt{2} = 2\sqrt{2}$ .

### Some square root no-noes

The following are common *mistakes* to be avoided:

**No no 4**  $\sqrt{x+y} \neq \sqrt{x} + \sqrt{y}$ ; that is, taking square roots is not distributive over addition. For example,  $\sqrt{9+16} = \sqrt{25} = 5$ , whereas  $\sqrt{9} + \sqrt{16} = 3 + 4 = 7$ .

Recall that the sum of the lengths of any two sides of a triangle must exceed the length of the remaining side. For this reason, this last “no no” is sometimes called the *triangle inequality* (can you see why?) and stated as

**Rule 17 (Triangle inequality)**  $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$  and equal only if  $x$  or  $y$  or both is (are) zero.

**No no 5**  $\sqrt{xy} \neq x\sqrt{y}$ ; that is, a multiplicative constant cannot be passed “through” the square roots operator. Recall, too, Rule 16.

## Equations of the form $ax^2 + bx + c = 0$ | A1.4

If a polynomial equation of the form  $ax^2 + bx + c = 0$  has a solution, then the solution or solutions are given by the [quadratic formula](#):

**Rule 18 (Quadratic formula)** If

$$ax^2 + bx + c = 0$$

has a solution, then its solution or solutions are given by the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (\text{A1.1})$$

Some quick observations:

1. Because one cannot take the square root of a negative number, we see that  $ax^2 + bx + c = 0$  can only have solutions if  $b^2 \geq 4ac$ .
2. The equation  $ax^2 + bx + c = 0$  will have *one* solution only if  $b^2 = 4ac$ . If  $b^2 > 4ac$ , then it will have two solutions.

Examples:

- $x^2 - 2x - 2 = 0$ . So solutions are

$$\frac{-(-2) + \sqrt{(-2)^2 - 4 \times 1 \times (-2)}}{2 \times 1} \text{ and } \frac{-(-2) - \sqrt{(-2)^2 - 4 \times 1 \times (-2)}}{2 \times 1},$$

which simplify to

$$\frac{2 + \sqrt{12}}{2} = 1 + \sqrt{3} \text{ and } \frac{2 - \sqrt{12}}{2} = 1 - \sqrt{3},$$

respectively.

- $x^2 + 2x + 1 = 0$ . Observe  $b^2 = 4ac$  (*i.e.*,  $2^2 = 4 \times 1 \times 1$ ), so there is only one solution:  $x = \frac{-b}{2a} = -2/2 = -1$ .
- $x^2 + 5x + 4 = 0$ . So solutions are

$$\frac{-5 \pm \sqrt{25 - 16}}{2} = \frac{-5 \pm 3}{2} = -1 \text{ or } -4.$$

## Lines | A1.5

A *line* has the formula  $y = mx + b$ , where  $m$  and  $b$  are constants and  $x$  and  $y$  are variables. This way of writing a line is also called *slope-intercept* form because  $m$  is the *slope* of the line and  $b$  is the *intercept* (the point on the  $y$  axis at which the line intersects the  $y$  axis). If  $m > 0$ , the line slopes up (*i.e.*, rises as its graph is viewed from left to right). If  $m < 0$ , the line slopes down (*i.e.*, falls as its graph is viewed from left to right).

Consider an expression of the form

$$Ax + By = C,$$

where  $A, B \neq 0$ , and  $C$  are constants. This, too, is an expression for a line, which we can see by rearranging:

$$\begin{aligned} By &= -Ax + C \\ y &= -\frac{A}{B}x + \frac{C}{B}. \end{aligned}$$

Hence,

**Rule 19** *An expression of the form  $Ax + By = C$  is equivalent to the line  $y = mx + b$ , where  $m = -A/B$  and  $b = C/B$ .*

### The line through two points

The line through two distinct points  $(x_1, y_1)$  and  $(x_2, y_2)$  is the line satisfying the equations:

$$\begin{aligned}y_1 &= mx_1 + b \text{ and} \\ y_2 &= mx_2 + b.\end{aligned}$$

Notice that if we subtract the first equation from the second, we get:

$$y_2 - y_1 = mx_2 - mx_1 = m(x_2 - x_1),$$

from which it follows that

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Substituting that back into the second equation, we get

$$\begin{aligned}y_2 &= \frac{y_2 - y_1}{x_2 - x_1}x_2 + b \\ y_2 - \frac{y_2 - y_1}{x_2 - x_1}x_2 &= b \\ \frac{y_2(x_2 - x_1)}{x_2 - x_1} - \frac{x_2(y_2 - y_1)}{x_2 - x_1} &= b \\ \frac{x_2y_1 - x_1y_2}{x_2 - x_1} &= b.\end{aligned}$$

To conclude:

**Rule 20** The line  $y = mx + b$  through the points  $(x_1, y_1)$  and  $(x_2, y_2)$  is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1} \text{ and } b = \frac{x_2y_1 - x_1y_2}{x_2 - x_1}.$$

### Parallel lines

If  $y = mx + b$  is a line and  $(x_1, y_1)$  is a point not on that line, we can find a line parallel to  $y = mx + b$  that goes through  $(x_1, y_1)$ . Call this parallel line  $y = Mx + B$ . A parallel line has the same slope, so  $M = m$ . But, then,  $B$  is readily found by subtraction:

$$B = y_1 - mx_1.$$

## Solving Rational Expressions

# A1.6

Consider an equation of the form

$$\frac{x-2}{x+5} = 6.$$

We can solve it by multiplying both sides by  $x + 5$ ,

$$(x + 5)\frac{x - 5}{x + 5} = 6(x + 5)$$

$$x - 5 = 6x + 30,$$

then rearranging,

$$-5 - 30 = 5x,$$

and then solving  $5x = -35$ , which yields  $x = -7$ .

More generally, consider the following:

$$\frac{Ax + B}{Cx + D} = \frac{Ex + F}{Gx + H},$$

where we seek to solve for  $x$  and the parameters  $A$ - $H$  are constants. To solve, multiply both sides by the two denominators (*i.e.*,  $Cx + D$  and  $Gx + H$ ):

$$(Cx + D)(Gx + H)\frac{Ax + B}{Cx + D} = (Cx + D)(Gx + H)\frac{Ex + F}{Gx + H}$$

which, simplifying, becomes

$$(Gx + H)(Ax + B) = (Cx + D)(Ex + F)$$

$$AGx^2 + (BG + AH)x + BH = CEx^2 + (CF + DE)x + DF;$$

hence,

$$(AG - CE)x^2 + ((BG + AH) - (CF + DE))x + BH - DF = 0.$$

The last expression is a quadratic in standard form and can be solved using the quadratic formula (*i.e.*, formula (A1.1) on page 172).

## Logarithms | A1.7

Often we need to solve equations of the form  $A^x = B$ , where  $A$  and  $B$  are constants and  $x$  is the unknown to be solved for. This section explores, *inter alia*, how such equations are solved.

### The Natural Logarithm

For reasons that we won't pursue here, the constant  $e$ , which we will define in a moment, arises a lot in mathematics. This constant is defined to be<sup>4</sup>

$$e = \sum_{t=0}^{\infty} \frac{1}{t!}, \quad (\text{A1.2})$$

---

<sup>4</sup>An alternative—but equivalent—definition is

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n.$$

This is why the formula for continuous compound interest is expressed in terms of  $e$ .

where  $t!$ —read  $t$  *factorial*—is defined as

$$t! = \begin{cases} 1, & \text{if } t = 0 \\ 1 \times \cdots \times t, & \text{if } t = 1, 2, \dots \end{cases}$$

Calculations show that  $e \approx 2.718281828$ . Because  $e$  is an irrational number, no exact decimal representation exists.

The number  $e$  is called the *base of the natural logarithm*. We define the function  $\exp(x) = e^x$ .

Observe that  $\lim_{x \rightarrow -\infty} e^x = 0$  and  $\lim_{x \rightarrow \infty} e^x = \infty$ . Moreover, it can be shown that  $e^x$  is increasing and continuous. Hence, for every  $y > 0$  there exists a unique  $x$  such that  $e^x = y$ .

### The Function $\ln(\cdot)$

The function  $\ln(\cdot)$  is defined as the inverse of  $\exp(\cdot)$ . That is,

$$\ln(\exp(x)) = \ln(e^x) = x.$$

The function  $\ln(\cdot)$  is called the *logarithm* or *log*.<sup>5</sup> Because  $\exp(\cdot)$  takes only positive values, it follows that  $\ln(\cdot)$  is defined for positive numbers only.

Recall that, for every  $y > 0$ , there exists a unique  $x$  such that  $e^x = y$ . Taking logs of both sides, we see that  $x = \ln(y)$ ; that is, in this case,  $x$  is the log of  $y$ .

**Rule 21**  $\ln(AB) = \ln(A) + \ln(B)$ .

**Proof:** Recall there exists a unique  $a$  and a unique  $b$  such that  $A = e^a$  and  $B = e^b$ . We thus have

$$\begin{aligned} \ln(AB) &= \ln(e^a e^b) \\ &= \ln(e^{a+b}) && \text{(Rule 3)} \\ &= a + b && (\ln(\cdot) \text{ is inverse of } \exp(\cdot)) \\ &= \ln(A) + \ln(B). \end{aligned}$$

■

**Rule 22**  $\ln(A/B) = \ln(A) - \ln(B)$ .

**Proof:** Define  $a$  and  $b$  as in the proof of Rule 21. Observe

$$\begin{aligned} \ln(A/B) &= \ln\left(\frac{e^a}{e^b}\right) \\ &= \ln(e^a e^{-b}) && \text{(Def'n of negative exponent)} \\ &= \ln(e^{a-b}) && \text{(Rule 3)} \\ &= a - b && (\ln(\cdot) \text{ is inverse of } \exp(\cdot)) \\ &= \ln(A) - \ln(B). \end{aligned}$$

---

<sup>5</sup>To be technical, we would say the log with respect to  $e$  (the base).



**Rule 23**  $\ln(A^\gamma) = \gamma \ln(A)$ .

**Proof:** Define  $a$  as in the proof of Rule 21. Observe

$$\begin{aligned} \ln(A^\gamma) &= \ln((e^a)^\gamma) \\ &= \ln(e^{a\gamma}) \quad (\text{Rule 4}) \\ &= a\gamma \quad (\ln(\cdot) \text{ is inverse of } \exp(\cdot)) \\ &= \gamma \ln(A). \end{aligned}$$



### Solving Equations in Exponents

Consider the motivating example, namely that we wish to solve the equation

$$A^x = B$$

for  $x$ . Using Rule 23, this is equivalent to

$$x \ln(A) = \ln(B).$$

Dividing both sides by  $\ln(A)$  yields

$$x = \frac{\ln(B)}{\ln(A)}. \quad (\text{A1.3})$$

## Checking Your Work | A1.8

One of the nice features of algebra is that it is often easy to check whether you've made a mistake. Two methods are (1) plug your answer back into the original problem and (2) check the result using arbitrary values.

As an illustration of the “plug the solution back in” method, consider the following example. Suppose you are asked to solve  $x^2 - 2x + 1 = 0$  and you work out a solution of  $x = 2$ . Is this correct? Well, you can check by plugging 2 back into the original equation and see if it satisfies the equality:

$$\begin{aligned} 2^2 - 2 \times 2 + 1 &= 4 - 4 + 1 \\ &\neq 0; \end{aligned}$$

hence,  $x = 2$  does *not* satisfy the equation and so *cannot* be the correct solution.

To illustrate the “arbitrary values” method, consider the following. Suppose you are asked to solve

$$\frac{x - K}{x} = 2 \quad (\text{A1.4})$$

( $K \neq 0$ ) and you get a solution of  $x = 2K$ . Is this correct? You could plug  $2K$  in and try to verify the equation, but an alternative is to try arbitrary values for  $K$  and see what happens. Suppose  $K = 1$ , then  $x = 2$ . The left-hand side of the original expression, expression (A1.4), would then be

$$\frac{2-1}{2} = \frac{1}{2} \neq 2;$$

hence, we see that  $x = 2K$  *cannot* be the correct solution. That is, logically, if we write  $x = 2K$ , what we're saying is that for *all* possible values of  $K$ , the solution to equation (A1.4) is  $x = 2K$ . We can falsify this claim—show that we've made a mistake—simply by finding some value for  $K$  for which it doesn't work.

## System of Equations

# A2

Often in economics we need to know the solution to a system of equations. For instance, we could have that demand is given by

$$Q_D = A - Bp$$

and supply is given by

$$Q_S = F + Gp,$$

where  $Q_D$  is quantity demanded as a function of price,  $p$ ,  $Q_S$  is quantity supplied as a function of price, and  $A$ ,  $B$ ,  $F$ , and  $G$  are non-negative constants. If the market is in equilibrium, then price must be such that  $Q_D = Q_S$ . To solve for that price, we need to solve the system of equations above.

## A Linear Equation in One Unknown

# A2.1

Recall that if  $x$  is an unknown variable and  $A$ ,  $B$ , and  $C$  are constants ( $A \neq 0$ ), then the equation

$$Ax + B = C$$

is solved by the following procedure:

1. Subtract  $B$  from both sides (equivalently, add  $-B$  to both sides). This yields

$$Ax = C - B.$$

2. Divide both sides by  $A$  (equivalently, multiply both sides by  $1/A$ ). This yields the solution,

$$x = \frac{C - B}{A}.$$

Observe this method can be used even if, rather than  $x$ , we have some invertible function of  $x$ ,  $f(x)$ . That is, consider the equation

$$Af(x) + B = C.$$

Employing the two steps above yields

$$f(x) = \frac{C - B}{A}.$$

Letting  $f^{-1}(\cdot)$  denote the inverse function (e.g., if  $f(x) = \sqrt{x}$  then  $f^{-1}(y) = y^2$ ), we can invert both sides to obtain

$$x = f^{-1}\left(\frac{C-B}{A}\right).$$

## Two Linear Equations in Two Unknowns | A2.2

Consider the system

$$Ax + By = C \quad \text{and} \quad (\text{A2.1})$$

$$Dx + Ey = F, \quad (\text{A2.2})$$

where  $A$ – $F$  are constants and  $x$  and  $y$  are the two unknowns that we wish to solve for. Recall, from Section A1.5, that expressions like (A2.1) and (A2.2) are formulæ for lines. Hence, solving this system is equivalent to finding the point,  $(x, y)$ , at which those two lines cross.

Because of this equivalence, it follows that, if the lines don't cross—are parallel—then no solution exists. Recall two lines are parallel if they have the same slope. From Section A1.5, the slopes of these two lines are  $-A/B$  and  $-D/E$  respectively. Hence, if  $A/B = D/E$ , there is no solution.<sup>1</sup>

Assume that  $A/B \neq D/E$ . Then a solution can be found by the method of substitution, which has the following steps:

1. Write the first equation in slope-intercept form:

$$y = -\frac{A}{B}x + \frac{C}{B}. \quad (\text{A2.3})$$

2. Substitute the righthand side of this last equation for  $y$  in the second equation (i.e., equation (A2.2)):

$$Dx + E\left(-\frac{A}{B}x + \frac{C}{B}\right) = F.$$

3. Combine terms in  $x$ :

$$\left(D - \frac{EA}{B}\right)x = F - \frac{EC}{B}$$

or, simplifying,

$$\frac{BD - EA}{B}x = \frac{BF - EC}{B}.$$

---

<sup>1</sup>To be technical, if it is also true that  $C/B = F/E$ , so the two lines have the same intercept, then they are the same line. In which case, one could say that there are an infinite number of solutions, because all points on the line satisfy the two equations.

4. Solve this last expression for  $x$ :

$$x = \frac{BF - EC}{BD - EA}. \quad (\text{A2.4})$$

5. Substitute for  $x$  in expression (A2.3) using the righthand side of expression (A2.4)

$$y = -\frac{A}{B} \times \frac{BF - EC}{BD - EA} + \frac{C}{B}$$

or, simplifying,

$$\begin{aligned} &= \frac{-A(BF - EC) + C(BD - EA)}{B(BD - EA)} \\ &= \frac{CD - AF}{BD - EA}. \end{aligned}$$

For example, recall the demand and supply equations with which we began this appendix:

$$\begin{aligned} Q &= A - Bp \quad \text{and} \\ Q &= F + Gp. \end{aligned}$$

The first equation is already in slope-intercept form (letting  $Q$  be our  $y$  and  $p$  be our  $x$ ). Substituting into the second equation yields:

$$A - Bp = F + Gp.$$

Solving for  $p$  yields:

$$\frac{A - F}{B + G} = p.$$

Finally, observe that if, instead of  $x$  and  $y$  in equations (A2.1) and (A2.2), we had had invertible functions of  $x$  and  $y$ , then we could still use this technique. That is, if, instead of  $x$ , we had  $f(x)$  and, instead of  $y$ , we had  $g(y)$ , then the substitution method would reveal

$$f(x) = \frac{BF - EC}{BD - EA} \quad \text{and} \quad g(y) = \frac{CD - AF}{BD - EA};$$

hence,

$$x = f^{-1} \left( \frac{BF - EC}{BD - EA} \right) \quad \text{and} \quad y = g^{-1} \left( \frac{CD - AF}{BD - EA} \right).$$



It is beyond a simple appendix to teach calculus fully. Here we seek to review the main points. The intention is that it be understandable by anyone with a good mastery of high school mathematics.

Calculus is divided into two parts. One, known as *differential calculus*, is concerned with the rates at which things change. For instance, marginal cost,  $MC$ , is the rate at which total cost changes as we increase production.

The second part, known as *integral calculus*, is concerned with areas under curves. For instance, the area under the demand curve (ignoring income effects) from 0 to  $x$  units is the total benefit consumers derive from the  $x$  units.

Recall the speed analogy in Section 3.4. We saw, there, that the speed at which you are travelling at a moment in time,  $t$ , is best estimated by

$$\text{speed} = \frac{D(t+h) - D(t)}{h},$$

where  $D(\cdot)$  gives your distance as a function of time travelled and  $h$  is some very small increment of time (*e.g.*, a small fraction of an hour, such as a second). Indeed, the smaller we can make  $h$ , the better our estimate will be. In fact, the ideal estimate is

$$\text{speed} = \lim_{h \rightarrow 0} \frac{D(t+h) - D(t)}{h},$$

where “lim” means the limit of that ratio as  $h$  approaches zero. For instance, if  $D(t) = St$ , then

$$\begin{aligned} \frac{D(t+h) - D(t)}{h} &= \frac{S(t+h) - St}{h} \\ &= \frac{Sh}{h} \\ &= S. \end{aligned}$$

Clearly, the last expression doesn’t depend on  $h$ , so the limit of it as  $h$  goes to zero is  $S$ .

**Definition 10** The derivative of a function  $f(\cdot)$  at  $x$  is

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (\text{A3.1})$$

assuming that limit exists.

For our purposes in 201A, we generally will assume that the limit exists.

**Definition 11** A function is said to be differentiable if a derivative exists for any point in its domain.<sup>1</sup>

As just noted, we will typically limit our attention to differentiable functions in 201A.

We denote derivatives in a number of ways. Specifically, the derivative of  $f(x)$  can be denoted as  $f'(x)$  (read “f prime of x”) or as  $df(x)/dx$ . If we have previously stated that  $y = f(x)$ , then we can also denote the derivative of  $f(\cdot)$  as  $dy/dx$ . For some functions, such as cost functions,  $C(\cdot)$ , and revenue functions,  $R(\cdot)$ , we use a prefix “M,” for *marginal*, to denote the derivative. That is, for example, the derivative of  $C(\cdot)$  is  $MC(\cdot)$ .

### Properties of Derivatives

**Note:** Proofs of the following rules have been included for the sake of completeness. Reading the proofs is completely optional. Do, however, read the rules themselves.

**Rule 24** If  $f(x) = K$ ,  $K$  a constant, for all  $x$ , then  $f'(x) = 0$  for all  $x$ .

**Proof:** The numerator of the fraction in expression (A3.1) is always zero. ■

**Rule 25** If

$$b(x) = \alpha f(x) + \beta g(x),$$

where  $\alpha$  and  $\beta$  are constants, then

$$b'(x) = \alpha f'(x) + \beta g'(x).$$

**Proof:** Observe (A3.1) can be written as

$$\alpha \frac{f(x+h) - f(x)}{h} + \beta \frac{g(x+h) - g(x)}{h}.$$

Now take limits. ■

**Rule 26** If  $b(x) = f(x)g(x)$ , then  $b'(x) = f'(x)g(x) + f(x)g'(x)$ .

<sup>1</sup>The *domain* of a function, recall, is the set of values for which the function is defined.

**Proof:** Observe

$$\begin{aligned} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} &= \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &\quad + \underbrace{\frac{f(x+h)g(x) - f(x+h)g(x)}{h}}_{+0} \\ &= g(x) \frac{f(x+h) - f(x)}{h} + f(x+h) \frac{g(x+h) - g(x)}{h}. \end{aligned}$$

Now take limits (note  $\lim_{h \rightarrow 0} f(x+h) = f(x)$ ). ■

**Rule 27 (Chain rule)** If  $b(x) = f(g(x))$ , then  $b'(x) = f'(g(x))g'(x)$ .

**Proof:** Observe

$$\begin{aligned} \frac{f(g(x+h)) - f(g(x))}{h} &= \frac{f(g(x+h)) - f(g(x))}{g(x+h) - g(x)} \times \frac{g(x+h) - g(x)}{h} \\ &= \frac{f(g(x) + \eta) - f(g(x))}{\eta} \times \frac{g(x+h) - g(x)}{h}, \end{aligned}$$

where  $\eta = g(x+h) - g(x)$ , so  $g(x+h) = g(x) + \eta$ . Observe that  $\lim_{h \rightarrow 0} \eta = 0$ ; hence, the limit of the righthand-side of the last expression is  $f'(g(x))g'(x)$  as claimed. ■

Note the chain rule can be applied repeatedly.

The following two rules we state without proof.<sup>2</sup>

**Rule 28**

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$$

**Rule 29**

$$\lim_{\varepsilon \rightarrow 0} \frac{\ln(1 + \varepsilon)}{\varepsilon} = 1.$$

**Rule 30** The derivative of  $e^x$  with respect to  $x$  is  $e^x$  (i.e.,  $de^x/dx = e^x$ ).

**Proof:** Observe

$$\frac{e^{x+h} - e^x}{h} = e^x \frac{e^h - 1}{h},$$

where we have used Rule 3 to factor out  $e^x$ . The result then follows from Rule 28. ■

**Rule 31** The derivative of  $\ln(x)$  with respect to  $x$  is  $1/x$  (i.e.,  $d\ln(x)/dx = 1/x$ ).

---

<sup>2</sup>The curious can consult any decent calculus text. A specific citation is §2.6 of D.W. Jordan and P. Smith's *Mathematical Techniques*, 2nd ed., Oxford: Oxford University Press, 1997.

**Proof:** Observe

$$\begin{aligned}
 \frac{\ln(x+h) - \ln(x)}{h} &= \frac{1}{h} \ln\left(\frac{x+h}{x}\right) \quad (\text{Rule 22}) \\
 &= \frac{1}{h} \ln\left(1 + \frac{h}{x}\right) \\
 &= \frac{1}{x\varepsilon} \ln(1 + \varepsilon) \quad (\text{making the substitution } h = x\varepsilon) \\
 &= \frac{1}{x} \times \frac{\ln(1 + \varepsilon)}{\varepsilon}.
 \end{aligned}$$

Because  $\varepsilon = h/x$ , it goes to zero as  $h$  goes to zero. Hence, the result follows from Rule 29. ■

**Rule 32** The derivative of  $x^z$  with respect to  $x$  is  $zx^{z-1}$  (i.e.,  $dx^z/dx = zx^{z-1}$ ).

**Proof:** Let  $f(y) = e^y$  and  $g(w) = z \ln(w)$ . Using Rule 23, we have

$$\ln(x^z) = z \ln(x) = g(x).$$

Because  $\exp(\cdot)$  and  $\ln(\cdot)$  are inverse functions, we have

$$\begin{aligned}
 x^z &= \exp(\ln(x^z)) \\
 &= e^{g(x)} \\
 &= f(g(x)).
 \end{aligned}$$

Hence, we can calculate the derivative of  $x^z$  using the chain rule:

$$\begin{aligned}
 \frac{dx^z}{dx} &= f'(g(x))g'(x) \\
 &= e^{g(x)}g'(x) \quad (\text{Rule 30}) \\
 &= e^{z \ln(x)} \frac{z}{x} \quad (\text{Rule 31}) \\
 &= e^{\ln(x^z)} \frac{z}{x} \quad (\text{Rule 23}) \\
 &= \frac{zx^z}{x} \quad (\exp(\cdot) \text{ and } \ln(\cdot) \text{ are inverses}) \\
 &= zx^{z-1} \quad (\text{Rule 3})
 \end{aligned}$$

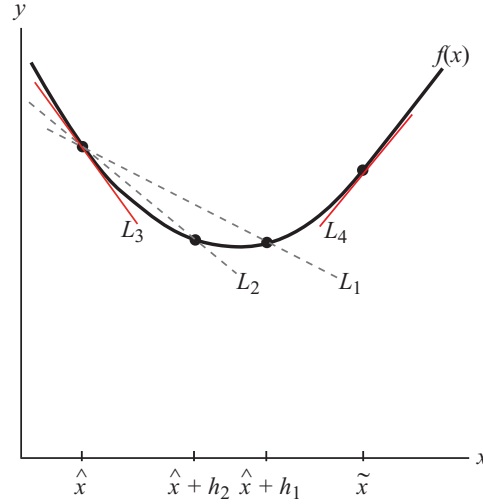
■

## Graphical Interpretation of Derivatives

## A3.2

Consider Figure A3.1. Consider the gray dashed line labelled  $L_1$ . Observe it goes through the points

$$(\hat{x}, f(\hat{x})) \quad \text{and} \quad (\hat{x} + h_1, f(\hat{x} + h_1)).$$



**Figure A3.1:** The derivatives of  $f(\cdot)$  at  $\hat{x}$  and  $\tilde{x}$  are the slopes of the lines  $L_3$  and  $L_4$ , respectively.

From Rule 20, the slope of that line is

$$\frac{f(\hat{x} + h_1) - f(\hat{x})}{(\hat{x} + h_1) - \hat{x}} = \frac{f(\hat{x} + h_1) - f(\hat{x})}{h_1},$$

which looks like the fraction in expression (A3.1) above. Likewise, the slope of line  $L_2$  is

$$\frac{f(\hat{x} + h_2) - f(\hat{x})}{h_2}.$$

As the figure indicates, as we make  $h$  smaller (go from  $h_1$  to  $h_2$ ), the slope of the resulting line will get closer to the slope of the red line ( $L_3$ ), which is just tangent to  $f(\cdot)$  at  $(\hat{x}, f(\hat{x}))$ . This, thus, provides a graphical interpretation of the derivative: *The derivative of  $f(\cdot)$  at a point  $x$ ,  $f'(x)$ , is the slope of the line just tangent to  $f(\cdot)$  at  $(x, f(x))$ .*

**Derivative:** *The slope of the line tangent to the function.*

Observe from Figure A3.1 that where the function is decreasing, for instance at  $\hat{x}$ , the tangent line has a negative slope; which means that the derivative of  $f(\cdot)$  at  $\hat{x}$  is negative. When the function is increasing, for instance at  $\tilde{x}$ , the tangent line has a positive slope; which means that the derivative at  $\tilde{x}$  is positive. This establishes:

**Rule 33** *If  $f'(x) > 0$ , then  $f(\cdot)$  is increasing at  $x$ . If  $f'(x) < 0$ , then  $f(\cdot)$  is decreasing at  $x$ . If  $f'(x) = 0$ , then  $f(\cdot)$  is neither increasing nor decreasing at  $x$ ; it is flat.*

## Maxima and Minima | A3.3

As noted previously, the slope,  $m$ , of the line going through points  $(x_1, y_1)$  and  $(x_2, y_2)$  is

$$m = \frac{y_2 - y_1}{x_2 - x_1}. \quad (\text{A3.2})$$

Let  $L(\cdot)$  be the function of this line; that is,  $L(x_1) = y_1$  and  $L(x_2) = y_2$ . Suppose we knew  $x_1$ ,  $y_1$ , and  $m$ , then we can rearrange expression (A3.2) as

$$y_2 = y_1 + m(x_2 - x_1).$$

Therefore, we have

$$L(x) = y_1 + m(x - x_1). \quad (\text{A3.3})$$

Given that the derivative is the slope of the line tangent to the point  $(x, f(x))$ —that is,  $m = f'(x)$ —expression (A3.3) gives us a way to approximate  $f(\cdot)$ . Specifically,

$$f(x) \approx \underbrace{f(x_1)}_{y_1} + \underbrace{f'(x_1)}_m (x - x_1).$$

The closer  $x_1$  is to  $x$ , the more accurate is this approximation.

Suppose we wish to determine where a function reaches its maximum and minimum; that is, where the function reaches its greatest and least values. Let's start with the maximum. Suppose the maximum occurs at  $x^*$ . Then  $f(x^*) \geq f(x)$  for all other  $x$ . Consider  $\varepsilon > 0$  but arbitrarily small. We know that  $f(x^*) \geq f(x^* - \varepsilon)$  and  $f(x^*) \geq f(x^* + \varepsilon)$ . Moreover, we have

$$\begin{aligned} f(x^*) &\geq f(x^* - \varepsilon) \\ &\approx f(x^*) + f'(x^*)((x^* - \varepsilon) - x^*) \\ &= f(x^*) - \varepsilon f'(x^*) \quad \text{and} \end{aligned} \quad (\text{A3.4})$$

$$\begin{aligned} f(x^*) &\geq f(x^* + \varepsilon) \\ &\approx f(x^*) + f'(x^*)((x^* + \varepsilon) - x^*) \\ &= f(x^*) + \varepsilon f'(x^*). \end{aligned} \quad (\text{A3.5})$$

Observe that (A3.4) implies  $0 \geq -\varepsilon f'(x^*)$  or  $\varepsilon f'(x^*) \geq 0$ . Expression (A3.5) implies  $0 \geq \varepsilon f'(x^*)$ . Putting them together we get

$$\varepsilon f'(x^*) \geq 0 \geq \varepsilon f'(x^*).$$

Given that  $\varepsilon > 0$ , the only way this can be true is if  $f'(x^*) = 0$ . We've established:

**Rule 34** A necessary condition for  $x^*$  to maximize the differentiable function  $f(\cdot)$  is that  $f'(x^*) = 0$ .

Note this is only a *necessary* condition.

Now let's consider the minimum. Suppose the minimum occurs at  $x^{**}$ . Then  $f(x^{**}) \leq f(x)$  for all other  $x$ . Consider  $\varepsilon > 0$  but arbitrarily small. We know that  $f(x^{**}) \leq f(x^{**} - \varepsilon)$  and  $f(x^{**}) \leq f(x^{**} + \varepsilon)$ . Moreover, we have

$$\begin{aligned} f(x^{**}) &\leq f(x^{**} - \varepsilon) \\ &\approx f(x^{**}) + f'(x^{**})((x^{**} - \varepsilon) - x^{**}) \\ &= f(x^{**}) - \varepsilon f'(x^{**}) \quad \text{and} \end{aligned} \tag{A3.6}$$

$$\begin{aligned} f(x^{**}) &\leq f(x^{**} + \varepsilon) \\ &\approx f(x^{**}) + f'(x^{**})((x^{**} + \varepsilon) - x^{**}) \\ &= f(x^{**}) + \varepsilon f'(x^{**}). \end{aligned} \tag{A3.7}$$

Observe that (A3.6) implies  $0 \leq -\varepsilon f'(x^{**})$  or  $\varepsilon f'(x^{**}) \leq 0$ . Expression (A3.7) implies  $0 \leq \varepsilon f'(x^{**})$ . Putting them together we get

$$\varepsilon f'(x^{**}) \leq 0 \leq \varepsilon f'(x^{**}).$$

Given that  $\varepsilon > 0$ , the only way this can be true is if  $f'(x^{**}) = 0$ . We've established:

**Rule 35** A necessary condition for  $x^{**}$  to minimize the differentiable function  $f(\cdot)$  is that  $f'(x^{**}) = 0$ .

Note, again, this is only a *necessary* condition.

We can interpret these last two results as follows. At the top of a hill or the bottom of a valley it is flat. In other words, at the very top, the derivative must be zero. Likewise, at the very bottom, the derivative must be zero as well.

The fact that a point at which the derivative is zero could be either a maximum or a minimum underscores that these rules are only necessary conditions.<sup>3</sup> A zero derivative neither implies a maximum nor a minimum.

### Sufficient Conditions

While Rules 34 and 35 tell us how to identify *candidates* for the maximum or minimum, they don't tell us whether our candidates are maxima or minima. For this reason we need sufficient conditions.

Recall Figure A3.1. Observe that, to the left of the minimum,  $f(\cdot)$  is decreasing; that is, if  $x^{**}$  is the minimum, then  $f'(x) < 0$  for  $x < x^{**}$ . Likewise, to the right of the minimum,  $f(\cdot)$  is increasing; that is,  $f'(x) > 0$  for  $x > x^{**}$ .

Similarly, if we were considering a maximum, then as we approach the maximum ("top of the hill") from the left, we must be climbing; that is, if  $x^*$  is the maximum, then  $f'(x) > 0$  for  $x < x^*$ . As move away from the maximum going to the right, we must be descending; that is,  $f'(x) < 0$  for  $x > x^*$ .

To summarize:

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<sup>3</sup>To be rigorous, we should note that there are some functions for which  $f'(x) = 0$  and  $x$  is neither a minimum nor a maximum. Such functions are not of concern, however, in 201A.

**Rule 36** If  $f'(x) > 0$  for all  $x < x^*$  and if  $f'(x) < 0$  for all  $x > x^*$ , then  $x^*$  maximizes  $f(\cdot)$ . If  $f'(x) < 0$  for all  $x < x^{**}$  and if  $f'(x) > 0$  for all  $x > x^{**}$ , then  $x^{**}$  minimizes  $f(\cdot)$ .

**Example 31:** As an example, suppose that profits as a function of output,  $x$ , equal

$$100x - x^2.$$

Let's find the level of output that maximizes profit:

1. Differentiate the profit function:  $100 - 2x$ .
2. A candidate for a maximum is any point at which  $100 - 2x = 0$ .
3. The only such point is  $x = 50$ .
4. Is 50 a maximum or a minimum?
5. Observe that if  $x < 50$ , then  $100 - 2x > 100 - 2 \times 50 = 0$  and if  $x > 50$ , then  $100 - 2x < 100 - 2 \times 50 = 0$ . So, by Rule 36,  $x = 50$  maximizes profit.

**Example 32:** As another example, suppose that  $f(x) = -e^{(x-5)^2}$ . What is its maximum (or minimum)?

1. Using the chain rule, we have  $f'(x) = -e^{(x-5)^2} 2(x-5)$ .
2. Setting that equal to zero, yields  $-e^{(x-5)^2} 2(x-5) = 0$ .
3. Solving for  $x$  yields  $x = 5$ .
4. Is  $x = 5$  a maximum or a minimum?
5. Recall that  $\exp(\cdot)$  is always positive, so  $-\exp(\cdot)$  is always negative. So  $f'(x)$  is positive for  $x < 5$  and  $f'(x)$  is negative for  $x > 5$ . So,  $x = 5$  is a maximum.

## Corner Solutions

So far we have been a bit loose with regard to one detail. Recall that the arguments that led to Rules 34 and 35 relied on comparing the extremum (*i.e.*,  $x^*$  or  $x^{**}$ ) to points to the left and right of the extremum. But what if the function isn't defined both on the left and right of the extremum? Then our arguments don't work fully.

For example, suppose you were asked to find the minimum of  $f(x) = \sqrt{x}$ . Because  $f(\cdot)$  is increasing and defined only for  $x \geq 0$ , we see that the minimum must occur at  $x = 0$ . But

$$f'(x) = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2\sqrt{x}} \quad (\text{Rule 32})$$

and, thus,  $f'(0) = \infty \neq 0$ . That is, 0 is, in this case, a minimum that does not satisfy the necessary condition Rule 35. This is an example of a **corner solution**—because the function can't pass the boundary (the “corner”) of the domain, it can be that the extremum occurs at the corner even though the derivative is not zero at the corner.

Hence, to our earlier analysis, we need to add the following caveat: *If the domain of  $f(\cdot)$  is bounded (e.g., as  $\sqrt{\cdot}$  is at zero), then one must also check whether the boundary points—the corners—are maxima or minima.*

## Antiderivatives | A3.4

Often we want to reverse the process of differentiation. That is, if given  $f'(\cdot)$ , we want to know  $f(\cdot)$ . To indicate taking an antiderivative we write  $\int f'(x)dx$ , where  $\int$ , the *integral sign*, denotes we wish to determine the antiderivative and the  $dx$  term indicates the variable with respect to which we wish to calculate it.

Consider, for instance,  $\int x dx$ . We want to know what function has as its derivative  $x$ . Reversing Rule 32, we see that  $f(x) = x^2/2$  has  $x$  as its derivative. But so, too, does  $g(x) = x^2/2 + K$ ,  $K$  a constant, because the derivative of a constant is zero. Hence, antiderivatives are only defined up to an additive constant.

Some antiderivatives. Observe these can all be verified by differentiating the righthand sides using the rules derived above.

- $\int e^x dx = e^x + K$ .
- $\int \frac{1}{x} dx = \ln(x) + K$
- $\int x^z dx = \frac{1}{z} x^{z+1} + K$  (provided  $z \neq -1$ ).
- $\int A dx = Ax + K$ ,  $A$  a constant.
- $\int (Af(x) + Bg(x)) dx = A \int f(x) dx + B \int g(x) dx$ ,  $A$  and  $B$  constants.

Sometimes we have an *initial condition* that ties down  $K$ . For example, suppose an advertising campaign generates 10,000 additional customers per month. Suppose the company doing the advertising has 20,000 consumers initially. How many customers will it have in  $T$  months? Observe that 10,000 is the derivative, and we want to find the antiderivative:

$$\int 10,000 dt = 10,000 t + K.$$

We need to find  $K$ . But note that at  $t = 0$  the company has 20,000 customers. So

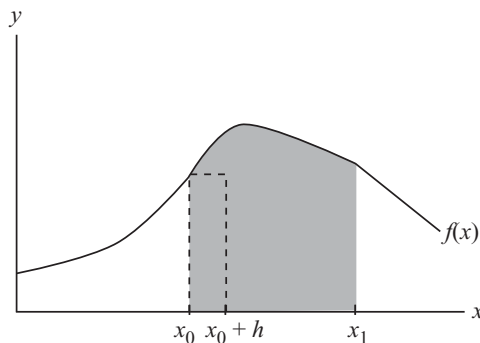
$$10,000 \times 0 + K = 20,000;$$

hence,  $K = 20,000$ . The number of customers in  $T$  months is, thus,

$$10,000 T + 20,000.$$

## Area under Curves | A3.5

Consider Figure A3.2. We are interested in determining the area in gray; that is, the area under  $f(\cdot)$  between  $x_0$  and  $x_1$ .



**Figure A3.2:** The gray area under  $f(\cdot)$  is  $\int_{x_0}^{x_1} f(x)dx$ . The dotted rectangle has width  $h$  and height  $f(x_0)$ .

To do this, define  $F(x)$  to be the area under  $f(\cdot)$  from 0 to  $x$ .<sup>4</sup> Observe that the gray area is, thus,  $F(x_1) - F(x_0)$ . But what is  $F(\cdot)$ ? To determine it, observe that the area under the curve between  $x_0$  and  $x_0 + h$ ,  $F(x_0 + h) - F(x_0)$ , can be approximated by the rectangle whose height is  $f(x_0)$  and whose width is  $h$ . That is,

$$F(x_0 + h) - F(x_0) \approx hf(x_0).$$

Moreover, this approximation is better, the smaller is  $h$ . But rearranging this last expression, we have

$$\frac{F(x_0 + h) - F(x_0)}{h} \approx f(x_0),$$

which becomes exact as  $h \rightarrow 0$ . But the value of the lefthand side as  $h \rightarrow 0$  is the derivative of  $F(\cdot)$  at  $x_0$ . In other words, we've shown  $F'(x_0) = f(x_0)$ . Because  $x_0$  was arbitrary, we have that

$$F'(x) = f(x) \tag{A3.8}$$

for all  $x$ . But this means if we take the antiderivative of  $f(\cdot)$  we'll get  $F(\cdot)$  up

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<sup>4</sup>As a technical matter, we could define  $F(x)$  to be the area under  $f(\cdot)$  from  $\underline{x}$  to  $x$ , where  $\underline{x}$  is such that no number in the domain of  $f(\cdot)$  is less than  $\underline{x}$ . As drawn in the figure, 0 can serve this role.

to an additive constant. That is,

$$F(x) = \int f(z)dz|_{z=x} + K,$$

where  $z$  is a dummy variable of integration that will be set equal to  $x$ .

Recall the gray area in Figure A3.2 is  $F(x_1) - F(x_0)$ ; hence,

$$\begin{aligned} \text{Area} &= F(x_1) - F(x_0) \\ &= \int f(z)dz|_{z=x_1} + K - \left( \int f(z)dz|_{z=x_0} + K \right) \\ &= \int f(z)dz|_{z=x_1} - \int f(z)dz|_{z=x_0} \\ &= \int_{x_0}^{x_1} f(z)dz, \end{aligned}$$

where the notation  $\int_{x_0}^{x_1}$  means that we're integrating between  $x_0$  and  $x_1$ .

**Example 33:** Suppose  $f(x) = 20 - 2x$  and consider the area beneath that curve between 4 and 8:

1. The antiderivative of  $20 - 2x$  is  $20x - x^2$ .<sup>5</sup>
2. The antiderivative evaluated at 8 is 96. The antiderivative evaluated at 4 is 64.
3. The difference is 32. So the area is 32.

Observe that

$$\int_x^x f(z)dz = F(x) - F(x) = 0;$$

that is, a region with zero width has zero area.

Define the function  $G(x) = \int_a^x f(z)dz$ , where  $a$  is a constant. Note, therefore, that  $G(x) = F(x) - F(a)$ . Observe that

$$\begin{aligned} \frac{d}{dx} \int_a^x f(z)dz &= G'(x) \\ &= F'(x) \quad (\text{because } G(x) = F(x) - F(a)) \\ &= f(x) \quad (\text{from expression (A3.8)}) \end{aligned}$$

We've thus proved the following.

**Theorem 1 (Fundamental theorem of calculus)** *The derivative of  $\int_a^x f(z)dz$  with respect to  $x$  is  $f(x)$ ; that is,*

$$\frac{d}{dx} \int_a^x f(z)dz = f(x).$$

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<sup>5</sup>As seen above, we can ignore any additive constant,  $K$ .



# Probability Appendices



## Fundamentals of Probability

# B1

This appendix reviews a number of fundamental concepts related to probability.

A *trial* (alternatively, experiment or observation) is a situation in which one *outcome* will occur out of a set of possible outcomes. For example, a flip of a coin is a trial and its possible outcomes are *heads* and *tails*. Other examples are:

- A toss of a die is a trial and its possible outcomes are the numbers one through six.
- A toss of a pair of dice is a trial and its possible outcomes are the 36 possible combinations of two die faces.
- The price of a barrel of crude oil a year from now is a trial and its outcomes are all non-negative dollar amounts.
- Whether a given manufactured good is defective is a trial and its possible outcomes are “defective” and “not defective.”
- The number of defective products in a production run of 1000 is a trial and its possible outcomes are the integers between 0 and 1000 (inclusive).

We are often interested in the *probability* that a given outcome of a trial will occur; that is, how likely that outcome is. For example, the probability of the outcome *heads* when a coin is flipped is  $1/2$ ; that is, we believe that it is equally likely the coin will land *heads* as it will *tails*. A belief, moreover, that is supported by our past experiences with flipping coins. Similarly, if ...

- ... the trial is the toss of a die, then the probability of getting a “four” is  $1/6$ .
- ... the trial is the toss of a pair of dice, then the probability of getting a “three” on the first die and a “two” on the second die is  $1/36$ .
- ... the trial is the toss of a cube, four sides of which are black and two sides of which are white, then the probability of getting a black side is  $4/6 (= 2/3)$ .

The greater the probability, the more likely that outcome is. Hence, for instance, in the last example the probability of getting a black side ( $2/3$ ) is greater than the probability of getting a white side ( $1/3$ ), reflecting that a black side is twice as likely to appear as a white slide.

There are certain rules that probability over the outcomes of a trial must satisfy. To illustrate these rules, let  $N$  denote the number of outcomes in the trial, let  $n = 1, 2, \dots, N$  index the possible outcomes, let  $\omega_n$  denote the  $n$ th outcome, and let  $P\{\omega_n\}$  denote the probability of the  $n$ th outcome. Then the rules are

1. For each outcome  $n$ ,  $P\{\omega_n\} \geq 0$ ; and
2.  $P\{\omega_1\} + P\{\omega_2\} + \dots + P\{\omega_N\} = 1$ . This can equivalently be expressed as

$$\sum_{n=1}^N P\{\omega_n\} = 1. \quad (\text{B1.1})$$

where  $\Sigma$  means sum and the expression is read as “the sum from  $n = 1$  to  $N$  of  $P\{\omega_n\}$ .”

The first rule is the requirement that probabilities be non-negative. The second rule is the requirement that if we add up the probabilities of all the possible outcomes, then this sum must equal one. If the probability of an outcome is zero, then its occurrence is impossible. If the probability of an outcome is one, then its occurrence is certain.

How outcomes get assigned probabilities is a tricky question, and one which has engendered much philosophical debate. For practically minded people, however, there are essentially three methods of assigning probabilities to outcomes. They are logically, experimentally, and subjectively.

Logic applies, for instance, in deciding that the probability of *heads* is  $1/2$ : It seems clear that *heads* and *tails* are equally likely, so—since probabilities sum to one— $P\{\text{heads}\} = 1/2$ .

Experimentally refers to probabilities that are drawn from data. For instance, experimentally applies to how seismologists determine that there is a .67 probability of a “major” earthquake in the Bay Area within the next thirty years:<sup>1</sup> The geological record reveals that major quakes tend to follow cycles. Currently, the Bay Area is in a stage of a cycle, which two out of three times in the past has ended with a major quake within the next thirty years.

Subjectively is a kind way to refer to probabilities that are guesses. For example, the probability that a brand new technology will prove popular in the market place is not a number that is readily derived logically or from past experience. The executives of the company launching this new technology, however, must make guesses about the probability in order to make good decisions about the product’s launch.

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<sup>1</sup>Ever noticed how they don’t put this kind of information in the promotional brochures for the Haas School?

# Events | B1.1

Often one is interested in combinations of outcomes known as *events*. An event is a collection—or *set*—of outcomes that satisfy some criteria. An event is said to occur if one of the outcomes that satisfies these criteria occurs. Suppose, for instance, the trial is the roll of a die, then an example of an event would be getting a number less than or equal to four. This event is the set of outcomes 1, 2, 3, and 4 and it occurs if a 1, 2, 3, or 4 is rolled. If the trial is the sexes of a couple's first two children, then an event is “the first child is a girl,” which is the set of outcomes (girl, girl) and (girl, boy). Another event for this trial is “the sexes of the two children are the same,” which is the set of outcomes (girl, girl) and (boy, boy).

As a rule, we will denote events by capital italic letters (*e.g.*,  $A$ ,  $B$ , etc.).

The probability of an event is the sum of the probabilities of the outcomes in that event. For example, the probability of rolling a number less than or equal to 4 is

$$\frac{2}{3} = P\{1\} + P\{2\} + P\{3\} + P\{4\}.$$

The probability of the first of a couple's two children being a girl is

$$\frac{1}{2} = P\{\text{girl, girl}\} + P\{\text{girl, boy}\}.$$

Formally, the probability of an event  $A$  can be expressed as

$$P(A) = \sum_{\omega \in A} P\{\omega\}, \quad (\text{B1.2})$$

where  $\omega \in A$  means  $\omega$  is one of the outcomes that make up  $A$  and the expression is read as “the sum over outcomes in  $A$  of the probabilities of those outcomes.”

Often one is interested in comparing the probabilities of events. *Equality* and *contain in* are two relations between events that are of interest. Two events,  $A$  and  $B$ , are equal—denoted  $A = B$ —if they contain the same outcomes. From the definition of the probability of an event, it follows that

**Proposition 1** *If  $A$  and  $B$  are events and  $A = B$ , then  $P(A) = P(B)$ .*

Note the converse is *not* true; that is, the fact that two events have the same probability does not imply they are the same event.

One event,  $A$ , is *contained in* another event,  $B$ —denoted  $A \subseteq B$ —if every outcome in  $A$  is also in  $B$ .<sup>2</sup> For example, the event “both of a couple's two children are girls” is contained in the event “their first child is a girl.” Or, for instance, the event “an even number is rolled” is contained in the event “a number greater than or equal to two is rolled.” If  $A \subseteq B$ , every outcome in  $A$  is also in  $B$ , but  $B$  may contain outcomes that are not in  $A$ ; hence, it follows that

<sup>2</sup>You may have previously seen the term “subset of” used for “contained in.”

**Proposition 2** *If  $A$  and  $B$  are events and  $A \subseteq B$ , then  $P(A) \leq P(B)$ .*

Again, the converse is *not* true; that is,  $P(A) < P(B)$  does not imply that  $A$  is contained in  $B$ . Proposition 2 does, however, imply that if  $P(A) > P(B)$ , then  $A$  cannot be contained in  $B$ .

Although Proposition 2 may seem obvious, the truth is that studies have shown that many people make mistakes in this area. For example, suppose I tell you that Linda grew up in a politically liberal household in San Francisco and that Linda herself was involved with a number of progressive causes while she was an undergraduate at Berkeley. Which of the following two statements is more likely to be true about Linda?

1. Linda is a bank teller; or
2. Linda is a feminist and a bank teller.

Most people choose statement 2, but that is incorrect:<sup>3</sup> the event Linda is a feminist and a bank teller is contained in the event Linda is a bank teller (feminist or not); or, to put it differently, all feminist bank tellers are bank tellers, but not all bank tellers are feminist bank tellers. So, from Proposition 2, we know that statement 1 is more likely than statement 2.

Often we want to know what the probability is that an event will *not* happen. This is equivalent to asking what is the probability one of the outcomes that do not make up the event will occur. If  $A$  is an event, let  $A^c$  be the event made up of the outcomes that do not make up  $A$  ( $A^c$  is called the *complement* of  $A$ ). From equation (B1.1), we know that

$$\begin{aligned}
 1 &= \sum_{n=1}^N P\{\omega_n\} \\
 &= \underbrace{\sum_{\omega \in A} P\{\omega\}}_{\text{sum over outcomes in } A} + \underbrace{\sum_{\omega \in A^c} P\{\omega\}}_{\text{sum over outcomes not in } A} \\
 &= P(A) + P(A^c).
 \end{aligned}$$

This in turn establishes

**Proposition 3** *The probability that an event  $A$  does not occur (alternatively, that event  $A^c$  does occur) is  $1 - P(A)$ .*

For instance, the probability that “at least one child is a boy in a pair of children” is most easily calculated by recognizing that this event is the complement of the event “both children are girls.” The probability that “both children are girls” is  $1/4$ , so the probability that “at least one child is a boy” is  $3/4$ .

Sometimes we want to know the probability that two events will both occur. The event that both events occur is, itself, an event. If  $A$  and  $B$  are two events,

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<sup>3</sup>This is what is called the *conjunctive fallacy*.

then the event that they both occur is denoted  $A \cap B$  (read “ $A$  intersection  $B$ ”). An outcome is contained in  $A \cap B$  if it is in *both*  $A$  and  $B$ . In other words, every outcome in  $A \cap B$  is in  $A$  and in  $B$ . It follows that

$$A \cap B \subseteq A \text{ and } A \cap B \subseteq B.$$

From Proposition 2, we can therefore conclude that

**Proposition 4** *If  $A$  and  $B$  are two events, then  $P(A \cap B) \leq P(A)$  and  $P(A \cap B) \leq P(B)$ .*

That is, the probability of both events occurring cannot exceed the probability of one event occurring (and the other event either occurring or not occurring). This is another way to view the “Linda” problem above. One event is “Linda is a bank teller” and another is “Linda is a feminist.” Statement 2—“Linda is a feminist bank teller”—is the intersection of these two events; hence, from Proposition 4, it follows that the probability of statement 2 cannot exceed the probability of statement 1.

Sometimes two events cannot possibly happen together (*e.g.*, the event “the next person I meet is a man” and the event “the next person I meet is pregnant”).<sup>4</sup> We denote this by writing  $A \cap B = \emptyset$ , where  $\emptyset$ —called the *null set*—denotes that the event  $A \cap B$  is impossible. To reflect the idea that  $\emptyset$  is impossible, we define  $P(\emptyset) = 0$ .

The probability of  $A \cap B$  is calculated by summing up the probabilities of all outcomes in  $A \cap B$  (*i.e.*, by employing formula (B1.2)—except substituting  $A \cap B$  for  $A$  in that expression).

In some situations, we want to know the probability that either event  $A$  or event  $B$  or both will occur. That is, we want to know the probability of an outcome occurring that is contained in either  $A$  or  $B$  or both. This defines a new event, which we denote as  $A \cup B$  (read “ $A$  union  $B$ ”). For example, if the trial is the roll of a die, then the union of the events “an even number is rolled” and “a prime number is rolled”<sup>5</sup> is the event “a number larger than one is rolled,” since the outcome one is the only outcome not in at least one of the two events (the even numbers are 2, 4, and 6, while the prime numbers are 2, 3, and 5). Note that if an outcome is common to both events it is only counted once in their union. For example, the union of “an even number is rolled” and “a prime number is rolled” is  $\{2, 3, 4, 5, 6\}$  and *not*  $\{2, 2, 3, 4, 5, 6\}$ . Because every outcome in  $A$  must, by definition, be in  $A \cup B$ , it follows from Proposition 2 that

**Proposition 5** *If  $A$  and  $B$  are events, then  $P(A) \leq P(A \cup B)$  and  $P(B) \leq P(A \cup B)$ .*

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<sup>4</sup>Bad Arnold Schwarzenegger movies notwithstanding.

<sup>5</sup>Recall a prime number is a whole number *larger* than one that is divisible only by one and itself (*e.g.*, 3 is prime—it is divisible only by 1 and 3—whereas 4 is not prime—it is divisible by 2 in addition to being divisible by 1 and 4).

That is, for example, the probability of rolling an even number cannot exceed the probability of rolling an even number, or a prime number, or both.

The event  $A \cup B$  may contain outcomes that are common to both  $A$  and  $B$ . Since we don't double count such events it follows that

**Proposition 6** *If  $A$  and  $B$  are events, then  $P(A \cup B) \leq P(A) + P(B)$ .*

We illustrate Proposition 6 as follows

$$\begin{aligned}
 P(A \cup B) &= \sum_{\omega \text{ in } A \text{ only}} P\{\omega\} + \sum_{\omega \text{ in } A \& B} P\{\omega\} + \sum_{\omega \text{ in } B \text{ only}} P\{\omega\} \\
 &\leq \underbrace{\sum_{\omega \text{ in } A \text{ only}} P\{\omega\} + \sum_{\omega \text{ in } A \& B} P\{\omega\}}_{P(A)} + \underbrace{\sum_{\omega \text{ in } A \& B} P\{\omega\} + \sum_{\omega \text{ in } B \text{ only}} P\{\omega\}}_{P(B)} \\
 &= P(A) + P(B).
 \end{aligned} \tag{B1.3}$$

From expression (B1.3), we can see that the problem with using  $P(A) + P(B)$  for  $P(A \cup B)$  is that it double counts the probabilities of the outcomes in both  $A$  and  $B$ ; that is, it double counts the probabilities of outcomes in  $A \cap B$ . It follows, then, that we can get the correct value for  $P(A \cup B)$  by subtracting  $P(A \cap B)$  from  $P(A) + P(B)$ . This yields

**Proposition 7** *If  $A$  and  $B$  are two events, then*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Two events are *mutually exclusive* if they cannot both occur; that is, if  $A \cap B = \emptyset$ . It follows from Proposition 7 that

**Proposition 8** *If  $A$  and  $B$  are two mutually exclusive events, then*

$$P(A \cup B) = P(A) + P(B).$$

Often we want to know the probability that one event will occur given that we know another event has occurred or will occur. For example, what is the probability that the next unit off the assembly line will be defective given that two of the last one hundred were defective? Or what is the probability that it will rain this afternoon given that it is cloudy this morning? These probabilities are called *conditional probabilities* and they are denoted  $P(A|B)$ —read “probability of  $A$  given  $B$ ”—where  $B$  is the event we know has occurred or will occur and  $A$  is the event in whose probability we are interested. For the example in which we ask what is the probability that it will rain this afternoon given that it is cloudy this morning,  $A$  is the event “rain in the afternoon” and  $B$  is the event “cloudy this morning.”

To better understand conditional probability, let the trial be the roll of a die and let  $A$  be the event “a six is rolled”<sup>1</sup> and let  $B$  be the event “an even number is rolled.” Suppose I secretly rolled the die and told you that I had rolled an even number (*i.e.*,  $B$  has occurred). What would you imagine the probability of my having rolled a six to be (*i.e.*, what is  $P(A|B)$ )? You would probably reason as follows: There are three even numbers—2, 4, and 6—each of which is equally likely; hence, the probability of a six having been rolled given that the number rolled is even is  $1/3$ . This reasoning is correct. Now suppose I asked you what the probability that I rolled a five is given that I have rolled an even number (*i.e.*,  $A$  is now “a five is rolled”). You would respond zero, since a five is not even and, so, is impossible given that the number rolled is even. Finally, suppose I reversed the situation and told you that I had rolled a six and asked you what the probability is that I have rolled an even number (*i.e.*,  $A$  is now “an even number is rolled” and  $B$  is “a six is rolled”). You would respond one, since six is an even number.

You should be able to verify that each of the probabilities you calculated above satisfy the following rule:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (\text{B2.1})$$

This is, in fact, the formula for conditional probability (provided  $P(B) > 0$ ).<sup>2</sup>

<sup>1</sup>This is also an outcome. An event, however, can contain only one outcome (in fact, some authors refer to outcomes as *simple events* for this reason).

<sup>2</sup>The case where  $P(B) = 0$  is of no interest, because then  $P(A|B)$  is the probability of  $A$  given that something impossible has occurred or will occur; that is, it is a nonsensical quantity.

Although the examples of conditional probability given above were trivial, conditional probability is often harder than it seems. For example, suppose I asked you what the probability is that a couple with two children has at least one boy given that you know they have at least one girl. Most people's intuition is that the answer is  $1/2$ . This, however, is wrong. To see why, let  $A$  denote the event "at least one child is a boy" and let  $B$  denote "at least one child is a girl." The goal is to calculate  $P(A|B)$ . Writing out  $B$ , we see that it is the outcomes (girl, girl), (boy, girl), and (girl, boy). Hence  $P(B) = 3/4$ . Writing out  $A$ , we see that it is the outcomes (boy, boy), (girl, boy), and (boy, girl). Hence  $A \cap B$  is the outcomes (girl, boy) and (boy, girl); hence  $P(A \cap B) = 1/2$ . Using (B2.1), we get  $P(A|B) = 1/2 \div 3/4 = 2/3$ ; that is, the probability of at least one boy given that you know the couple has at least one girl is  $2/3$ . The reason most people's intuition fails is that they *incorrectly* reason that since one child is a girl, they wish to determine the probability that the other child is a boy. What they are forgetting is that there are three ways for a couple to have at least one girl—see  $B$ —two out of three of which involve the couple also having one boy.

In Section B1.1 we saw that the *unconditional probability* (i.e., given no additional information) of at least one of a couple's two children being a boy was  $3/4$ . Given the information that at least one child is a girl, we saw that the conditional probability that they have at least one boy is  $2/3$ , which is less than  $3/4$ . What has happened is that, upon receiving the information that at least one child was a girl, we *revised* or *updated* our beliefs about the probability that the couple has at least one boy. In general, we say that a person is revising or updating his or her beliefs about an event when he or she uses new information to calculate a conditional probability. The probability of the event prior to receiving the information is called the *prior probability* and the probability of the event after receiving the information is called the *posterior probability*. Note that the posterior probability need not be lower than the prior probability: if, for instance, the event was the probability that a couple with two children has two boys, then you would update your beliefs upon learning that at least one of the children was a boy from  $1/4$  (the prior probability) to  $1/3$  (the posterior probability). As you will see, calculating revised or updated beliefs is one of the primary uses of conditional probability.

The rules from Appendix B1 hold true for conditional probabilities:

**Proposition 9** *Let  $\omega$  be an outcome and  $A$ ,  $B$ , and  $C$  be events. Assume  $P(B) > 0$ . Then the following are all true.*

- (i)  $P\{\omega|B\} \geq 0$  and  $P(A|B) \geq 0$ .
- (ii)  $\sum_{\omega \in B} P\{\omega|B\} = 1$ .
- (iii) If  $A = C$ , then  $P(A|B) = P(C|B)$ .
- (iv) If  $A \subseteq C$ , then  $P(A|B) \leq P(C|B)$ .
- (v) The probability that event  $A$  does not occur given  $B$  is  $1 - P(A|B)$ ; in other words,  $P(A|B) + P(A^c|B) = 1$ .

(vi)  $P(A \cap C|B) \leq P(A|B)$  and  $P(A \cap C|B) \leq P(C|B)$ .

(vii)  $P(A \cup C|B) \geq P(A|B)$  and  $P(A \cup C|B) \geq P(C|B)$ .

(viii)  $P(A \cup C|B) = P(A|B) + P(C|B) - P(A \cap C|B)$ , from which it follows that

(a)  $P(A \cup C|B) \leq P(A|B) + P(C|B)$ ; and

(b)  $P(A \cup C|B) = P(A|B) + P(C|B)$  if  $A \cap B$  and  $C \cap B$  are mutually exclusive events.

Note that we can rewrite (B2.1) as

$$\begin{aligned} P(A \cap B) &= P(A|B) \times P(B); \text{ or, reversing the roles of } A \text{ and } B, \\ P(A \cap B) &= P(B|A) \times P(A). \end{aligned}$$

## Independence | B2.1

In the previous section we noted that the difference between  $P(A)$  and  $P(A|B)$  reflected what learning that event  $B$  had occurred or would occur told us about the probability that event  $A$  would occur. If  $P(A) < P(A|B)$ , then learning  $B$  caused us to increase the probability with which we believed  $A$  would occur. If  $P(A) > P(A|B)$ , then learning  $B$  caused us to decrease the probability with which we believed  $A$  would occur. What if  $P(A) = P(A|B)$ ? Then  $B$  tells us nothing about the probability that  $A$  will occur. In this last case, we say that  $A$  and  $B$  are *independent* events.

We know that if  $A$  and  $B$  are independent, then

$$P(A) = \frac{P(A \cap B)}{P(B)} = P(A|B).$$

Rearranging the first equality, we see that

$$1 = \frac{P(A \cap B)}{P(A)P(B)}.$$

From which we may conclude

**Proposition 10** *If  $A$  and  $B$  are independent events, then  $P(A \cap B) = P(A) \times P(B)$ .*

Note  $P(A \cap B) = P(A) \times P(B)$  only if  $A$  and  $B$  are independent. If they are *not* independent, then this equality will not hold.

## Bayes Theorem | B2.2

In this section, we study one of the most important rules for updating probabilities, namely Bayes Theorem. We begin with some additional definitions.

Let  $\Omega$  (read “omega”) be the set of all possible outcomes. That is, every outcome is contained in  $\Omega$  (i.e.,  $\omega \in \Omega$  for all outcomes  $\omega$ ). From this, it follows that every event is also contained in  $\Omega$  (i.e.,  $A \subseteq \Omega$  for all events  $A$ ). Finally, since all outcomes are in  $\Omega$ , it follows from (B1.1) that  $P(\Omega) = 1$  (which can be interpreted as “something is certain to happen”).

We say that a list of events are *exhaustive* if collectively they contain all the possible outcomes. This is denoted by  $A_1 \cup A_2 \cup \dots \cup A_T = \Omega$ , where  $A_t$  denotes one of  $T$  events in the list of events.

**Proposition 11** *Let  $A_1, \dots, A_T$ , and  $B$  be events. Suppose that  $A_1, \dots, A_T$  are mutually exclusive and exhaustive, then*

$$P(B) = P(A_1 \cap B) + \dots + P(A_T \cap B).$$

Recall that  $P(A \cap B) = P(B|A) \times P(A)$ . From this fact and Proposition 11, the following proposition follows:

**Proposition 12 (Law of Total Probability)** *Let  $A_1, \dots, A_T$ , and  $B$  be events. Suppose that  $A_1, \dots, A_T$  are mutually exclusive and exhaustive, then*

$$P(B) = P(B|A_1) \times P(A_1) + \dots + P(B|A_T) \times P(A_T). \quad (\text{B2.2})$$

Let  $A_1, \dots, A_T$  be mutually exclusive and exhaustive events. We know that

$$P(A_t|B) = \frac{P(A_t \cap B)}{P(B)}. \quad (\text{B2.3})$$

We also know that  $P(A_t \cap B) = P(B|A_t) \times P(A_t)$  and that  $P(B)$  is given by (B2.2). Substituting these two expressions into (B2.3) yields:

**Theorem 2 (Bayes Theorem)** *Let  $A_1, \dots, A_T$ , and  $B$  be events. Suppose that  $A_1, \dots, A_T$  are mutually exclusive and exhaustive and that  $P(B) > 0$ , then for any of the  $T$  events,  $A_t$ ,*

$$P(A_t|B) = \frac{P(B|A_t) \times P(A_t)}{P(B|A_1) \times P(A_1) + \dots + P(B|A_T) \times P(A_T)}.$$

As a first illustration of Bayes Theorem consider the following example:

**Example 34 [Paradox of the Three Prisoners]:** Once upon a time, there were three prisoners: 1, 2, and 3. Each prisoner was kept in a separate cell and the prisoners could not communicate with each other. The prisoners knew that in the morning one of them would be executed, while the other two would be set free. Although the authorities knew

which prisoner would be executed, the prisoners did not. Suppose each prisoner assumed she had a  $\frac{1}{3}$  probability of being executed. Suppose that Prisoner 3 could not wait the night to find out whether she was to be executed. She called a guard over and asked him “am I to be executed?”

The guard replied, “you know that tradition forbids me from telling you that.”

“Well, then,” said Prisoner 3, “could you at least tell me which of the *other* two prisoners will be set free?”

“Since there is no information in that, I don’t see why not. Prisoner 1 will be set free,” said the guard, who never lies.

Prisoner 3 sat back in her cell and reflected, “well, then, it’s between Prisoner 2 and me. But, wait, this means I have a 50% chance of being executed! Woe is me!” As Prisoner 3 sank into depression, it suddenly occurred to her that had the guard told her Prisoner 2 would be set free, she still would have calculated her probability of being executed as being  $\frac{1}{2}$ . “Wait a minute,” she thought, “that means that regardless of what the guard told me, my probability of being executed was  $\frac{1}{2}$ ; which means that my initial probability of being executed should have been  $\frac{1}{2}$  instead of  $\frac{1}{3}$ . I must have made a mistake somewhere!”

Indeed, Prisoner 3 has made a mistake. To see what her mistake was, as well as what the true probability of her execution is, let’s employ Bayes Theorem. To do so, we need one more assumption: Suppose that if both Prisoners 1 and 2 are to be set free, then the probability is  $\frac{1}{2}$  that the guard tells Prisoner 3 that Prisoner 1 will be released and the probability is  $\frac{1}{2}$  that the guard tells Prisoner 3 that Prisoner 2 will be released. In terms of the formula above, we can think of  $A_t$  as the event “Prisoner  $t$  will be executed” and we can think of  $B$  as the event “Prisoner 3 is told Prisoner 1 will be set free.” The question we seek to answer is what is the probability that Prisoner 3 is to be executed given that Prisoner 1 is to be set free; that is, what is  $P(A_3|B)$ ? The data we have been given to answer this question can be expressed as

1.  $P(A_1) = P(A_2) = P(A_3) = \frac{1}{3}$  (the *unconditional* probability of Prisoner  $t$  being executed is  $\frac{1}{3}$ ).
2.  $P(B|A_1) = 0$  (if Prisoner 1 is to be executed, then Prisoner 3 will *not* be told that Prisoner 1 will be set free—recall the guard never lies).
3.  $P(B|A_2) = 1$  (if Prisoner 2 is to be executed, then Prisoner 3 will *certainly* be told that Prisoner 1 will be set free).
4.  $P(B|A_3) = \frac{1}{2}$  (if Prisoner 3 is to be executed, then, by assumption, the probability that the guard tells Prisoner 3 that Prisoner 1 is to be set free is  $\frac{1}{2}$ ).

Plugging these data into Bayes Theorem, we have that  $P(A_3|B)$  equals

$$P(A_3|B) = \frac{\frac{1}{3} \times \frac{1}{2}}{0 \times \frac{1}{3} + 1 \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{3}} = \frac{1}{3}.$$

That is, having learned that Prisoner 1 will be set free, the probability that Prisoner 3 will be executed is  $\frac{1}{3}$ . As the guard said, the knowledge that Prisoner 1 will be freed is uninformative with respect to whether Prisoner 3 will be executed (note this also means that it is independent of whether Prisoner 3 will be executed). Prisoner 3's mistake was to forget that the guard would always tell her that one of her fellow prisoners would be released regardless of who was to be executed. In other words, asking *which* of her fellow prisoners will be released is equivalent, in terms of estimating her own probability of execution, to asking will one of her fellow prisoners be released. Since only one prisoner will be executed, this second question cannot yield an informative response; hence, neither can her original question.

The Paradox of the Three Prisoners illustrates one way Bayes Theorem can be useful; namely showing that what seems like information is not really information. Another way Bayes Theorem can be useful is in showing that what seems uninformative is actually informative; as the following example illustrates.

**Example 35 [The Monty Hall Problem]:** This example is *based* on an old American television game show called *Let's Make a Deal* hosted by a man named Monty Hall.<sup>3</sup> At one stage of the game, a contestant is asked to choose among three closed doors (numbers 1, 2, and 3). Behind one of the doors is a valuable prize (*e.g.*, a new car). Behind two of the doors are worthless joke prizes (*e.g.*, a goat). The contestant does not know which door conceals the valuable prize, but he does know that the probability that the valuable prize is behind any given door is  $\frac{1}{3}$ . After choosing one of the three doors, Monty Hall has one of the *unchosen* doors opened to reveal one of the two joke prizes (Monty Hall knows which door conceals the valuable prize). Monty Hall then asks the contestant if he would like to switch his choice from the door he chose originally to the other closed door. If the contestant switches, then he gets the prize behind the door to which he switches. If he doesn't switch, then he gets the prize behind the door he originally chose. Assuming the contestant wants to maximize his probability of winning the valuable prize, should he switch or should he stay with the door he originally chose?

Most people when confronted with this problem answer that it doesn't matter whether the contestant switches or not—the probability of winning the valuable prize will be the same; that is, seeing a door opened is uninformative. This, however, is incorrect: By switching the contestant *doubles* his probability of winning! Let's use Bayes Theorem to see why. Before doing so, we need one further assumption: if both the doors that the contestant did *not* select have joke prizes behind them, then Monty Hall is equally likely to open one as the other (remember Monty Hall only opens an unchosen door and only a door that has a joke prize behind it). For concreteness, suppose that the contestant chooses door #1 and Monty Hall opens door #2 to reveal a joke prize. Let  $A_t$  denote the event “the valuable prize is behind door number  $t$ ” and let  $B$  denote the

---

<sup>3</sup>We make no claims that the actual game show was played in the manner described here. This is simply a well-known puzzle.

event “Monty Hall opens door #2.” We want to know the probabilities that the valuable prize is behind door #1 or door #3 (since Monty Hall opens door #2 to reveal a joke prize, we know the valuable prize is not there); that is, we want to know  $P(A_1|B)$  and  $P(A_3|B)$ . Indeed, since  $P(A_1|B) = 1 - P(A_3|B)$ , we need to determine only  $P(A_3|B)$ . The data we are given are

1.  $P(A_1) = P(A_2) = P(A_3) = \frac{1}{3}$ .
2.  $P(B|A_1) = \frac{1}{2}$  (recall Monty is equally likely to open door #2 as he is to open door #3 if both conceal joke prizes).
3.  $P(B|A_2) = 0$  (recall Monty only opens a door that has a *joke* prize behind it).
4.  $P(B|A_3) = 1$  (same reason).

Putting all this into Bayes Theorem yields

$$P(A_3|B) = \frac{1 \times \frac{1}{3}}{\frac{1}{2} \times \frac{1}{3} + 0 \times \frac{1}{3} + 1 \times \frac{1}{3}} = \frac{2}{3};$$

hence  $P(A_1|B) = \frac{1}{3}$ . As claimed, the contestant doubles his probability of winning the valuable prize by switching.



## Random Variables and Expectation

# B3

Often we are interested in the payoffs associated with different outcomes. For instance, suppose you and a friend decided to wager a dollar on the outcome of a coin toss: heads you win a dollar; tails you lose a dollar. From your perspective, the outcome *heads* is associated with the payoff \$1 and the outcome *tails* is associated with the payoff -\$1. Traditionally, such payoffs are called *random variables* and are denoted by the *function*  $x(\cdot)$ , which associates each outcome,  $\omega$ , with some payoff. For example, one would write  $x(\text{heads}) = 1$  and  $x(\text{tails}) = -1$ .

In general, one can dispense with keeping track of the outcomes; thus, instead of writing  $x(\omega)$ , one can just write  $x$ ; and instead of writing “the probability of getting  $x(\omega)$  is  $P\{\omega\}$ ,” one can just write “the probability of getting  $x$  is  $P(x)$ .” In a sense, one can just think of payoffs as being outcomes.

The payoffs, since they are numbers, can be ordered from smallest to largest. Hence, if there are  $N$  possible payoffs, we would list them as  $x_1, x_2, \dots, x_N$ , where it is to be understood that if  $m < n$ , then  $x_m < x_n$ .

Taking advantage of this ordering, we can write the probability of  $P(x_n)$  as just  $p_n$ . If there are  $N$  possible payoffs, then the list of probabilities  $p_1, \dots, p_N$  is called the *density* associated with the random variable (payoff)  $x$ .

## Expectation

# B3.1

When faced with an uncertain situation, say the gamble described in the previous section, we are often interested in knowing how much we can expect to win or how much we might win on average. This notion is captured in the concept of *expectation*. The *expectation of a random variable* is denoted  $\mathbb{E}\{x\}$  and is defined as

$$\mathbb{E}\{x\} = p_1 \times x_1 + \dots + p_N \times x_N \quad (\text{B3.1})$$

for a random variable with  $N$  possible payoffs. In words, the expectation of a random variable is the sum, over the possible outcomes, of the product of each payoff and its probability.

One way to think about expectation is that the expectation of  $x$  is exceedingly close to your *average*<sup>1</sup> payoff if you were to repeat the uncertain situation

---

<sup>1</sup>Recall the average of a set of numbers is the sum of the numbers in the set divided by how many numbers are in the set.

a large number of times.<sup>2</sup> Another way to think about expectation is that the expectation of  $x$  is the “best guess” as to value that  $x$  will take; where “best guess” means the one with the least error.

As an example, the expected value of the gamble in which you receive \$1 if a coin lands heads but you pay \$1 (receive -\$1) if the coin lands tails is

$$\frac{1}{2} \times (-1) + \frac{1}{2} \times 1 = 0$$

dollars.

As a second example, suppose that your ice cream parlor makes \$1000 on a sunny day and \$500 on a rainy day. Suppose that the probability of rain tomorrow is  $\frac{1}{5}$ , then you can expect to make

$$\frac{1}{5} \times 500 + \frac{4}{5} \times 1000 = 900$$

dollars tomorrow.

One can also take the expectation of a function of a random variable. For instance, suppose that you bet \$1000 on the toss of a coin: heads you win \$1000 and tails you lose \$1000. Suppose that you must pay 14% income tax on your winnings and you cannot deduct your losses from your income taxes. Then, from your perspective, you care about the following function of  $x$ :

$$f(x) = \begin{cases} x, & \text{if } x \leq 0 \\ .86x, & \text{if } x > 0 \end{cases} \quad (\text{B3.2})$$

because it gives your winnings in *after-tax* dollars. Your expected after-tax winnings are

$$\mathbb{E}\{f(x)\} = \frac{1}{2} \times (-1000) + \frac{1}{2} \times .86 \times 1000 = -70$$

dollars.

The general rule for the expectation of a function of a random variable is

$$\mathbb{E}\{f(x)\} = p_1 \times f(x_1) + p_2 \times f(x_2) + \cdots + p_N \times f(x_N). \quad (\text{B3.3})$$

Note that it is generally *not* true that  $\mathbb{E}\{f(x)\} = f(\mathbb{E}\{x\})$ ; that is, it is generally not true that the expectation of the function equals the function of the expectation.

Finally, some standard definitions concerning random variables and their expectations:

**Definition 12** *The mean of a random variable is its expectation.*

---

<sup>2</sup>In fact, it can be proved that as the number of replications tends to infinity, the expectation will be arbitrarily close to the average you actually get. This is called the *Law of Large Numbers*.

**Definition 13** The variance of a random variable is the expectation of the square of its deviation from its mean. That is, the variance of  $x$ —denoted  $\text{Var}(x)$ —is

$$\begin{aligned}\text{Var}(x) &= p_1 \times (x_1 - \mathbb{E}\{x\})^2 + \dots + p_N \times (x_N - \mathbb{E}\{x\})^2 \\ &= \mathbb{E}\{h(x)\},\end{aligned}$$

where  $h(x)$  is the function  $(x - \mathbb{E}\{x\})^2$ .

**Definition 14** The (population) standard deviation of a random variable is the square-root of its variance; i.e.,  $\sqrt{\text{Var}(x)}$ .

**Definition 15** The  $r^{\text{th}}$  central moment of a random variable is the expectation of its deviation from its mean raised to the  $r^{\text{th}}$  power; that is,

$$p_1 \times (x_1 - \mathbb{E}\{x\})^r + \dots + p_N \times (x_N - \mathbb{E}\{x\})^r.$$

Note that the variance of a random variable is also its  $2^{\text{nd}}$  central moment.

## Distributions | B3.2

The [distribution](#) of a random variable  $x$  is a function that gives the probability that  $x \leq v$  for each value  $v$ . If  $F(v)$  is the distribution function for the random variable  $x$ , then  $F(v)$  is defined as

$$F(v) = \sum_{\{n|x_n \leq v\}} p_n,$$

where  $\{n|x_n \leq v\}$  is read as “the set of indices such that the value  $x_n$  is less than or equal to  $v$ .” For example, if

$$\begin{aligned}x &\in \{1, 2, 4, 7, 10\}; \text{ and} \\ p_n &= \frac{n}{15},\end{aligned}$$

then

$$\begin{aligned}F(3) &= \frac{1}{15} + \frac{2}{15} = \frac{1}{5}; \\ F(6.99) &= \frac{1}{15} + \frac{2}{15} + \frac{3}{15} = \frac{2}{5}; \text{ and} \\ F(7) &= \frac{1}{15} + \frac{2}{15} + \frac{3}{15} + \frac{4}{15} = \frac{2}{3}.\end{aligned}$$

Since it is impossible to get an  $x$  less than 1,  $F(v) = 0$  for  $v < 1$ . Since we must get an  $x$  less than or equal to 10,  $F(v) = 1$  for  $v \geq 10$ .



### Continuous distributions

Although so far we have defined probability in terms of discrete outcomes, we can define them in terms of continuous outcomes. Basically, this means considering distribution functions that are differentiable. Let  $G(v)$  be a differentiable distribution function. The derivative of a differentiable distribution function is called the *density function*. Let  $g(v)$  denote the density function; that is,

$$g(v) = \frac{d}{dv} G(v).$$

From the fundamental theorem of calculus, it follows that

$$G(v) = \int_{x_{\min}}^v g(x) dx,$$

where  $x_{\min}$  is the smallest possible value of  $x$ .

For random variables with differentiable distribution functions, their mean and variance are defined as follows:

$$\begin{aligned} \mathbb{E}\{x\} &= \int_{x_{\min}}^{x_{\max}} x g(x) dx; \text{ and} \\ \text{Var}(x) &= \int_{x_{\min}}^{x_{\max}} (x - \mathbb{E}\{x\})^2 g(x) dx, \end{aligned}$$

where  $x_{\max}$  is the largest possible value of  $x$ .

Two commonly used differentiable distribution functions are the normal and the uniform.

The **normal** distribution has a distribution function equal to

$$\int_{-\infty}^v \sqrt{\frac{1}{2\sigma\pi}} \exp\left(-\frac{1}{2\sigma^2} [x - \mu]^2\right) dx,$$

where  $\sigma$  (read “sigma”) is the standard deviation of the distribution, where  $\mu$  (read “mu”) is the mean of the distribution, where  $\pi$  is the constant pi (i.e., 3.1416...), and where  $\exp(z)$  means raise the base of the natural logarithm (i.e.,  $e \approx 2.7183$ ) to the  $z^{\text{th}}$  power. The density function for the normal distribution is

$$\sqrt{\frac{1}{2\sigma\pi}} \exp\left(-\frac{1}{2\sigma^2} [x - \mu]^2\right).$$

The **uniform** distribution from 0 to  $Y$  has a distribution function equal to

$$\int_0^v \frac{1}{Y} dx = \frac{v}{Y}.$$

---

*Its density function is  $\frac{1}{Y}$ . The mean of the distribution is  $Y/2$ . Its variance is  $Y^2/12$ .*

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